

UNIVERSITY OF CALGARY

Gamified Design Configurator: A Performance-Aware, Data-Driven Framework for Enabling
Informed Decision-Making by Non-Experts in Early-Stage Design

by

Arman Khalilbeigi Khameneh

A THESIS

SUBMITTED TO THE FACULTY OF GRADUATE STUDIES
IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
DEGREE OF DOCTOR OF PHILOSOPHY

GRADUATE PROGRAM IN ENVIRONMENTAL DESIGN

CALGARY, ALBERTA

JULY, 2025

© Arman Khalilbeigi Khameneh 2025

ABSTRACT

This thesis addresses the challenge of enabling non-expert designers to make informed decisions by balancing preferences with technical performance in early-stage architectural design. A gamified decision-making framework is developed, integrating real-time feedback on structural, environmental, and cost metrics. Using a mixed-method approach, the study combines a within-subject experiment comparing feedback conditions, iterative prototyping, and evidence-based behavioral analysis of interaction logs. The proof-of-concept framework enabled non-expert designers to make design configurations.

While the importance of user-centered approaches in architecture is widely recognized, the majority of existing tools and methodologies are still geared toward expert users, necessitating technical knowledge and fluency in domain-specific software or performance metrics. Non-experts are frequently excluded from early-stage design processes because performance data is inaccessible and design interfaces are complex. Current design tools prioritize expert control while providing limited opportunities for iterative learning, exploration, and decision transparency for lay participants.

Results demonstrated that participants receiving feedback achieved a 51.19% average improvement across seven performance metrics (e.g., structural displacement, operational carbon), with latent learning effects reducing decision-making time consistency by 28% in later design attempts. The framework achieved an excellent usability score (SUS=81.3), validating its efficacy as a pedagogical and practical tool. The framework has enhanced players' spatial navigation and increased the range of decisions and exploration of the design landscape for more than 70 percent of users. Most importantly, the efficacy of encapsulated feedback is proven, and the framework enabled participants to make better-performing choices across metrics without reading them explicitly. Several trends in user profiling, time, and choices are also reported. The findings of this study demonstrate that the proposed gamified, performance-aware framework effectively enables informed decision-making among non-expert users during early-stage architectural design. Also, it proved that real-time feedback significantly impacts decision-making behavior through micro learning and that encapsulated, multi-metric feedback successfully guides users toward higher-performing design outcomes without overwhelming them. The framework also enhanced participant engagement, usability, and behavioral consistency across tasks. By linking game mechanics and behavioral theory to design workflows, this work contributes a scalable model for evidence-based design education and a technical blueprint for gamified feedback systems in participatory architecture.

Keywords: Gamification, informed decision-making, non-expert engagement, serious games, evidence-based design.

LIST OF IMAGES

Figure 1: Some of the earliest efforts in HCI and design-for-all theory in the 70's (Negroponte 1970).....	7
Figure 2: Earliest efforts in developing problem-aware partners of design (Friedman 1980; Negroponte 1970).....	16
Figure 3: Sample size (required number of interactions).....	57
Figure 4: Conceptual overview diagram.....	60
Figure 5: Stills form pre-alpha tests on different modes of interactions.....	63
Figure 6: Backend services handling geometrical and structural constraints	64
Figure 7: The framework development stages: Alpha version	65
Figure 8: The framework development stages: Beta version.....	66
Figure 9: Stills from various stages of tutorials (user p14).....	67
Figure 10: Stills of the Main version and experiment, extracted from actual footage of a user interaction.....	69
Figure 11: User participation setting.....	72
Figure 12: Retrofitted exoskeleton design-space, generated by users of the framework.....	73
Figure 13: Comparison between two conditions of the within-subject study; left: nIDM, right: IDM	74
Figure 14: A few sample rows of the collected data or log, in their raw form -csv.....	76
Figure 15: Flowchart of various services comprising the framework.....	77
Figure 16: Results and content of post-game SUS usability questions, followed by ethical consent information	83
Figure 17: Standard deviation and mean of decision time between conditions	87
Figure 18: Decision time distribution comparison between conditions.....	88
Figure 19: Decision time mapped by interactions per participant	89
Figure 20: Decision time mean change in both conditions per user	90
Figure 21: Users' mean decision time comparison. Top: All interaction, Middle: Removing 2% outliers, Bottom: Removing 20% outliers	91
Figure 22: DTime comparison in different time ranges.....	93
Figure 23: Micro-learning effect (mean slope comparison) of two sample participants in both conditions	94
Figure 24: Frequency of feedback in different categories	97
Figure 25: Distribution of encapsulated feedback for both conditions.	98
Figure 26: Improvement in three categories of feedback between conditions in the last three decisions	100

Figure 27: Improvement in each metric (C1 to C7).....	102
Figure 28: Binary comparison for improvement in each metric (C1 to C7).....	103
Figure 29: Binary comparison for improvement in each metric (C1 to C7) per participant.....	104
Figure 30: Distribution of metrics per condition	107
Figure 31: Each pair displays the final design configurations of one participant at the end of two conditions: Left: receiving feedback, Right: not receiving feedback.	109
Figure 32: Stills from exoskeleton configurations	110
Figure 33: Mean values of the last three decisions across all criteria, per condition – 5% outliers reduction	112
Figure 34: Mean values change trend of last three decisions across all criteria, per condition	113
Figure 35: Distribution of players' focal point; In each section, the bottom pair is for IDM (with feedback) and the top for nIDM. In each pair, the right hemisphere represents players facing forward.	117
Figure 36: Detailed view of selected participants' navigation behavior: left shows users' positions, right shows users' focal points.	118
Figure 37: Distribution of players' movement while configuring design, relative to the design subject. In each pair, the left item is for the nIDM condition, and the right is for IDM.	119
Figure 38: IPC: Socket, An overview of the main components and functions	234

LIST OF TABLES

Table 1: Some of the most recent commercial tools, for engagement of non-experts (classification scheme inspired by Weber et al., 2022).....	10
Table 2: Some features of design frameworks that support decision-making	17
Table 3: Methods of public engagement with criteria to evaluate their acceptance and efficiency (derived from (Rowe and Frewer 2000)	21
Table 4: Theories that encircle human-environment interaction; and their possible contribution to GIDM	35
Table 5: Features of the proposed research and the relevant supporting theories.....	41
Table 6: Definitions of key terms	48
Table 7 :Research data and Acquisition methods and analysis tools. (Research data categorization derived from Hensel and Vujovic (2022).	56
Table 8: Summary of framework developments across versions.....	71
Table 9: Statistical measures and definitions.....	85
Table 10: Basic statistics and distribution measures of Decision-Time per condition	86

Table 11: Number of design configurations and user movements per participant and condition.....	114
Table 12: Statistical description of design configuration and user movement	118
Table 13: Design exploration ranges from average configuration per participant and condition.	121

LIST OF ABBREVIATIONS

DT – DDT	Decision Making Time – Decision to Decision Time
EBD	Evidence Based Design
FEA - FE	Finite Element Analysis
GIDM	Gamified Informed Decision Making
HCI	Human Computer Interaction
HRI	Human Robotics Interaction
IDM	Informed Decision Making
nIDM	Non-Informed Decision Making
PAR	Participatory Action Research
PD	Participatory Design

OVERVIEW OF KEY FRAMEWORK ELEMENTS

To aid the reader's orientation, this page provides a concise overview of the core components of the proposed Gamified Informed Decision-Making (GIDM) Framework. The information below summarizes these components and their roles within the framework. Each component is elaborated in detail in subsequent chapters, but this high-level overview is provided here to clarify how the elements interconnect and support the research questions.

Component	Description	Function	Chapters Covered
Exoskeleton Configurator	A 3D rule-based interface for configuring a retrofit façade using spatial components	Provides the interaction environment where design decisions are made and evaluated	3, 4, 5
Informed vs. Non-Informed Conditions (IDM / nIDM)	Two experimental modes: IDM provides feedback after each interaction; nIDM provides none	Enables comparison of behavior and performance with and without access to feedback	4, 5
Feedback Encapsulation (ENC1-ENC3)	Simplifies multi-metric performance data into user-friendly visual cues	Supports understanding and decision quality without exposing raw simulation outputs	3, 4, 5, 6
Relative Labeled Feedback	Shows performance changes compared to the previous decision using directional labels	Helps users iteratively improve by making comparative effects perceptually visible	5
Design Space (Game Environment)	Bounded, rule-based and feedback driven environment implemented in Unreal Engine	Utilized to enable exoskeleton configuration	3, 4, 5
Performance Metrics (C1–C7)	Structural, environmental, and cost/fabrication-based indicators	Drive feedback logic internally; also a subject of study withing different framework iterations	5
Turn-Based Interaction	Participants complete iterative decisions per condition	Allows behavior change and effect of feedback use to emerge over time	2, 4
Behavioral Logging	Logs decision time, feedback usage, navigation, and configuration data	Enables empirical analysis of user learning and behavioral trends	3, 4, 5
Theoretical Foundations	Informed Decision-Making, Decision support Systems, HCI, gamification	Grounds framework design in interdisciplinary concepts from related theories	1, 2

STRUCTURE AND OVERVIEW OF THIS THESIS

This thesis begins by outlining its background and context in Section One, which elaborates on the three main areas: informed decision-making, gamifying design, and the significance of a performance-aware, data-driven holistic approach. Chapter Two expounds on the theoretical aspects supporting this research, covering our current state of including non-experts and design tools, as well as state-of-the-art tools aimed at involving non-expert users at different stages. Chapter Three briefly explains the research structure by defining the study's variables and detailing the research methods and methodology. Chapter Four is significant because it not only reports the framework developments but also explains the experiment and data collection details. Chapter Five is dedicated to analysis, revisiting the research question and hypothesis. Chapter Six outlines the contributions of this thesis. Chapter Seven provides insights into future research directions and risks involved. Three appendices follow, reporting the user interviews, technical codes, raw log data, and scripts for controlling the results, and list of publications stemming from different aspects of this research.

CONTENTS

Abstract	ii
List of Images	iii
List of Tables	iv
List of Abbreviations	vi
Overview of Key Framework Elements	vii
Contents	ix
1 Introduction	1
1.1 Background and Context	2
1.2 Gamifying Decision Making	2
2 Theorithical Background and Literature Review	4
2.1 A Brief Review of Historical and Contemporary Frameworks Aiming to Enable Non-experts to Make Informed Decisions	6
2.2 Non-expert Involvement in Design Process	11
2.3 Current Design Tools and Frameworks	16
2.4 Current Methods of Non-expert Engagement: Methods and Challenges	20
2.5 Serious Games for Informed Decision Making	26
2.6 Theoretical Framework	34
2.7 Reflections on Experiment Design: A Single-player, Multi-turn, Within-subject Experiment	43
2.8 Discussion	46
3 Research Scope and Objectives	48
3.1 Research Gap	50
3.2 Aim and Objective	51
3.3 Variables	52
3.4 Research Methodology	53
3.5 Research Methods	54
3.6 Limitations	57
3.6.1 Sample Size	57
3.6.2 Limits on Data Transfer	58

3.6.3	Lack of Prior Studies in the Research Area	58
4	The Framework Architecture and Experiment Design	59
4.1	Overview	59
4.2	Gamification Strategies	60
4.3	Pre-alpha Tests	63
4.4	Alpha Version	64
4.5	Beta Version	65
4.6	Main Version	68
4.7	Data Collection procedure	72
4.8	Data Refinement	75
4.9	Technical Framework	76
4.9.1	Feedback Measures	79
5	Analysis	82
5.1	Post-game Questionnaire	82
5.2	Time	84
5.2.1	Decision-Time: a Closer Examination	88
5.3	Feedback Data	96
5.3.1	Feedback Data: a Closer Examination	101
5.4	Final Design Configuration	109
5.4.1	Design Configurations: a Closer Examination	114
6	Contributions	122
6.1	Revisiting Research Questions and Hypotheses	124
7	Future Research Directions	127
7.1	Theoretical Extensions	127
7.2	Technical Advancements	128
	References	132
	Appendix I: Participants' Feedback	146
	Appendix II: R/Python Code for hypothesis testing	150
	Appendix III: Publications	235

1 INTRODUCTION

Buildings have been proved to have a direct impact on our health. The wellbeing of building inhabitants is adversely affected by poor building performance (Keeling et al. 2012; Kanthila et al. 2021). According to a recent energy flow report by EIA in US, residential and commercial buildings in 2021 account for over 38% of energy consumption. Among various elements of a building, User's interaction with a building's facade is extremely important. Since the building envelope separates the indoor and outdoor worlds, it plays a crucial role in environmental comfort. The importance is evident in both architectural and urban contexts because of its exposure to a broader public. It is one of the first structural elements to encounter lateral loads and is constantly subjected to thermal loads and harsh external environmental conditions. It is perhaps the most visually interacted element of the building from the user's perspective. As a result, a profound study of user-facade interaction is necessary. However, one of the biggest challenges in this field is that several multidisciplinary design forces are involved in the design of the envelope. Recent studies in user-facade interaction cite the necessity for a multidisciplinary approach and classify the key actors in the field as social sciences and environmental psychology, building science and physics, and human-computer interaction and computer science (Azar, O'Brien, Carlucci, Hong, Sonta, Kim, Andargie, Abuimara, El Asmar, et al. 2020; Luna-Navarro et al. 2020). These studies conclude that one of the major gaps in research relates to the lack of a holistic approach to user-facade interactions with interactive scenarios (D'Oca et al. 2017; von Grabe 2016; Luna-Navarro et al. 2020). This holistic approach is naturally enriched with a variety of information, which points to the next pillar of this research: informed decision-making -IDM. IDM in design refers to the process by which individuals—regardless of their expertise—make choices based on accessible, relevant, and comprehensible information. In architectural contexts, where decisions often carry spatial, environmental, structural, and economic implications, the ability to understand and weigh performance-related data is crucial. However, traditional design tools and workflows are often inaccessible or overly technical for non-expert participants, creating a disconnect between users' needs and the design solutions they help shape. This challenge becomes especially significant in early design stages, where foundational decisions are made and the potential for impact is highest. Addressing this gap requires solutions that not only convey performance insights effectively but also empower users through intuitive, interactive, and engaging design environments.

1.1 Background and Context

Architectural design—particularly in its early phases—is characterized by uncertainty, complexity, and multidimensional decision-making. While design decisions made at this stage carry long-term implications for environmental performance, cost, structure, and usability, the tools and systems used to support these decisions are often tailored for expert users with specialized knowledge. As design becomes increasingly data-driven, the question arises: how can we structure decision-making environments so that non-expert designers can actively participate, contribute, and make sense of performance trade-offs? This thesis investigates that question by proposing an informed decision-making framework that empowers non-experts through feedback-rich, gamified environments. It situates this inquiry at the intersection of human-computer interaction, behavioral theory, and computational design, while responding to persistent shortcomings in conventional design tools and participatory platforms. The following sections lay the foundation for this exploration—first by examining the gamification of decision-making processes, and then by unpacking the dynamics of cross-agency collaboration in informed design systems.

1.2 Gamifying Decision Making

Design involves multiple disciplines working together at different stages. There are often conflicting requirements and expectations between the various parties that negatively affect both the outcome and the performance (Du, Shi, et al. 2018). In other words, a comprehensive space plays a crucial role in enabling effective communication, enhancing design performance, and coordinating multiple design factors. A game engine can mediate between the static automaton of digital design and the dynamism of human decision making. This is due in part to its dual nature, which benefits from gaming and its engaging characteristics as well as realistic and immersive aspects that augments players' perception by feedback. In the same way that software interfaces bridge the gap between humans and computers, game engines can serve a similar purpose but with extended capabilities. This mediator is where the two – the designer and the design system – can work and complement each other. Because in this space the two entities coexist and communicate. The complex nature of the interface matches the complexity of the human and computer relationship (Hookway 2014). This threshold space provides a platform for interaction among participants as well as the opportunity to communicate data to them. Moreover, these platforms offer a communication platform that allows multiple users to participate and make decisions in various scenarios, enabling more efficient informed co-design and ease of data transfer and analysis (Sebenius 2009; Poplin and Vemuri 2018; Guyot and Honiden 2006). They also allow for engaging with various audiences: from novice interdisciplinary beneficiaries to designers, fabricators, contractors and different stakeholders in the design and fabrication process. Mediator spaces are vital, and given the multidisciplinary nature of architecture, it is possible to

use game engines to rectify the deficiencies of the current design environment (explained in section 2.3). It is estimated that the game industry will have more than three billion users in 2023 (Finances-online, 2022) and have demonstrated their utility in many aspects. First, the nature of today's games, is defined by tradeoffs, strategies, efficient use of resources and the interaction between individual and collective choices. These concepts originated from game theory and interface theory but have been overlooked by today's design tools and platforms (Sebenius 2009; Tan 2014). Also, due to the development of computational tools available to developers, the data produced by various modelling and analytic tools can be embedded in games or transferred to them. Third, the user-centred values in architecture and participatory design frameworks that designers have used to approach other stakeholders through surveys, questionnaires, board games, sorting games, and so on can also be effectively incorporated into these immersive games. To conclude, game engines can enable interaction, give a thorough understanding of user behavior, and be integrated with digital design tools to assess design performance. Hence, this study proposes a novel method using game engines and a theoretical framework to address those queries.

2 THEORITICAL BACKGROUND AND LITERATURE REVIEW

The design of our built environment profoundly affects human behavior, well-being, and environmental performance. Yet, early-stage design decisions—where the greatest impact can be made—are often limited to expert stakeholders. Non-expert designers, including students, emerging professionals, or interdisciplinary collaborators, are frequently excluded from critical decision-making due to the complexity of tools, lack of transparent performance data, and insufficient support for intuitive understanding.

Over the past decades, widely accessible computers with significant processing power have changed the design discipline. Today, architects and designers have access to various software and hardware that can be used to analyze and model almost any aspect of the building. In contrast, very few methods or tools have been developed to enable them to consider various sources of data during their design; the simulation-based and analytical tools have shortcomings that make them not usable effectively in early design stages (briefly explained in section 2.3).

Taking the preferences of occupants, for example, there are simulation tools and conventional methods to estimate how end-users may behave in certain settings. Non-specialist decision makers in the early design stages need to consider these preferences. However, they often lack sufficient feedback or metrics to understand this discrepancy effectively. Examining the reasons behind this discrepancy can be beneficial: Conventional social science research utilizes an analytic method in which hypothetical possibilities of relating physical environment elements to measurable behavioral and psychological phenomena are evaluated. This research methodology is based on identifying an outcome measure and assessing how it is influenced by changing environmental factors (Vischer 2008). As a result, these studies hypothesize a one-to-one relationship between physical characteristics of the space and occupant behavior, such as the effect of ventilation on people's health (Milton 2000) or the effect of circulation on fostering small conversation between employees (Lee, Lee, Research, et al. 2021). One of the key sources of uncertainty in simulations of a building's performance is occupant behavior. This has caused several modelling techniques to be created and integrated to accurately capture the intricate, stochastic nature of human-building interaction (Mahecha Zambrano, Filippi Oberegger, and Salvalai 2021). Tools for building performance simulation are used to enhance decision-making in the latest stages of design. However, there is usually a discrepancy between the expected and actual performance of buildings. The main reasons for this discrepancy between anticipated and actual outcomes can be linked to the designers' assumption and oversimplification about occupants. Users are characterized as uniform and passive agents whose interaction with a building are

limited; they do not actively engage with the built environment. As a result, it is recognized that assumptions about occupant behavior, context variations, and differences between design and built are the key causes for these inaccuracies (Azar et al. 2020; D’oca et al. 2019; Shi et al. 2019; Mahecha Zambrano, Filippi Oberegger, and Salvalai 2021). To address this issue, numerous modelling techniques have been developed over the past forty years, with the goal of capturing the complex nature of behavior, the diversity of occupants, and the two-way interaction between occupants and their built environment (Carlucci et al. 2020; Azar et al. 2020). But building performance evaluations often uses rule-based models or deterministic scheduling (O’Brien et al. 2016). Understanding user dynamics, behavior patterns, and socio-technical interactions is essential not only for post-occupancy analysis (Shen, Newsham, and Gunay 2017; Sonta, Simmons, and Jain 2018). but also for structuring informed design environments. While non-expert designers cannot be expected to model behavioral probabilities explicitly, their decisions can be guided through feedback mechanisms that encapsulate these dynamics. For instance, design frameworks can abstract such patterns into interactive rules or trade-offs, enabling users to account for likely behaviors and performance impacts without technical simulation knowledge.

Numerous studies affirm that almost any kind of user input is one of the most significant sources of uncertainty in building performance (Saeidi et al. 2018; Nord et al. 2018; Jia, Srinivasan, and Raheem 2017; De Wilde 2014). However, conventional assessment tools often fail to represent this behavior meaningfully during the design phase. Sonta et al. (2018) note that most current methods overlook occupant dynamics, limiting their ability to offer insight into realistic activity outcomes or behavior-driven design feedback.

Moreover, traditional simulation techniques are resource-intensive—relying on preconfigured heuristics or extensive sensory data collection (Azar, O’Brien, Carlucci, Hong, Sonta, Kim, Andargie, Abuimara, El Asmar, et al. 2020). While appropriate for high-resolution modeling, these approaches are inaccessible to non-expert designers and incompatible with iterative early-stage workflows. Importantly, many agent-based models oversimplify occupant behavior, treating agents as deterministic and environment-independent (Lee, Lee, Zhang, et al. 2021). In contrast, early design tools aimed at inexperienced users should embed reactivity and behavioral nuance indirectly, by integrating performance feedback that reflects user-driven variability.

It can be concluded that: 1) the design process of actual buildings cannot be effectively scaled or deployed using current approaches (Azar et al. 2020; Sonta, Simmons, and Jain 2018; Jia, Srinivasan, and Raheem 2017); 2) Including end users in the design process, whether in post-design evaluation and optimization or in the early pre-construction design stage, is difficult due to the complexity of human behavior and preferences. Hence, moving away from post-occupancy data extraction for feeding into design tools toward

the earlier integration of real-time user information is of great importance. Returning to the scope of this thesis, although the target users of the frameworks are not end-buyers, the research gap is similar. Additionally, the body of knowledge emphasizes the importance of taking action at the early stages of design. It also highlights the significance of using actual first-hand data and incorporating it back into the design process.

In this thesis, such behavioral complexity is not simulated in full fidelity, but is instead operationalized through encapsulated performance feedback and scenario testing. The goal is not to predict occupant behavior, but to ensure that non-specialist designers can understand and respond to its design implications—increasing the likelihood of producing resilient, responsive, and human-centered spatial configurations.

Returning to the starting gap, it was established that in a field like end-user preferences, the stochastic nature of the issue and the complexity of human behavior are significant obstacles that prevent early designers from embedding that data. Oversimplification and assumptions by designers about knowing the end users' preferences have proven to be inaccurate. However, this research proposes methods for all non-expert beneficiary groups to make design configurations. Therefore, there is no need to decode the black box of preferences. The non-expert is empowered to make their own choices, enriching the design and steering it from the start.

2.1 A Brief Review of Historical and Contemporary Frameworks Aiming to Enable Non-experts to Make Informed Decisions

The connection between architecture practice, participatory design, and evidence-based methods spans over the last 50 years. However, in recent years, architectural practices have a common trait: they engage with non-experts innovatively and actively involve them in the design process through tangible interventions (Luck 2018). Design software experience is not tailored for broader participation, which raises the question of how to empower non-experts to actively participate in making design decisions. Non-technical stakeholders' participation has been a long-standing issue, calling for a new mediating or intervention space. Historically, some of the earliest efforts in design automation and non-expert involvement focused on automated floorplan generation (Friedman 1971); Although the intentions and methods in this field have evolved, the question remains a valid discourse today. Perhaps one of the earliest examples is the work of Yona Friedman in the 1970s. He aimed to incorporate end-user preferences into generating housing layouts using a novel interaction medium (Friedman 1980; 1971; 1996). With the development of computers and their role in the design discipline, new types of configurators emerged. For instance, Cedric Price's

Generator was a voxel-based spatial configurator developed in 1984 (Price 2003). Those early examples mainly used pre-computed options, yet they envisioned the process through a modular kit of parts in a bottom-up process (Weber, Mueller, and Reinhart 2022a). As early as 1977, Negroponte's concept of Soft Architecture Machines sparked ideas about user involvement and how computers can play a role in aiding this relationship (Negroponte 1970). Influenced by the work of Yona Friedman, he aimed to redesign the design process to one in which architects become translators of users' requirements and wishes (Purcell 1977; Friedman 1980). He and Guy Winzafel introduced Architecture-by-yourself on top of Yona's ideas, an experiment in computer-aided design that uses the medium of computer graphics for a general populace. They define their goal as allowing people to design their homes without either a human intermediary or a middle machine creating whole solutions for them (Winzafel and Negroponte 1976). During the 1970s and 1980s, other scholars pursued different frameworks, such as parametric shape grammar and graph representations (Weber, Mueller, and Reinhart 2022b; Koning and Eizenberg 1981; Stiny 1980).



Figure 1: Some of the earliest efforts in HCI and design-for-all theory in the 70's (Negroponte 1970)

Interestingly the challenges they report match the research gap that is still valid today. They report that participants' "... greatest difficulties involved a clear understanding of "next steps." Their previous experience enabled them to select favorites from a set of plans or perspectives, but they had no idea how to go about creating plans of their own. Nor could they evaluate given plans against their own stated objectives. When shown problems in or shortcomings with plans they liked, they could "explain away" neither the plans nor their criteria" (Weinzapfel and Negroponte 1976). That statement completely affirms the necessity of augmenting users' perception. Another notable point is that the YONA framework starts from a graph representation and evolves into a bubble diagram. Eventually, the authors proposed converting to more detailed drawings as their next step. This underlines the design hierarchy and acknowledges the role of a facilitator. Although they aim to eliminate any outside agency, the platform itself is a great example of digital mediation with the capability to give feedback. Cedric Price proposed the "Generator": a reconfigurable voxel-based spatial unit that visitors could use to create different layouts. During the 1970s and 1980s, other scholars also pursued different frameworks, mainly based on abstractions such as parametric shape grammar and graph representations (Weber, Mueller, and Reinhart 2022b; Koning and Eizenberg 1981; Stiny 1980). All of these efforts played a significant role in today's approach.

Then, with the emergence of more rigorous computational approaches, some of the first and most effective attempts in this field were made on an urban scale; Involving non-designers in the design process has been attempted in a number of physical model-based projects. The BIG U project by BIG (Bjarke Ingels Group), submitted in response to the REBUILD BY DESIGN competition held in June 2013 by The Hurricane Sandy Rebuilding Task Force, is an illustration of how to use a model (and proposals) to engage the community (BIG (Bjarke Ingels Group) 2013; U.S. Department of Housing and Urban Development 2013). Also Play the City and the city games by Ekim Tan are examples of using Lego-like modules in a game where participants, both experts and non-experts, can gather in an interactive game session that helps resolve disputes and make decisions on urban projects (Tan 2014; 2018). Although using a physical model can improve a participant's perception, using scaled-down modules may lead to lack of perception dealing with size and scale among the non-experts. But more importantly, the missing element is the feedback data, informing players of the performance or consequences of their choices. The City Science group at the MIT Media Lab's project on "responsive cities" or "smart cities" by Kent Larson addresses this deficiency (MIT Media Lab 2021). Some annotation and data - the pedestrian behavior - is projected onto the physical model using a projector in real-time to help the player understand how their decisions may affect the real world. Once more, the issue of immersion and comprehension caused by bird-eye point of view and scale differences are unexplored. They all aim to follow the same principle of empowering non-experts (Parvin

2013; Madrazo, Sicilia, and Cojo 2009). A similar schema can be observed after narrowing the scope to tools focused on non-expert designers' interaction with design in the early stages, with only a few cases providing at least one type of performance feedback. These tools are designed for Non-technical stakeholders offering mainly massing information as well as basic predictive analysis, such as TestFit (Harness and Griege 2022) Kope (Thorley, Flynn, and Gidei 2022) Forma/SpaceMaker (Autodesk 2022), Digital Blue Foam (Weijenberg and Patel 2016), Delve (SideWalk Lab 2020), Finch3D (Wallgren 2020), iCD (Integrated Environmental Solutions Limited 2016), DesignBuilder (DesignBuilder Software Ltd 2010), Cove (Cove.Tool 2018), Sefaira (Trimble Inc 2012), SimScale (SimScale Lab 2012), MetricMonkey (Parametric Monkey Pty Ltd. 2019) and Archilogic (Archilogic AG 2016). With most of these efforts, mutual shortcomings and difficulties are reported. For example, researchers developed a framework in West Helsinki to facilitate facade refurbishment strategies between contractors and occupants. They reported that written documents were the main source of misunderstanding, creating conflicts among the parties (Saari 2008). Dalsgaard and Halkov (2010) aimed to explore urban participation in decision-making related to urban media facades. They utilized workshops, presentations, and physical models to interact with stakeholders. Among the challenges they reported were unfamiliar interfaces, aligning stakeholders, and balancing interests. The challenge of lacking a common language can also be seen in the research of Di Mascio and Dalton (2017), who aimed to establish a dialogue between designers and citizens using serious games. They claim that "engaging people as co-designers in any preliminary design phases would be greatly beneficial. However, because of their lack of expertise in the architectural domain, a better option would be to engage them in even earlier design phases". Therefore, it was chosen to communicate using 3D models instead of verbal content and the participation was divided into smaller controlled tasks (Di Mascio and Dalton 2017). Jang and Kim aimed to design frameworks for interactive media facades. Facing the challenges of user participation, they are inclined towards using continuous (multiturn) participation (Jang and Kim 2016). In conclusion, the medium's significance and early intervention's importance are accentuated, which will be refined in the following sections.

Table 1: Some of the most recent commercial tools, for engagement of non-experts (classification scheme inspired by Weber et al., 2022)

Name	Type of participants	Use case	Citation
TestFit	Real estate developers, Non-technical stakeholders, Contractors	Feasibility study, Massing, High-level plan layout generation	(Harness and Griege 2022)
Kope (Matterlab)	Suppliers, Contractors, Non-expert designers	Incorporation of products and suppliers	(Thorley, Flynn, and Gidei 2022)
Autodesk Forma (Formerly SpaceMaker)	Non-technical stakeholders, designers, Planners	Conceptual design exploration, Massing, Basic predictive analysis	(Autodesk 2022)
Digital Blue Foam	Non-technical stakeholders, Real estate developers, Planners	Feasibility study, Massing, Basic predictive analysis	(Weijenberg and Patel 2016)
Delve	Real estate developers,	Massing, Basic predictive analysis	(SideWalk Lab 2020)
Finch3D	Non-expert designers,	Massing, High-level plan layout generation	(Wallgren 2020)

In recent years, high-profile architectural practices have begun to explore immersive configurators as a means of democratizing design through popular gaming platforms. A notable example is Zaha Hadid Architects’ Re:Imagine London, released as a mode within Epic Games’ Fortnite Creative (UEFN). In Re:Imagine London, participants select among six building typologies—walkway, structure, park, commercial, office, and residential—and place voxels within a predefined 3D grid to generate parametric structures that replicate iconic London landmarks (“‘Re:Imagine London’ in Fortnite, ZHA Collaborates with Epic Games – Zaha Hadid Architects,” n.d.). Expertise in Unreal Engine and modular geometry enables ZHA to craft a cooperative, multiplayer environment in which hundreds of players can simultaneously negotiate urban intensification scenarios, experimenting with sustainable, walkable, mixed-use waterfront developments. The configurator emphasizes open-ended exploration and collective participation, allowing users to experience the complexities of urban planning in a shared, gamified landscape. While Re:Imagine London marks a disruptive innovation in the domain of collective

configurational design, from a research perspective this thesis proposes a more controlled environment with a closed-ended scenario to provide a reliable foundation for future studies, which can in turn inform and enhance such large-scale efforts. It must also be noted that Re:Imagine London provides no form of performance feedback to participants and is therefore designed for a different target group and set of objectives than those addressed by this research.

In conclusion, this section provided a review of non-expert involvement in the design process. It spans from early examples in the 1970s to recent software development in the 2020s.

2.2 Non-expert Involvement in Design Process

This section explores the literature surrounding the involvement of Individuals without domain expertise in design decisions. However, such statements require careful deliberation: numerous groups benefit from a building without comprehensive knowledge of all its aspects. There is substantial debate over the existence of a hypothetical group possessing enough knowledge in all sectors of a building—a group expert in financial aspects, construction procedures, socio-economic aspects of the built environment, design and many more aspects. The preceding argument highlights the fields where design affects its surroundings, end-users, and stakeholders. Addressing whether one or more stakeholder groups have the essential knowledge to make informed design decisions can be a complementary research area. However, within this research's scope, it can be broken down into smaller inquiries. By referring to the body of knowledge, studies can examine each of these non-expert groups and their current involvement in the design process. For instance, studies focus on end buyers and their current level and tools of involvement in the design process. Another segment examines tools for land developers and investors to determine if they are active decision-makers using current design methods and tools. Another research line explores how architectural designers can benefit from the knowledge of BIM experts and so forth. Naturally, all stakeholders have different, sometimes contradictory, demands. More importantly, when efforts are made to align their opinions, it becomes clear that each group individually is non-expert or at least a not all-knowing party. Therefore, an underlying theme connecting these pieces of literature is the intention of involving broader groups of stakeholders, who are non-experts in other fields, in the complex task of design. When Yona Friedman introduced his architecture machine, did he intend it for end-buyers, investors, land developers, or early designers? The author believes a more significant and fundamental question needs to be addressed first. Efforts like those of Yona Friedman and Nicholas Negroponte, to the most recent examples, mark the pitch for this game; This perspective is less discussed. When new interfaces are introduced, such as Friedman's custom keyboard or Smart Labs' Lego building house, they are possible solutions to the

fundamental question: if non-expert stakeholders are to be involved in the design process, how and to what extent can they be active decision-makers? The next few paragraphs denote different literature from that perspective, exploring if there is a unifying method or methods to involve different non-experts in decision-making and what those possible solutions or recommendations are.

Many studies on game design and education suggest that the two are linked in a meaningful way (Valls et al. 2016; Westera et al. 2008; Chien 2002; Radford 2000; Woodbury, Shannon, and Radford 2001)." The main objective of applying games is to engage learners in complex problem spaces that mimic real-world situations, without importing unwanted constraints and risks of the real world" (Westera et al. 2008) According to Chien, one of the most beneficial aspects of games is that they can be used to simulate rules and limitations during the design process (Chien 2002) Games have many advantages in design education, including the cyclical nature of play, immersive experience, resemblance to workplace situations, higher level of engagement, the ability to provide multiple levels of feedback, and the multiplayer or supervised aspect of gameplay. Therefore, the essence of games lies in a player's immersion in a game within a set of rules (Radford 2000) The similarities and differences between game elements and design learning steps are explored in Woodbury et al. (2001) with a metaphorical analogy, describing both as a "balance of forces" as well as the fact that both have an "embodiment" aspect. Design education and gaming emphasize the importance of "rules" by noting that the design process is a platform where multiple forces interact simultaneously and sometimes contrastingly. Hence, a gaming environment offers an excellent opportunity for participants to explore the superimposition of the forces. Additionally, the embodiment aspect of games, which occurs through more thorough three-dimensional navigation coupled with annotated data, leads to a more comprehensive understanding of the design subject (Woodbury, Shannon, and Radford 2001).

To strengthen the pedagogical grounding of the GIDM framework, four established learning and motivation theories have been integrated. First, Kolb's Experiential Learning Cycle posits that learning occurs through a four-stage process of concrete experience, reflective observation, abstract conceptualization, and active experimentation (Kolb 1984). In the GIDM configurator, each turn constitutes a concrete experience in which users make a design choice. The encapsulated feedback delivered immediately afterward fosters reflective observation, prompting users to consider what changed and why. Over successive rounds, these reflections give rise to implicit heuristics—an abstract conceptualization—which participants then test through further interaction, closing the loop and accelerating learning. Second, Vygotsky's Zone of Proximal Development emphasizes that learners achieve optimal growth when tasks are pitched just beyond their current capabilities but supported with appropriate scaffolding (Eun 2019). In the GIDM framework, rule enforcement (preventing invalid configurations) and simplified visual cues serve as scaffolds that guide

non-expert users through design challenges they could not tackle unaided. By constraining the design space and highlighting only the most salient performance changes, the system maintains engagement while preventing cognitive overload, thus keeping participants within their proximal development zone. Third, Dweck's concept of Growth Mindset describes how individuals who believe their abilities can improve with effort demonstrate greater resilience and willingness to experiment when facing challenges (Yeager and Dweck 2020). By emphasizing incremental gains rather than fixed outcomes, the platform encourages users to persist through trial and error and to view setbacks as informative rather than discouraging. Finally, Csikszentmihalyi's Flow Theory argues that deep engagement and optimal performance arise when challenge and skill levels are balanced, goals are clear, and feedback is immediate and unambiguous (Csikszentmihalyi 1990). Users enter flow by focusing on the current design task within a bounded, interactive space, receiving clear indications of success or areas for improvement in real time. This immersive state sustains motivation and fosters the iterative exploration necessary for informed, high-performance design outcomes.

Together, these four theories provide a robust pedagogical and psychological foundation for the GIDM framework. They explain not only why users learn and engage more effectively through gamified, feedback-rich interaction, but also how each component—from encapsulated feedback to turn-based structure—contributes to learning trajectories that mirror well-established educational models. As a result, game engines offer a variety of benefits not just to users with no design background but also to professionals throughout various disciplines who want to gain a deeper understanding of the design process.

Literature reviews point to another key missing link: While the tools mentioned above offer automation capabilities in floor and mass generation, they do not offer a comprehensive approach, as the data they provide on their feedback is limited, and the user experience is not tailored for non-specialists. Given the complexity of architectural projects, it is necessary to reciprocally improve both the productivity of construction processes and the energy and resource efficiency of construction systems. Therefore, achieving a genuine informed decision-making process requires interdisciplinary research that spans architecture, structural engineering, building physics, manufacturing, systems engineering, humanities and social sciences (Knippers et al. 2021)—in a review of methods and applications of floorplan generation, Weber, Mueller, and Reinhart (2022b) followed frameworks and software from 1970 to 2022, emphasizing the need for future research to take a holistic approach and highlight the importance of considering design and fabrication limitations, structural performance, and environmental performance comprehensively (Weber, Mueller, and Reinhart 2022a; Rogeau et al. 2022). In the architectural design of interior spaces, a comparable body of knowledge aims to find connections between design features and users' preferences (Banaei, Ahmadi, and Yazdanfar 2017; Karakas and Yildiz 2020; Vannucci, Gori, and Kojima 2014). It's a testament to the fact that the design of our built environment is a significant determinant of human preferences, underlining the need for thoughtful and human-centric design. Finally, studies on user-facade interaction identify fields of social sciences, environmental psychology, building science and physics, human-computer interaction and computer science as the primary disciplines involved in this field. These studies also highlight the significant research gap: the absence of a comprehensive approach to evaluating end-users through interactive scenarios (D'oca et al. 2019; von Grabe 2016; Luna-Navarro et al. 2020).

Alexander (2006) notes that a design's functionality is not universal, but context-dependent, shaped by time, user intent, and environmental conditions. Thus, evaluating the usability of a design tool or generated outcome must account for these shifting parameters. Leaman and Bordass (2010) push this further by framing usability as a form of “productivity,” where the focus is not merely on enabling function but enhancing the quality and quantity of what users can accomplish. While their focus is on post-occupancy performance, the underlying concepts—personal control, system responsiveness, and contextual adaptation—are essential in early-stage design tools as well.

Non-expert engagement is not solely about shaping future outcomes, but also about helping individuals recognize that they have meaningful choices. Within the context of this thesis, this notion aligns with the idea that novice designers—when equipped with appropriate tools—can develop an awareness of viable design alternatives and make augmented decisions early in the design process. Similarly, Leong and Iversen (Leong and Iversen 2015; Iversen, Halskov, and Leong 2010) emphasize the value of engaging users in

reflecting on what is important to them. Rather than focusing on collective negotiation, as in many participatory design models, this thesis adopts a more structured stance: supporting individual reflection and agency through the design of decision-making environments. The transformation in participants' ways of thinking—central to Leong and Iversen's view—is here interpreted as a shift from intuitive or aesthetic preferences to performance-aware design reasoning, enabled by real-time feedback and goal-oriented interaction.

This approach also acknowledges that user input must be carefully coupled with computational systems that manage complexity and provide guidance. As Lo, Schnabel, and Moleta (2017) observe, end-users are diverse, and their design needs often fall outside standard parameters. Traditional tools are ill-equipped to accommodate this diversity—particularly for non-experts—due to their technical opacity and rigid workflows. Thus, design environments must be flexible, responsive, and perceptually accessible.

In this context, game-based frameworks offer a compelling structure for informed decision-making. While inherently interactive and engaging, these systems also enforce a set of rules, constraints, and goals, making them particularly suitable for guiding users through complex decisions. As Clark, Tanner-Smith, and Killingsworth (2016) explain, serious games use the motivational and cognitive strengths of gameplay not simply to entertain, but to train, educate, and inform. In architectural design, such systems can provide structured freedom: the designer defines the environment and performance goals, while the non-expert user engages through meaningful, feedback-rich interactions that shape the design outcome.

2.3 Current Design Tools and Frameworks

The notion of involving non-experts in design is nearly 50 years old. In the 1970s, the issues of computer-aided architectural design and user participation were both very active dialogues. Game-engine-enabled design platforms were seen as "problem-aware partners of design" rather than problem-solvers or optimizers. It seems that the pace of technical developments and the accessibility of computational tools in designers' hands changed the path of the two. User participation evolved into more face-to-face interactions and surveys, while computational tools also took a different path.

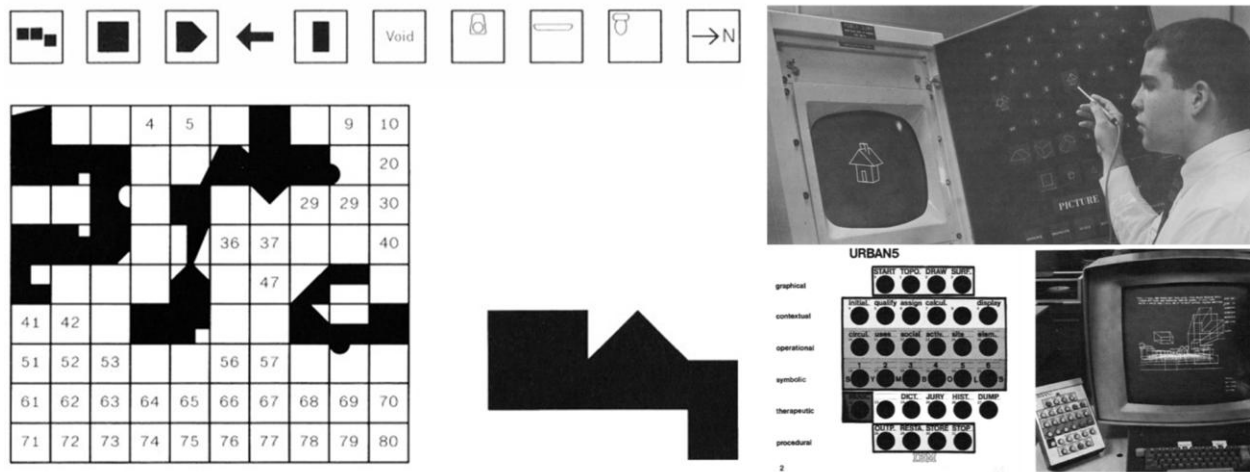


Figure 2: Earliest efforts in developing problem-aware partners of design (Friedman 1980; Negroponte 1970).

Today the most used design tools can be broadly divided into three categories:

1. Computer Aided design tools which enable automation and facilitation of drawing
2. Generative Design Tools which enable more profound exploration of design space and have some development and analytical capabilities
3. Simulation-based tools where in the latest design stages, performance of design can be evaluated

Regardless of the target audience for each category, some evaluation criteria or must-have features for design software can be extracted from the literature. These features or attributes are drawn from a variety of references (Selker 2005; Hewett 2005; Yamamoto and Nakakoji 2005; Resnick and Silverman 2005; Shneiderman 2000; Myers, Hudson, and Pausch 2000; Resnick et al. 2005; Nasirova et al. 2011) as essential

properties that any tool aiming to support creative design must possess. In this context, creativity simply means the ability to create unfamiliar combinations from familiar elements. Those features are listed in Table 2.

Table 2: Some features of design frameworks that support decision-making

Feature	Computer Aided Design Tools	Generative/Algorithmic Design Tools	Simulation-based Design Tools
Easy to try out, easy to undo	☑	☒	☒
Self-revealing	☐	☒	☒
Flexible	☐	☑	☐
Pleasurable and fun to use	☒	☐	☐
Supporting exploration	☒	☑	☐
Talks back to the user	☒	☒	☑
Augmenting users' decision	☒	☒	☑
Supporting Iteration	☒	☑	☐
Low Threshold, high ceiling	☒	☒	☒
Supporting Collaboration	☑	☐	☑
Supporting Open Interchange	☑	☑	☐
Supporting Evaluation	☒	☒	☑
Supporting User Perception	☒	☒	☒

Given the theoretical explorations of this thesis, many other attributes can be added to this list, such as transparency, direct and indirect user assessment, or being engaging and immersive. However, even with the current metrics of literature, shortcomings of our current design tools are obvious. For example, CAD software are relatively easy to use but they lack performance feedback of any kind and also cannot profoundly engage users' perception about the subject. Algorithmic tools on the other hand are far better in

design exploration but in terms of what is called low threshold high ceiling which means the tool must be able to be used by as many people as possible but with extra capabilities for experts, in that sense these tools are hard to learn for some designers. However, the “easy to undo” principle which notes that the user must have a clear understanding that to change one part of the outcome which exact part in the process needs to be changed in these tools are not clear. In complex algorithms, when the output is not as desired, it is unclear which segment or decision was the main reason. Lastly as simulation-based tools depend on data transfer to external frameworks, it adversely affects the real-time feedback needed to intuitively develop the design because of the efforts needed to transfer data.

On a lower level, besides the general deficiencies of every category, each one has some particular unique shortcomings as well: For example, some research indicating that having two windows for process and result, which is common in these tools, is creating constant movement of the focus point of designer and more importantly as the complexity grows gradually, designers' perception fails to keep up with changes in the model (also called change blindness effect).

Also, the effort that can be defined as one overtime plus resources to evaluate performance of any design alternative prevents these tools to maintain a real-time relation with the user and is prone for oversimplification in analysis. To conclude, the shortcomings of our current state-of-the-art tools as designers provide further evidence of the significance of proposing novel frameworks for design.

Each of these categories contributes value in specific ways, yet none fully supports the cognitive, perceptual, and learning needs of non-expert designers—especially when it comes to making decisions based on performance data. Some of the reasons for this inadequacy are:

1. Cognitive Load and Complexity of Interface

Non-expert users often face excessive cognitive load due to complex interfaces, nested functionalities, and inconsistent interaction paradigms. Studies in cognitive psychology and HCI consistently show that high cognitive demand impairs learning, decision quality, and task persistence (Norman 2016). CAD platforms may allow rapid drafting, but they offer limited semantic feedback—providing visual representation without informing users about implications of their choices. Generative tools, in turn, require mastery of rule systems or scripting environments, creating a high barrier to entry and steep learning curves (Myers, Hudson, and Pausch 2000).

2. Absence of Meaningful Feedback Loops

Feedback is central to informed decision-making. Yet, few current tools provide timely, contextual, and comparative performance feedback that can guide user behavior. Simulation tools often require external plug-ins, cumbersome data transfers, or post-processing steps, breaking the continuity of the design-feedback loop. This temporal disconnect reduces the user's ability to connect cause and effect, which is critical for developing intuition and design literacy (Schön 2017; Donald 1993).

3. Opaque Causality and the 'Black Box' Problem

Many generative or parametric tools suffer from opaque causality—users may generate a form, but cannot trace which decision led to which outcome, especially when complex interdependencies are involved. This problem is further exacerbated in simulation environments that rely on abstract parameters, often leaving users unable to deconstruct or adjust their inputs meaningfully. As a result, users become passive operators, dependent on presets, rather than active decision-makers.

4. Misalignment with Non-Expert Mental Models

Design tools are typically created by and for experts, reflecting expert mental models and assumptions. However, non-experts approach design with different heuristics, goals, and expectations. Studies on novice cognition (Vrcelj Božić et al. 2025a) suggest that experts and novices not only differ in knowledge quantity but also in how they represent and interpret problems. Current tools rarely provide scaffolding for novices—such as cues, progressive disclosure, or analogy-based reasoning—to help them structure design logic. In this sense, the tools fail not only in usability but also in epistemic alignment.

5. Fragmentation of Function and Workflow

Tools are often modular—CAD for geometry, analysis tools for simulation, separate platforms for environmental, structural, or cost evaluations. For non-experts, switching between tools, managing file formats, or synchronizing models becomes a source of friction and error. This fragmented workflow discourages experimentation and diminishes the immediacy needed to understand the real-time consequences of design moves.

6. Lack of Support for Learning and Iteration

Non-experts learn best through iterative exploration—by trying, failing, adjusting, and receiving feedback. Yet most tools are not designed as learning environments. They lack scaffolds such as performance visualizations, narrative support, or step-by-step guidance. Moreover, undo functions are often linear and

shallow, preventing users from retracing logic across decision layers. This hinders reflective learning and inhibits the development of design heuristics necessary for making informed trade-offs.

7. Insufficient Engagement and Motivation

Gamification literature (Deterding et al. 2011; Legaki et al. 2021) emphasizes that elements like feedback, reward, progress tracking, and immersive interaction increase motivation and focus—especially among novice users. Traditional design tools, by contrast, offer little intrinsic motivation beyond task completion. They do not reward understanding, insight, or improvement, and their visual language often fails to stimulate engagement or narrative immersion.

2.4 Current Methods of Non-expert Engagement: Methods and Challenges

In the context of this research, participation is closer to active decision-making processes. Therefore, a definition similar to that of Beierle and Cayford (2002) in their “Democracy in Practice” book, can be defined as: Non-expert participation is a collection of mechanisms intentionally instituted to involve the public or their representatives in making informed administrative decisions.

The level of Involvement varies for different methods of participation. On one end of the spectrum, the public is mostly informed through one-way communication with encapsulated or representative information. More profoundly, there can be mechanisms, such as focus groups or questionnaires, that facilitate two-way communication. The highest degree of involvement is to provide the public with a degree of decision-making authority (Rowe and Frewer 2000). Some of the most commonly used methods for non-expert engagement are referendums, public hearings, focus groups, surveys and questionnaires, consensus conferences, citizen jury panels, and citizen advisory committees. For the scope of study, the methods of governmental section and policy making are merged into their closest similar method. The methods commonly used in environmental psychology and design are mentioned in Table 3 with criteria to evaluate the acceptance or efficiency of each method.

Table 3: Methods of public engagement with criteria to evaluate their acceptance and efficiency (derived from (Rowe and Frewer 2000))

criteria to evaluate the acceptance or efficiency of each method.	Referenda (Direct Questioning)	Negotiated rule making consensus conferences	Public Survey and Questionnaire	Public advisory committee	Focus groups
Potential for early involvement	Variable	Variable	High	High	High
Influence on final decision	High	High	Unclear/variable	Variable	Indirect/variable
Transparency of the process	High	Moderate	Variable	Low	Low
Resource availability	Low	Low	Low	High	Low
Task definition	High	High	Low	High	Variable
Structured decision making	Low	Low	Low	Variable	Low
Cost-effectiveness	Variable	Low	Variable	Variable	High

For example, in the public advisory format, the structure of decision-making is likely to be biased by facilitators. Additionally, the direct influence on the final result is unclear, as there is a mediating level by experts that interpret the information. Or the task definition in the referendum is highly clear, but the resource accessibility is relatively low, meaning the broad target audience is likely to not have the necessary context information or enhanced unbiased interpretations of it. In focus groups, people may be reluctant to engage in time-consuming face-to-face activities, particularly if they are uncertain of the practical results that their contribution could make (J. Abelson et al. 1995). Furthermore, the structure of decision-making is unclear and not transparent, but the cost-effectiveness is moderate due to mediation and negotiation rules.

Abelson et al. (2003) in a critical review explores means and reasons behind two-way interaction between decision makers and public as well as deliberation or negotiation between public. “Confirming that Much of the current emphasis on participation methods is also a response to the prevailing view that methods used in the past are no longer appropriate for current decision-making processes or for a more educated, sophisticated and less deferential public”. According to the authors Other challenges of conventional methods, such as surveys, public hearings, focus groups, polls, and citizen panels, are:

- Empirical studies of this method using pre- and post-deliberation polls suggest that participant views do change as a result of the consensus conferences although it is unclear precisely how this occurs.
- The information flow and learning is unidirectional (i.e., from decision-maker to participant) rather than a two-way information exchange as idealized by the principles of participation.
- The participants wish to be given an opportunity on an equal footing with professionals, with equal time and equal opportunity for questioning.
- Questions have arisen regarding the amount of data presented and the speed and time with which people must understand and evaluate it.

They conclude that “public finds these processes stimulating and informative. Whether deliberative processes lead to improved or even different decisions is not yet known. Their future promise, however, lies in the ability for those constructing them to overcome numerous challenges to their legitimacy driven by an increasingly demanding and discerning public” (Julia Abelson et al. 2003).

Yet perhaps the biggest question is, what is the outcome of these efforts? As we mentioned, if the mapping of engagement to the actual result is unclear to the participant, they are reluctant to engage. This gets more complicated in the building industry, where multiple sources of information need to be merged with public decisions. Therefore, it is difficult to structure the learned outcomes and find a universal method to convert them to practical steps.

However, there are some suggestions for this challenge. Webler (1995; 2000) suggests two principles of fairness and competence. The goal of "fairness" necessitates the equal distribution of opportunities for meaningful participation in all aspects of the process, such as agenda setting, establishing procedural rules, selecting information and expertise to inform the process, and assessing the validity of claims. The "Competence" goal is mainly concerned with processing content. By accessing and understanding data, a competent process is completed effectively, and the necessary knowledge of the topic is obtained. Competence also requires that appropriate procedures be used to select the knowledge that will be taken

into account in the process (Webler and Tuler 2000; Webler 1995; Julia Abelson et al. 2003). The problems discussed in this section, such as the importance of understanding the conveyed information before making decisions, affirm the significance of the research question. Furthermore, complications such as reluctance in face-to-face surveys and the desire to see the direct results of choices can be addressed with the proposed medium of game engines. This section showed that while current methods are able to engage people at different levels, however, the next process of interpreting that information by experts into practical steps or guidelines is unclear and a point for future development.

Collaborative problem solving in Gamified informed decision-making frameworks

Today, distributed communication among people is possible due to the information infrastructure and the average users' access to comparatively high processing power in personal computers or other devices. In many instances, the collective behavior that results from this widespread digital interaction is a reliable source for problem-solving. For example, Phylo, a Multiple game, is one of the most powerful tools available today for studying the evolution and function of DNA, RNA, and protein sequences. or InnoCentive, an Open Innovation Marketplace that connects corporations looking for solutions to major problems with a diverse network of people. InnoCentive reports an 80% success rate in discovering a solution (Kwak et al. 2013; Wazoku, n.d.; Nelson and Stolterman 2014).

Consider Kahoot! As a real-time quizzing platform, Kahoot! demonstrates how simple, immediate feedback loops and competitive elements dramatically improve engagement, recall, and knowledge acquisition in classrooms (Wang and Tahir 2020). Learners receive instant confirmation of correct or incorrect answers, which accelerates their reflective cycle and embeds new concepts. Similarly, Codecademy's interactive coding exercises convert abstract syntax and logic into playful, problem-solving modules. Users learn to program by writing real code, seeing immediate validation (or error messages), and iterating until a task is complete. This iterative experimentation mirrors core aspects of design thinking: hypothesize, test, reflect, and refine. Empirical studies show that these platforms foster self-efficacy and sustained motivation by breaking down large skill domains into bite-sized challenges with clear goals.

Even at the level of real-world innovation networks, for instance InnoCentive, (Wazoku, n.d.) uses gamified problem-posting and reward systems to crowdsource solutions to scientific problems. Their model—posing successive challenges, offering milestones, and rewarding progress—demonstrates that well-structured incentives and feedback can guide distributed networks of non-expert contributors toward highly technical breakthroughs.

This important factor needs to be thoroughly examined. Some research suggests a moderating entity or a mediation space regulating the votes and decisions to deal with irrelevant data and misbehaving users. While others contend that the collective's capacity for self-management or self-organization can produce complex solutions even -or specially- in the absence of a higher authority. However, both sides acknowledge that having everyone involved in a project with the same level of access and authorities do *not* assist in ensuring a satisfactory outcome. To ensure the viability of the findings, a thorough analysis of the participation's method, technique, as well as the compilation plot and rules and regulations is required. acknowledging that everyone has the potential to design does not mean that every user must be involved in every project at every stage (Light and Luckin 2008). For instance, some platforms like Tygron Engine provide varying levels of access to their various stakeholder categories. All participants have access to fundamental 3D data, but not all assets are playable by all groups. Non-playing stakeholders who act as facilitators or regulators to help resolve player conflict also exist ("Tygron Geo-Design Platform," n.d.). Only with a carefully moderated framework can the complexity resulting from synergistic interactions between users be realized; This perspective may seem contradictory to motivation of this thesis, however, this position is not at odds with the concept of informed decision-making as proposed. Rather than advocating for indiscriminate or fully open-ended access, this research supports **targeted, context-aware engagement**—where novice designers are empowered to make meaningful, situated decisions through structured interfaces and curated feedback. The principle of informed decision-making acknowledges the diversity of roles, competencies, and goals among users, and it recognizes that effective participation depends on **alignment between user capabilities and system affordances**. Platforms like the Tygron Engine exemplify this stratification, where stakeholders are granted differentiated levels of interaction based on their role, while still contributing to a shared design ecosystem. Similarly, the framework proposed in this thesis introduces **moderated complexity**, where users engage within controlled design tasks, supported by real-time performance feedback and decision scaffolds. This ensures that their contributions are both cognitively manageable and substantively valuable, without overwhelming the design process or compromising outcome viability. In this sense, **informed decision-making operates within a bounded but meaningful design space**, preserving the benefits of user engagement while avoiding the pitfalls of unstructured participation.

User intervention needs to be coupled with automated computational design tools. "Having no outside agency intervene is a fundamental idea to make possible emergence of this collective wisdom. In other words, judgment depends on the accumulation of the experienced consequences of choices made in complex situations" (Nelson and Stolterman 2014). The process of this collective endeavor is equally

important just as the outcome. Not only does it give people voices and realization of possible alternatives, but it also enables them to negotiate the values that are important to them. *"It empowers stakeholders and allows them to feel connected to the design process ...[it] is about negotiating values realized through participation. Values in the design process are seen as an ethos that respects people's democratic rights in that the people whose activity and experiences will ultimately be affected most directly by a design outcome ought to have a substantive say in what that outcome is"* (Iversen, Halskov, and Leong 2010). The complexity resulting from synergistic user interactions can be realized only with careful moderation.

Game engines allow us to explore a design system that enables us to capture the diversity and complexity of different users at different stages of design. Using game engines in design is a collaborative venture. In the design domain, games -also referred to as serious games- are engaging, participative, interactive, performance-aware processes. The designer remains at the center of the design process, creating the environment, establishing design schemes, considering technological limitations, incorporating them into features and rules, creating various scenarios, and enabling user preferences in the form of interactions to be embodied into the design product.

It is significant to note that the proposed approach fundamentally differs from efforts like WikiHouse or BarCode housing in terms of pre-engineering or integrating technical data into the design process (Parvin 2013; Madrazo, Sicilia, and Cojo 2009). WikiHouse offers a great foundation for open-source housing, and the system provides options for layouts, but the user experience is not tailored for non-experts. BarCode fills this gap by giving participants graphs tools to employ, and then automatically created algorithms create the design. The participant is not fully involved in the design process in either case. The spatial cognition of those mediums is different from how non-designers perceive space. Furthermore, during the process, they are not informed of the performance or consequences of their choices. These two factors, "Augmenting users' perception" (of design) and "informing their decision", are the main elements shaping the proposed framework.

2.5 Serious Games for Informed Decision Making

Immersive virtual environments in this context are used in their broader meaning and are not necessarily referring to virtual, augmented, or mixed reality mediums. Studies indicate that regardless of the tool or medium of participation (whether it's a head-mounted display, a projection, or a screen), the primary factor affecting levels of presence, believability, and perception is the content of the simulation (Schwind et al. 2019; Baus et al. 2011; Baños et al. 2004). Therefore, in the context of this thesis, '*immersive*' refers to a photo-realistic physical environment developed using game engines that are augmented with information and interaction tools and are similar to the spatial experience of real-world situations. The medium for this study is personal computers and conventional screens.

Studies in spatial cognition and computation have investigated qualitative spatial representation and reasoning techniques in light of how humans abstract from geometrical details when interacting with spatial information. Participants can (and prefer) think about and communicate using qualitative relations when solving various product design problems, particularly during earlier design stages. This mode of engaging with the design process contrasts with exact numerical values, which is common in today's design tools (Schultz, Bhatt, and Borrmann 2017). Further, early design choices determine up to 70% of the product's production cost. Optimizing manufacturing parameters can only address 30% of those (Kleban et al. 2001). Therefore, it can be concluded that the early design stages provide the most significant opportunity for a collaborative, data-driven intervention.

Game Engines in Architectural Ecosystem:

The recent arrival of game engines into the domain of architectural design demonstrates their potential to be used as an extended design platform with new functionalities. Game engines in building engineering allow various types of data to be integrated into these platforms with an interactive experience tailored to each stakeholder. Due to their interactive data capabilities, they may be effective tools for involving and instructing non-expert and assisting designers.

Game environments and corresponding software platforms offer enhanced graphics, scenario-based design, character design, and embedded artificial intelligence. Whilst building information modelling (BIM) is a collaborative method that facilitates data exchange and management between different disciplines. An increasing amount of information is required for complex and dynamic projects within the architecture, engineering, and construction (AEC) industry. Scholarly publications from 2006-2016 demonstrate this

focus and trend. Eiris and Gheisari (2017) grouped the results of AEC publications into: simulation, education, training, and visualization categories according to their purpose (Eiris and Gheisari 2017), concluding that collaboration and communication are among the most studied aspects of integrating immersive technologies in the construction industry. Previously, researchers combined building information data with game engines. Results indicate that the variability in game engine features, such as providing different layers and scenarios, as well as a variety of points of view, aids in a better understanding of building information and improved collaboration in the design and construction fields (W. Yan, Culp, and Graf 2011; Du et al. 2018).

Further experiments involve synchronizing data during changes in the model and elevating the environment to provide immersive experience. As a result, immersive environments are said to be conducive to involving building users in creative innovation processes together with building designers, as well as for capturing and formulating end-user needs and requirements regarding buildings and their functionality (G. Edwards, Li, and Wang 2015; Christiansson et al. 2011; Shiratuddin and Thabet 2011). Through collaboration within immersive environments, everyone involved in the design process can access information about the design and can iteratively influence the design from early on. Additionally, procedural modelling in game engines offers data-driven capabilities and emergent or unpredictable complexity. They often result in synergistic interactions with immersive environments that take advantage of their ability to handle dynamic forms, rule-based paradigms, and real-time data flow (Lin and Hsu 2017).

One of the most beneficial aspects of game engines is that they can simulate rules and limitations during the design process (Chien 2002). Using game engines has many advantages in design processes, including the cyclical nature of play, immersive experience, resemblance to workplace situations, higher level of engagement, the ability to provide multiple levels of feedback, and the multiplayer or supervised aspect of gameplay. In any design process, various parameters interact simultaneously and sometimes contrastingly. Hence, a gaming environment offers an excellent opportunity for participants to explore the superimposition of the forces. Additionally, the embodiment aspect of the gamified experience, which occurs through more thorough three-dimensional navigation coupled with annotated data, leads to a more comprehensive understanding of the design subject (Woodbury, Shannon, and Radford 2001). As a result, game engines offer a variety of benefits not just to users with no design background but also to professionals throughout various disciplines who want to gain a deeper understanding of the design process.

Involving a diversified [non-expert] group of people in the design of environments is essential to create inclusive, cohesive and sustainable locations (Ingrassia et al. 2020). Therefore, it is necessary to examine available methods to tackle challenges associated with engagement tactics. Since the 1960s, gaming has

been suggested to promote informed decision making by creating cooperative environments that facilitate interpersonal engagement. Evidence-based Design processes have been widely adopted in the realm of design to support decision-making. This approach is examined in this section.

Nisha and Nelson define evidence-based decision-making as *integrating individual expertise and external evidence through systematic research*. In their search for proposing evidence-informed decision-making for participatory urban design, the authors introduce a 5D approach, consisting of five steps: Diagnose, Define, Data, Design, and Decide (Nisha and Nelson 2012). A similar five step agenda is presented by Pezzica et. al. (2017). To propose an informed decision-making process for public spaces, they emphasize the iterative nature of design. They start with defining significant attributes, then modeling construction and data extraction take place. In stage three, the model is validated through direct observations, and in stage four, the design is developed iteratively. The last step is a final design audit and validation test. they assert that “An effective framework for supporting the design of successful [...] spaces must consider the non-procedural nature of the design process and allow for a certain freedom in deciding the sequence of analysis and design. Creative processes are not linear and often lack clear direction. [...] designers learn about design problems through the development and the critical assessment of solutions, rather than through a separate analysis of the problem itself” (Pezzica, Lopes, and Paio 2017). Similarly other authors have suggested using exhaustive Monte Carlo simulations in a two-step iterative design procedure: the first step requires preparation by the modeler, through which is followed by a multi-collaborator session. The authors of the study suggested that prior to involving multiple stakeholders in a decision-making process, it is important to consider several quantitative and qualitative factors as well as conflicting performance measures. Taking a comprehensive approach can help to ensure that the design is of high performance and is well-balanced (Østergård, Jensen, and Maagaard 2017). This reinforces the value of utilizing mixed methodology for this research. The role of the modeler in their research highlights the same concept as a facilitator or game moderator.

The constructive role of a facilitator during program design is undeniable; It is suggested that collective decision-making should be administered by a higher role, such as a facilitator (Buss, Del Bianco, and Keck 2013). The idea of an expert facilitator can also be seen in “U_CODE Urban Collective Design Environment” as a part of Horizon 2020 EU Agenda. Authors argue that “common-use participatory formats either follow a highly diffuse way of decision making or fall back onto limited methods such as voting or tally sheets”. They define seven steps for construction projects, starting from the *initiation* and ending in the *integration* phase. A key element of their strategy is to ensure communication between organizers and authorities at first two stages. Beginning from the third phase, co-designing with public

involvement and voting would start. Then, the mediator or expert takes over the process of analysis and sorting, and develops the project (Stelzle, Jannack, and Rainer Noennig 2017). An educational approach focusing on the social aspects of collaborative design decision-making, observing design students, shows that collaboration emerges at the end of a three-stage process; It has been suggested that once an individual or group of individuals come up with the initial concept, the entire team can use this to further refine and hone the idea before settling on a final alternative (Campbell, Roth, and Jornet 2019). The implication is that alignment occurs when several ideas emerge and there is enough time for all participants to absorb it which results in more focused and channelized solutions.

Rastegar et al. (2022) take Functional Scenario Analysis (FSA), an EBD-based technique formulated in the SimTigrate Design Lab, to understand the intricate qualitative requirements of stakeholders in a quantitative manner. In Functional Scenario Analysis the following steps are suggested: A) Developing functional scenario (for example, asking participants to find a place to read book, or find a room in a building) B) Developing metrics and criteria for analysis (for example, the number of errors they made or total time for task completion) C) Analyzing each case study D) comparison across cases E) Extracting design features (Rastegar et al. 2022).

In a recently published book titled “Public, Play, Space” more than 30 projects focused on informed placemaking is studied (Ingrassia et al. 2020); Some of the main characteristics that can be traced among those focusing on decision-making are:

- From *Corporate growth serious game* by Nick Apeldorn (2020), Bera, Netherlands:
 - Importance of token-based hierarchy (Player who generates most money or points wins) and freedom of play (Players are allowed to make new rules as long as they can persuade others to conform) in a board game for urban growth.
 - Authority of a game leader or judge.
- From *Keryon City* by Bernardo Monechi, Enrico Ubaldi, Vittorio Loreto (2018), Paris, France:
 - Significance of recording the problem-solving process of participants in order to train machine learning models.
- From *Metrogame* by MSLab, Metrohub (2018), Kuala Lumpur, Malaysia:
 - Employing facilitators providing scenarios of negotiation
 - Utilizing another form of token-based hierarchy by giving player a limited number of interventions

To conclude this subsection, serious games focused on decision making have been studied; The following key points can be extracted from the literature:

1. Digital mediator is essential: The role of facilitator or mediator is highlighted in the body of knowledge. (This is embodied in this research as backend processes that facilitate valid design configurations)
2. Turn-based, Iterative engagement is a valid approach: Participants can be given more than one opportunity for intervention, which leads to more holistic outcomes with less antagonism.
3. Scenario-based task definition by experts: More accurate design features can be extracted by studying the behavioral metrics and decisions of users when engaged in carefully designed tasks and objectives of various kinds, ranging from simpler selection tasks to more complex design tasks. (which in this study is interoperated as a clear scenario of an exoskeleton design task)
4. Decision hierarchy: It is not practical for all parties to participate at all stages, so different tools for engagement are needed at each stage, and the amount of information revealed to the user varies accordingly.

Gamifying decision-making has also been suggested to improve the accuracy of judgmental forecasting and the decision-making process for laypeople and experts (Legaki et al. 2021). Scholarly publications from 2006-2016 demonstrate this focus and trend; Eiris and Gheisari (2017) grouped benefits from the publications in this domain into enhancements in simulation, education, training, and visualization (Eiris and Gheisari 2017). Complimentary results indicate that the variability in this approach aids in a better understanding of building information and improved collaboration in the design and fabrication fields (W. Yan, Culp, and Graf 2011; Du et al. 2018). Moreover, adding performance data to this gamified approach can bring about more advantages; Performance-aware augmented systems for informed decision-making are conducive to involving building users in creative innovation processes together with building designers, as well as for capturing and formulating end-user needs and requirements regarding buildings and their functionality (G. Edwards, Li, and Wang 2015; Christiansson et al. 2011; Shiratuddin and Thabet 2011; Lin and Hsu 2017). The embodiment aspect of the gamified experience, which occurs through more thorough three-dimensional navigation coupled with annotated data, leads to a more comprehensive understanding of the design subject (Woodbury, Shannon, and Radford 2001).

Moreover, promising results in bias mitigation and deeper engagement using gamified experiments are reported; Legaki et al. (2021) report that gamified applications improve learning outcomes positively and mitigate heuristics and biases in informed decision-making. Their findings align with much other research

that supports the idea that serious games strengthen knowledge and mitigate cognitive biases. Several reasons contribute to this, including i) Intrinsic motivation in games, ii) Possible intervention potentials by the game moderator, iii) Psychological affordances, iv) Using strategies such as points, levels, challenges, storytelling, leaderboard, augmented communication, and achievement-based tasks (Nah et al. 2014; Dichev and Dicheva 2017; Jacobson and Dargue 2019; Yan, Culp, and Graf 2011; Du, Zou, et al. 2018). As for effective information communication, adding v) adaptive learning systems and vi) intelligent tutoring or supervision has proven more advantageous than traditional learning systems (Koivisto and Hamari 2019; Dunbar et al. 2014).

Thus far, it is estimated that, To conclude, A) gamifying could eventually lead to further improvements in the accuracy of informed decision-making; B) Earlier interventions are proven to be more effective in practice; C) Adding performance data to the game-enabled platform could lead to more efficient means of capturing and understanding end-user needs and requirements and an effective strategy to quantify some qualitative measures of human behaviour; D) User engagement can happen in a turn-based cyclical manner involving multiple subtasks; E) Possible moderation intervention, enabled by gamifying methods and led by experts can facilitate the process by increasing transparency of the decision-making process.

The topic of transparency in decision-making is significant because it is tied to other factors such as readability and understandability. Informed decision-making enables transparency in the design process. The individual's "right to be informed" or "right to explanation" is a critical issue in many respects, including ethical and practical (Wachter et al., 2017). Researchers have affirmed the enormous promising potential of transparency in decision-making. They have also expressed concerns about its mechanisms and applicability. Wachter et al. (2017), composed these debates and concluded that the alleged right to explanation would inevitably necessitate data controllers to clarify how such mechanisms reach decisions. They also suggested that the lack of clarity could be addressed by introducing an *external* auditing mechanism for automated decision-making or by setting internal auditing requirements for data controllers (Wachter, Mittelstadt, and Floridi 2017); A game-enabled farmwork can take the role of a mediator. For this novel design mediator, the transparency of decision-making is linked to its understandability and usability. Understanding decision-making can be defined as the ability of a user to identify the attributes that trigger the system's actions and predict how changes in the situation can result in different outcomes for the system (Weld and Bansal 2018). According to Guidotti et al. (2019), there can be two solutions for improving 'users' understanding of decision-making systems. The first solution involves describing how the black box works, while the second solution aims to explain the decisions without requiring an understanding of the general workings of the system. In fields dealing with data, the second definition is more applicable.

Confirming the validity of the second approach, Chen et al. (2019) aimed to explore how describing the backend algorithm affects the decision-making process of non-experts. After conducting an online experiment with nearly 200 participants, they made a significant discovery: "...Surprisingly, we also find that 'users' trust in algorithmic decisions are not affected by the explanation interface or their level of comprehension of the algorithm" (Cheng et al. 2019). Results from the body of knowledge in transparency of gamified informed decision-making can be categorized into four main propositions:

1. Readability and Intelligibility: Making sure the user can comprehend the environment, navigate, make decisions, and comprehend the consequences of those decisions is essential when creating a design platform. The difficulty non-designers may have in comprehending the size and scale of 3D objects and spaces further complicates the situation. Numerous studies have made fragmentary recommendations; For example, people comprehend the first-person first-eye perspective better than the bird's-eye or top view. Or the positive impact of realistic environmental features has on a person's sense of immersion. Or even on a smaller scale, findings show that non-designers perceive an object's width and depth more accurately than they do its height and elevation. These factors need to be considered in the development of mediated digital spaces so that the surroundings and task remain believable and understandable to the participant.

2. Reversibility: According to research in the field of human-computer interaction (HCI), when using a digital system for decision-making, people tend to prefer a reversible system that allows them to undo or reverse the outcome if they are not happy with their actions. They favor systems that let them make mistakes and offer them solutions or more chances.

3. Design System as an Intelligent Collaborator: Users, particularly those working with complicated software or platforms, tend to interact with the system more effectively when they are not given many options, and they do not have to make all the choices. Platforms that allow players to skip some decisions while still offering options at every decision have been found to be more effective. More users have mentioned feeling frustrated and confused when they must start with a bad option that must be changed into a better option. Users prefer to begin the process with an acceptable option already determined and available to them so they can build on top of it. Meaning that the system itself must be able to make recommendations and let users skip some of the choices. A typical player might also find it confusing to be asked to turn a bad or underperforming option into a good one. They would feel more at ease moving toward their preferred course of action after beginning with a "good enough" solution.

4. Understanding the Augmented Information: In order to make sure a non-expert player understand the information, The platform should be carefully designed when dealing with data, taking into account the

following principles of a) abstraction b) encapsulation c) recasting and remapping the data. This will ensure that a non-expert player can understand the information. The term "abstraction of data" (or 'data encapsulation the framework) refers to a process of only ever providing the user with relevant data. The player can avoid confusion and relieve frustration by hiding the remaining data. Encapsulation is the process of grouping data into packets or groups that are labelled and have readable graphics for the user. Recasting and remapping the data aims to improve the platform's usability for the users. Even though a common person can understand that a bar chart filling up or the number of stars rising indicates that an action had a positive impact, they will become perplexed if they are presented with multiple data that need be maintained within different acceptable ranges.

In conclusion, user preferences and choices have long been considered in the field of design. Informed decision-making can potentially increase transparency, accountability, and efficiency (Bell and Reed 2022). Exploring the body of knowledge resulted in clear suggestions for framework development, all of which are used in this thesis experiment.

2.6 Theoretical Framework

Under the broad term of *psychological impacts of design elements*, various studies have been examined in the previous section. For example, open spaces might evoke feelings of happiness and creativity, while cramped, poorly lit areas could lead to discomfort or anxiety (Martínez-Soto et al. 2013; Banaei et al. 2017). Similarly, under the term *cognitive mapping* (Evans et al. 1980), examples aiming to study the effect of visual cues and mental images in increasing spatial awareness are introduced (Kalantari et al. 2022). In the field of *Sensory engagement and emotional responses*, Particularly, with a focus on gamifying, there is significant importance placed on sensory engagement in participation dynamics (Lieberoth 2015). For instance, the use of natural elements and lighting in design has been shown to reduce stress and enhance well-being. Further theories such as *psychological affordances* and *cultural and social context* (Maier, Fadel, and Battisto 2009; Young and Cleveland 2022) have also been considered by the author. Nonetheless, the author aims to investigate lower-level, specifically directed terminology. Thus, in the next subsection the theories are selected to focus specifically on decision-making as analysis of human emotions is beyond the scope of this research; The aim is to obtain recommendations and applications from each theory and contextualize them for the proposed research (as seen in Table 4).

"If decision making were easy to understand or easy to do, there would be no need for such elaborate research efforts" (quoted in Sirakaya & Woodside, 2005). It is recognized that the existing approaches for comprehending human interaction within the constructed environment cannot be completely reliable (Azar, O'Brien, Carlucci, Hong, Sonta, Kim, Andargie, Abuimara, El Asmar, et al. 2020; Sonta, Simmons, and Jain 2018; Jia, Srinivasan, and Raheem 2017). Incorporating end users into the design process is a challenging task, whether it be during post-design evaluation and optimization or in the early stages of pre-construction design. This difficulty arises from the intricate nature of human behavior and preferences. The complexity of the human decision-making system is one of the primary reasons for discrepancies between actual outcomes and predicted results. (Nord et al., 2018; Sonta, Simmons, and Jain, 2018; Jia, Srinivasan, and Raheem, 2017; de Wilde, 2014). In this section the author has introduced various theories about human-environment interactions with the main objective of determining what can be drawn from them specifically for adoption in the domain and scope of this research. Therefore, when explaining the limitations of each method, it is not intended to reject a theory but rather to simply explain why it may or may not be suitable for application in the proposed research.

Table 4: Theories that encircle human-environment interaction; and their possible contribution to GIDM¹⁻²

Theory (or model)	Description
Health Belief Model (HBM) ~1950s (Janz and Becker 1984; Champion and Skinner 2008)	The theory suggests that human behavior is directed by two factors:1. The desire to avoid harm or recover from it. 2. The belief that a specific course of action can be effective. Both factors are influenced by humans' perceptions of the benefits and barriers associated with each decision.
	Limitations
	- It assumes that everyone has equal access to information. - It does not account for a person's preconceived ideas and habitual decisions.
	Applications or recommendations for the proposed research
	In addition to suggestions to improve the human perception of choices, two more recommendations are mentioned:1. Cue to action: a stimulus that prompts action. 2. Self-efficacy: the level of a person's confidence in their ability to successfully perform a behavior.
Theory (or model)	Description
Theory of Planned Behavior (TPB)	The aim is to understand an individual's intention to engage in a behavior at a certain time and space. It assumes that actions occur based on motivation (intention) and ability (behavioral control). It uses six constructs to explain a person's actual control over their behavior. Two of which can be adopted in the proposed research.

¹ Edwards, in his highly cited book titled "Theory of Decision Making," challenged the accuracy of older theories such as "riskless choice" and "risky choices" (W. Edwards 1954). As a result, those theories were not included in the table. Also, the exclusion of theories with limited applicability within specific fields, such as brand buyer theory (Howard and Sheth 2001), is based on variations in the scope of study.

² The limitations and approximate dates are derived from (LaMorte 2022).

(started as theory of reasoned action) ~1980s Quoted in (Conner and Armitage 1998)	Limitations - It does not account for variables such as uncertainty or previous failed attempts to perform certain behaviors. - It assumes that behavior is the result of a linear decision-making process and that it does not change over time. - The movement between intent and action is not explained in this theory.
	Applications or recommendations for the proposed research
	The TPB suggests that there are two constructs or factors that can enhance the decision-making process: 1. Power factor; and 2. Behavioral control. Perceived power factor and perceived behavioral control are associated with a person's perception of having power over each factor, as well as their perception of the ease or difficulty of performing the behavior of interest.
Theory (or model)	Description
Diffusion of Innovation Theory (DOI) ~ 1960s (Rogers et al. 2014)	Although it was originally used to describe how an idea or a product gains popularity and disperses, it has since been adopted to explain how a person gradually changes their approach over time. It introduces five constructs to understand how people adapt and perform actions differently.
	Limitations - It works better to adopt new behaviors or make new decisions instead of ceasing older ones; it may seem that both are necessary for users to adapt to a new system or method. - Unclear suggestions on individual ability and resources to adopt the new behavior assumes that everyone will take the same course of action in the same situations.
	Applications or recommendations for the proposed research

	An effective strategy to appeal to early adopters is to include how-to information on implementation, focusing on the "how" rather than the "why" of the behavior being promoted. An effective strategy to appeal to the "early majority," who are average individuals with less inertia to change, is to show them evidence of effectiveness or demonstrate the results before they begin to engage. Later, groups with high inertia to change or bound by tradition will eventually be pushed by the majority and engage. Simplicity of use and observability are among the top five main factors that determine the adoption of an idea or a system.
Theory (or model)	Description
Social Cognitive Theory (SCT) ~1980s (Bussey and Bandura 1999; Bandura 1989)	SCT originated from Social Learning Theory (SLT) in 1960 by Albert Bandura. Pointing out the dynamic and reciprocal interaction between individuals, their environment, and their behavior, this approach considers both external and internal reinforcements and influences. Among the five constructs of this theory, reciprocal determinism refers to the responses to stimuli to achieve goals. It acknowledges the importance of observational learning, where individuals can witness an action and reproduce those behaviors.
	Limitations
	- The theory is broad spanning but may be too complex to implement in its entirety. - The extent to which each factor (human, environment, and action) influences one another is unclear.
	Applications or recommendations for the proposed research
	In the Social Cognitive Theory (SCT), the role of reinforcements is recognized. These reinforcements can be self-initiated or come from the environment, and

	they can be negative or positive. This concept is associated with a reciprocal nature in the model, affirming the important relationship between reinforcements and anticipated consequences. In simple terms, the theory proposes a series of reinforcement and informing processes for all participants, including the individual, environment, and actions. This dynamic interplay among the person, behavior, and environment is emphasized.
Theory (or model)	Description
The Transtheoretical Model (TTM) (also called the Stages of Change Model), ~1970s (DiClemente 2005; Prochaska and Prochaska 2019; Spencer et al. 2006)	TTM (Transtheoretical Model) focuses on individuals' decision-making and intentional changes. TTM functions as a model, rather than a theory, that conceptualizes change through a cyclical process.
	Limitations
	- It is unclear in this model how many cycles can lead to a change in decision making. - Using questionnaires has proven to be a debatable means of transferring information.
	Applications or recommendations for the proposed research The proposed six stages of each cycle are designed for habitual decisions and, therefore, not aligned with the research intentions of the author. However, it also introduces ten significant processes; According to TTM (Theory of Planned Behavior), the following factors have a great impact on decision-making: - Consciousness raising involves increasing awareness and providing information about available choices. - Triggering emotional arousal, whether positive or negative, is known as multisensory engagement. - Self-appraisal and environment appraisal are similar processes that involve rewarding the gamified environment and recognizing corresponding changes after each decision.

	- Reinforcement management: A positive system for rewarding desired decisions. - Stimulus control involves engineering the environment to include cues and reminders that support effective decision-making.
Theory (or model)	Description
Social Norms Theory (SNT) ~1986 (Perkins 2006; Berkowitz 2004)	The theory is based on the idea that understanding the environment and influencers may be more important than focusing on the individual itself. The theory implies the need for an interactive environment, although in its early days, this was not technologically possible. It proposes interventions to correct misconceptions about the norms.
	Limitations
	- There is ambiguity regarding the amount of data that must be transferred and how it can be measured based on different participants.
	Applications or recommendations for the proposed research SNT provides six suggestions that are highly aligned to the feedback format of the proposed research. According to SNT, effective communication involves the following steps: 1. Assessing and refining the data; 2. selecting and normalizing it; 3. pre-testing to ensure it is well received; 4. choosing the appropriate medium of communication; 5. ensuring the right amount or comprehensible dosage; and 6. implementing a feedback mechanism to ensure it is understood as intended.
Theory (or model)	Description
Utility Theory (EUT)	The foundation of utility theory lies in the assumption of rationality, which allows for the assessment of decision outcomes based on the value individuals place on them. Within this framework, decisions can be comprehended by

(also called Expected Utility Theory) ~1979 (Fishburn 1979)	categorizing them according to a hierarchy of utility associated with various outcomes.
	Limitations
	- Explanations exclusively revolve around the reward system, neglecting to consider other significant social and environmental factors.
	Applications or recommendations for the proposed research
	To ensure that decisions are not arbitrary, a framework should propose clear and concrete incentives.

After a brief introduction to the theories that shape our understanding and guide our decision-making, it is important to cross-reference them to the features of the proposed research. This will help elaborate on how each feature is supported by one or more theories. Table 5 illustrates how the attributes of gamified frameworks are based in various behavioral theories.

Table 5: Features of the proposed research and the relevant supporting theories

Features / attributes of the proposed framework	Theories from Table 4 to support each feature
Engaging in turn-based cyclical participation in multiple subtasks	TTM, SNT
Utilizing game-based affordances and reinforcements such as points, tokens, and performance evaluation feedback	HBM, SCT, TTM, EUT
Possible moderation intervention by experts	HBM
Augmented and adaptive learning through storytelling and enhanced tutoring	HBM, DOI, TTM
Augmented multimedia communication method	TPB, SCT
A realistic and responsive environment that reacts to decisions	TPB, DOI, SCT, TTM, SNT

When John Howard and Jafish N. Sheth attempted to synthesize a theory explaining decision-making, they set forth assumptions that they claimed were necessary in any attempt to explain choices. The assumptions are as follows: 1. Choices are rational and are made within the limits of one's cognitive and learning capacities, as well as within the constraints of limited information. 2. Researchers assume that decision making is systematic, and therefore it can be observed in standardized ways. Researchers assume that if decision-making is systematic, it is caused by events, stimuli, or other factors. This input is then processed by a system, resulting in a decision as the output. The aim is to remap the input to the desired outputs. Theories, on the other hand, must be assumed to be positive rather than normative (Howard and Sheth 2001). This means that they should focus on value-free concerns about what is, rather than taking a value-based approach to what should be. These assumptions hold true in the realm of informed decision-making. To sum up, this section offers an overview of theories in the field of environmental behavior that can provide valuable insights into how the proposed framework can be developed to promote an impartial and informed decision-making process. Lastly, the characteristics of the proposed system are based on well-established frameworks to guarantee their effectiveness and practicality. Also, the lessons derived from each theory or model are extracted and refined to be implemented in the research. This ensures that valuable insights are gained and applied effectively to improve outcomes in subsequent endeavors.

In Conclusion, various theories and models in the field of human-environment interaction can be used to gain insight into the attributes that an informed decision-making framework must have. The Health Belief Model (Janz and Becker 1984; Champion and Skinner 2008) suggests the importance of a cue to action or stimulus. A similar concept, under the title of the role of reinforcement and dynamic feedback of anticipated results, is acknowledged in Social Cognitive Theory (Bussey and Bandura 1999; Bandura 1989). The Theory of Planned Behavior or Reasoned Action associates the power of making choices to perform the behavior of interest (Quoted in Conner and Armitage 1998). The diffusion of innovation theory emphasizes the importance of immediately demonstrating choices' effectiveness and consequences (Rogers et al. 2014). The Transtheoretical or Stages of Change Model also mentions the importance of self and environment appraisal in increasing awareness and making informed decisions (DiClemente 2005; Prochaska and Prochaska 2019; Spencer et al. 2006). This argument is further refined in Social Norms Theory, adding that choosing the appropriate medium of communication, ensuring the right amount or comprehensible dosage, and implementing a feedback mechanism to ensure it is understood as intended are equally important (Bussey and Bandura 1999; Bandura 1989). Similarly, the Utility Theory states that a framework should propose clear and concrete incentives to ensure that decisions are not arbitrary (Fishburn 1979). It can be concluded that: A) A framework must be capable of providing positive, negative, and neutral feedback in self-earned or system-given formats; B) The informing process must follow the reinforcement process, explaining what happened and why; C) Communication would be more effective if conveyed through multimedia rather than just written text; D) Cues and reminders can be an effective support for decision-making; E) There must be a control system that checks if the information is conveyed in understandable portions; F) Pre-testing can be an effective mechanism to check the viability of the previous steps; G) The demonstration of consequences for each choice must be immediate and understandable.

It is important to note that the above findings are also acknowledged when exploring the complementary field of games and interface theory. This research also draws from foundational theoretical work on game environments and interface mediation. For instance, Johan Huizinga's concept of the "magic circle", introduced in *Homo Ludens* (1938), refers to the creation of a special, bounded domain where players act under specific rules that differ from everyday life. In this research, the configurator functions as a "magic circle" not metaphorically but structurally—it enforces rule-based actions, restricts invalid decisions, and embeds consequences through visual and performance feedback. Thus, the decision-making space is transformed into a bounded, learnable arena where uncertainty is reduced through exploration and feedback. Similarly, Hookway's *Philosophy of the Interface* (Hookway 2014) provides a valuable lens to understand

the role of the configurator as an epistemic mediator. The interface does not present objective reality but selectively filters it (for instance, translating raw feedback data into cues that are perceptually meaningful to the user which is elaborated in section 4.9.1 is based on this approach). This mediating role is central to the encapsulated feedback structure, which replaces simulation output with interpretable, context-aware signals. These conceptual foundations justify using a game engine not only for rendering or interaction but also for systematically mediating agency and feedback, making theoretical principles operable in a real-time design environment. In conclusion, the key points mentioned can be reached from different starting points, such as behavioral theories or theories on interfaces and games.

2.7 Reflections on Experiment Design: A Single-player, Multi-turn, Within-subject Experiment

In this section, answers to three common questions regarding the experiment are elaborated. First, why is the experiment single player as opposed to multi-player. Second, how the moderator can address the possible lack of computer proficiency in participants. Third, why is the experiment designed as within-subject versus between-subject design:

The debate over single-player versus multiplayer approaches in game-enabled informed decision-making platforms centers on the timing and structure of user interactions (synchronous vs. asynchronous), and the implications for data collection, engagement, and decision-making validity. While multiplayer, synchronous participation may seem more dynamic, single-player, turn-based games are equally valid and can, in many cases, provide a more *controlled* and rigorous framework for informed decision-making research. There are several reasons contributing to this statement. Such as

1. *Cognitive Load and Decision-Making Quality*: Informed decision-making requires participants to process information, reflect, and make choices based on feedback. In single-player, turn-based games, participants can engage with the design environment *at their own pace*, which helps in managing cognitive load. Studies suggest that reducing cognitive pressure leads to more thoughtful and accurate decisions. Turn-based systems allow players to process feedback without the potential distractions or time constraints that multiplayer settings might impose. For instance, Social Cognitive Theory highlights the importance of reinforcement and reflection during decision-making processes (Bussey & Bandura, 1999), which is more easily managed in a single-player environment.

2. *Reinforcement Learning and Decision Efficacy*: Reinforcement learning theoretical models suggest that the timing of feedback is crucial to improving decision-making skills. Single-player games allow for feedback to be delivered immediately following each action, ensuring that participants have the opportunity to understand the consequences of their choices before moving forward. This aligns with theories like the

Diffusion of Innovations (Rogers, 2014), which emphasize the importance of demonstrating the immediate effectiveness of decisions (as opposed to delayed team feedback). Turn-based approaches thus enable iterative feedback, enhancing decision efficacy through repeated and controlled exposure to information.

3. *Data Validity and User Behavior Insights*: Single-player, asynchronous games enable controlled environments that limit external variables (such as *peer influence* or *competitive behavior*), providing cleaner, more individualized datasets. The Health Belief Model (Champion & Skinner, 2008) underscores that decisions are often influenced by perceived control and individual risk assessment, both of which can be skewed in multiplayer settings. In a single-player environment, researchers can isolate the individual decision-making process, gaining clearer insights into how users interact with the system, which would otherwise be confounded by group dynamics in multiplayer scenarios.

4. *Multiturn Participation and Bias Mitigation*: A single-player, turn-based framework facilitates multiturn participation, a key feature in improving decision-making accuracy and mitigating biases. Research in Serious Games has shown that repeated exposure to decision-making tasks in controlled environments help mitigate heuristics and biases (Legaki et al., 2021). Players in single-player games can pause, reflect, and reengage with the task without the pressure of synchronous group dynamics, reducing the likelihood of rushed or suboptimal decisions driven by competitive or social influences.

5. *Gamified Engagement and User Motivation*: Gamification research shows that intrinsic motivation—driven by personal achievement, mastery, and exploration—is often higher in single-player environments where users can progress at their own pace (Nah et al., 2014). Multiplayer environments, while fostering collaboration, can introduce competitive or performance-related stress, which may detract from the user's focus on the decision-making task itself. Turn-based participation can maintain engagement through *achievement-based progression*, allowing users to feel in control and supported by system feedback at every step.

6. *Controlled Moderation and Feedback Management*: Single-player frameworks also lend themselves to more controlled feedback mechanisms. As outlined in this research, one of the primary challenges in participatory design is ensuring that feedback is digestible and manageable for users (D'oca et al., 2019). A turn-based system can curate the timing and presentation of feedback, ensuring that users are not overwhelmed with too much information at once. This aligns with Utility Theory (Fishburn, 1979), which argues that decisions are best made when feedback is clear and comprehensible.

To Conclude, while multiplayer games offer valuable real-time interaction, the single-player, turn-based approach is not only valid but, in many cases, advantageous for *research purposes*. By reducing cognitive

load, offering immediate and reinforced feedback, and providing cleaner data collection environments, single-player frameworks can yield more precise insights into user behavior. This approach aligns with established cognitive and behavioral theories and offers a robust foundation for the kind of informed decision-making processes central to research. This perspective is supported by a body of knowledge spanning serious games, cognitive psychology, and decision-making theory, making single-player games a scientifically sound method for gamified framework research.

The next debate over the experiment design may address how the author ensured the computer proficiency of the participants. Simply put, the question is whether the participants were acquainted with computer software in general to perform the experiment's tasks. This debate has three responses. First, the usability test scored higher than the top 10% of the industry benchmark, falling into the category of *excellent* usability based on the SUS test—please refer to Section 5.1 for the detailed report. Moreover, since all participants were early designers, they had completed at least 60 units in a Canadian bachelor design-related curriculum (or its equivalent), ensuring they had interacted with at least one design software. Lastly, the framework offered a standalone tutorial. Nonetheless, the data refinement procedure in Section 4.7 explained that based on tutorial time, possible outlying inputs are refined later. Therefore, in this experiment, there is no need for prior computer proficiency tests.

The last debate may address the choice of designing the main experiment as a within-subject study as opposed to a between-subject study. First, while the literature in HCI and HRI supports the validity of this approach, there is a significant statistical benefit: increased statistical power due to reduced error variance. In a within-subject design, each participant serves as their own control, eliminating inter-individual variability from the error term in the statistical model. Because the same participants experience both conditions (with and without feedback), the variance due to individual differences is controlled, leading to greater sensitivity in detecting true effects. Secondly, a within-subject study can lead to better comparison of subjective experiences and learning effects. The research involves participants experiencing and comparing two feedback modes (feedback vs. no feedback), likely evoking subjective judgments (e.g., perceived usefulness, clarity). Having participants try both conditions allows them to directly compare experiences, enabling more nuanced feedback and helping isolate the influence of feedback on decision-making behavior and design outcomes. Moreover, trends in individual learning or adaptation patterns can be observed, which are central to evidence-based and iterative design workflows. Therefore, the main experiment is conducted as a within-subject experiment.

2.8 Discussion

The theoretical framework presented in this study highlights the complex and often fragmented nature of supporting informed decision-making in early-stage design—particularly for non-expert designers. While the literature provides a rich historical and interdisciplinary foundation, spanning architecture, cognitive psychology, environmental behavior, and game studies, it also reveals several recurring limitations and unresolved tensions.

First, early efforts in design participation made significant strides toward empowering users through novel tools and representations. However, they often relied on preconfigured options or static visualization methods that limited user agency and failed to provide meaningful performance feedback. These limitations remain prevalent in many of today’s commercial configurators, which prioritize automation and feasibility over clarity and understanding. Tools like TestFit, Forma, and Digital Blue Foam, while technically sophisticated, still present steep learning curves, limited transparency, and generalized user models—falling short of enabling true informed decision-making for non-experts.

Second, the reviewed literature reinforces that non-expert designers are not passive recipients of design but active agents capable of engaging with performance trade-offs—if systems are properly structured. This distinction is critical. Unlike participatory frameworks focused on the general populace, this research targets a more specific user group: early-stage designers with limited domain expertise but high motivation to engage meaningfully in the design process. For this group, the key challenge is not access to options, but access to intelligible, context-aware feedback that augments perception and supports reflection.

Third, an important theoretical insight emerges from the review of behavioral and cognitive models. Across frameworks like the Health Belief Model, Theory of Planned Behavior, and Social Cognitive Theory, recurring themes include the importance of feedback timing, perceived control, reinforcement, and cognitive scaffolding. These elements align well with principles from game design and suggest that gamification is not simply a stylistic choice but a cognitive strategy—one that can enhance learning, reduce bias, and structure decision-making in a controlled and digestible manner.

Moreover, the analysis of non-expert engagement methods underscores the limitations of traditional participatory mechanisms. Approaches like surveys, citizen panels, or consensus workshops often fail to deliver two-way communication, struggle with information overload, or rely too heavily on expert mediation. In contrast, gamified, simulation-based environments allow users to interact directly with the

design system, receive immediate performance feedback, and adjust their actions over multiple turns—leading to a more nuanced and personal understanding of the consequences of their decisions.

In conclusion, the theoretical insights outlined in this section validate the need for a new category of tools—ones that go beyond data visualization and automation to actively mediate meaning, support decision learning, and scaffold performance literacy for non-expert designers. The reviewed models and precedents converge on a shared understanding: that decision-making is not a one-time event, but a layered, iterative, and perceptually rich process. These insights directly inform the structure and rationale of the gamified framework proposed in this research.

3 RESEARCH SCOPE AND OBJECTIVES

In the context and scope of this thesis, Table 6 expounds the key terms.

Table 6: Definitions of key terms

Definitions
Games and Serious Games (game-enabled digital environment): “We describe computer games in terms of being interactive [...], based on a set of agreed rules and constraints [...], and directed toward a clear goal that is often set by a challenge [...]. In addition, games constantly provide feedback, either as a score or as changes in the game world, to enable players to monitor their progress toward the goal [...]. In speaking of a serious game, we mean that the objective of the computer game is not to entertain the player, which would be an added value, but to use the entertaining quality for training, education, health, public policy, and strategic communication objectives” (quoted in Clark, Tanner-Smith, and Killingsworth 2016)
Framework: Generally, a framework is a conceptual structure, model, or set of guidelines that provides a systematic way to organize and analyze complex data, concepts, or phenomena. Frameworks can take various forms and are used to establish a theoretical foundation for research, guide data collection and analysis, or provide a structure for interpreting and presenting research findings (Creswell and Creswell 2018; Maxwell 2013). In the context of this research, a system is referred to that connects, creates, and describes links between parts (research data, method, methodology, and analysis) towards a cohesive whole (research results).
Framework usability: “The extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.” (quoted in Moreno-Ger et al., 2012). Framework usability is closely related to “task completion” in HCI and HRI.

Participation:

the process of ‘non-expert’ individuals, acting as ‘value consultants’, to combine technical facts with their values into a set of conclusions and recommendations’ (derived from Beierle, 1999). In the context of this thesis, a more accurate term for this interpretation is informed decision-making.

Evidence-based decision making:

integrating individual expertise and external evidence through systematic research (Nisha and Nelson 2012). Non-expert participation is a collection of mechanisms intentionally instituted to involve the public or their representatives in making informed administrative decisions (Beierle and Cayford 2002).

Informed decision making:

In the scope of this thesis, Informed Decision-Making is defined as a process wherein individuals make choices based on comprehensible information, enabling them to evaluate alternatives, predict outcomes, and align decisions with personal or contextual goals and values. There are four key characteristics to this definition: 1) Information Availability: The user is provided with necessary, relevant, and evidence-backed information. 2) Information Comprehension: The user can understand and process the information (considering cognitive load, literacy, and presentation format). 3) Autonomy: The decision is made freely without coercion or manipulation of any higher agency. 4) Goal Oriented: The decisions are made under a narrative or objective. 5) Outcome Awareness: The user anticipates and is made aware of the consequences and trade-offs of their choices.

Also, The term "**Non-expert designer**" (the target user group) with no or limited proficiency in building science or building physics can be vague; For further clarification on the topic, the audience is positioned into Bloom’s six-stage taxonomy of ‘Cognitive Skill Levels’ and Dreyfus's five-stage Model of ‘Skill Acquisition Stages’ (Dreyfuss and Dreyfus 1980; Mayer 2002). In Bloom’s levels, the bottom three categories are named Remembering, Understanding, and Applying, which refer to participants who recall facts and concepts, can explain ideas, and use their knowledge in new situations. The participants fall into the first three categories of Bloom's taxonomy. In Dreyfus's categorization, the first three of five categories are named Novice, Advanced Beginner, and Competent. These are defined, respectively, as users who follow rules without contextual judgment, those who recognize situational patterns, and those who can

prioritize tasks based on goals. The participants fall into the first three categories of Dreyfus's skill acquisition stages. The explanation above helps position the participants in the related taxonomy. However, clear rules were in place to define the range of participants practically. For the validity and repeatability of the study, the term early designer is defined in the scope of this thesis with inclusive and exclusive rules: *Inclusion*: Participants must have completed two years or less of formal education/training in design (e.g., architecture or equivalent). *Exclusion*: Participants with six months or more of professional experience in roles requiring building performance analysis (e.g., energy modeling, structural engineering). Therefore the target group of the experiment includes non-expert designers, interdisciplinary collaborators, emerging professionals in complementary fields such as building modelers, etc. A pre-screening process of participant acquisition and self-reported information ensures adherence to this range, which was also monitored by the author during data collection.

3.1 Research Gap

In the previous section, the following gaps or significant areas needing more research have been identified: the lack of a holistic approach in the design process, especially concerning the significance of building facades, and the importance of using data-driven, informed decision-making frameworks enabled by game engines. Based on the recent trend of using game engines, there are potential benefits in collective problem-solving and the positive effects of gamifying design tasks.

While the significance of informed decision making and user-centered approaches in architecture has been widely acknowledged, most existing tools and methodologies remain oriented toward expert users, requiring technical knowledge and fluency in domain-specific software or performance metrics. Non-experts are often excluded from early-stage design processes due to the inaccessibility of performance data and the complexity of design interfaces. Moreover, while recent platforms offer automation and performance evaluation capabilities, they rarely provide feedback in a format that supports comprehension and agency for non-specialist users. Current design tools tend to prioritize expert control, offering limited affordances for iterative learning, exploration, or decision transparency for lay participants.

Additionally, although gamification has been applied in adjacent fields such as education, training, and behavioral change, its potential in architectural design tools—particularly as a mechanism for enabling informed decision-making—remains underexplored. Theoretical models from human-computer interaction and cognitive psychology emphasize the importance of feedback, motivation, and reinforcement in shaping decision-making behavior, yet these insights are seldom operationalized in participatory architectural frameworks. The absence of a cohesive, interdisciplinary approach that merges real-time feedback,

performance awareness, and intuitive interaction design represents a critical gap in both research and practice. This study addresses these gaps by proposing and evaluating a gamified, data-driven design framework tailored for non-expert designers. It investigates how encapsulated feedback across multiple performance domains can influence user behavior, support informed choices, and enhance engagement in early-stage design tasks—offering a new model for inclusive and intelligible participation in architectural decision-making.

3.2 Aim and Objective

Based on the identified research gap, the aim of this study is to *investigate how gamified, real-time feedback systems can support informed decision-making among non-expert designers during early-stage architectural design*. The study seeks to evaluate whether feedback, when delivered through encapsulated and comparative metrics, can meaningfully influence user behavior, design performance, and engagement without requiring prior technical knowledge.

The aims lead to defining the research objectives as 1. To design and implement a gamified decision-making framework that provides real-time performance feedback across structural, environmental, and cost-related criteria. 2. To assess the impact of feedback on decision-making time, behavioral consistency, and learning trends in a controlled within-subject experiment. 3. To evaluate whether non-expert users can make better-performing design decisions using encapsulated and relative feedback, despite not being exposed to detailed technical data. 4. To validate the usability and comprehensibility of the feedback mechanisms through standardized tools (e.g., SUS), interaction logs, and post-experiment interviews. And 4. To explore how gamification strategies affect engagement, comprehension, and the ability of participants to align design preferences with performance goals.

Thus, the following research questions are formulated:

RQ1: *What are the mechanics and characteristics of a game-enabled informed decision-making framework to help non-expert designers make design configurations?*

RQ2: *How does the presence of real-time feedback influence the decision-making behavior of non-expert designers in early-stage architectural design tasks?*

RQ3: *Can encapsulated performance feedback across structural, environmental, and cost metrics help non-expert users make better-informed design decisions without explicit technical knowledge?*

RQ4: *What is the effect of gamification strategies on the engagement, comprehension, and decision-making consistency of early designers within a participatory design framework?*

For the evidence-based experiment design, the aim is to test the following two hypotheses:

H1: *Participants receiving real-time performance feedback will demonstrate significantly different decision-making times compared to those who do not receive feedback.*

H2: *Gamified, encapsulated feedback can steer non-expert designers toward better-performing design configurations across multiple metrics without overwhelming them with technical complexity.*

3.3 Variables

This study examines variables across three categories:

- Independent Variables (IVs):
 - Primary IV: Presence/Absence of Real-Time Feedback (binary: "informed" vs. "non-informed" conditions).
- Dependent Variables (DVs):
 - Behavioral Metrics: Decision Time, Design Configuration, and keystroke dynamics.
 - Performance Metrics: Seven design performance metrics encapsulated into three categories of structural, environmental and cost (see section 4.9.1 for detailed definitions).
- Control Variables:
 - Participant Factors: Design proficiency level (all participants were early designers with no expertise in building physics), Computer proficiency
 - Environmental Factors: Lab setup consistency (hardware, session timing, etc.)

It is important to note that in the analysis section, the first round of statistical analysis is conducted pairwise. For example, the initial examination involves the effect of receiving feedback (IV) on decision-making time (DV), which is elaborated in section 5.2. In that case, after demonstrating the difference between the two feedback groups on decision-making time, a separate section (5.2.1) examines trends in performance metrics to see if they can explain the effect as a mediating variable. Therefore, in each section of chapter 5,

the role of some variables is adjusted to elaborate on their relationships. These variables collectively operationalize the study's core inquiry.

3.4 Research Methodology

Stemming from the identified research gap and objectives, a **mixed-method with triangulation** methodology is proposed. This approach has a triangulated structure; The three elements constituting the proposed research methodologies are: **A. Experimental Design research**, **B. Correlational Research**, and **C. User-centered Evidence-based Design** approach. The reason behind selecting this triangulation is rooted in several factors. First, as this research partially falls under evidence-based behavioral research, it deals with both qualitative and quantitative data. Therefore, the evidence-based research methodology, with its tools and methods, is necessary for this exploration. Second, since the data requires exploratory or confirmatory examination of the research objectives derived from an *experiment*, the principles of experimental design research are needed for framework development, usability testing, and results examination. Third, due to various qualitative and quantitative analytical processes needed to establish possible relations between variables, correlational research is used. To elaborate, correlational research principles enable the author to examine the effect of the dependent variable -feedback on the independent variable -informed decision-making, with user behavior as a mediating variable. This division of variable study stems from the experimental research methodology, and the correlational methodology can be used for its confirmation. To elaborate, This study adopts a pragmatic mixed-methods methodology, grounded in the need to holistically evaluate how gamified feedback systems influence non-expert decision-making in early-stage architectural design. The methodology integrates three complementary strands: (A) experimental design research to isolate causal relationships between feedback and behavioral outcomes, (B) correlational analysis to identify patterns linking dynamics to design performance, and (C) user-centered, evidence-based design (EBD) to iteratively refine the framework's usability and pedagogical efficacy. Rooted in behavioral theory (e.g., Social Cognitive Theory, Utility Theory) and participatory design principles, this triangulated approach balances quantitative rigor (e.g., controlled within-subject experiments, non-parametric statistics) with qualitative depth (e.g., thematic analysis of user feedback) to address the multidimensional nature of informed decision-making. By embedding iterative cycles of alpha/beta prototyping, the methodology acknowledges the complexity of human-computer interaction, prioritizing ecological validity through real-world task simulations while maintaining experimental control over confounding variables. The synthesis of these strands enables a nuanced exploration of how and why gamified feedback steers non-experts toward performance-aware design choices, bridging theoretical insights from serious games, behavioral psychology, and architectural pedagogy.

The triangulated mixed methodology of the proposed research calls for using various research methods and tools, which are explained henceforth.

3.5 Research Methods

The novel mixed methodology of this thesis requires using various research methods and tools. Broadly, two main methods bridge the different parts of this research, as explained hereafter. However, specific tools are also used at each stage, which are elaborated afterwards.

1. *Computer-based Simulation and Generative Research*: The research is done primarily through computer-based simulations taking advantage of the computational power in mathematical calculations and fast-paced analysis. As a research method, using generative design exercises, participants can explore their visions, dreams, needs, and desires, creating rich information that can be used to develop concepts. The notion aligns with the generative design agenda in computational design. It addresses the question holistically and collectively, leading to many design alternatives. Various criteria can then be used to evaluate each option, leading to selecting the best candidates.

2. *Digital Prototypological Research*: Heisel, inspired by methods of biomimetics, describes this method as “joined from the terms” “prototype” and “typology” prototypology represents a full-scale application. It means an experiment and proof in itself to effectively and holistically discover all connected aspects and address unknowns of a specific question, yet at the same time is part of a bigger and systematic test series of such different typologies with similar characteristics yet varying parameters” (Heisel and Hebel 2019). The study uses a digital prototype (i.e., a variety of design configurations), followed by feedback and evaluation. The idea of rapid prototyping in digital fabrication, which enables partial tests and experiments before full-scale construction, is adopted here in a digital context.

Apart from the aforementioned two, other research tools were used at each stage; At the Experiment Design, the main research method is (3) *Within-Subject Controlled Experiment*: with presence/Absence of real-time feedback (structural, environmental, cost) as Independent Variable (IV). The Dependent Variables (DV) are feedback conditions. And mediating variables are Behavioral Metrics (e.g. Decision time (DTime), interaction frequency, configuration changes), Performance Metrics (e.g. Scores across 7 criteria (e.g., structural displacement, operational cost). In this stage (4) *Statistical Tools* are used to explain the correlational or causal effect. Methods used include but are not limited to (4.1) Mann-Whitney U Test: Compare DTime distributions between feedback conditions. (4.2) Cohen’s Kappa: Assess redundancy in feedback categories. (4.3) Linear Regression: Model micro-learning trends (DTime vs. attempt number).

Moreover, many descriptive comparisons are made based on variance, mean and other distribution measures.

At the Correlational Analysis stage, in order to examine possible relations between interaction patterns (e.g., time spent on feedback review) and final design performance, other statistical tools are used such as (4.4) Pearson's r to link hover time on environmental feedback with solar gain improvements. (4.4) t-test, f-test and ANOVA with repeated measure.

For the User-Centered Evaluation (Qualitative) state, usability testing tools are used. Primarily, (5) *System Usability Scale (SUS)* is used which is a standardized 10-item survey to quantify usability (Grier et al. 2013; Lewis 2018). Also (6) Interviews have been carried out as *Post-Session Semi-Structured Interviews*.

In Iterative Framework Development phase (7) *Action Research Cycles* was utilized. This included: Alpha Testing: Identify usability pain points (e.g., feedback comprehension), Beta Testing: Test redesigned UI/UX (e.g., comparative feedback vs. absolute values) and Main Version: Validate stability via A/B testing (e.g., tutorial effectiveness).

In addition to the type of data and how it can be analyzed, the source of the data is also a critical factor that must be taken into account when considering data analysis. There are four types (or sources) of research data: observational data, experimental data, simulation data, and derived or compiled data. The four categories and their features are shown in Table 7.

Table 7 :Research data and Acquisition methods and analysis tools. (Research data categorization derived from Hensel and Vujovic (2022)).

Research Data	Acquisition Method	Research Method/Tool	Data Category
Observational Data	Obtained through observation of participant by the researcher during different versions and post session interviews.	Human observation,	Mixed
	Obtained through a record of activities, behaviors, and interactions of player inside the framework.	Surveys, Framework log (Recorded Realtime), - Methods 3,4,5,6,7	Quantitative
Experimental data	Collected using active intervention of the researcher or interactions and prompts by the framework.	Statistical analysis - Methods 4,5	Quantitative
Simulation Data	Collected based on users' interaction log.	Computer-based simulation, Generative Methods - Methods 1,2,3,7	Mixed
Derived/ Compiled Data	Creating new data, based on compilation or augmentation of the existing data.	Data-set processing, Data Augmentation and Data synthesize methods - Methods 1,3,4,5,7	Quantitative

To conclude, seven research tools were utilized to explore the relationships between the variables and address the research question.

3.6 Limitations

The technical development of the proposed framework may face limitations, ranging from technical issues to scope and a lack of prior studies. These limitations are addressed in the section.

3.6.1 Sample Size

The study employs approaches from the field of human-computer interactions to calculate sample size. A priori power analysis is rarely used in human-computer interaction research since it requires knowing the variance in a sample and the difference in means on the dependent variable (effect size) before collecting data (MacKenzie 2013). As a result, the recommended procedure is to review the available literature (MacKenzie 2013). If a comparable study reports statistically significant results with a certain number of participants, then using that number of participants is a reasonable decision (Arif 2015). Arif compiles a report on literature in user studies in HCI and notes that for novice users, 11 to 14 participants are needed.

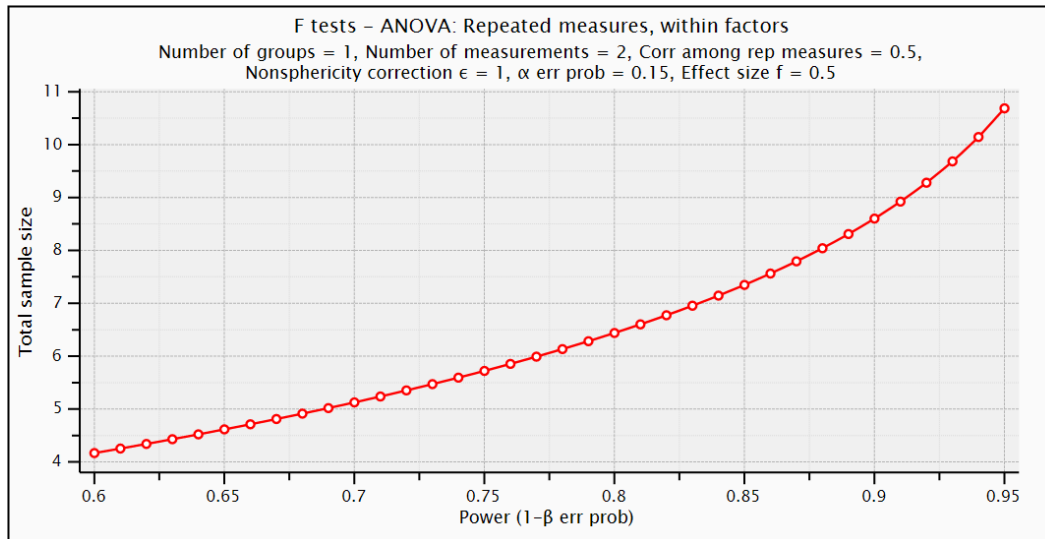


Figure 3: Sample size (required number of interactions)

However, in this thesis, a more accurate procedure is used. Based on the G*Power a priori test (Figure 3), 11 experiences are required. Since each participant experiences two conditions, at least 22 participants are needed to establish possible significant results.

3.6.2 Limits on Data Transfer

Data transfer is probably the most challenging part of creating a design platform inside a game engine. There are two key reasons for the challenge. First, the rendering structure in the game industry requires a considerable amount of precomputing before gameplay begins. Consequently, these platforms provide limited tools for creating or modifying geometric data during the game. The design process and providing visual feedback to the user take place during the game. Bridging between the two is, therefore, a significant challenge. In the study, methods are studied to purge, compress, and optimize data generated in design software to be exported and read by the game engine. At this stage, computational resources are required in terms of time, memory, and processing power and they are related to the final intended detail level in the environment. Additionally, these platforms cannot perform environmental or structural analyses. Therefore, the metadata, together with geometrical data, comprise the information that needs to be read by the hosting engine. As a result, the second problem is maintaining real-time interaction with the user. Increasing the level of details in the study and the architectural choices will result in higher volumes of data sent and received. In addition, alternative protocols and methods will be tested for faster and more convenient data transfer.

3.6.3 Lack of Prior Studies in the Research Area

Despite many previous studies in the field, the comprehensive approach of this study has limited precedence. There has been research on two or three parameters, such as including BIM data or implementing energy consumption and behavior into game engines. In this study, however, various data will be analyzed, and methods explored to accommodate multiple contradictory design choices at the same time. Hence, the scarcity of similar studies can be a limitation.

4 THE FREAMEWORk ARCHITECTURE AND EXPERIMENT DESIGN

Revisiting the aim and objective of this thesis, it must be emphasized that the goal is to propose an interactive, augmented framework based on indirect behavioral evidence that empowers non-designers to make informed design decisions. Therefore, the experiment is designed as a design configuration task for non-expert designers at early design stages. The indirect user behavior analysis feeds back into the development process and provides solid evidence on the usability of such frameworks. Thus, the design task chosen is the design of an exoskeleton: the framework is designed to engage non-experts in designing a retrofitting solution for a historical building in a downtown building in Calgary, CANADA. The design task is to redesign the facade of the building and add livable spaces attached to the original structure. The design subject is an exoskeleton with customizable balconies that each participant can engage in. An exoskeleton is proven to have numerous environmental advantages (Weber, Mueller, and Reinhart 2022b), as well as minimizing the repurposing efforts

This section provides an overview of the experiment in 4.1, followed by developments and iterations of the framework in 4.3, 4.4, 4.5. After discussing lessons from each version, the main experiment is outlined in section 4.6. The complete process of the main experiment and user experiences are explained in section 4.7. Lastly, performance measures and feedback data are defined in section 4.9.

4.1 Overview

With the development of the framework and the achievement of a stable version through revisions from alpha and beta experiments, data collection began. The following section explains the data collection procedure and results. However, it is beneficial to revisit the conceptual framework, which forms the main and subsidiary hypotheses. A simplified flowchart (Figure 4) explains that there are two treatments in the experiment: A) receiving performance feedback while making design decisions, referred to as the ‘informed’ group, and B) not receiving any feedback information, referred to as the ‘non-informed’ group. The experiment is designed as a within-subject study, meaning each participant experiences both treatments (measures to prevent different biases are addressed subsequently). Hence, the interactions of the two groups regarding performance criteria, mediated through the design gamified platform, are the subject and deeper analysis in this section. As explained in earlier sections, non-direct user assessment plays a crucial role in this study. Therefore, the interaction log, which holds the behavior, is divided into smaller measures (Figure 4). Measures such as interaction times, order effects, keystroke dynamics, configuration uncertainties, and

many others are extracted from the raw log of interactions and analyzed accordingly in the following section.

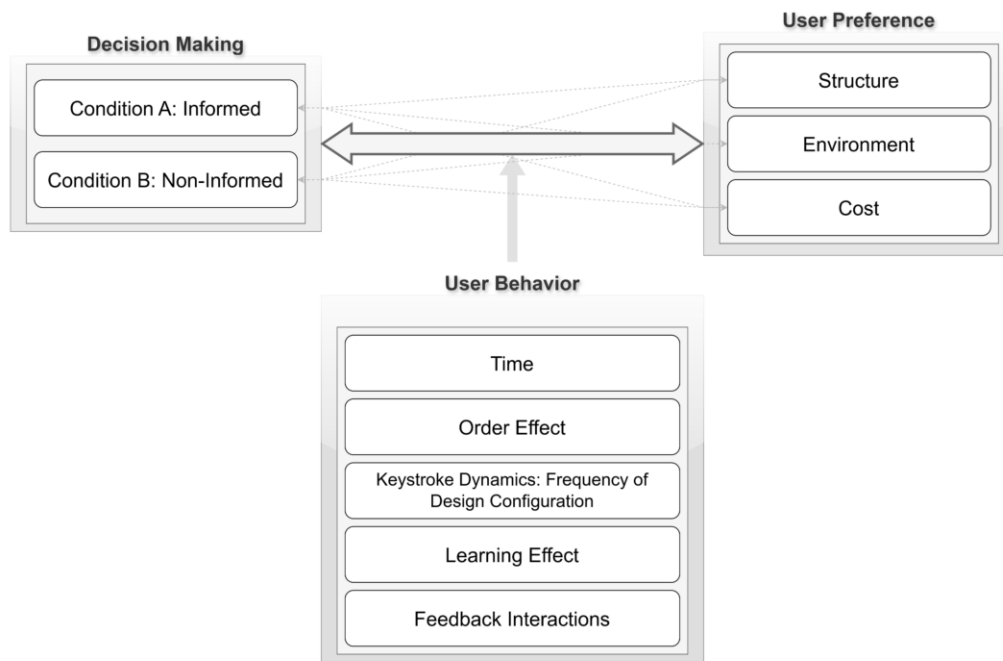


Figure 4: Conceptual overview diagram

In other words, performance measures are always dependent variables (DV), while the two conditions of feedback are independent variables (IV). At a lower level, when comparing the effect of user behavior on this connection, user behavior acts as a mediator variable. This means it is explored *how* it affects the relationship between IV and DV.

4.2 Gamification Strategies

In the previous sections, the importance and effect of serious games on different aspects of design were explained briefly. However, this section aims to explicitly shed light on which game-based or gamified strategies are embedded in the system. The first category of suggestions, briefly explained in previous sections and based on behavioral theories, is quoted here again, followed by complementary guidelines from serious games.

To design a framework for engaging non-expert designers in the early design stages, the proposed framework benefits from:

1. A holistic approach that engages users with various sources of data, such as structural, environmental, and cost-related information.
2. The comprehensive approach is augmented with prompt multitype feedback information such as numerical, nominal and visual cues.
3. Cues to action or stimulus.
4. Ability to immediate demonstration of choices' effectiveness and consequences
5. Self and environment appraisals
6. Appropriate medium of communication, ensuring the right amount or comprehensible dosage, and implementing a feedback mechanism to ensure it is understood
7. Capability of providing positive, negative, and neutral feedback in self-earned or system-given formats
8. Following the reinforcement process: explaining what happened and why
9. Effective communication through multimedia rather than just written text
10. Reminders that support decision-making
11. Pretesting, which can be an effective mechanism to check the viability of the previous steps

Also based on a recent classification of gamification strategies (Vrcelj Božić et al. 2025b), the following point were also used:

12. Use of visually enhanced feedback and psychological affordances and feedback especially in tutorial and exploration phases (for example, in the process system, the user received visual reinforcement at the end of each tutorial section, such as changes in lighting conditions and visual effects like small fireworks)
13. Storytelling, where the whole experiment has a clear objective (in the proposed case, it was explained that you are the buyer of an office building and can make decisions regarding an exoskeleton, allowing users to walk around their office)
14. Having points and levels in the experiment, which helps engagement of the users
15. Small puzzles especially in learning phase
16. Small lessons with narratives, embodied in the proposed framework as an informative tutorial board, explained what an exoskeleton is and its benefits.

Those 16 principles were all embedded in the framework. The next section provides more detail on the implementation process of those principles.

A design workflow using a game engine is proposed and developed. Unreal Engine (Epic Games 2021) is the host environment for this workflow. Tools for structural and environmental design compatible with

Rhinoceros (Robert McNeel & Associates 2020) are utilized to provide analytical data. Geometrical constraints also can be modelled algorithmically within Rhino's visual programming environment Grasshopper™. Other pipeline elements are coded in Python 2.7 and its implementation for the .NET framework IronPython (IronPython Contributors, Microsoft Corporation, and Python Software Foundation 2013; Python Software Foundation 2010).

Maintaining a real-time engagement with the user is one of the primary challenges in the cross-software framework. Higher pauses or response time, according to studies, may disturb decision-making. In other words, a successful design process requires an uninterrupted flow of information (Brady 1986; Jones and Reinhart 2018). Since this is related to the amount of information conveyed, data types and formats must be examined after each development step. Another issue involves the lack of previous similar studies to determine sample size (MacKenzie 2013).

4.3 Pre-alpha Tests

The framework evolved through several versions, with almost every aspect changing, from modes of interaction to functionalities and features. Three main versions are explained succeeding sections: alpha, beta, and the main version. In each iteration, users were involved, and developments occurred based on their feedback.

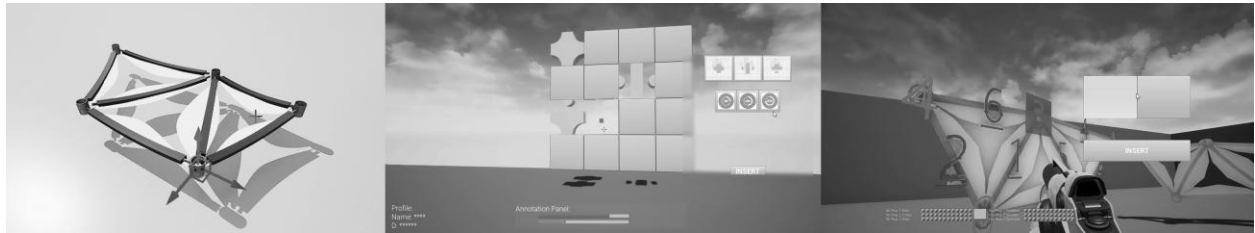


Figure 5: Stills form pre-alpha tests on different modes of interactions

However, in the pre-alpha stage, various methods were tested for interaction and feedback formats (Figure 5). Also, a key development that remained in the final version is the architecture of the backend service, which handles different aspects of design configurations. For the mode of interaction, methods such as placeholders prompting pop-up selection menus, menu hierarchies, and selection with projectiles were examined.

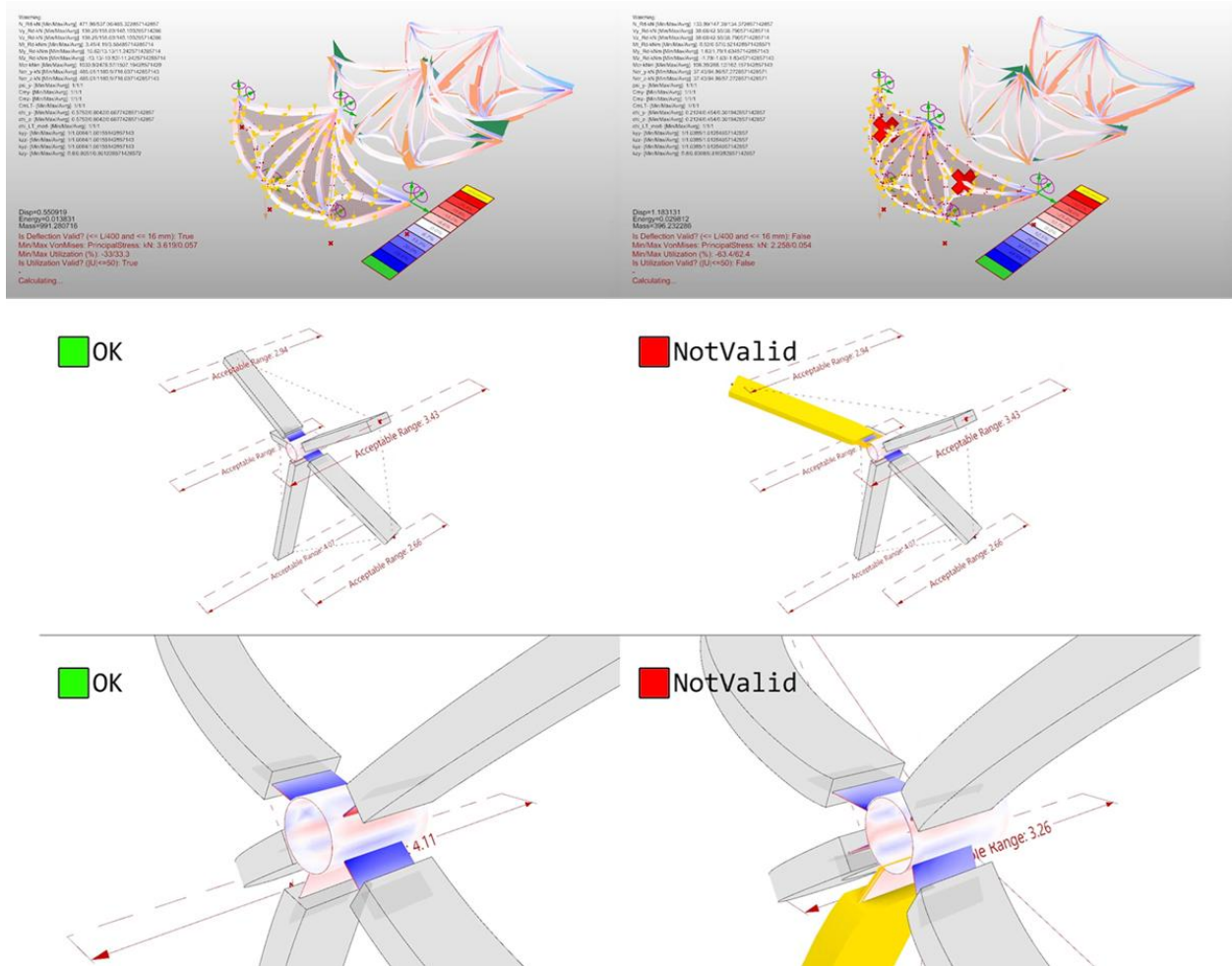


Figure 6: Backend services handling geometrical and structural constraints

For the backend process, algorithms were developed to retrieve the allowable modification range based on geometrical constraints. Additionally, analytical algorithms were created to check structural performances, and if configurations failed to generate a valid option, they automatically snapped to the closest valid one (Figure 6).

4.4 Alpha Version

The most important takeaway from the alpha version (Figure 7) was that users struggled to comprehend feedback data when presented as plain text, numbers, or bar charts. Additionally, the complexity and extent of numerous alternatives prevented the backend algorithms from knowing the bounds of all possible options to remap the results. This meant each configuration could not be positioned in a range showing the best to worst performing possibilities. Meanwhile, the option of benchmarking was under examination to overcome this. However, more tests showed that users have a tendency toward comparative results rather

than absolute values, wanting to know if their recent actions performed better or not, regardless of the overall absolute results. Therefore, in the beta version, a comparative option for feedback was added while keeping the option to see a full detailed report available.



Figure 7: The framework development stages: Alpha version

Another key factor that changed the course of development was the mode of interaction. In the alpha version and earlier versions, interaction between the user and the object in the 3D space of the game happened through projectiles and then pointers. However, tests showed that users were more familiar with the function of a 3D pointer. In 3D design software, the default mode of interaction allows the mouse cursor to work in both a 2D layout and 3D space simultaneously. This means it can select menus on a 2D layout and, when used in 3D space, select objects under the cursor after ray tracing. Therefore, in later versions, the mouse's function as the main means of interaction was developed to be more like conventional 3D software. This behavior in HCI is known as a 3D user interface (3DUI).

4.5 Beta Version

The beta version (Figure 8) also included an AI assistant aware of the experiment's context and key bindings. It was designed to help users find keys if needed and respond to questions about the experiment's objective. However, beta testers showed limited need for interaction with this feature. Consequently, in the latest iteration, the most frequently asked questions were interpreted into a comprehensive tutorial experience. To limit the number of dependent variables, this feature was not transferred into the main experiment.

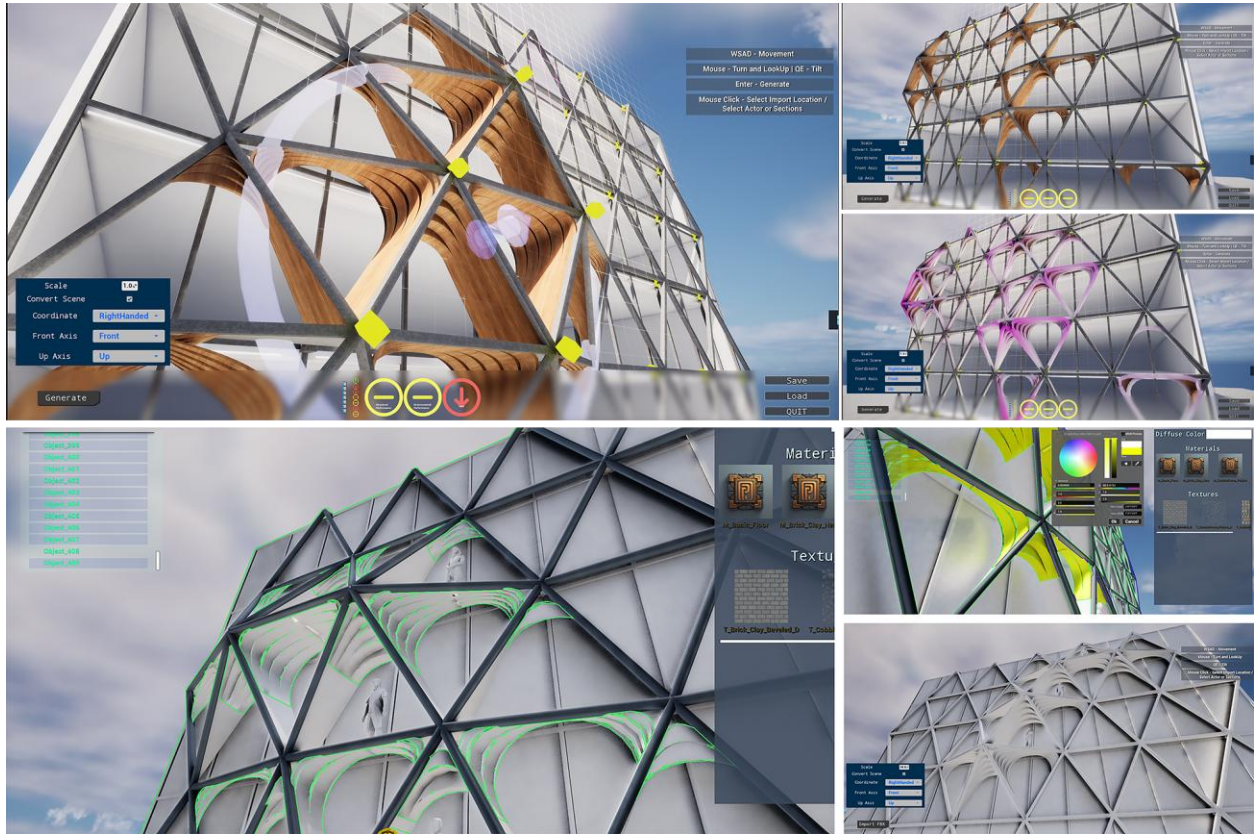


Figure 8: The framework development stages: Beta version

The beta version also allowed users to change the color and texture of the generated geometry. Again, testers showed limited interest in using that option. Therefore, to simplify the experience and variables, this feature was not migrated into the main experiment.

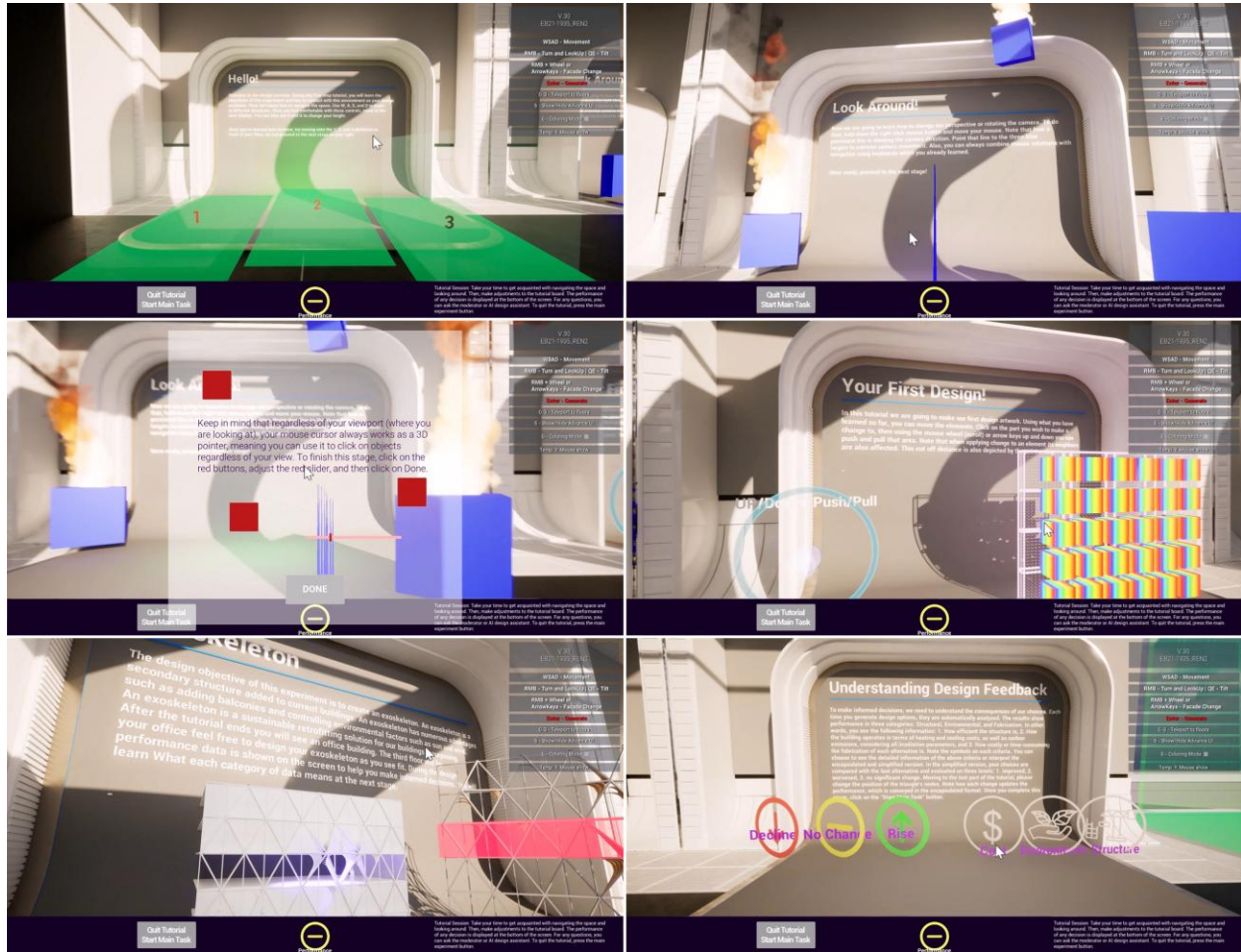


Figure 9: Stills from various stages of tutorials (user p14)

Another key finding from the beta test was the need for an elaborate tutorial session to teach interaction step-by-step instead of providing a list of all possible key bindings and functions. Questions asked by testers led to small lessons designed to be delivered at different stages of the tutorial session. Thus, one of the key takeaways from the beta version was designing a whole new level of tutorial and learning at the beginning of the experiment (Figure 9).

4.6 Main Version

The main version (Figure 10) incorporated all the lessons learned from previous iterations and included major revisions. It began with a comprehensive eight-step tutorial. The first three steps taught users how to navigate the space and select objects through the 3D user interface. The fourth step was a small quiz designed to teach users the game's mechanics and how they can interact with and modify objects. The fifth step introduced the concept and definition of an exoskeleton. The next step presented the narrative and objective of the experiment. Step seven, titled "Understanding the Feedback Information," introduced the types of feedback, their definitions, and their effects on the building, as well as the icons and colorings of the experiment. The last step was another quiz designed to test all the functions users need to interact with the experiment. It was a simpler version of the interface and events of the main objective. After completing this section at their desired pace, a two-minute recap video explained all the functions and objectives one last time. The video was annotated and explained all the interface sections. At the user's discretion, by pressing a button, the main experiment would begin. At this point, all interactions were logged in the backend. Each participant changed the design configuration as they pleased. Each user experienced both conditions of receiving and not receiving feedback in a randomized order, as explained in the data collection procedure. Each condition ended at the user's discretion whenever they were satisfied with their decisions.



Figure 10: Stills of the Main version and experiment, extracted from actual footage of a user interaction

Not only was this the most technically advanced version with functionalities like viewing thumbnails of structural FE results and a detailed environmental report, but it was also more physically accurate and advanced. Surrounding details and furniture were added, and interaction could happen both from inside and outside the building. The user could navigate a fully designed building situated in its proper context with adjacent cityscapes.

To clarify the evolution of the platform across its alpha, beta, and main versions, the Table 8 outlines the distinguishing features, number of test users, goals, observed issues, and subsequent changes at each development stage.

Table 8: Summary of framework developments across versions

Version	Key Features	Number of Users	Goals of Testing	Key Issues Found	Changes Implemented
Pre alpha	Basic Environment development in UE; Simple Numerical Feedback; Pick and choose task	4	Basic usability and interaction comprehension test; Validate backend integration with Rhino/Grasshopper; Communication protocol check	Issues with precomputation approach; Issues with immersion due to pop-ups; Slow backend response times	Replaced feedback mechanism from precomputation to real-time analysis and trained ML algorithms
Alpha	Selection mechanism change from projectiles to 3D click; Feedback format change to bar charts	6	Test spatial navigation and core interaction model; Test comprehend of absolute feedback and charts	Users struggled with navigation; misunderstood improvement direction of metrics	added visual cues for available moves; enabled full camera freedom; Optimized communication
Beta	Design task redefined to exoskeleton configuration; High level interactions introduced (Field manipulation instead of member change); Added performance logging	11	Evaluate effect of absolute and charted feedback on comprehension and behavior	Users found raw/bar metrics confusing; decision-making became trial-and-error; Inconsistent engagement with feedback; Users confused with Task description	Replaced metrics with encapsulated visual feedback (zone & relative); Simple tutorial Added; Feature request added such as teleportation and design from interior space
Main	Extended Tutorials added; Cue to action added and Treatment conditions added; Framework finalized with stable feedback and interface; Scalability enhancements for future metrics; Documentation of behavioral insights for longitudinal studies	31 (24)	Validate framework efficacy in IDM vs. nIDM conditions; Measure usability (SUS) and performance improvements; Analyze behavioral patterns	N/A (finalized version); behavior and performance logged and analyzed	N/A; Future research directions (see Chapter 7)

4.7 Data Collection procedure

Participants arrived at the lab for integrative design at the University of Calgary. The environment was kept private during all sessions, with only two people present: the author as the moderator and the participant. The participant was then given a hard copy of the ethical approval [by the Conjoint Faculties Research Ethics Board (CFREB) at the University of Calgary with Ethics ID REB21-1935 REN1-2-3]. After reading it, the moderator explained the process and the study's objective, answering any questions.

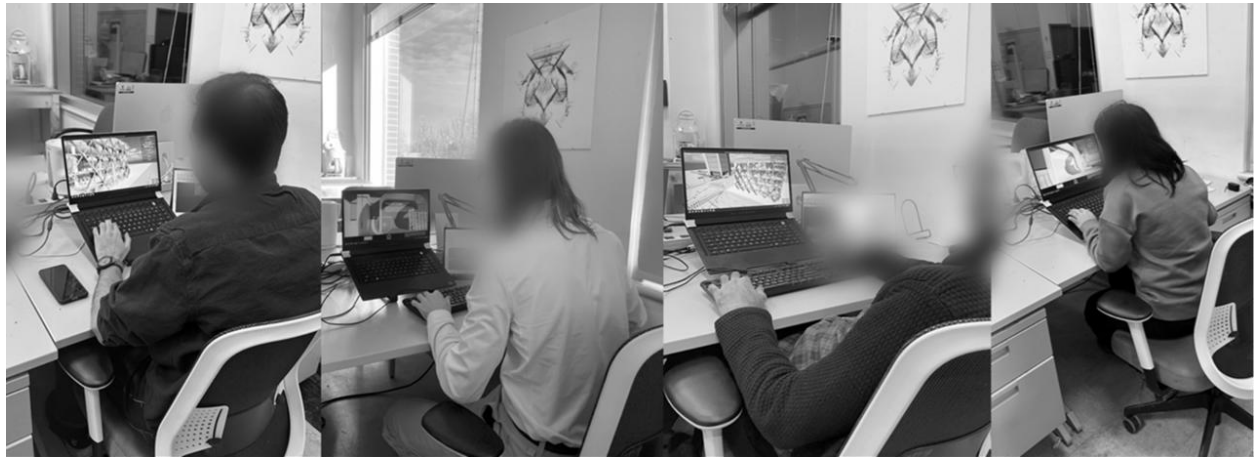


Figure 11: User participation setting

Participants were guided to sit in front of a conventional PC, similar to an office or school setting. A separate keyboard was connected and placed in front of the moderator to ensure the interaction remained uninterrupted (Figure 11³). Each session began with a prompt acknowledging the ethics requirement, asking the participant to click "I agree" or "exit." Then, the participant started the interactive tutorial, a step-by-step guide to learning keyboard events, session objectives, game mechanics and wording of the platform. The moderator was always present to answer questions but otherwise remained unobtrusive. After successfully completing all tutorial steps, which were self-explanatory and self-paced, the participant clicked "start main design task," which displayed a short wrap-up video of necessary information and session objectives. After the two-minute mandatory video, the main experiment began.

³ Photographing was done with selected participants only, with separate written consent and agreement for demonstration purposes.

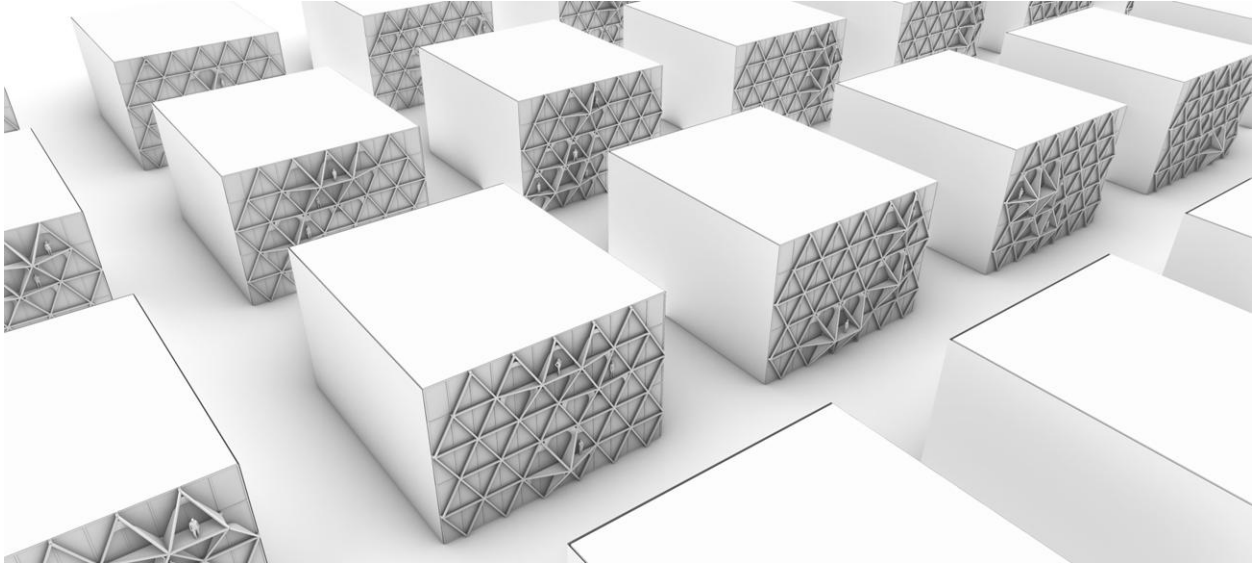


Figure 12: Retrofitted exoskeleton design-space, generated by users of the framework

Each participant designed the exoskeleton facade (Figure 12) twice, consecutively; As the experiment was within-subject, all participants experienced both conditions of receiving feedback and not receiving feedback. To address possible carryover, learning, and order biases, the order of receiving feedback was randomized. Half the participants received feedback during their first design and none during their second, while the other half experienced the reverse. Participants ended the session at their discretion, either due to satisfactory design or reaching a desirable alternative (Figure 13). With that cue, the second treatment began, following the same principles.

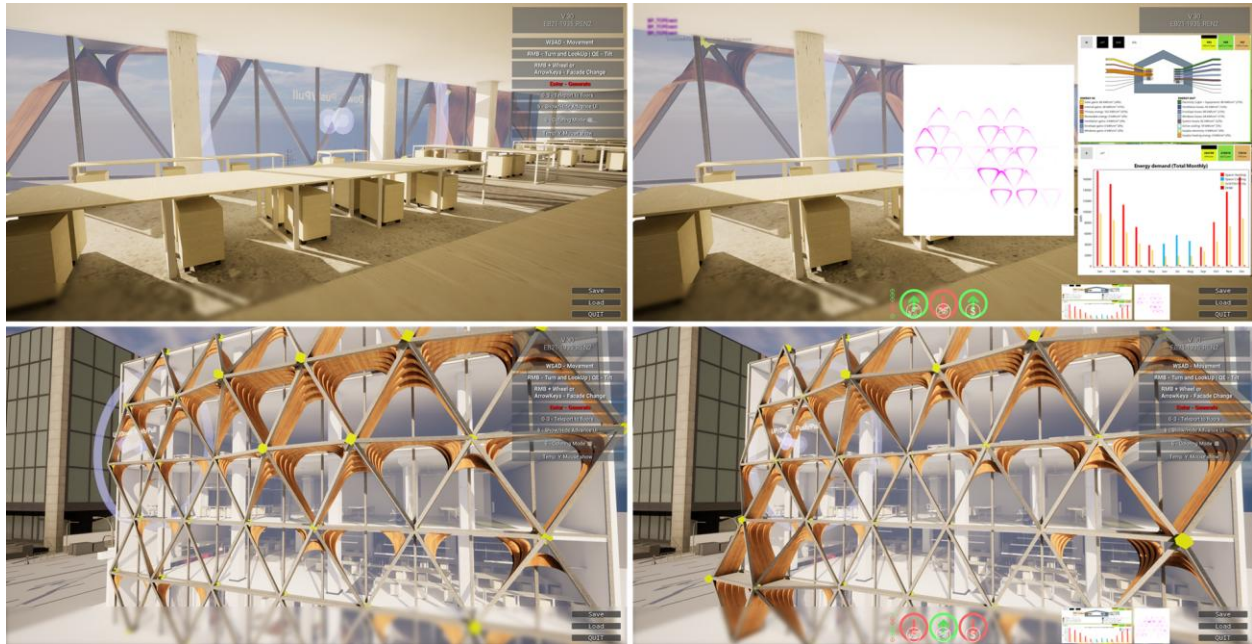


Figure 13: Comparison between two conditions of the within-subject study; left: nIDM, right: IDM

After both designs ended, the participant clicked "end session," saving their data and redirecting them to the last step of the usability study. The usability study (SUS) was conducted as an online form. At the end, the moderator explained they could reach out if interested in the results or if they had any post session concerns.

A voluntary exit informal interview was conducted after completing all previous steps. Participants shared their extended opinions, detailed comments, and feature requests, which the moderator recorded and reflected in the next chapter (See Appendix I for the transcripts).

As the literature suggests, the mechanics of design configurations are designed high-level. This means that the user selects the parts of the facade they intend to change through the 3DUI. By pressing up and down or using the mouse scroll, they make changes to that part. The framework handles the rest, such as the effect of changes on neighboring segments and ensuring the changes remain within valid, constructible bounds.

4.8 Data Refinement

The author, serving as the session moderator, observed the entire experiment and participant behavior. Based on this, the input of five participants were dismissed.

The total interaction time of 12 participants in the Alpha and Beta versions, plus the first 8 participants in the main experiment (including Tutorial Time), formed the benchmark of 28.53 minutes in total interaction time. In the main experiments, the rest of the participants received a visual pop-up message at 60% of 28.53 minutes, reading, "You're doing great! You've now passed 60% of the average participation time. Feel free to continue exploring at your own pace—there's no rush, and you can take as much time as you need." If participants exceeded 200% of that time, their data was dismissed. For this case, no participation was dismissed. The same logic was applied during tutorial time, which averaged 7.47 minutes for Alpha, Beta, and the first eight participants. If a participant exceeded 15 minutes for the tutorial, it was noted. This occurrence did not happen.

Because of the presence of the moderator and the fact that they were free to ask questions if needed, two participants asked extensive questions about the experiment design principles. The author's immediate response or postponing the conversation until after the experiment could both potentially create biases. Therefore, if the irrelevant conversation took more than 25% of 28.53 minutes, the data was dismissed. In this case, the participation of two individuals was dismissed.

Since the data collected at different times, the data input from one participant collected in the evening was dismissed due to fatigue and/or low computer proficiency.

As the data collection occurred both online and in person, the challenge of online latency or ping was also considered. Participants were asked to perform a simple latency test at the beginning, which involved reading the time (in seconds) on the screen. If the delay exceeded the average generation analysis time of 4.54 seconds (=average decision time among early participants), the session was discarded, which resulted in the removal of one participant's records. Otherwise, the data was recorded until further T-tests could examine possible differences between the groups.

If the participant hadn't finished both treatments in one sitting, the session was meant to be discarded. This did not happen. If the participant failed to design anything or reach a desirable alternative, the session was planned to be erased and the framework to be adjusted. This did not happen.

All participation, whether online or in person, occurred between 9:00 AM and 5:00 PM at the participants' local time, using the same machine with the same settings.

[illegible]

Figure 14: A few sample rows of the collected data or log, in their raw form -csv

According to the ethics approval, no personal data that was not anonymous or traceable back to participants was collected (Figure 14). Exceptional pictures of sample participants were taken with separate agreements for presentation purposes. However, 21 participants voluntarily provided an email address and agreed that the moderator could reach out again if needed. After removing the aforementioned data, 24 records were selected for analysis.

4.9 Technical Framework

Possibly one of the biggest challenges in developing holistic informed decision-making frameworks lies in the technical platforms or software that need real-time connections to one another. At the top level, there is the challenge of inter-process communication (IPC), which involves determining how and through which protocols the services communicate. In a conventional design pipeline, one design software hosts all the segments, algorithms, or plugins, such as one code for structural analysis, another for environmental analysis, and so on. However, when the host environment is a game engine, communication can become problematic. Design environments are not intended for interaction management, so even with coding, their architecture does not record every interaction event (Considering that for an indirect user assessment experiment, the author needed a time-bound log with predefined metadata such as state of the other elements). On the other hand, game engines are far more developed in providing various tools for designing custom interaction events. Therefore, when selecting a game engine as the host environment, the rest of the processes must constantly be within reach of the host, making constant communication between the pieces inevitable—at least in the research and development domain. Of course, all dependencies can be coded from scratch and embedded in one platform, but this approach sacrifices the agility of fast-paced experiment designs and makes iterative development very expensive. Among many different IPC methods, lossless network communication was selected in this research due to its lightness, ease of implementation, reliability, and minimal interference with task/thread management of different parties. A bi-directional socket connection was developed by the author for this purpose, enabling communication between generative, analytical, and host segments.

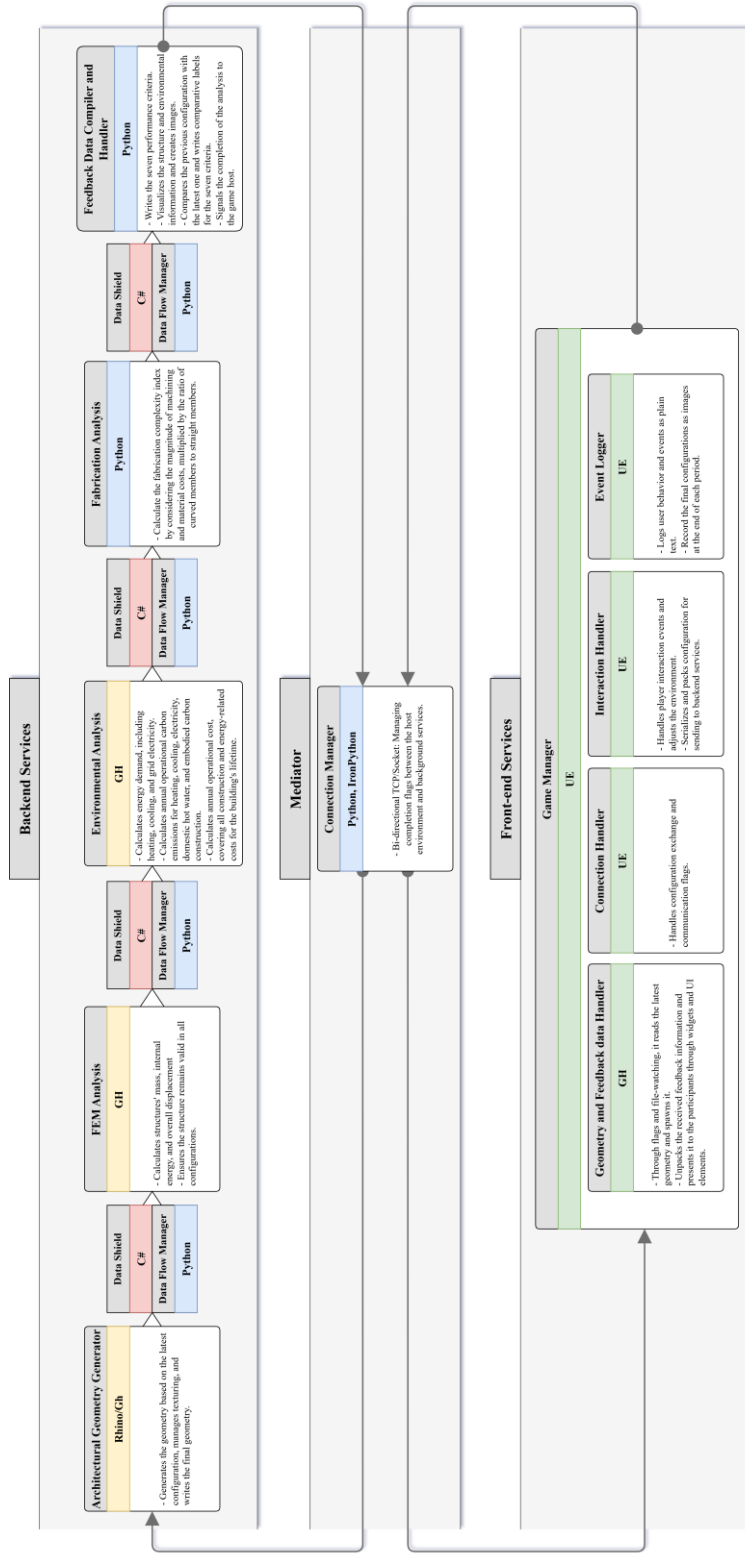


Figure 15: Flowchart of various services comprising the framework

However, on a deeper level, there is a fundamental difference between the architecture of a design platform and that of a game engine. A game engine, prioritizing performance, precompiles all necessary data and packs it before runtime. At runtime, it efficiently manages these resources. In contrast, even pioneering generative design platforms do not follow this logic. There is no distinction between the precompiling stage and runtime in design software. Design software, at least at the generation level, produces the outcome in a single computation. In node-based algorithmic platforms, this is typically referred to as Down/UP streams in microservices: Whenever a new user input is received, all the downstream nodes are recomputed sequentially until the final outcome is achieved. This presents a major challenge in reducing generation time. For example, let's say a decision triggers a change in the state of the generative parameters. As it triggers a recomputation flag in all its downstream segments, the design software calculates all the new changes and eventually outputs the results. This is problematic when some parts are more time-consuming and have lower priority: In this research, the geometrical result is of the highest importance and must be generated swiftly so the player can see the immediate effect of their decision. However, processing the finite-element results and energy consumption is computationally heavier and requires more processing time. A design software's architecture calculates all of them together and responds once at the end. However, to maintain real-time experience, some analytical parts can be delayed. This issue is addressed using three strategies. First, using multitasking and multithreading whenever possible. There are limitations to this approach as both design software and game engines are not open-source or adjustable to this degree. The second approach is to replace final accurate results with good-enough faster results. In early design stages, results within an acceptable range can steer design in the right direction without needing highly accurate and detailed technical reports. The third method is to create a data dam or data shield within the generative processes, meaning there are checkpoints that force the design software to retrieve results even when the whole process is not completed yet. This requires low-level manipulation of the back-end design software. Figure 15 shows that after geometry generation is completed and after each category of performance data is completed, another algorithm forces the design service to report the results at those stages. Combined, these three methods contribute to reducing calculation time because all information is presented as soon as it is ready in the designed order. It must be added that this approach is most probably slower than recreating all necessary services from scratch inside one game engine host, but it requires extensive time to recreate all the accredited analytical tools again, which is not feasible within the scope of this research. Another possible solution could be training machine learning algorithms to provide targeted and trained guesses on each performance criterion. This was the case for the alpha test version of the platform (which is also presented and published in the first paper, P1). However, considering the training time, when comparing the lite analysis and an ML outcome, the processing time did not vary significantly.

Therefore, in favor of the scalability of the research and faster experiment development, the machine learning approach was discarded in favor of lite semi-final analysis.

4.9.1 Feedback Measures

The backend processes of the framework analyzed each design configuration across seven different measures: C1. Maximum displacement [cm] of each load-case at endpoints and midpoints of elements. C2. Change in internal elastic energy [kNm] for each load-case, computed as the scalar product of external nodal loads and nodal displacements. C3. Mass of the exoskeleton in [kg]. C4. Annual operational cost for cooling, heating, electricity, and domestic hot water, considering the building's expected lifetime. C5. Annual operational carbon emissions for heating, cooling, electricity, and domestic hot water, plus annualized embodied carbon emissions from building construction, considering the expected building lifetime. C6. Annual solar gain. C7. Fabrication complexity: Material needed combined with machining time. Environmental metrics (C4-C6) are measured using tools developed by ETH Zurich for early design stage analysis (Elesawy et al. 2018), and the finite element analysis is conducted using FEA tools from Bollinger+Grohmann™ (Preisinger et al. 2014) for C1 to C3. The author analyzes the fabrication complexity measure (C7); The fabrication complexity is defined by author as a composite performance metric that quantifies the cost, effort, and resources required to realize a digitally designed component through robotic fabrication processes. It integrates material consumption and machining time as primary dimensions, with weighted contributions based on their relative impact on system efficiency, energy consumption, and decision feedback loops in human-robot collaboration.

Let:

- M : Total material volume or mass required [kg]
- T : Total machining/fabrication time [h]
- ω_m : Weight assigned to material usage (0.5 in main version)
- ω_t : Weight assigned to machining time (0.5 in main version)

$$FC = \omega_m \times \frac{M}{M_{ref}} + \omega_t \times \frac{T}{T_{ref}}$$

Where:

- T_{ref}, M_{ref} : Reference values (max values across options)

These constraints are designed infallibly or fault-proof, meaning that any choices resulting in an unconstructable and invalid option are automatically snapped to the closest valid option. This decision is based on the body of knowledge, which indicates that A) any obstacle in an experiment that causes it to fail has a negative effect on the validity of the result (Jones and Reinhart 2018); B) Debugging an invalid option is non-linear. There are numerous ways to undo problematic decisions, and research shows that participants should always have a clear and straightforward path to go back and correct choices that they do not deem fit (Brady 1986; Jones and Reinhart 2018); C) Similar user-centred frameworks with a non-limiting strategy, where users can freely generate design options, have been shown to reduce the usability of spaces and the constructability of the outcome (Anam and Tan 2023). This general rule is sometimes referred to as "to improve performance, make decisions or present choices that are less likely to result in errors" (McFarlane and Latorella 2002); D) Another critical factor is regarding how the information is presented to the user; In a comprehensive systematic review spanning 35 years of proceedings from the ACM Conference on Human Factors in Computing, Hornbæk et al. (2019) examined over 4000 papers intending to shed light on the styles of interaction. They categorized 515 interaction styles into eleven categories, with the one with the highest occurrence and viability described as 'direct' and 'multimodal'. Other complex multidimensional styles that incorporate widgets and drop-down menus that first gained attention with the emergence of cellphones evolved and gave way to qualities such as 'smooth', 'simple', and 'natural' in the following years.

In the alpha version of the framework, all seven measures were displayed in bar charts. Feedback from our alpha testers indicated that our audience (which are early designers, who are not building physics experts) struggled to interpret each category. Partially because some measures improve with decreasing values, like structural mass, while others are better performing when increased, like solar gain (or at least are not one-directional and are dependent on the user preference). After testing data remapping and different visualization charts, the beta test showed that a combination of data encapsulation and the option to view detailed reports was better understood by users. From the beta version onward, all measures in each category were encapsulated into one measure, resulting in three categories: structure, environment, and fabrication (These are referred to as ENC1, ENC2, and ENC3, respectively, henceforth.). During encapsulation, each category received an overall score labeled as an improvement, disimprovement, or no significant change. For instance, if a new design configuration improved two or more of the three structural measures, it was labeled as an improvement. The same logic applied to the other categories.

The final question was whether to show absolute measure values or relative amounts. The beta test showed higher engagement when results were presented relatively, compared to previous configurations. Thus, the seven measures were downsized to three and shown in a comparative format, with an option for users to view the full detailed report of absolute values. Hence, in the main version, the user received three distinct types of feedback: improvement, neutral, or disimprovement. They also had the option to visually see an image of the entire structure's displacement and access a detailed report of the environmental and fabrication information by hovering the mouse over the lower interface menu.

5 ANALYSIS

In this section, when a subsection is marked as observation, it highlights an interesting area for further analysis. Observations provide clues that lead to subsequent definitions and analysis. No conclusions are drawn from observation sections. Observations identify areas for future research, depict *trends* in data and call for more work with a different scope. All findings at the end of each section are based on solid statistical analysis, not on observations.

Subsections in this chapter are formatted according to the validation process. First, it examines whether users were able to understand and interact non-randomly with the framework and comprehend the design task. Section 5.1 addresses this question. The second question is whether there is a meaningful difference in decision time between feedback conditions, meaning whether participants behaved differently (regardless of the cause) with and without receiving feedback. Section 5.2 responds to this question. The third question is whether the selection and encapsulation of feedback data were meaningful. Specifically, was the platform able to convey distinct information to users, or was there correlation or redundancy in the conveyed information? Section 5.3 answers this question. After addressing these first three questions, user behavior can be analyzed.

5.1 Post-game Questionnaire

Each participant received a standard usability test at the end of the session. Therefore, before moving to the outcome analysis, the results are reported in this section. The question, along with the schematic results, is shown in Figure 16.

1. The System Usability Scale Standard Version. Please Rate your experience with the framework.

● Strongly disagree ● Disagree ● Neutral ● Agree ● Strongly agree

I think that I would like to use this system frequently.

I found the system unnecessarily complex.

I thought the system was easy to use.

I think that I would need the support of a technical person to be able to use this system.

I found the various functions in this system were well integrated.

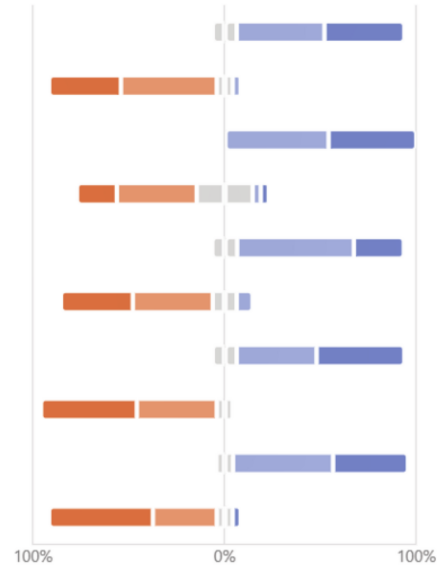
I thought there was too much inconsistency in this system.

I would imagine that most people would learn to use this system very quickly.

I found the system very cumbersome to use.

I felt very confident using the system.

I needed to learn a lot of things before I could get going with this system.



2. The moderator informed me that this experiment has ethics approval from Conjoint Faculties

Research Ethics Board (CFREB) at the University of Calgary with Ethics ID **REB21-1935 REN1-2-3**. I was also informed about the privacy, confidentiality, data security, voluntary participation, benefits, and purpose of the study. Therefore, by clicking "Agree," I attest that I have been informed of the above information and have received answers to any questions I asked. I consent to take part in the study.

● Agree
● Decline

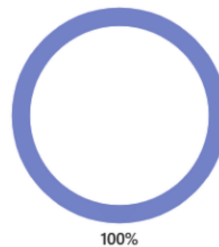


Figure 16: Results and content of post-game SUS usability questions, followed by ethical consent information

All responses collected over 185 days of data collection, with an average of 2:14 minutes spent filling out the questionnaire, are combined. The collective results show that:

Descriptive Statistics:

- Mean SUS Score: 81.3
- Median: 82.5
- Standard Deviation: 10.2
- Minimum: 52.5
- Maximum: 97.5
- Interpretation:

Based on industry benchmarks:

- **81.3 → Excellent usability:** The average SUS score of **81.3** exceeds the industry benchmark of **68** and ranks in the **top 10%** of systems.
 - The average of **81.3** places your system in the "**Excellent**" category.
 - Adjective rating scale (Bangor et al., 2009): → the system is perceived as having "**Excellent**" usability by most participants.
 - **The system demonstrates excellent overall usability**
 - Users found the system **easy to use** (Q3: 3.48/4) and felt **confident** using it (Q9: 3.30/4).
 - Most users felt the system was **quick to learn** (Q7: 3.39/4).
 - The distribution is slightly skewed toward high scores, with most participants rating usability between 76.25 and 88.75.
 - The lowest score (52.5) is still above the threshold of "poor" (<50), suggesting no major usability failures.
-

Since the overall usability results are promising, this indicates that users understood the task and their interactions were intentional, not random. Therefore, the results are analyzed in the following sections.

5.2 Time

In this section, only the time of the interactions is analyzed initially. This means that the decision-to-decision time (DTime) is examined, referring to the time spent on each configuration or the time between two changes. First, the decision time, or DTime, is examined across all participants, regardless of their condition (receiving feedback or not receiving feedback), and then per participant. Finally, the same steps are examined within the two conditions. Some definitions used in this section are explained in Table 9.

Table 9: Statistical measures and definitions

Measure	Definition/Use case
Standard Deviation	Measures the spread of data around the mean. Larger values indicate more dispersion.
Variance	The square of standard deviation. Measures the average squared deviation from the mean.
Skewness	Measures the asymmetry of the distribution. Positive values indicate right-skewed data.
Kurtosis	Measures the 'tailedness' of the distribution. Higher values indicate more extreme outliers.
Interquartile Range	The range between the 25th and 75th percentiles. More robust to outliers than range.
Coefficient of Variation	Standard deviation divided by mean, expressed as a percentage. Useful for comparing variability across different scales.
Quartile Coefficient of Dispersion	Measures relative dispersion. Useful for comparing spread across different scales.
Z-scores	Number of standard deviations each value is from the mean. Useful for identifying outliers.

Considering all Design DTime values across participants and conditions, the basic statistics are as follows (excluding Tutorial Event, etc.):

Basic Statistics:

- Count: 4292
- Mean: 5.60
- Median: 0.74
- Standard Deviation: 14.81
- Variance: 219.28

Table 10: Basic statistics and distribution measures of Decision-Time per condition

Definition/Use case	Feedback on	Feedback off
Basic Statistics:		
Count	2582	1710
Sum	13771.51	10276.53
Mean	5.33	6.01
Median	0.84	0.66
Standard Deviation	13.09	17.08
Variance	171.25	291.68
Distribution Measures:		
Skewness	7.59	6.85
Kurtosis	91.18	63.45
Range	243.62	242.89
Interquartile Range	4.25	3.69

Statistics of the collected DTime divided into condition groups can be seen in Table 10 and Figure 17.

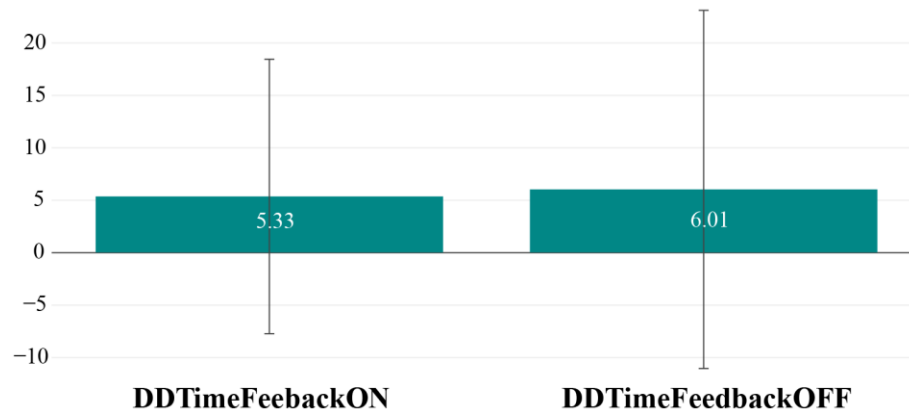


Figure 17: Standard deviation and mean of decision time between conditions

Therefore, the first question is: Are they significantly different? This means, is the decision-making time different when feedback was on versus off? For this comparison, a non-parametric Mann-Whitney U test is used, which is appropriate for comparing two independent samples with skewed distributions. The results are as follows:

The results of the descriptive statistics showed that the *DTime-FeedbackON* group had higher values for the dependent variable ($Mdn = 0.84$) than the *DTime-FeedbackOFF* group ($Mdn = 0.66$).

A Mann-Whitney U test was conducted to compare scores between *DTime-FeedbackON* and *DTime-FeedbackOFF*. For the given data a Mann-Whitney U-Test showed that the difference between *DTime-FeedbackON* and *DTime-FeedbackOFF* with respect to the dependent variable was statistically significant, $U = 2094417$, $n1 = 2582$, $n2 = 1710$ $p = .004$. Thus, the null hypothesis (there is no difference among the two) was rejected. And it is proven that there is a difference.

Plotting these values also confirms the difference visually (Figure 18). Even though their means are close, Set B has a slightly wider spread and more density in the mid-range (e.g., 5–30), explaining the statistically significant difference found in the Mann-Whitney U test.

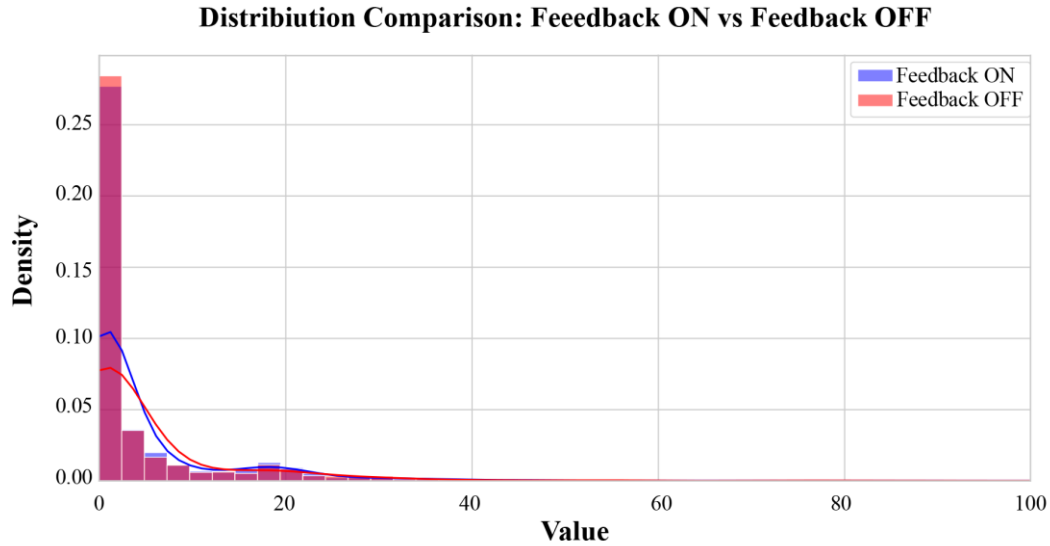


Figure 18: Decision time distribution comparison between conditions

So far, there is only evidence supporting a change in decision-making time across all participants within two feedback groups. However, this evidence is imprecise and must be examined further. It must be noted that in this section, we are not distinguishing between the types of feedback or whether the player chose to spawn the configuration. We are only comparing the decision-making pace between two conditions and have established that they are statistically significantly different.

To conclude, the decision time has been significantly affected by the two feedback treatments; The group with no-feedback has a higher mean value (6.01s) compared to the feedback condition which has a mean of 5.33s. This suggests that, on average, participants took longer to make decisions when they didn't have feedback. However, it's interesting to note that while the mean is higher in the no-feedback condition, the median is actually lower (0.66s vs 0.84s). This suggests the presence of some extreme values (outliers) that are pulling the mean up in the no-feedback condition. Therefore, a closer examination is conducted in the next section.

5.2.1 Decision-Time: a Closer Examination

Mapping the decision time for all participants by interactions (Figure 19) depicts possible trends in decision time. Therefore, the change rate of decision time at different stages of design is further examined.

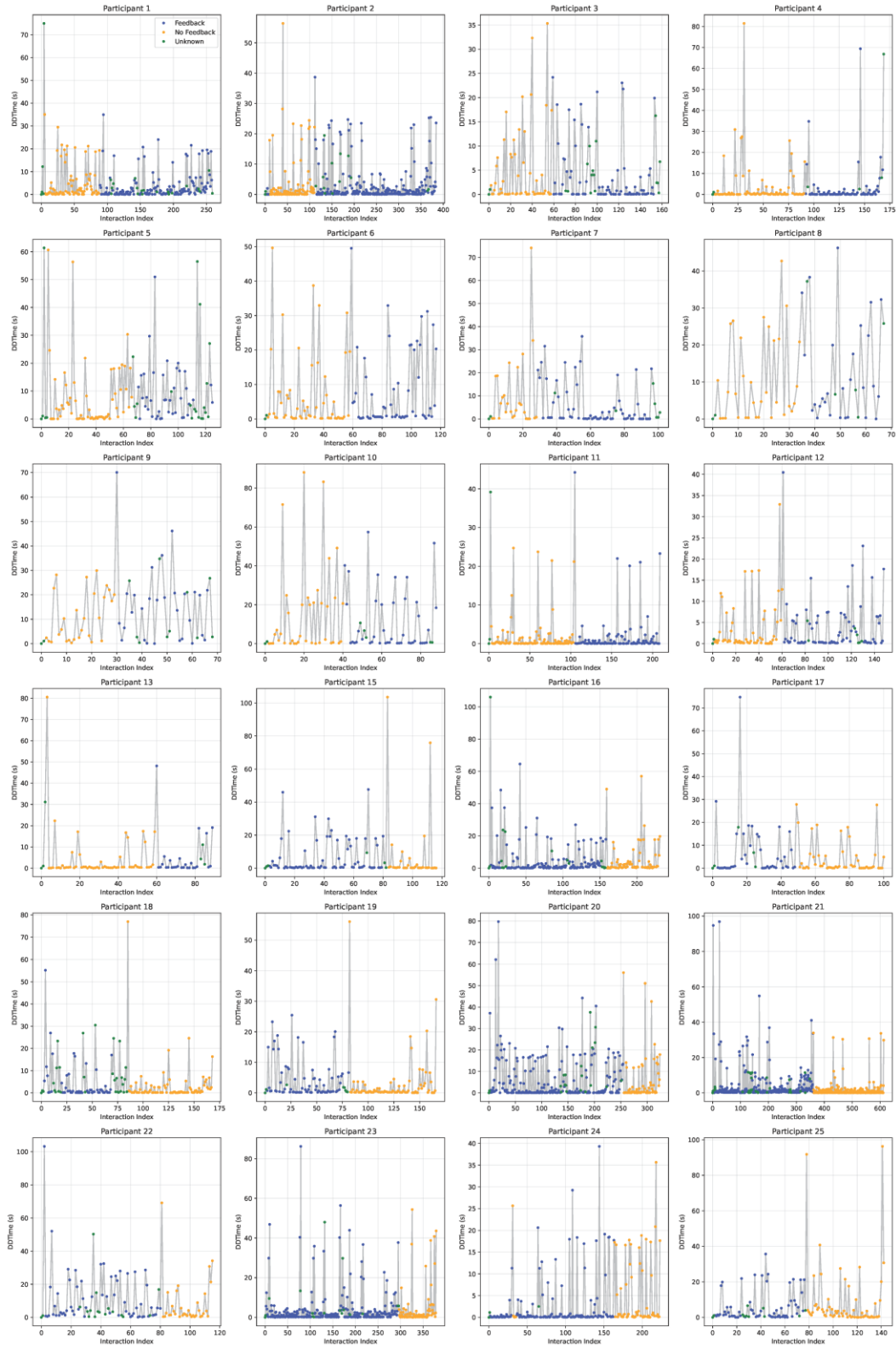


Figure 19: Decision time mapped by interactions per participant

Examining the mean values for each participant confirms the findings of the previous sections: Receiving feedback does not necessarily make decision-making faster in overall, possibly due to contemplation on the data. The Figure 20-Figure 21 visually confirm this statement.

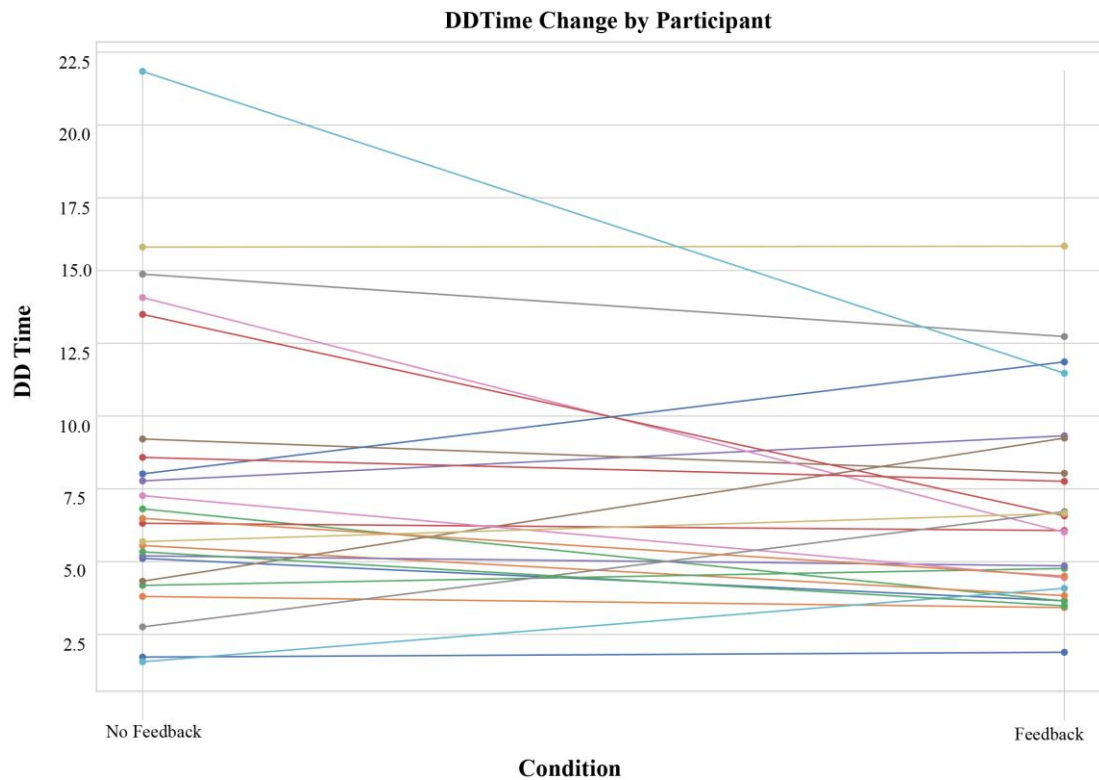


Figure 20: Decision time mean change in both conditions per user

However, this does not rule out the possibility of finding micro patterns of learning and adjustments. Our overall understanding is based on the mean value. Yet, there might be micro patterns that, for instance, enable participants with feedback to make consistently shorter decisions towards the end⁴. This means that the effect of receiving feedback can be hidden in micro scale of decision time.

⁴ For example, let's say Runner A, with no coach, has lap times of 10, 15, 12, 20, and 30, while Runner B, with a coach has lap times of 10, 40, 15, 12, and 11. The mean of the second runner is higher (17.60) than the first (17.40), meaning that overall, the runner with no coach has better laps. However, the effect of learning and improving consistently is observable in the runner with a coach, especially in the latter half (15 → 12 → 11)

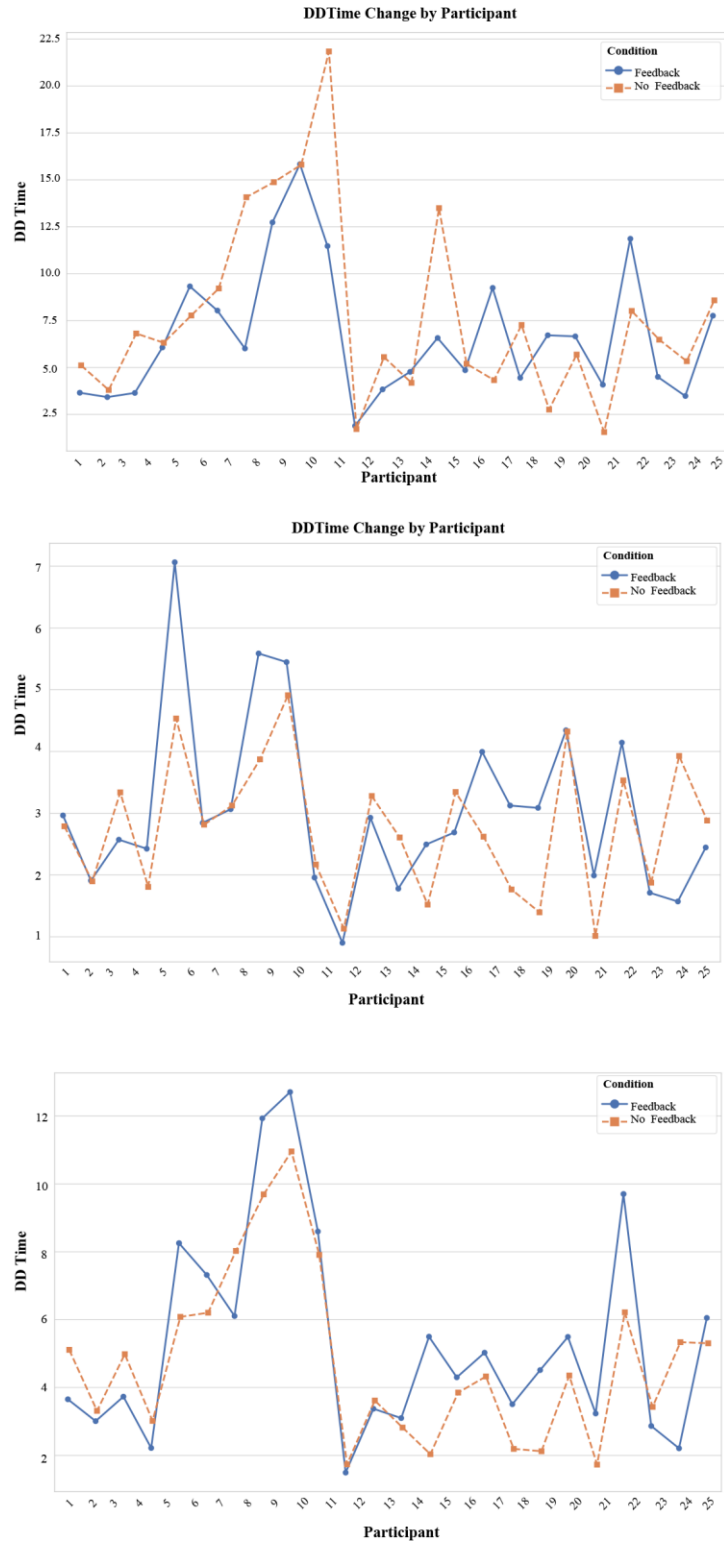


Figure 21: Users' mean decision time comparison. Top: All interaction, Middle: Removing 2% outliers, Bottom: Removing 20% outliers

Observation I: When truncating the DTime by 1% and 10% on each end, more consistency emerges. This means that omitting the interaction outliers makes the underlying pattern clearer (Figure 21). While there are differences between the conditions, they are relatively minor: I1. The feedback condition shows slightly better consistency relative to its mean (lower coefficient of variation). I2. The feedback condition shows slightly less skewness. However, the feedback condition has slightly more variability in its central range (higher interquartile range). These differences are small enough that we cannot conclusively say that one condition is substantially more consistent or less skewed than the other entirely. Nonetheless there is another complimentary observation.

Observation II: Another interesting observation can be made regarding the decision-making timing. If we compare the last three and last ten interactions of each group per participant, we see a more meaningful relationship. II1. Average DTime Comparison: the average DTime of the last 3 attempts in feedback condition is 10.73s versus in no feedback condition is 12.03s. the average DTime of the last 10 attempts in feedback condition is 7.51s versus in no feedback condition is 9.68s. This shows that the feedback has led to faster decision making with an accentuated effect over 10 attempts. In other words, the 10-attempt analysis shows better overall performance in making faster decisions than the 3-attempt analysis. II2. DTime Consistency (Standard Deviation): this measure show how variable decision making was under different conditions. Lower values indicate more consistent decision times. Figure 22 depicts that again the difference between the SD's in two conditions is much higher over the later and longer period (last 10 attempts).

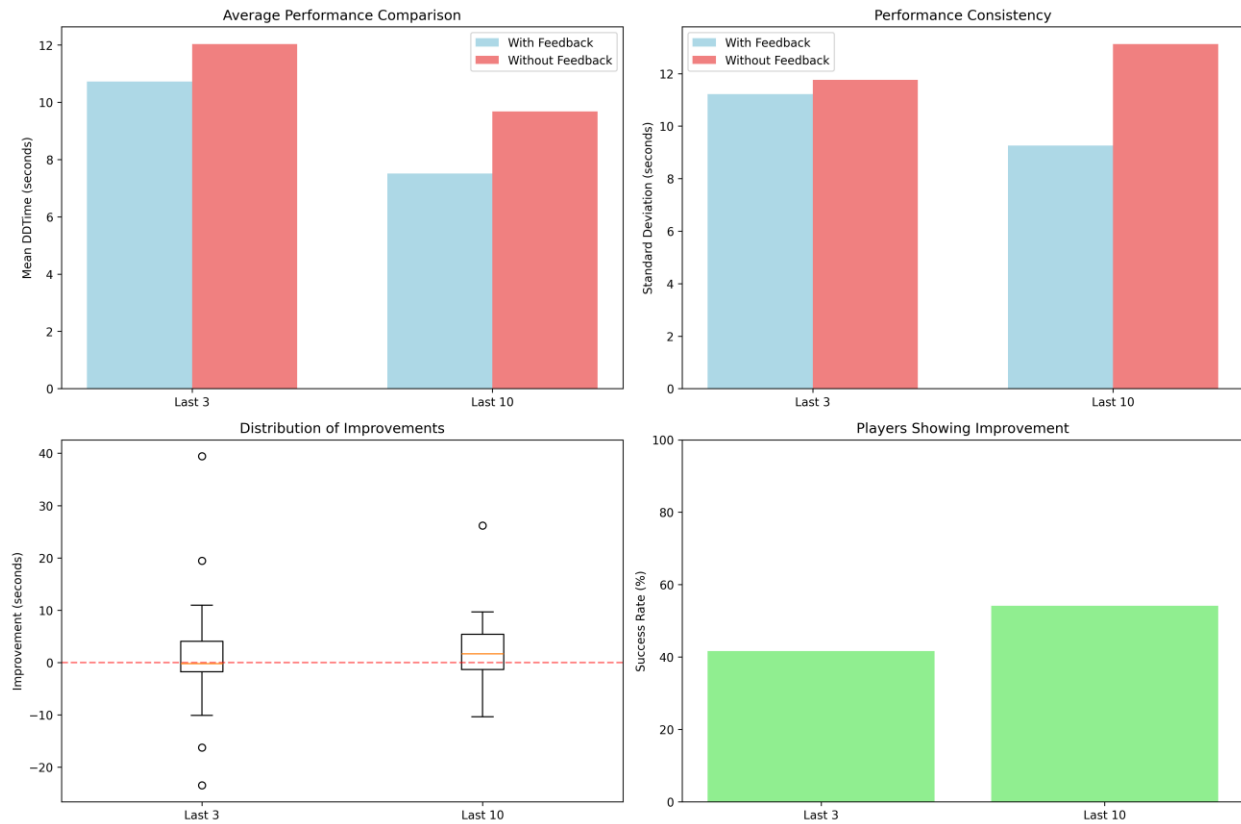


Figure 22: DTime comparison in different time ranges

Based on the two later observations, a deeper study of micro steps in the DTime is conducted (Figure 23). This measure is named Micro Learning: In the context of this section – which is examining the decision time only among conditions, "learning" is defined and measured as: Faster decision making over time. the primary metric for that is Decrease in DTime across attempts. When a player is "learning," it means that they are 1. Making faster decisions in subsequent attempts and 2. Showing consistent reduction in DTime. The negative slope in the linear learning rate indicates this improvement. For example: -0.633 s/attempt means each attempt is about 0.6 seconds faster than the previous one.

This analysis examined learning rates (slopes) of DTime across attempts where negative slopes indicate improvements (faster decisions over time). In summery for Feedback condition: Mean slope = -0.0235 (SD = 0.0878) and for No Feedback condition: Mean slope = 0.0557 (SD = 0.1314). for this comparison, t-statistic: -1.8460 and p-value: 0.0778 are reported and the mean difference (Feedback - No Feedback) is -0.0791 . This is also affirmed by looking at the DTime of two sample participants. The negative mean slope for the feedback condition indicates that when receiving feedback, participants consistently reduced their

decision-making over time. In contrast, the positive slope for the no-feedback condition suggests that the rate of decision-making did not change meaningfully when no information was conveyed. However, since the p-value for this comparison requires more data and is not completely aligned with the main hypothesis of this research, it is reported only as a trend in the data.

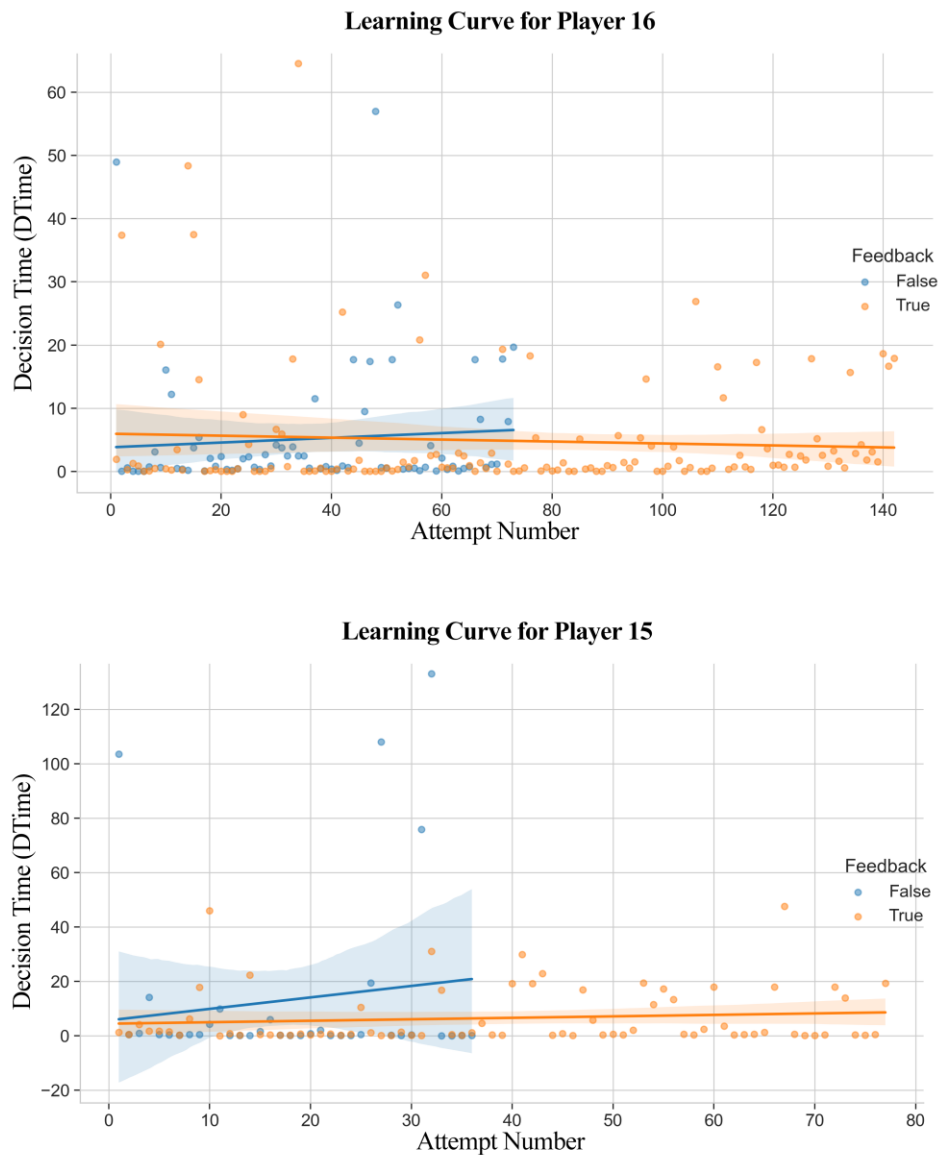


Figure 23: Micro-learning effect (mean slope comparison) of two sample participants in both conditions

To conclude this section, by examining decision-making time only, between two conditions, per and across participants, the following hypotheses are approved, and trends are extracted.

H1. There is a significant difference in decision-making time between the condition where feedback was shown and the no-feedback condition.

Regarding data trends,

T1. Receiving feedback has a relatively lower coefficient of variance, interpreted as a steadier reduction in decision-making fluctuation. Comparing the last 10 to the last 3 attempts confirms this more strongly, suggesting that more players have been able to make consistent decisions over longer periods. While this calls for deeper analysis of learning stages for this group, it can be confirmed that

T2. Learning is visible and is a time-consuming process when receiving feedback. The mean slope of learning confirms that the effect is not random, and improvement in making decisions shorter and steadier is occurring among those who received feedback.

Hence based on H1, T1 and T2:

- **Receiving feedback has a statistically proven effect on decision-making time.**
 - **Receiving feedback, when considered across all participants, has made decision-making faster (lower mean).**
 - **The effect of feedback on the user is latent, deferred or incubating; it emerges in the later attempts of each participant more strongly.**
 - **Patterns of micro-learning indicate improvement in reducing decision time and increasing its consistency.**
-

5.3 Feedback Data

Regarding the encapsulation format, some descriptive results are reflected below.

	Disimprovement	Improvement	Neutral	Invalid
Structure Feedback	182	96	182	0
Environmental Feedback	207	249	4	0
Cost Feedback	279	174	7	0

First, studies are needed to determine if the three categories are redundant, meaning they could be merged or downsized. It's important to note that this is fundamentally different from an association test among the three. Two categories can be associated but still convey different information and therefore not be redundant. For this purpose, Cohen's Kappa test is used. Cohen's Kappa is a statistical measure that assesses inter-rater agreement for categorical variables. It indicates how much two raters (or observers) agree beyond what would be expected by chance. The results indicate that Agreement Analysis (Cohen's Kappa) for Structure vs Environment is Poor agreement ($\kappa = -0.3218$). for Structure vs Cost there is Fair agreement ($\kappa = 0.3881$) and for Environment vs Cost there is again Poor agreement ($\kappa = -0.6694$). This indicate that these pairs actually disagree more than would be expected by chance, confirming they capture very different aspects of feedback. *This confirms that the encapsulation of data succeeded in conveying meaningfully different information.*

Interestingly, when studied across all participants, the Sankey Figure 24 shows the frequency of labels in each category, indicating that the most improvement is in the environmental category, while the most decline is in cost. Aligned with detailed interviews with users, they reported that as the environment was physically correct with accurate lighting and timing conditions, they could intuitively follow growth in that parameter.

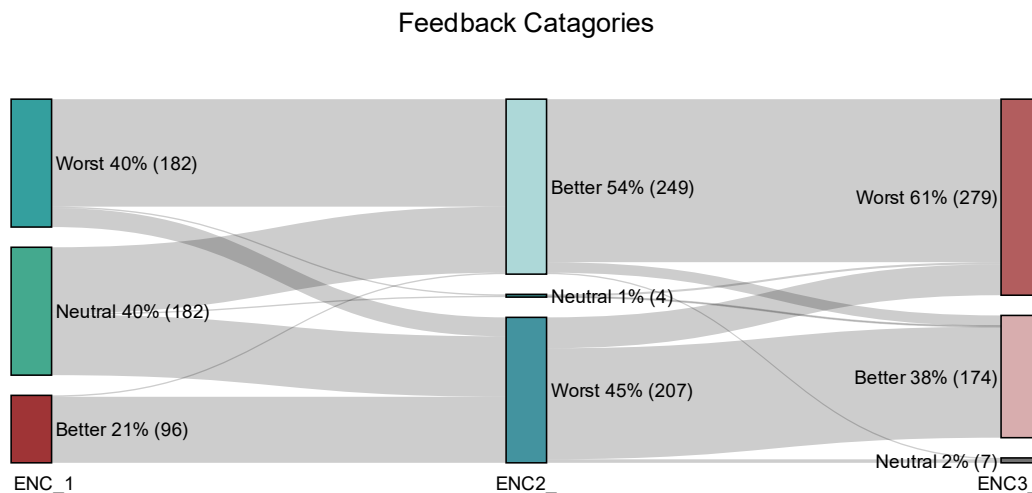


Figure 24: Frequency of feedback in different categories

While structural feedback shows more evenly distributed results, it can be concluded that most players improved the environmental category more easily and prioritized cost less compared to the other two categories. Another interesting trend in the data is the alignment of structural and cost feedback. When structure rates as "Better," cost almost always agrees (97.9%). Similarly, when structure rates as "Worst," cost also usually agrees (95.1%). *An interesting interpretation of that could be that a well-performing structural choice has also made the whole exoskeleton cheaper (more than 90% of the time). Also, choices that did not improve structurally, were also Costly ones.*

To conclude, despite the interesting association between the category of cost and structure, the three types of feedback proved to have no significant redundancy. Therefore, they convey different information and cannot be downsized. Another significant difference reveals itself when the encapsulated feedback is shown as stacked bars for both conditions (Figure 25). In all categories of choices, receiving feedback is shown to affect user behavior. As this is a cumulative diagram, it cannot be concluded from here if receiving feedback necessarily leads to decisions with a higher improvement rate. That is why this statement is further examined in detail in the following section. However, the stacked diagram is another testament to the change in behavior when encountering feedback.

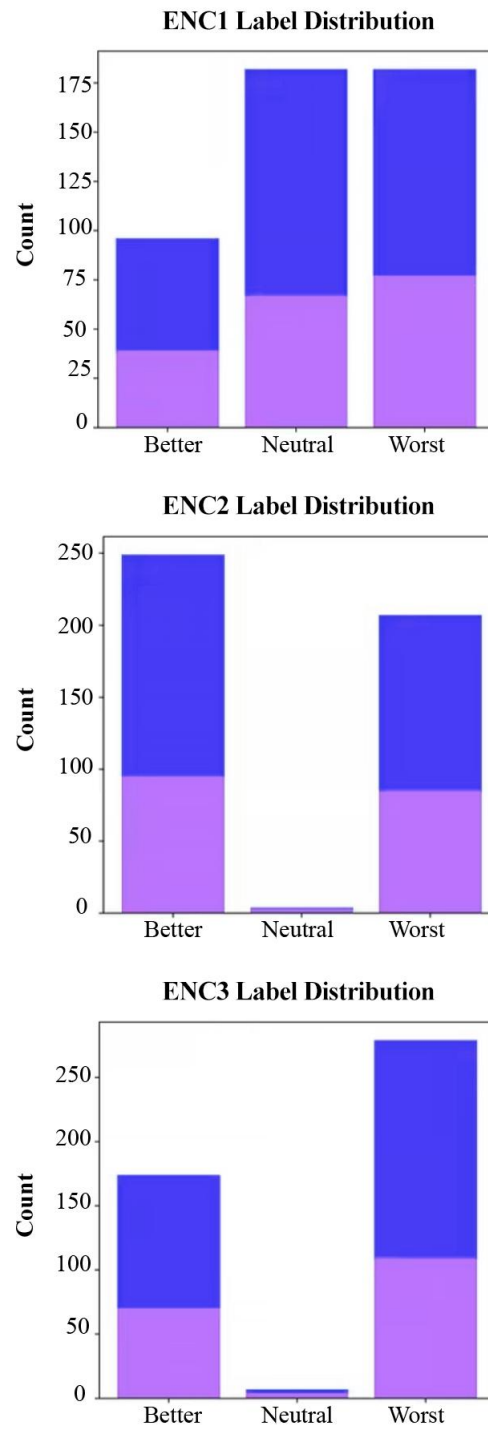


Figure 25: Distribution of encapsulated feedback for both conditions.

Another fact that supports the above claim numerically is that users were able to make better-performing choices in all three encapsulated categories, as shown in Figure 26. This figure demonstrates that, excluding 5 percent outliers and considering the last three decisions in each condition, receiving feedback led to a decrease in structure metrics (a decrease from C1 to C3 indicates better performance), a decrease in environmental metrics (indicating reduced energy consumption and carbon emissions, and higher solar gain - C4 to C6), and a reduction in fabrication complexity (C7). Therefore, in all three categories, the mean and whiskers provide solid proof that users understood the information and used it to make higher-performing choices.

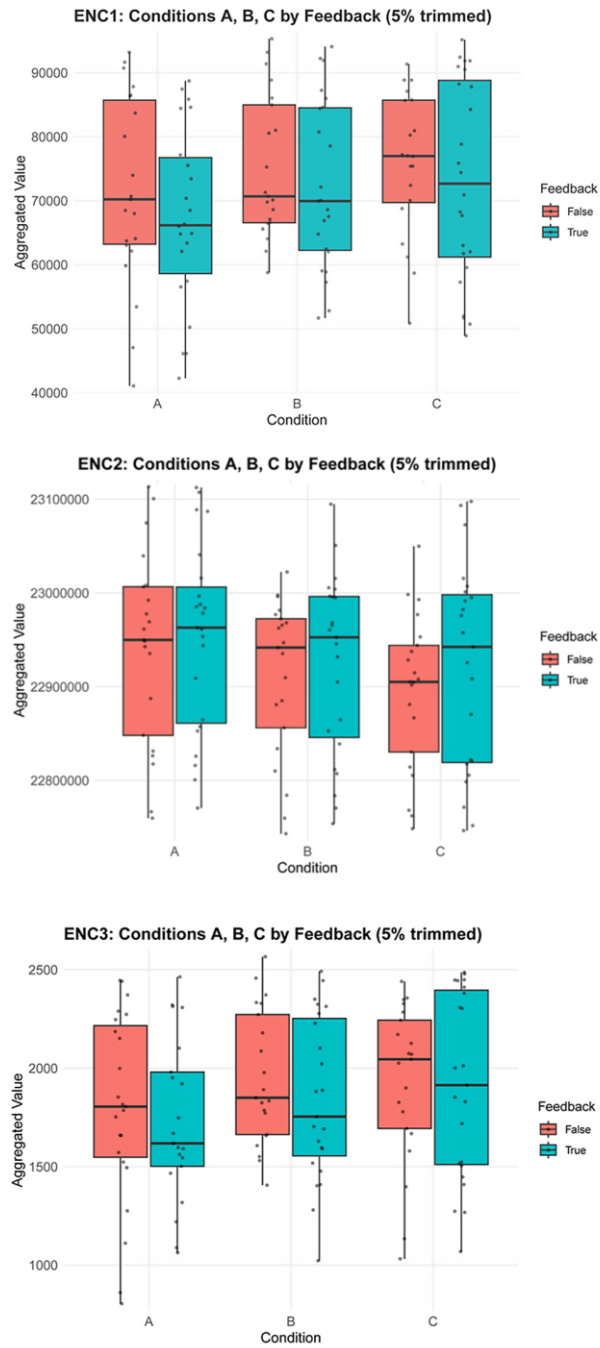


Figure 26: Improvement in three categories of feedback between conditions in the last three decisions

5.3.1 Feedback Data: a Closer Examination

So far, we have established that: 1. Receiving feedback affects decision-making time. 2. The effect of feedback as micro-learning is latent and becomes more visible towards the end. 3. Participants received three different categories of feedback with no redundancy. Therefore, an interesting deeper examination could be conducted to address the effect of feedback on the underlying seven measures. Notably, the seven measures were not presented explicitly, but the player was free to read them in a detailed report or look at the encapsulated format. Therefore, an indirect effect of feedback on measures is analyzed.

Figure 27 shows the percentage of change in all measures. Remarkably, the intensity of color affirms that each participant (each row) has visually changed their choice behavior after receiving feedback and has made more performative changes. If this comparison is mapped in a binary diagram, Figure 28 is generated, confirming the change in decision profiling before and after receiving feedback (Green cells: The participant performed better with feedback than without feedback, White cells: The participant did not improve with feedback).

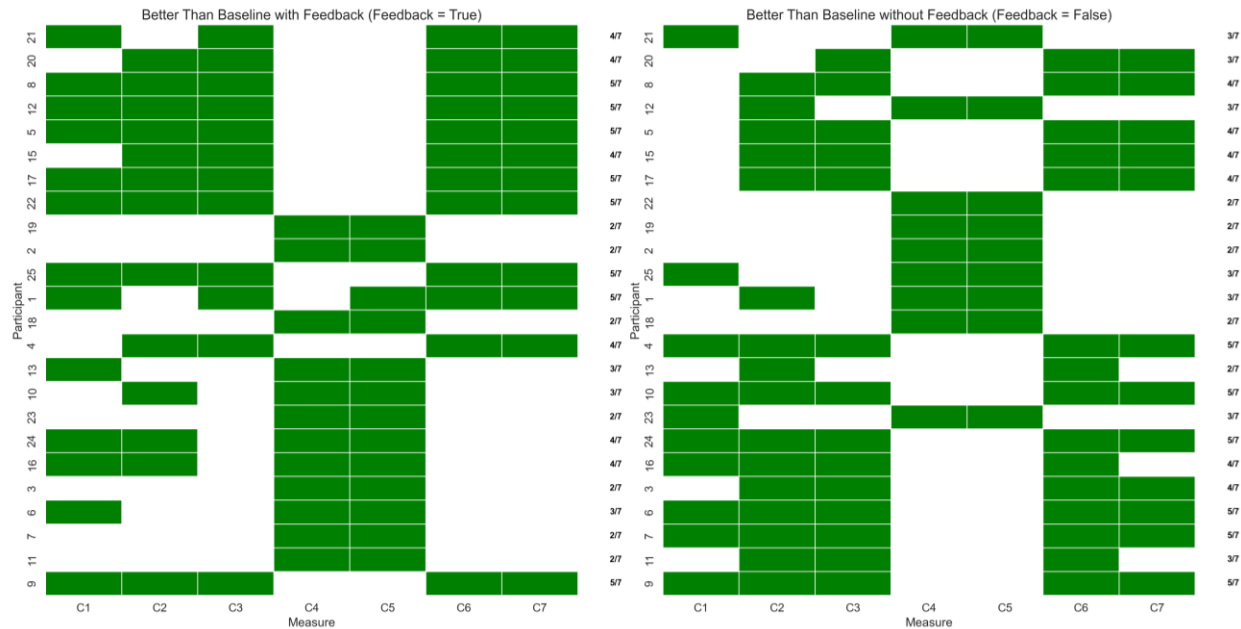


Figure 28: Binary comparison for improvement in each metric (C1 to C7)

While the change in decision-making profiles is apparent, a deeper examination can also be considered. Previous diagrams used either percentage (deviation from baseline) or mean (for binary comparison). However, since each participant has made several design configurations and the effect of feedback is established as latent, only the last decisions for each participant in each condition are examined at a lower level. Figure 29 depicts this comparison per participant based solely on their last two decisions in each category. The effect of receiving feedback is evident from the diagram, as well as intensified differences compared to considering all choices. This means that the final choices in both conditions are much more distinct, and the feedback effect is intensified towards the end.

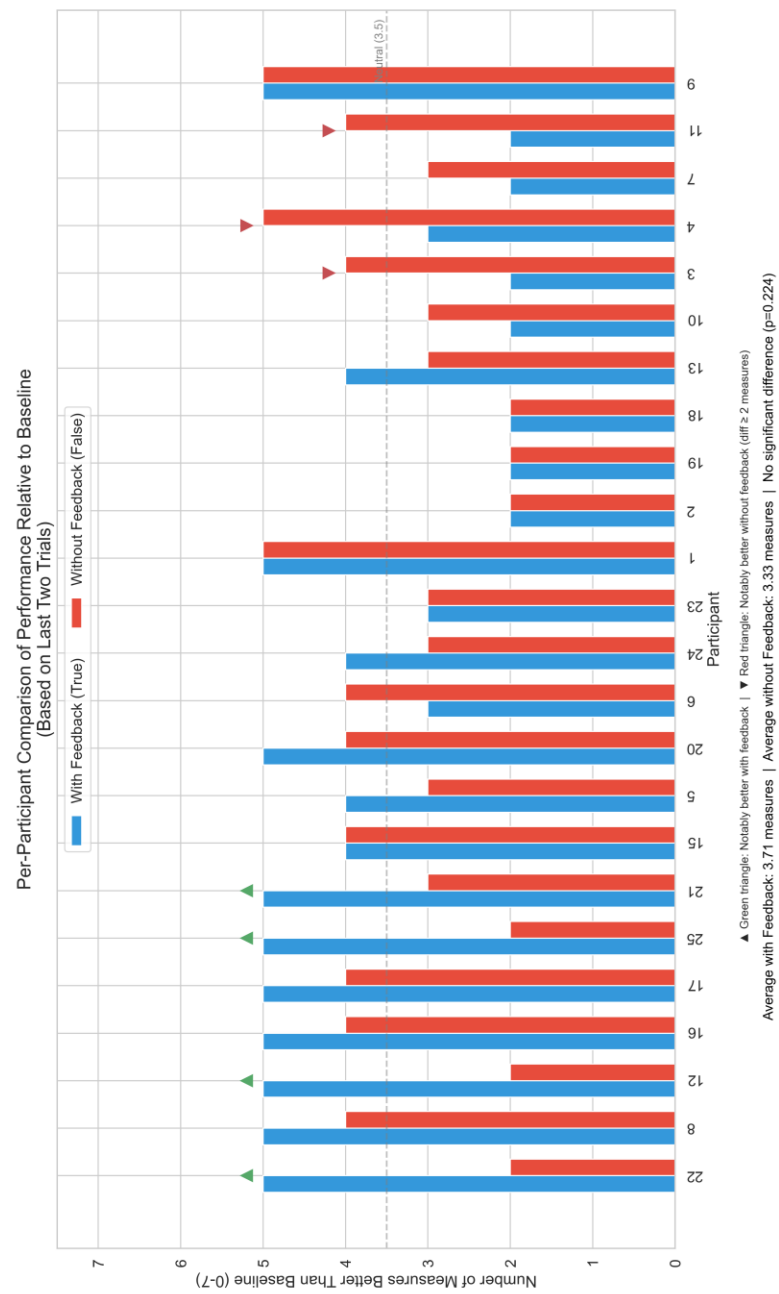


Figure 29: Binary comparison for improvement in each metric (C1 to C7) per participant

While the change in decision-making profiles is apparent, a deeper examination can also be considered. Previous diagrams used either percentage (deviation from baseline) or mean (for binary comparison). However, since each participant has made several design configurations and the effect of feedback is established as latent, only the last decisions for each participant in each condition are examined at a lower level. Figure 29 depicts this comparison, per participant based solely on their last two decisions in each category. The effect of receiving feedback is evident from the diagram, as well as intensified differences compared to considering all choices. This means that the final choices in both conditions are much more distinct, and the feedback effect is intensified towards the end. Which means in a binary comparison, except for five participants (~20%), the majority of participants (~80%) were able to make decisions that were better in an equal or greater number of metrics.

Other performance change statistics⁵:

Measure	Mean Change (%)	Median Change (%)	Min (%)	Max (%)

C1	-0.90%	1.85%	-50.58%	28.66%
C2	-19.84%	1.14%	-186.34%	62.40%
C3	-1.60%	6.64%	-68.62%	36.85%
C4	-0.12%	-0.22%	-4.13%	3.24%
C5	-0.04%	-0.09%	-0.94%	0.80%
C6	1.24%	1.28%	-19.06%	34.82%
C7	-2.65%	5.58%	-96.18%	45.22%

C1 | -0.90% | 1.85% | -50.58% | 28.66%

C2 | -19.84% | 1.14% | -186.34% | 62.40%

C3 | -1.60% | 6.64% | -68.62% | 36.85%

C4 | -0.12% | -0.22% | -4.13% | 3.24%

C5 | -0.04% | -0.09% | -0.94% | 0.80%

C6 | 1.24% | 1.28% | -19.06% | 34.82%

C7 | -2.65% | 5.58% | -96.18% | 45.22%

Average improvement percentage across all measures: 51.19%

This suggests that, on average, feedback had a positive effect on performance.

Measures where majority of participants showed improvement: C3, C6, C7

Measures where majority of participants showed less improvement: C1, C2, C4, C5

⁵ Note: This analysis focuses on individual-level improvement patterns and may differ from group-level statistical significance results.

The difference in mean values, as shown in Figure 30, is another meaningful measure: a higher mean difference between conditions indicates a more significant change in user preferences. For example, when feedback was shown, there was a greater change in the Cost (C7) metric. This suggests that participants, upon receiving feedback, made more substantial changes in cost, prioritizing this metric⁶.

⁶ To make claims on whether this prioritization was intentional, influenced by the user's inclinations, or a byproduct of receiving feedback is beyond the scope of this research. This study reports which metric experienced the most change as a result of receiving feedback. The author does not make claims about the *reasons* behind users' intentions.

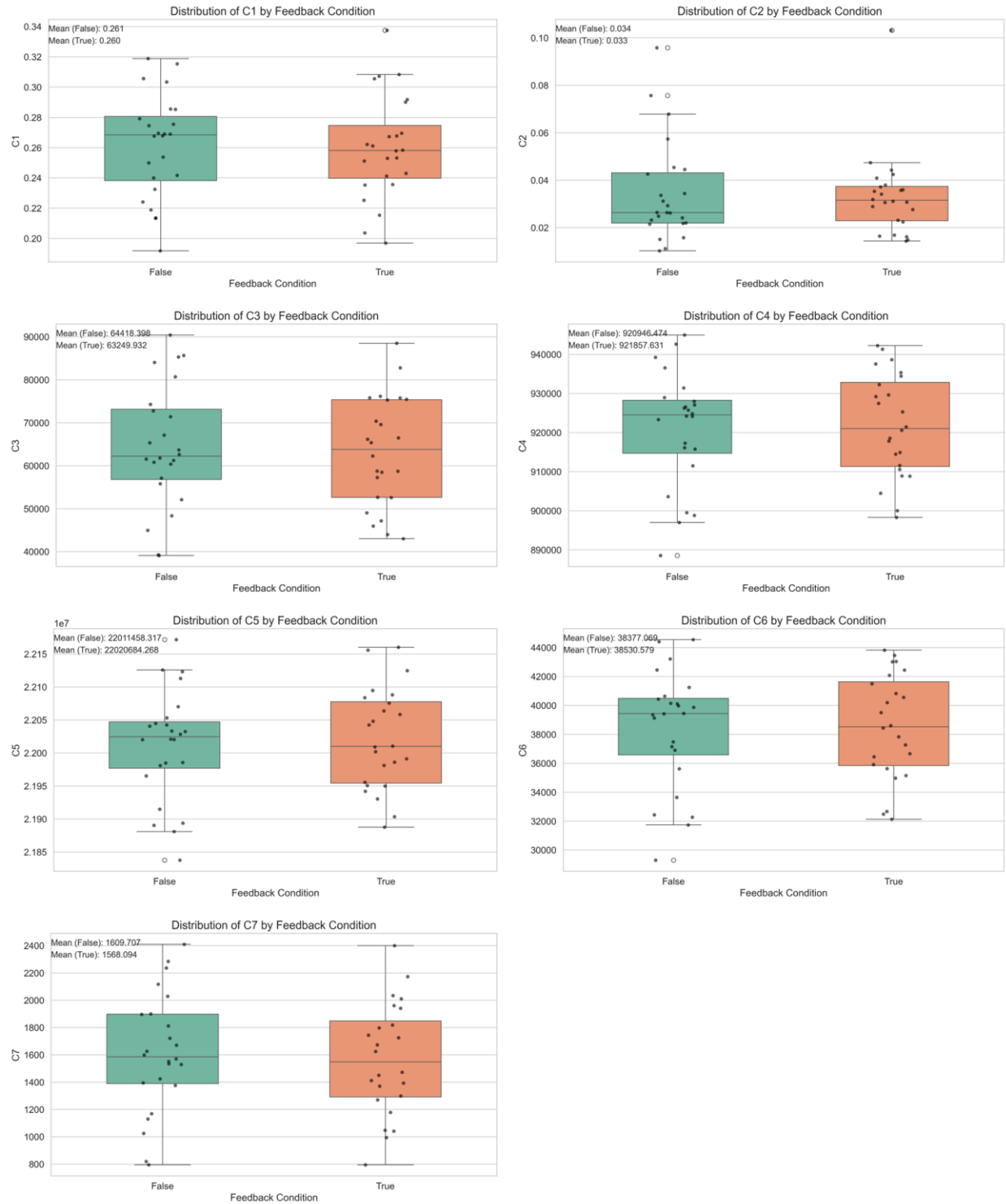


Figure 30: Distribution of metrics per condition

-
- *When the seven metrics are sorted by the change in their mean value, it can be concluded that the top three metrics with the most changes among users are: 1. Cost (62.5% improvement), 2. Weight of the structure (58.3%), and 3. Solar gain (54.2%). The least changed is carbon emission (41.7%).*
 - *Most users preferred the alignment of the structure and cost, indicating that they favored improvements in these two categories over the environment. 41.7% of participants improved in 5-7 measures. 20.8% of participants improved in 3-4 measures and 37.5% of participants improved in 0-2 measures (Figure 27)⁷*
 - Most participants (62.5%) showed either consistent improvement or consistent decline
 - Only about 1/5 of participants (20.8%) showed mixed results.
 - The relatively even split between positive and negative responders suggests that feedback had different effects on different participants
-

Regarding the event of viewing the detailed report, the player can do so by hovering over the corresponding thumbnail on the bottom dashboard. The following descriptive data are extracted: 1. Frequency Per Participant: There were 266 total occurrences across all participants. Most active participants in using this option 35 and 29 times and the least numbers are 1 and 3. And as for the Timing Analysis, the event occurred throughout sessions but with a clear pattern:

- Early Phase: Highest concentration of activity
- Many events clustered in the first few minutes (0.1-4.0)
- Middle Phase (minutes 4-12): Moderate, steady usage
- Late Phase (minutes 12+): Scattered, occasional usage

This can be interpreted to mean that toward the end of the session, most users preferred to use the encapsulated and relative format rather than looking at the detailed report. However, as the occurrences are low in repetition, further analysis of this is a possible direction for future research.

⁷ This analysis focuses on individual-level improvement patterns and may differ from group-level statistical significance results.

5.4 Final Design Configuration

The framework automatically saves a screenshot of each participant's final design configuration at the end of each treatment. Figure 31 and Figure 32 show the geometry of these configurations.

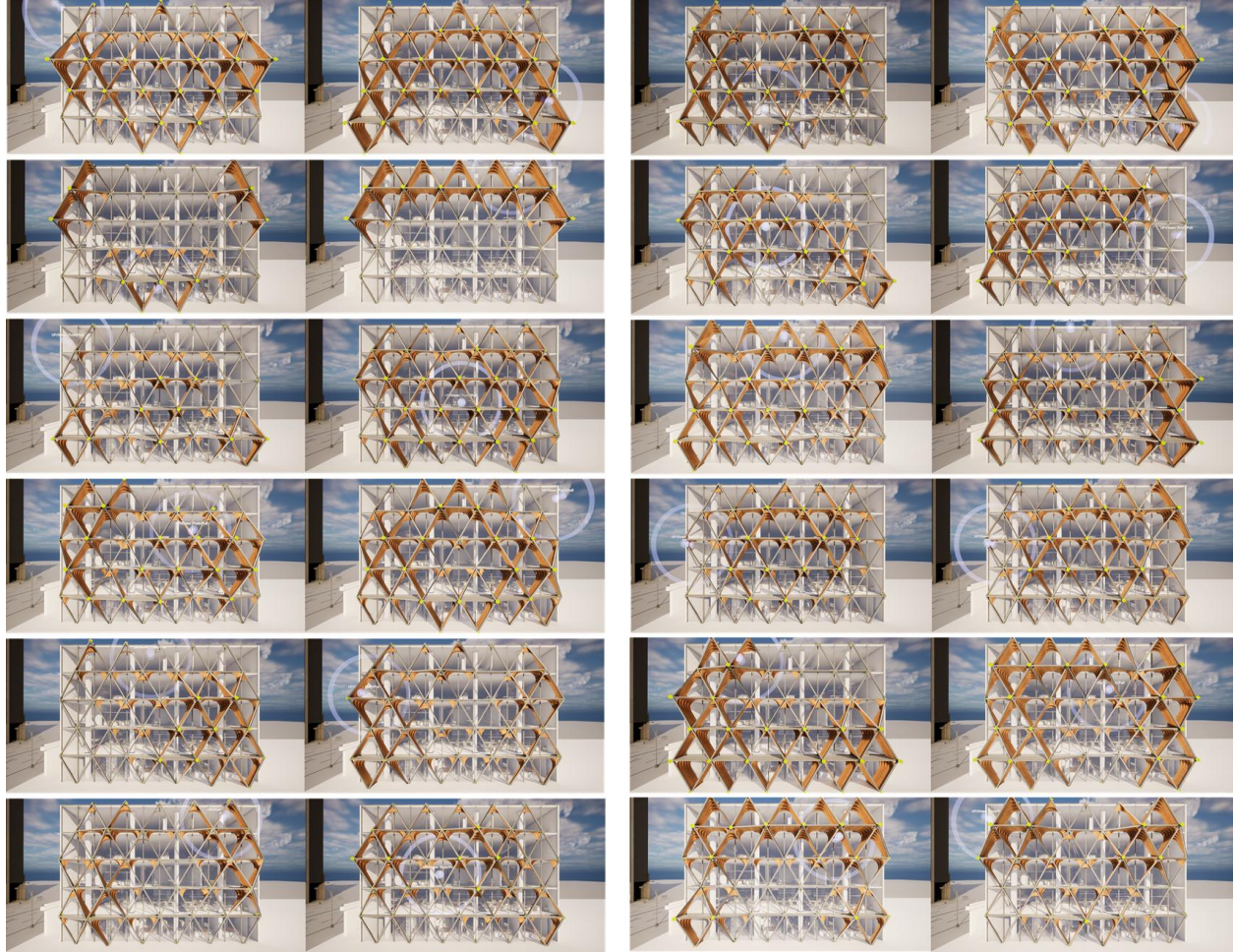


Figure 31: Each pair displays the final design configurations of one participant at the end of two conditions: Left: receiving feedback, Right: not receiving feedback.

Given the analysis of decision time in Section 5.2, along with observations of micro-learning and comparisons between the last three and the last ten decisions, a valid question arises: Was the framework able to change the direction of design? Specifically, did receiving feedback help the non-expert audience arrive at a better-performing design configuration or affect their decisions? To explore this, we will examine

each participant's last design configurations against its seven design measures. It must be noted that users had *not* seen those values explicitly but encountered them in an encapsulated and gamified format.

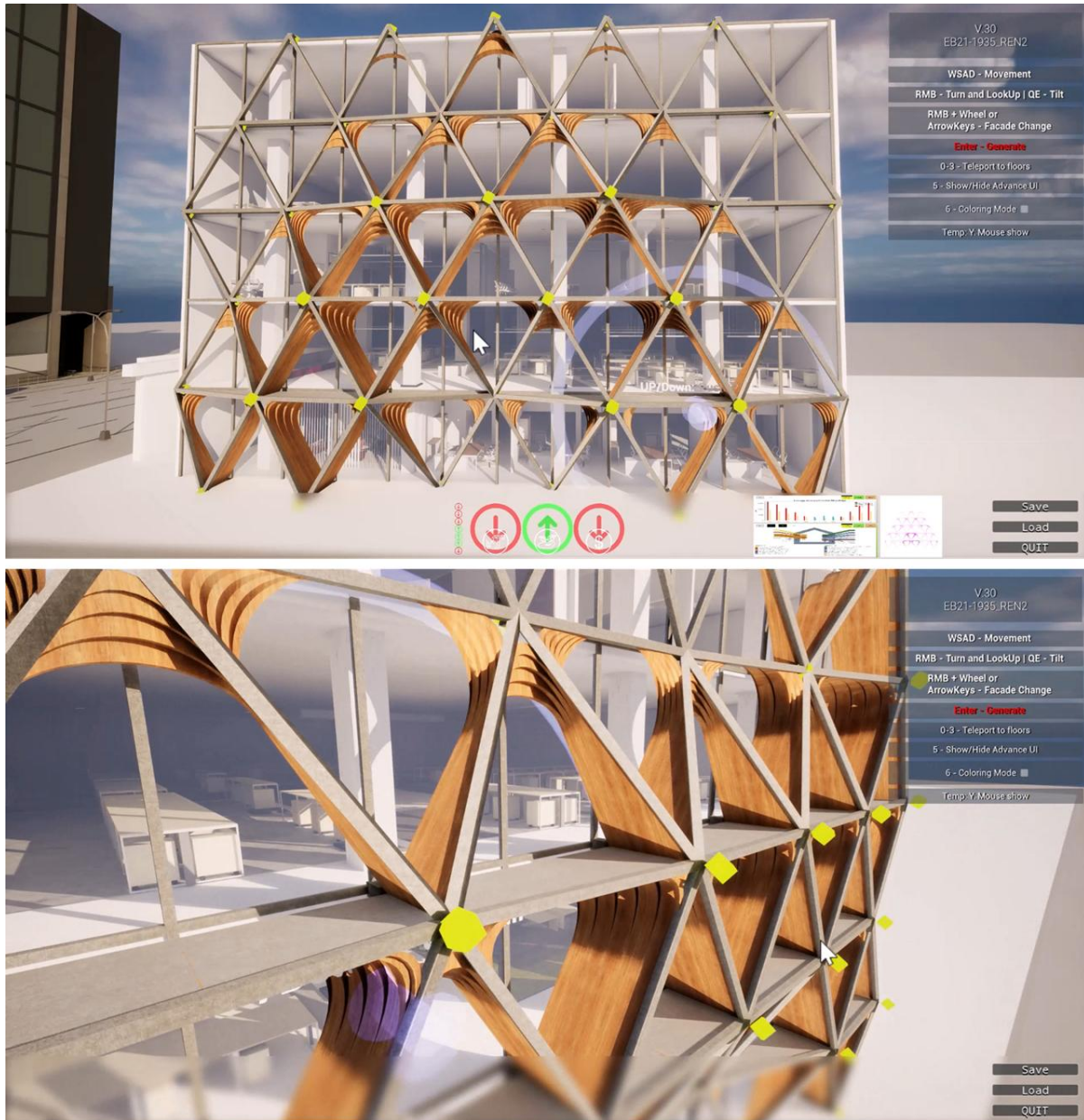


Figure 32: Stills from exoskeleton configurations

Figure 33 displays the mean values of the seven criteria across all participants' last three decisions (*_third is the last, *_second is second to last, etc.). For example, examining the C7_First, _Second, _Third, which is the Cost measure in the last three decisions, shows that receiving feedback has a clear effect on the final decision. The difference in means is increasing, indicating that users who received feedback adjusted their decisions to lower costs and achieve positive feedback in the corresponding category.

Note that Figure 33 provides solid numerical proof that in 5 out of 7 metrics, players chose higher-performing options during their last three decisions. For example, in C7, which relates to fabrication complexity and cost, the box chart clearly indicates that players aimed to reduce costs. The only exceptions to this trend are carbon emissions and annual operational costs, or simply the energy bill. Exploring the reasons behind this is beyond the scope of this research. Was it due to players giving less priority to these compared to structure or cost? Or was it more intuitive for players to understand the improvement logic in structure and cost compared to energy consumption and carbon emissions? However, this thesis can explain this using the logic of encapsulation: by revisiting Figure 24, which showed that participants improved all three categories, the reason is revealed. The range of C4 and C5 is far less than C6, meaning that changes in the exoskeleton have limited numerical impact (for instance, extreme changes in the exoskeleton created a 1.5% change in energy consumption), whereas the numerical impact of solar gain due to the shading effect of the design is far greater. Therefore, it seems that changes in shading were an intuitive response by non-experts to improve environmental performance. They were able to enhance encapsulated environmental performance as a whole, but a large portion of this is due to the drastic change in solar gain, which also compensated for the less significant changes in the other two areas.

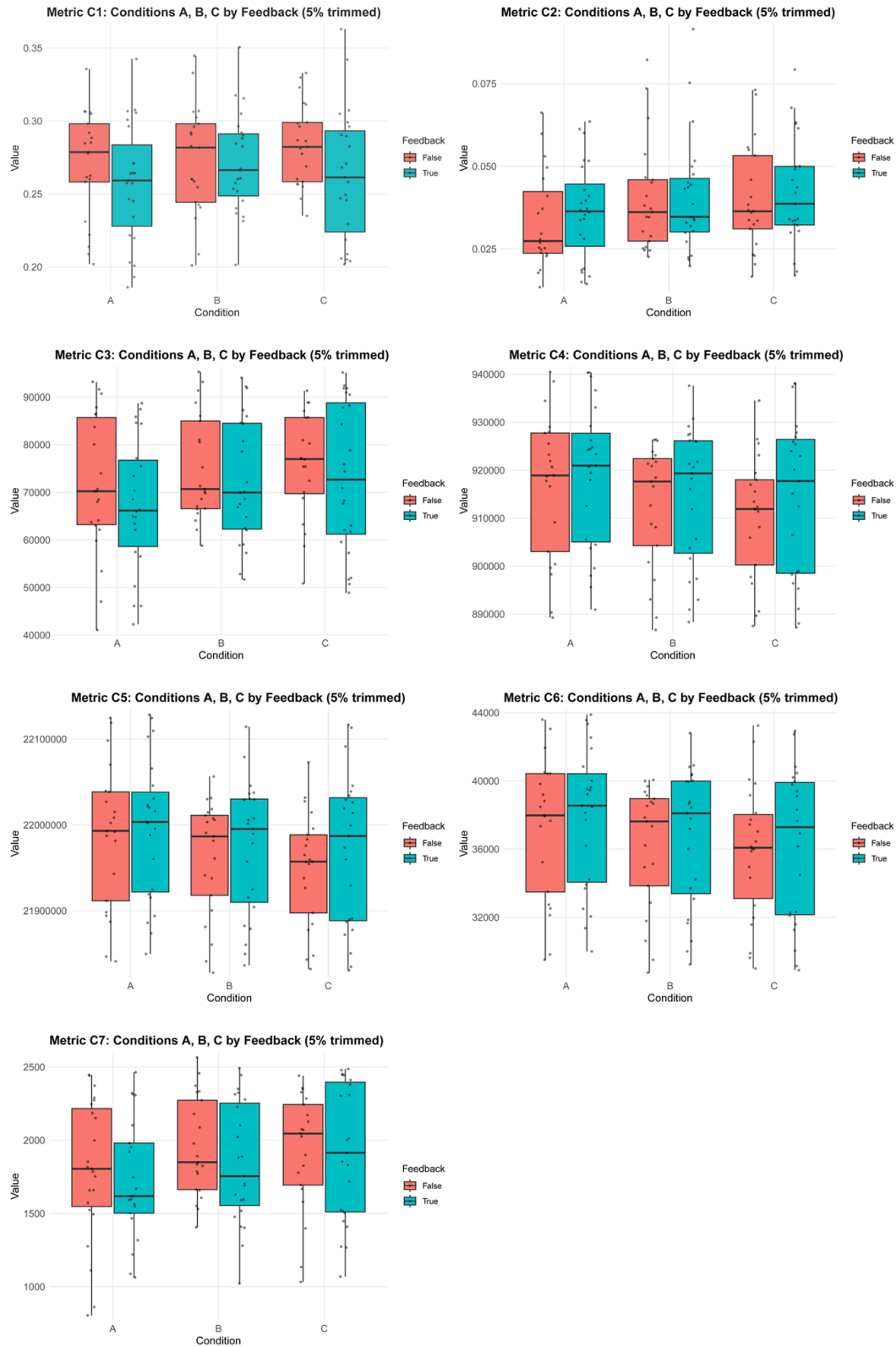


Figure 33: Mean values of the last three decisions across all criteria, per condition – 5% outliers reduction

Another diagram that helps with understanding this is Figure 34. The change in the slope of the last decisions clearly confirms the effect of receiving feedback and making *informed* decisions.

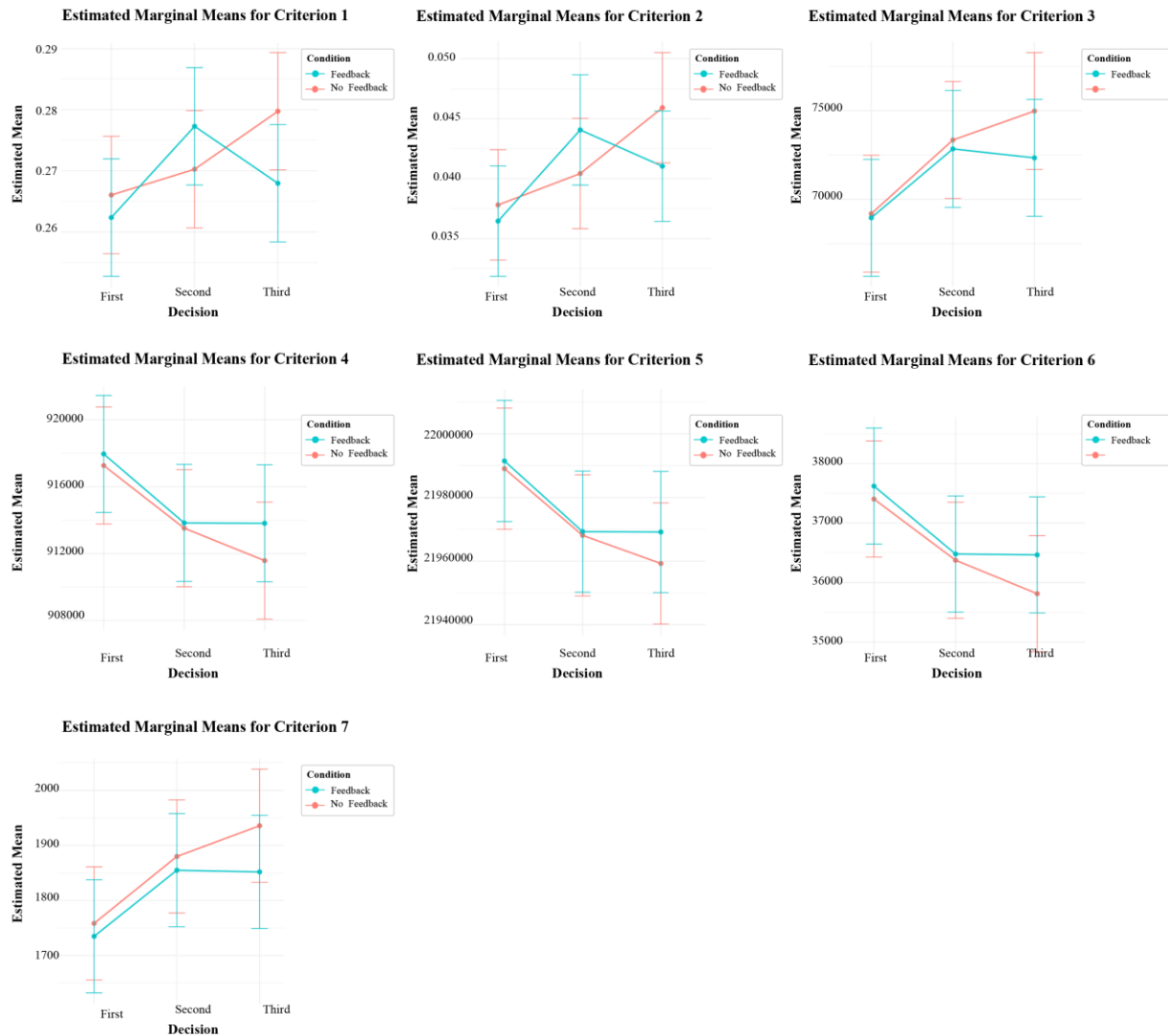


Figure 34: Mean values change trend of last three decisions across all criteria, per condition

Proving the significant effect of feedback marks the most important contribution and confirms the second part of this research's hypothesis; That is, by using gamified relative feedback, users were able to achieve a better-performing design configuration. Interestingly, they did not see the seven criteria individually

but received labeled, encapsulated, and relative feedback, which they were free to ignore. However, the platform, with its gamified strategies, was able to steer their design towards better-performing options while maintaining their preferences.

5.4.1 Design Configurations: a Closer Examination

In this section, the relationship between design alterations in both conditions is examined in more depth. Starting with selecting all the design events in the log, a subset of data is extracted. In this subset, the first possible correlation can be among the players' positions, focal points (users' front vectors), and condition groups. For example, it can be asked whether receiving feedback has caused users to navigate and explore more or look at the design differently. This can lead to further interesting research on profiling users and how they interact with design.

Table 11: Number of design configurations and user movements per participant and condition

Player	Condition	Design Alteration	Position Change - Sum	View Change - Sum
1	IDM	152	10428.63	5.08
1	nIDM	84	18226.50	11.06
2	IDM	269	28424.33	12.87
2	nIDM	102	5858.48	2.52
3	IDM	85	3638.68	1.64
3	nIDM	58	2319.38	1.53
4	IDM	75	0.00	0.20
4	nIDM	91	9466.76	8.15
5	IDM	40	320.00	0.00
5	nIDM	62	1038.30	0.28
6	IDM	59	268.00	0.27
6	nIDM	58	496.91	0.22

7	IDM	66	4514.09	1.97
7	nIDM	28	1059.55	0.44
8	IDM	28	70.00	0.00
8	nIDM	34	827.52	0.56
9	IDM	29	2880.92	1.39
9	nIDM	29	1413.54	0.25
10	IDM	43	80.00	0.85
10	nIDM	41	3089.50	1.74
11	IDM	105	0.00	0.00
11	nIDM	102	4453.50	2.46
12	IDM	81	250.00	6.46
12	nIDM	60	7038.16	9.54
13	IDM	27	1028.57	0.76
13	nIDM	57	7590.27	4.22
15	IDM	77	1254.14	0.22
15	nIDM	36	1615.26	0.50
16	IDM	142	9786.92	4.12
16	nIDM	73	1063.63	0.40
17	IDM	45	9693.89	4.75
17	nIDM	52	4123.99	2.48
18	IDM	55	1330.51	1.43
18	nIDM	86	4344.15	5.44
19	IDM	75	17281.83	5.42
19	nIDM	85	4119.88	1.24
20	IDM	236	14099.75	7.81
20	nIDM	70	5581.61	2.08

21	IDM	324	26664.29	14.69
21	nIDM	253	380.00	0.08
22	IDM	63	25216.04	6.87
22	nIDM	35	8083.81	5.96
23	IDM	281	40.00	0.00
23	nIDM	84	2405.44	0.90
24	IDM	157	6480.39	2.73
24	nIDM	65	8447.91	5.38
25	IDM	68	1654.20	1.58
25	nIDM	65	12820.49	11.39

Table 11 presents the subset that serves as the foundation for all examinations in this section. Returning to the question of how receiving feedback affects users' positions and focal points, Figure 35 shows the mapping of each user's focal point per condition. Note that the sphere represents the participant's field of view. Since the forward vector is drawn from the standing position, when the player makes design changes from inside the building while looking outside—looking back at the facade from inside—the back of the sphere is marked (because the facade is considered the origin of the coordinate system). Conversely, when the player stands in front of the facade, the forward half of the sphere is marked.

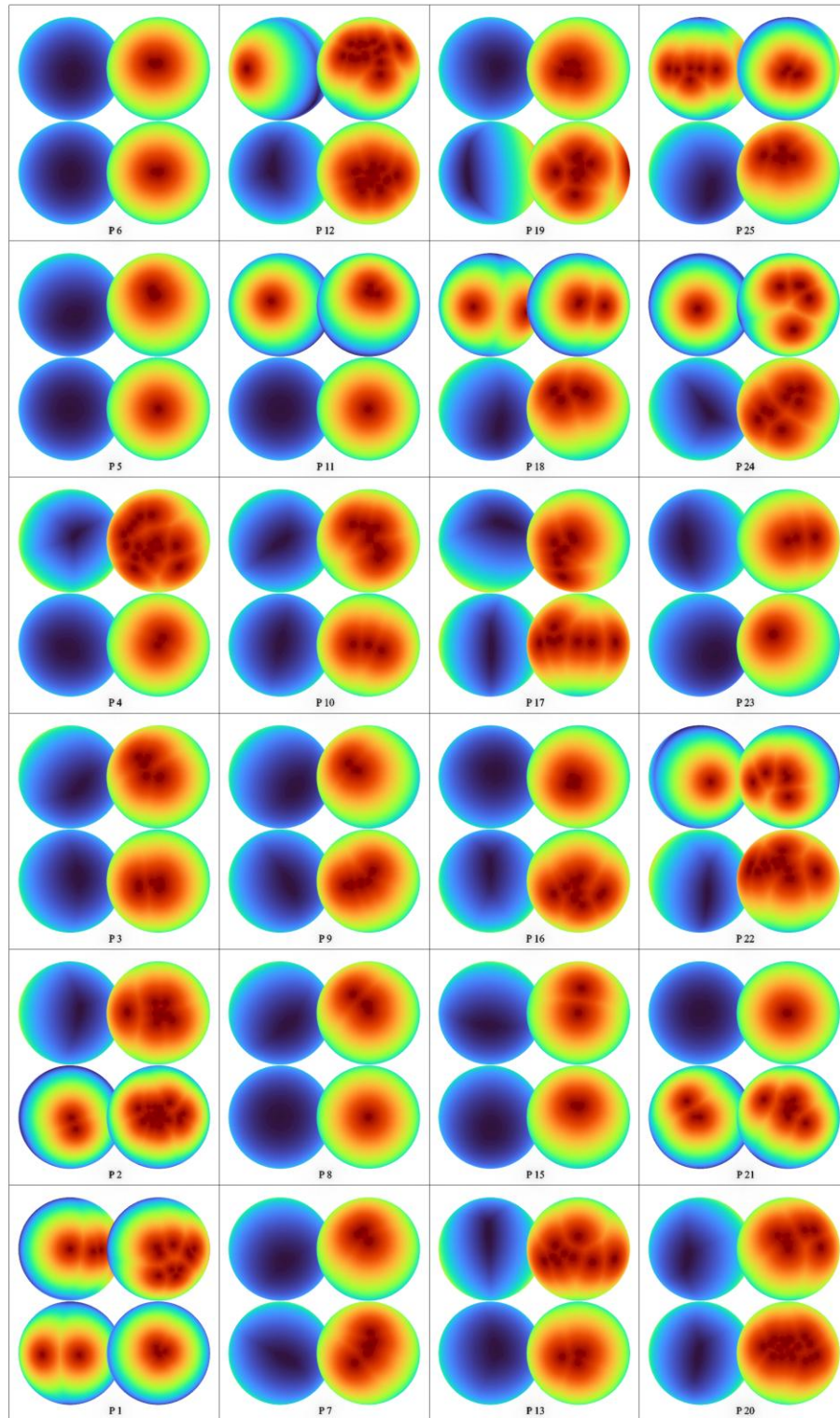


Figure 35: Distribution of players' focal point; In each section, the bottom pair is for IDM (with feedback) and the top for nIDM. In each pair, the right hemisphere represents players facing forward.

Another significant contribution of this thesis is evident in the number of design changes. As shown in Table 11, and Table 12, *users made significantly more design decisions when they received feedback. This is a testament to how the framework has enhanced user engagement with the design task, leading them to make more decisions and engage more* (statistically significant based on Table 12).

Table 12: Statistical description of design configuration and user movement

Measure	Feedback ON (mean $\pm$ SD)	Feedback OFF (mean $\pm$ SD)	t-statistic	p-value	Significant?
Design Alteration Count	107.6 $\pm$ 86.3	71.3 $\pm$ 44.6	2.79	0.0103	Yes (p < .05)
Unique Configurations	92.0 $\pm$ 79.3	60.9 $\pm$ 43.7	2.75	0.0114	Yes (p < .05)
Total Position Change	6 891.9 $\pm$ 9 093.9	4 827.7 $\pm$ 4 366.8	1.04	0.3090	No (p > .05)
Total View Change	3.38 $\pm$ 4.03	3.28 $\pm$ 3.58	0.09	0.9303	No (p > .05)

The players who received feedback moved around the design more and viewed it from different positions. This conclusion is based on the increased sum of player movements from Table 12. This is also visually depicted by mapping the players movement in Figure 36 and Figure 37.

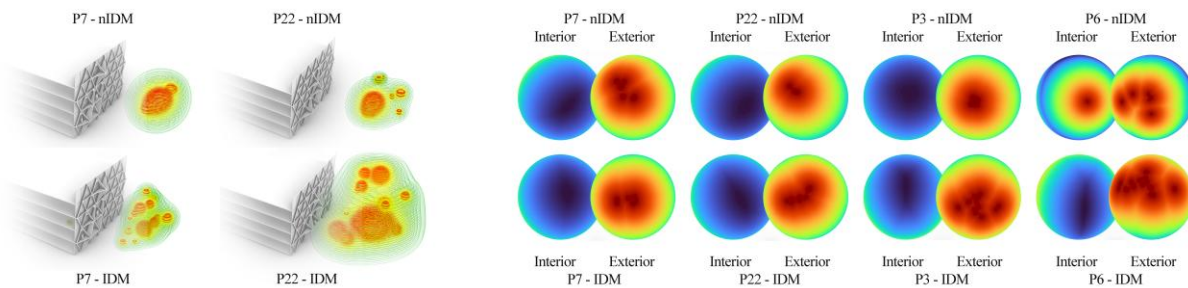


Figure 36: Detailed view of selected participants' navigation behavior: left shows users' positions, right shows users' focal points.

More precisely, receiving feedback has no significant effect on the angle of the user's interaction, only on its position (due to change in mean). *Therefore, the framework enhances spatial engagement with the design by encouraging a wider range of exploration and movement while receiving feedback.*



Figure 37: Distribution of players' movement while configuring design, relative to the design subject. In each pair, the left item is for the nIDM condition, and the right is for IDM.

Based on the earlier finding that receiving feedback enhances the range of user navigation, a complementary question arises: Does receiving feedback enhance the range of visited design configurations? In other words, do players tend to explore more varied alternatives when they receive feedback? Note that this is fundamentally different question. It is already established that the frequency or count of design decisions has increased, meaning the framework has enhanced the tendency for engagement. Nonetheless, this is now a deeper inquiry into the nature of each decision rather than their total count. It asks whether, regardless of the total design changes, the choices are more varied: This is an inquiry about the distribution of choices, not the frequency.

To explore this, identifiers for each design decision in the log are examined. The identifier is a string that stores which design configuration was generated by the user. The author takes an approach similar to standard deviation calculation. This means that for each participant, the list of identifiers forms a matrix where the rows are decisions. Transposing that matrix reveals changes per element. Averaging these movements, results in an average or standard configuration for each player. Then, the difference between each decision and the player's standard or average configuration is calculated. The greater this difference, the more varied the choices are, indicating more extensive design exploration.

For each participant, the average configuration and sum of differences can be seen in Table 13. Darker rows indicate an increase in exploration when feedback was shown. Therefore, *in 70.8 percent of the participants, receiving feedback enhanced the range of design exploration. In other words, the framework has enabled them to have a more extensive design landscape exploration with more variant choices.*

Table 13: Design exploration ranges from average configuration per participant and condition.

	Standard Configuration Identifier (Rounded/With Feedback)	Standard Configuration Identifier (Rounded/Without Feedback)	Differences (With Feedback)	Differences (No Feedback)	Enhanced Exploration
1	5545111545111143111122413222554332455	5135415534113343211321312222414232143	2361.15782	2006.761927	TRUE
2	555225511113511213541535553555411211	2224316663446513446611316611336621633	2523.789449	2675.345215	FALSE
3	4134413522445511545531215431112511511	1112214412333442332111313121314431323	2508.577904	2476.470714	TRUE
4	11111111441112411144111155111123355	122121213311231311222111111111111234	2449.712536	3027.130961	FALSE
5	211312222221122213542213443123322211	111111451311155111132111311213541231	2439.211647	2999.141632	FALSE
6	4111533433415331316511333541113221121	222211212213311212431322231333111111	2569.838741	1589.386455	TRUE
7	2114315545333531231352316544126424433	4116213514212243112213315411115411311	2436.493731	1852.731641	TRUE
8	433231422112112112132212222311331	122111111511212515222225511333411111	2569.724616	1483.398195	TRUE
9	212311332211121214421213531222121111	211111231111134111133112111113211111	2565.901913	1018.714277	TRUE
10	4111444314343312221111414651434651261	212221321222222212121113331112431111	2600.401114	1565.294629	TRUE
11	2134116311216641222451341333353132332	511121524322254122431322221321231333	2537.187727	1839.631941	TRUE
12	2223212321212211112243421143433433433	4555525112514323115333313141112333324	2362.13947	2088.840783	TRUE
13	4114435313112441414231116411114431122	31115421211432251521555132111113121	2472.116415	1951.709124	TRUE
15	2222113314113241121131111332121122	43311111511254411255131124332111144	2161.302983	1977.094318	TRUE
16	4213224332415431114311214211214521422	231323342211221221441121341123521421	2147.511444	1467.204317	TRUE
17	3221323111213111213211412211424111111	43333331111111211361323363225431342	1686.438862	1670.811757	TRUE
18	411432341423244122211141511514551555	6566354446444255554216525111545111111	2055.325733	2822.908822	FALSE
19	4225223515313254113334412334434525455	3333111336111155111156333166433425455	2274.817433	2327.321582	FALSE
20	211132121432234522323112433112353155	434112121411115111143512243442224244	2093.756068	2018.879859	TRUE
21	222221332311223311223322333333344344	32224144234454334555214455213335223	2019.383302	2179.570318	FALSE
22	3232112322111222211221432122223221232	4543221324415533215511312111442341143	1277.177418	2270.100651	FALSE
23	232322443432445543444333333443333333	2122212312442232435222134422112441134	2162.226761	1950.566016	TRUE
24	111421155521221112221212222212333123	211152422324222111321311442533143111	2288.885913	1924.448725	TRUE
25	4222222211221122422112415311114331112	525441312443545534345445431434443134	63675.48599	43318.67577	TRUE

6 CONTRIBUTIONS

Contributions of this research can be broadly divided into three categories. First, it established the theoretical foundations for developing an informed decision-making framework, rooting itself in decision-making and behavioral theories. It also provided clear suggestions for developing such frameworks based on those theories. In addition, other novel contributions from the framework's developments are formulated and proven by the usability test, such as the validity of feedback encapsulation and user inclination towards both relative and overall feedback, which is also supported by user comments.

The second category of contributions involves proposing a functional framework, addressing challenges, and conducting data collection. The framework underwent multiple iterations and has significance in its services, communications, and processes.

The third category is the statistical analysis of the collected data, which shows significant changes in user behavior and design steering. This novel contribution sheds light on how such frameworks can empower non-expert designers to make better-performing choices while maintaining their preferences.

The combination of the three layers of this research has enabled this work to:

- Providing factual evidence rather than preferences or self-reported data,
- Improving insight into user-design interactions,
- Proposing a prototypical, functional, technical framework,
- Contribution to higher-level argument and broader outlook,
- Proposing a novel and scalable research design.

To conclude, this thesis contributes to the body of knowledge at the intersection of informed decision-making, data-driven design, and user-centered design. This area has been explored with a novel evidence-based methodology. Several technical and theoretical contributions have been reported, collectively contributing to three levels of impact.

At the highest level, this thesis contributes to an overarching statement; Although this statement might have been presumed in the design/art field, the thesis sheds light on it and helps clarify its scope: Specifically, the statement is, if non-expert designers are equipped with feedback in a data-driven

framework, they can engage with it effectively, balancing performance and preference. This was proven through changes in decision-making time and detailed reports on metric changes.

On the second level, the thesis contributes to the field of serious games and their enabling effect on education and informed decision-making. This is achieved through multiple rounds of developing a functional technical framework that uses gamification strategies, novel feedback conveyance method and user interaction modes. The experiment design and trials explored options for these aspects, resulting in a functional proof of concept.

At the third level, this thesis has opened horizons for future research based on its statistical analysis, which not only supports the overarching statement but also outlines a clear roadmap for future evidence-based research by providing solid foundational information and early analysis.

6.1 Revisiting Research Questions and Hypotheses

This section reflects on the initial research questions and hypotheses, evaluating how the outcomes of this study addressed each point through both theoretical grounding and empirical analysis. The combination of a game-enabled framework, behavioral metrics, and mixed-method evaluation allowed for a comprehensive exploration of non-expert decision-making in design environments.

1 RQ1: What are the mechanics and characteristics of a game-enabled informed decision-making framework to help non-expert designers make design configurations?

This research identified and implemented key mechanics—such as encapsulated, multi-metric feedback, fault-proof design generation, achievement-based progression, and multimedia cues—within a gamified environment. The framework incorporated elements derived from behavioral models (e.g., Health Belief Model, Theory of Planned Behavior, Utility Theory), ensuring decision transparency, comprehensibility, and engagement. The developed system was curated and moderated to ensure a smooth decision-making experience for non-experts, successfully addressing RQ1.

2 RQ2: How does the presence of real-time feedback influence the decision-making behavior of non-expert designers in early-stage architectural design tasks?

Statistical analysis confirmed that real-time feedback significantly influenced decision-making behavior. Participants exposed to feedback showed notably different decision times, with a clear trend of micro-learning and increased behavioral consistency over time. This indicates that feedback not only changed how participants made decisions but also contributed to the development of internal heuristics during task repetition, successfully answering RQ2.

3 RQ3: Can encapsulated performance feedback across structural, environmental, and cost metrics help non-expert users make better-informed design decisions without explicit technical knowledge?

Yes. The use of relative and comparative feedback (rather than raw numerical data) enabled non-experts to make statistically better design choices across multiple performance domains (while maintaining the option of seeing the full report of analysis). The study recorded a improvement across seven performance metrics, despite participants not having access to the underlying technical detail. This confirms that encapsulated performance feedback is both effective and accessible, directly addressing RQ3.

4 RQ4: What is the effect of gamification strategies on the engagement, comprehension, and decision-making consistency of early designers within a participatory design framework?

Gamification elements—such as points, feedback loops, pre-testing, and cyclical subtasks—were found to enhance engagement and comprehension. The System Usability Scale (SUS) scores, along with observed learning behavior and decreased undo rates, point toward greater user confidence, consistency, and understanding over time. The framework's achievement-based flow and real-time visual feedback made the design process more intuitive and enjoyable, successfully responding to RQ4.

And for revisiting the hypotheses, it can be concluded that:

5 H1: Participants receiving real-time performance feedback will demonstrate significantly different decision-making times compared to those who do not receive feedback.

This hypothesis was supported. Empirical evidence showed a significant variation in decision-making time between feedback and no-feedback conditions. Participants with feedback engaged in faster yet more deliberate decision-making, suggesting increased confidence and understanding driven by iterative feedback.

6 H2: Gamified, encapsulated feedback can steer non-expert designers toward better-performing design configurations across multiple metrics without overwhelming them with technical complexity.

This hypothesis was also validated. Despite not receiving raw data, participants in the feedback condition made decisions with more improvements, and usability scores indicated they were not overwhelmed by the feedback mechanisms. The layered and intuitive delivery of feedback ensured accessibility and clarity, fulfilling H2.

Together, these findings underscore the value of a gamified, real-time feedback framework in enabling non-expert participation in early-stage architectural design. The research questions and hypotheses were not only addressed, but also enriched through the synthesis of theoretical grounding, system development, and empirical validation.

7 FUTURE RESEARCH DIRECTIONS

The future direction of this research can focus on adaptability, customization, and scalability. Building on the foundations established in this dissertation, future work may pursue two complementary trajectories—one centered on advancing the theoretical underpinnings of GIDM, and the other focused on extending the technical capabilities of the configurator.

7.1 Theoretical Extensions

Although this research has demonstrated that encapsulated, multi-metric feedback reliably guides non-expert users toward higher-performing design outcomes, several open questions remain regarding the evolutionary dynamics of learning and engagement over time. First, the procedural complexity scaling inherent in game design warrants systematic study: as users internalize each layer of feedback, new dimensions might be progressively unlocked, mirroring the escalation of challenge found in commercial serious games. Empirical investigation should examine how staged introduction of additional metrics influences retention, motivation, and design diversity. As users gain proficiency, static feedback may limit their ability to explore advanced design considerations. Future research may investigate procedural complexity scaling as this approach draws on the concept of latent learning observed and reported in this thesis.

Second, the current framework treats all users uniformly, yet some theories suggest that individual differences in prior knowledge, spatial skills, and self-efficacy will shape learning trajectories. Future research could develop and validate adaptive feedback algorithms that tailor both the content and pacing of performance cues to the learner’s profile to optimize scaffold timing and complexity. This direction is based on the fact that user profiling revealed diverse decision-making patterns, suggesting that personalized adaptivity could enhance outcomes. Further investigations may explore adaptive learning curves, where the framework adjusts feedback presentation, interface prompts, or task difficulty based on individual user profiles. For instance, users with faster decision times could receive more complex design tasks, while those struggling with navigation might receive enhanced spatial cues.

Finally, while this dissertation’s within-subject study provides high-resolution insight into short-term decision making, it does not capture longitudinal engagement or the decay of acquired design heuristics over time. A future program of repeated-play studies—spanning days or weeks—could elucidate how participants revisit and refine earlier strategies, whether performance improvements plateau or continue, and how feedback retention interacts with real-world design transfer. Such longitudinal data would also

permit analysis of motivational constructs—flow, self-efficacy, and growth mindset—in a sustained, real-use context.

Moreover, another line of research could explore how changing the design and demographics domain could be managed, particularly in interdisciplinary contexts like urban planning or product design. This will involve studying how the framework fosters inclusivity across diverse user groups, building on its scalability

7.2 Technical Advancements

On the technical front, the findings here provide a stable interface and fixed feedback rules, but gamified systems benefit from dynamic evolution. A primary next step is to integrate procedural complexity scaling directly within the engine: for example, unlocking new feedback layers (e.g., detailed cost breakdowns or structural safety margins) only after users demonstrate proficiency on earlier metrics. This approach requires building a modular feedback pipeline that can ingest user performance logs and adjust which cues are presented in real time. Complementing this, the system may adopt personalized experience and/or interface to adjust challenge difficulty or feedback granularity; The framework may incorporate adaptive interface algorithms that adjust feedback and task complexity based on user performance.

In accordance with the previous point, another line could be Cloud-Based Scalability. The framework's current TCP socket communication ensures reliable data transfer, but scaling to larger audiences or urban design tasks requires cloud-based infrastructure. This will enhance accessibility and support collaborative design across distributed teams, building on the framework's participatory goals.

Integration with emerging technologies could also be explored. Different mediums like AR/VR could be suggested to provide comparable data. More context-specific rules may be considered, such as implementing building codes or autonomous code checking during the configuration phase.

Finally, extending the platform to longitudinal engagement models will involve implementing user-account management, persistent progress tracking, and session-to-session state preservation. Coupled with periodic reassessment of baseline performance, this would allow the system to detect and respond to skill decay or emerging expertise. Integrating analytics dashboards for both researchers and users could further enable meta-reflection and self-directed learning. For future research in this field, an emphasis on portability can be suggested as a starting point. This means that if the capabilities developed here are packaged and managed internally, instead of relying on inter-process communication, it could alleviate some challenges in reaching out to other domains.

REFLECTIONS ON THE RISKS OF FUTURE RESEARCH DIRECTIONS

A question can arise regarding the long-term effect of gamification in design. One might ask whether the positive effects of gamification in learning are lasting and whether it relates to decision-making or is scalable to the complete design pipeline; Using gamification to engage stakeholders in design is a new and less explored topic. The answer to the first question can be sought in related fields like education and health. For example, Ratinho and Martins (2023) conducted a systematic review in higher education, examining over 500 studies on gamification with a longitudinal perspective. They affirm that even in that domain, there is limited knowledge about the long-term effects of using game-based strategies to build user capabilities. They report that even in the longest available study, from 2010 to 2022, spanning three years, it was assumed that as the novelty effect decayed, interest in engagement might reduce. Contrarily, they found that the positive effects of gamification in education were still observable throughout the three years. They note, “Declines in motivation occur especially after long periods of exposure to a gamified design, when the novelty effect disappears, and users lose interest in the gamification process... There is, however, evidence that complex gamified designs can reduce the influence of the novelty effect, affording students higher levels of motivation, even with longer exposure to gamification. In this review, the studies with the most complex gamification strategies revealed increases in motivation despite being longitudinal studies. This may be due to advanced gamified designs having specific conditions (e.g., alternate strategies, groups, and tasks) which readily facilitate gameful experiences and can thereby contribute to higher levels of motivation and engagement.”(Rodrigues et al. 2022).

This is also affirmed by other studies, such as another systematic review by Rodrigues et al. (2022), which concludes: “Interestingly, gamification’s impact shifted to an uptrend between six and ten weeks after the start of the intervention, partially recovering its contribution naturally. Thus, we found empirical evidence supporting that gamification likely suffers from the novelty effect but also benefits from the familiarization effect, which contributes to an overall positive impact on users.”

Therefore, this debate over the efficacy of gamification over time points to a novel future direction, which is also mentioned in Chapter Seven. A possible solution, based on the aforementioned literature review, to mitigate this is to emphasize the need for adaptability and customization in interaction systems based on user profiles.

Returning to the first part of the question, while acknowledging the call for further studies on the long-term effects of gamification in decision-making, a promising direction has already emerged: the need for adaptability and customization in rewards and interaction systems.

The second part of the question asks to what extent informed decision-making is related to the process of designing a building. This approach suggests that designing is more complex than simple configurational exploration and that decision-making can be an oversimplification in this context; Architectural design is undeniably a multifaceted, socio-technical process—far more than simply choosing massing options or toggling parameters. Yet at its core, design remains a structured form of problem solving, and there is clear precedent for harnessing gamified, feedback-driven environments to build capability in complex domains.

These examples illustrate two important points:

1. Structured feedback and incremental challenges significantly enhance learning, even in complex domains.
2. Problem solving—breaking a large, ambiguous goal into discrete, solvable tasks—is central to both education and design.

Another discussion is the Oversimplification Risk; It is fair to argue that configurational exploration and decision-making, as operationalized in this thesis, represent only one slice of architectural design. Focusing solely on a gamified exoskeleton configurator could be viewed as reducing design to pattern-matching, thereby overlooking the rich web of social, cultural, and technical considerations; However, there is a counterargument for this: “A Middle Way: Problem-Solving as a Core Module”; Rather than attempting to replace comprehensive design workflows, frameworks like the one proposed here can be seen as specialized decision-support modules that excel at the problem-solving subtask of multi-criteria trade-off. In practice:

1. Scoping Complex Problems: By bounding the design domain (e.g., exoskeleton façade), the system isolates a critical problem and makes it tractable for non-experts.
2. Accelerating Heuristic Development: Adaptive feedback helps users internalize performance heuristics that would otherwise require specialized training.
3. Integrating into Broader Workflows: Such modules can be embedded within larger BIM or participatory design platforms—serving as “design drills” that sharpen specific skills before or alongside broader conceptual work.

In conclusion, informed decision-making platforms are not a panacea for the entire edifice of architectural design; rather, they are powerful accelerators for problem-solving components within that process. Just as other similar examples do not claim to teach every aspect of their domains but excel at targeted learning

tasks, a gamified configurator can deliver measurable gains in decision quality, user confidence, and design literacy—without pretending to replace architects or supplant the full richness of design practice.

Which leads to another question: at which stage of design can the proposed framework be most beneficial? Game engines offer two essential qualities for informed decision-making in design: perceptual engagement and systemic mediation. As stated in the thesis, they are “rule-bound, feedback-rich environments” that enable users to explore design options with immediate, intuitive feedback. The gamified interface, featuring mechanics like rules and trade-offs, encourages intuitive exploration of the design space, resulting in over 70% of users increasing their exploration.

However, they may not be appropriate for *all* design tasks—especially highly linear, document-based workflows. Their strength lies in configurational, exploratory, and spatial design domains. In this context—early-stage, performance-sensitive spatial configuration—they are ideal. Game engines are ideal for early-stage design, where conceptual exploration and user engagement are critical, but specialized software may be needed for detailed engineering phases.

Another issue can be the reduction of information dimensions when feedback is encapsulated. Encapsulated feedback reduces cognitive load and prevents overwhelming non-expert users with technical data. As shown in this thesis, users improved performance across seven metrics without being exposed to raw values. This validates the idea that well-designed abstraction can be more effective for decision-making than unfiltered data. However, as the complexity of the problem increases or with different user profiles, there may be a need to reintroduce the feedback’s initial layers and report it in a detailed and non-relative manner. This is a valid point, yet the solution is already embedded in the framework, which has the capability to present multiple layers of information. This presents a path for future work, where, based on user behavior, the framework can provide different levels of information. It must be mentioned again that within the scope of this study, the efficacy of encapsulated feedback was proven effective in leading the design in a high-performing direction while maintaining system usability and not overwhelming users. However, this provides the groundwork for the next step, which will build upon the dosage, type, and timing of feedback information and its association with user behavior or type.

REFERENCES

- Abelson, J., J. Lomas, J. Eyles, S. Birch, and G. Veenstra. 1995. "Does the Community Want Devolved Authority? Results of Deliberative Polling in Ontario." *CMAJ: Canadian Medical Association Journal* = *Journal de L'association Medicale Canadienne* 153 (4): 403–12. <https://www.ncbi.nlm.nih.gov/pmc/articles/pmid/7634217/?tool=EBI>.
- Abelson, Julia, Pierre Gerlier Forest, John Eyles, Patricia Smith, Elisabeth Martin, and Francois Pierre Gauvin. 2003. "Deliberations about Deliberative Methods: Issues in the Design and Evaluation of Public Participation Processes." *Social Science & Medicine* 57 (2): 239–51. [https://doi.org/10.1016/S0277-9536\(02\)00343-X](https://doi.org/10.1016/S0277-9536(02)00343-X).
- Alexander, Keith. 2006. "The Application of Usability Concepts in the Built Environment." *Journal of Facilities Management* 4 (4): 262–70. <https://doi.org/10.1108/14725960610702947>.
- Anam, Nadia, and Linus Tan. 2023. "A Human-Centric Approach to a Design-to-Fabrication Process: A Case of Homeless Housing Design in Melbourne, Australia." *Proceedings of the 28th Conference on Computer Aided Architectural Design Research in Asia (CAADRIA) [Volume 2]* 2 (June):461–70. <https://doi.org/10.52842/CONF.CAADRIA.2023.2.461>.
- Archilogic AG. 2016. "Archilogic." 2016. <https://www.archilogic.com/>.
- Arif, Ahmed Sabbir. 2015. "Predicting and Reducing the Impact of Errors in Character-Based Text Entry," May. <https://yorkspace.library.yorku.ca/xmlui/handle/10315/28170>.
- Autodesk. 2022. "Autodesk Forma." Autodesk. 2022. <https://www.autodesk.com/products/forma/overview?term=1-YEAR&tab=subscription>.
- Azar, Elie, William O'Brien, Salvatore Carlucci, Tianzhen Hong, Andrew Sonta, Joyce Kim, Maedot S. Andargie, Tareq Abuimara, Mounir El Asmar, et al. 2020. "Simulation-Aided Occupant-Centric Building Design: A Critical Review of Tools, Methods, and Applications." *Energy and Buildings* - 110292 224 (October):1–17. <https://doi.org/10.1016/j.enbuild.2020.110292>.
- Azar, Elie, William O'Brien, Salvatore Carlucci, Tianzhen Hong, Andrew Sonta, Joyce Kim, Maedot S. Andargie, Tareq Abuimara, Mounir el Asmar, et al. 2020. "Simulation-Aided Occupant-Centric Building Design: A Critical Review of Tools, Methods, and Applications." *Energy and Buildings* 224 (October):110292. <https://doi.org/10.1016/J.ENBUILD.2020.110292>.
- Banaei, Maryam, Ali Ahmadi, and Abbas Yazdanfar. 2017. "Application of AI Methods in the Clustering of Architecture Interior Forms." *Frontiers of Architectural Research* 6 (3): 360–73. <https://doi.org/10.1016/J.FOAR.2017.05.002>.
- Banaei, Maryam, Javad Hatami, Abbas Yazdanfar, and Klaus Gramann. 2017. "Walking through Architectural Spaces: The Impact of Interior Forms on Human Brain Dynamics." *Frontiers in Human Neuroscience* 11 (September). <https://doi.org/10.3389/FNHUM.2017.00477>.
- Bandura, Albert. 1989. "Human Agency in Social Cognitive Theory." *American Psychologist* 44 (9): 1175–84. <https://doi.org/10.1037/0003-066X.44.9.1175>.

- Baños, R.M., C. Botella, M. Alcañiz, V. Liaño, B. Guerrero, and B. Rey. 2004. "Immersion and Emotion: Their Impact on the Sense of Presence." *CyberPsychology & Behavior* 7 (6): 734–41. <https://doi.org/10.1089/cpb.2004.7.734>.
- Baus, Oliver, Stephane Bouchard, Veronique Gougeon, and Francois-Xavier Roucaut. 2011. "Comparison of Anxiety in Response to Virtual Spiders While Immersed in Augmented Reality, Head-Mounted Display, or CAVE-like System." *Journal of CyberTherapy and Rehabilitation* 4 (2): 171–75. <https://doi.org/10.1089/CYBER.2009>.
- Beierle, Thomas C. 1999. "Using Social Goals to Evaluate Public Participation in Environmental Decisions." *Review of Policy Research* 16 (3–4): 75–103. <https://doi.org/10.1111/J.1541-1338.1999.TB00879.X>.
- Beierle, Thomas C., and Jerry Cayford. 2002. *Democracy in Practice: Public Participation in Environmental Decisions*. *Democracy in Practice: Public Participation in Environmental Decisions*. 1st ed. New York: Routledge. <https://doi.org/10.4324/9781936331017/DEMOCRACY-PRACTICE-THOMAS-BEIERLE>.
- Bell, Karen, and Mark Reed. 2022. "The Tree of Participation: A New Model for Inclusive Decision-Making." *Community Development Journal* 57 (4): 595–614. <https://doi.org/10.1093/CDJ/BSAB018>.
- Berkowitz, Alan D. 2004. "The Social Norms Approach: Theory, Research, and Annotated Bibliography."
- BIG (Bjarke Ingels Group). 2013. "THE BIG U – Rebuild by Design." 2013. <https://rebuildbydesign.org/work/funded-projects/the-big-u/>.
- Brady, James T. 1986. "A Theory of Productivity in the Creative Process." *IEEE Computer Graphics and Applications* 6 (05): 25–34. <https://doi.org/10.1109/MCG.1986.276789>.
- Buss, Andrew, Ashley Del Bianco, and Lindsey Keck. 2013. "Democratizing Design: Using Collaboration and Consensus Building as a Principle of Good Government." *International Journal of Design Management and Professional Practice* 6 (1): 47–59. <https://doi.org/10.18848/2325-162x/cgp/v06i01/38641>.
- Bussey, Kay, and Albert Bandura. 1999. "Social Cognitive Theory of Gender Development and Differentiation." *Psychological Review* 106 (4): 676–713. <https://doi.org/10.1037/0033-295X.106.4.676>.
- Campbell, Chris, Wolff Michael Roth, and Alfredo Jornet. 2019. "Collaborative Design Decision-Making as Social Process." *European Journal of Engineering Education* 44 (3): 294–311. <https://doi.org/10.1080/03043797.2018.1465028>.
- Carlucci, Salvatore, Marilena de Simone, Steven K. Firth, Mikkel B. Kjærgaard, Romana Markovic, Mohammad Saiedur Rahaman, Masab Khalid Annaqeeb, et al. 2020. "Modeling Occupant Behavior in Buildings." *Building and Environment* 174 (May):106768. <https://doi.org/10.1016/J.BUILDENV.2020.106768>.
- Champion, Victoria L., and Celette Sugg Skinner. 2008. "The Health Belief Model." In *The Health Belief Model. Health Behavior and Health Education: Theory, Research, and Practice*, edited by Karen Glanz, Barbara K. Rime, and K. Viswanath, 4:45–62. San Francisco: Published by Jossey-Bass A Wiley Imprint.
- Cheng, Hao-Fei, Ruotong Wang, Zheng Zhang, Fiona O'Connell, Terrance Gray, F. Maxwell Harper, and Haiyi Zhu. 2019. "Explaining Decision-Making Algorithms through UI." In *Proceedings of the 2019*

- CHI Conference on Human Factors in Computing Systems*, 1–12. New York, NY, USA: ACM. <https://doi.org/10.1145/3290605.3300789>.
- Chien, Sheng-Fen. 2002. “Games and Play Design Gaming, Designing Games Learning Design through Game Playing and Game Making.” *Proceedings of ECAADe 2002*, 28–33. <http://mail.ntust.edu.tw/~schien/>.
- Christiansson, Per, Kjeld Svidt, Kristian Birch Pedersen, and Ulrik Dybro. 2011. “USER PARTICIPATION IN THE BUILDING PROCESS.” *Journal of Information Technology in Construction - ISSN 1874-4753* 16:309–35. <http://www.itcon.org/2011/20>.
- Clark, Douglas B., Emily E. Tanner-Smith, and Stephen S. Killingsworth. 2016. “Digital Games, Design, and Learning: A Systematic Review and Meta-Analysis.” *Review of Educational Research* 86 (1): 79–122. https://doi.org/10.3102/0034654315582065/ASSET/IMAGES/LARGE/10.3102_0034654315582065-FIG3.JPEG.
- Conner, Mark, and Christopher J. Armitage. 1998. “Extending the Theory of Planned Behavior: A Review and Avenues for Further Research.” *Journal of Applied Social Psychology* 28 (15): 1429–64. <https://doi.org/10.1111/J.1559-1816.1998.TB01685.X>.
- Cove.Tool. 2018. “Cove.” 2018. <https://cove.tools/>.
- Creswell, John W., and J. David Creswell. 2018. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. 5th ed. Sage Publications.
- Csikszentmihalyi, Mihaly. 1990. *Flow: The Psychology of Optimal Experience*. https://www.researchgate.net/publication/224927532_Flow_The_Psychology_of_Optimal_Experience.
- Dalsgaard, Peter, and Kim Halskov. 2010. “Designing Urban Media Façades: Cases and Challenges.” *Conference on Human Factors in Computing Systems - Proceedings* 4:2277–86. <https://doi.org/10.1145/1753326.1753670>.
- DesignBuilder Software Ltd. 2010. “DesignBuilder.” 2010. <https://designbuilder.co.uk/>.
- Deterding, Sebastian, Dan Dixon, Rilla Khaled, and Lennart Nacke. 2011. “From Game Design Elements to Gamefulness: Defining ‘Gamification.’” In *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments, MindTrek 2011*. <https://doi.org/10.1145/2181037.2181040>.
- Dichev, Christo, and Darina Dicheva. 2017. “Gamifying Education: What Is Known, What Is Believed and What Remains Uncertain: A Critical Review.” *International Journal of Educational Technology in Higher Education* 14 (1): 1–36. <https://doi.org/10.1186/S41239-017-0042-5/TABLES/14>.
- DiClemente, Carlo C. 2005. “Conceptual Models and Applied Research: The Ongoing Contribution of the Transtheoretical Model.” *Journal of Addictions Nursing* 16 (1–2): 5–12. <https://doi.org/10.1080/10884600590917147>.
- D’oca, Simona, H Burak Gunay, Sara Gilani, and William O’Brien. 2019. “Critical Review and Illustrative Examples of Office Occupant Modelling Formalisms.” *Building Serv. Eng. Res. Technol* 40 (6): 732–57. <https://doi.org/10.1177/0143624419827468>.

- D'Oca, Simona, Chien Fei Chen, Tianzhen Hong, and Zsafia Belafi. 2017. "Synthesizing Building Physics with Social Psychology: An Interdisciplinary Framework for Context and Occupant Behavior in Office Buildings." *Energy Research & Social Science* 34 (December):240–51. <https://doi.org/10.1016/J.ERSS.2017.08.002>.
- Donald, Merlin. 1993. "Précis of Origins of the Modern Mind: Three Stages in the Evolution of Culture and Cognition ." *Behavioral and Brain Sciences* 16 (4). <https://doi.org/10.1017/s0140525x00032647>.
- Dreyfuss, Stuart E., and Hubert L. Dreyfus. 1980. "A Five-Stage Model of the Mental Activities Involved in Directed Skill Acquisition." *Operations Research Center*, no. February.
- Du, Jing, Yangming Shi, Zhengbo Zou, and Dong Zhao. 2018. "CoVR: Cloud-Based Multiuser Virtual Reality Headset System for Project Communication of Remote Users." *Journal of Construction Engineering and Management* 144 (2): 04017109. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001426](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001426).
- Du, Jing, Zhengbo Zou, Yangming Shi, and Dong Zhao. 2018. "Zero Latency: Real-Time Synchronization of BIM Data in Virtual Reality for Collaborative Decision-Making." *Automation in Construction* 85 (January):51–64. <https://doi.org/10.1016/j.autcon.2017.10.009>.
- Dunbar, Norah E., Claude H. Miller, Bradley J. Adame, Javier Elizondo, Scott N. Wilson, Brianna L. Lane, Abigail Allums Kauffman, et al. 2014. "Implicit and Explicit Training in the Mitigation of Cognitive Bias through the Use of a Serious Game." *Computers in Human Behavior* 37 (August):307–18. <https://doi.org/10.1016/J.CHB.2014.04.053>.
- Edwards, Gareth, Haijiang Li, and Bin Wang. 2015. "BIM Based Collaborative and Interactive Design Process Using Computer Game Engine for General End-Users." *Visualization in Engineering* 3 (1): 1–17. <https://doi.org/10.1186/S40327-015-0018-2/FIGURES/13>.
- Edwards, Ward. 1954. "The Theory of Decision Making." *Psychological Bulletin* 51 (4): 380–417. <https://doi.org/10.1037/H0053870>.
- Eiris, Ricardo, and Masoud Gheisari. 2017. "Research Trends of Virtual Human Applications in Architecture, Engineering and Construction." *Journal of Information Technology in Construction (ITcon)* 22:168–84. <http://www.itcon.org/2017/9>.
- Elesawy, Amr, Stefan Caranovic, Justin Zarb, Prageeth Jayathissa, and Arno Schlueter. 2018. "HIVE Parametric Tool - A Simplified Energy Simulation Tool for Educating Architecture Students." *Kepeczynska-Walczak, A, Bialkowski, S (Eds.), Computing for a Better Tomorrow - Proceedings of the 36th ECAADe Conference - Volume 1, Lodz University of Technology, Lodz, Poland, 19-21 September 2018, Pp. 657-666* 1:657–66. <https://doi.org/10.52842/CONF.ECAADE.2018.1.657>.
- Eun, Barohny. 2019. "The Zone of Proximal Development as an Overarching Concept: A Framework for Synthesizing Vygotsky's Theories." *Educational Philosophy and Theory* 51 (1): 18–30. <https://doi.org/10.1080/00131857.2017.1421941;WGROU:STRING:PUBLICATION>.
- Evans, Gary W., John Fellows, Marion Zorn, and Kim Doty. 1980. "Cognitive Mapping and Architecture." *Journal of Applied Psychology* 65 (4): 474–78. <https://doi.org/10.1037/0021-9010.65.4.474>.
- Fishburn, Peter C. 1979. *Utility Theory for Decision Making*. New York: John Wiley and Sons, INC.
- Friedman, Yona. 1971. "The Flatwriter: Choice by Computer." *Progressive Architecture*, March, 98–101. <https://usmodernist.org/PA/PA-1971-03.pdf>.

- . 1980. *Toward a Scientific Architecture*. MIT Press. <https://mitpress.mit.edu/9780262560191/toward-a-scientific-architecture/>.
- . 1996. “Structures Without Rules and Their Implementation in Architecture.” *International Journal of Space Structures* 11 (1–2): 85–91. <https://doi.org/10.1177/026635119601-214>.
- Grabe, Jörn von. 2016. “The Systematic Identification and Organization of the Context of Energy-Relevant Human Interaction with Buildings—a Pilot Study in Germany.” *Energy Research & Social Science* 12 (February): 75–95. <https://doi.org/10.1016/j.erss.2015.12.001>.
- Grier, Rebecca A., Aaron Bangor, Philip Kortum, and S. Camille Peres. 2013. “The System Usability Scale.” *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, September, 187–91. <https://doi.org/10.1177/1541931213571042>.
- Guidotti, Riccardo, Anna Monreale, Salvatore Ruggieri, Franco Turini, Fosca Giannotti, and Dino Pedreschi. 2019. “A Survey of Methods for Explaining Black Box Models.” *ACM Computing Surveys* 51 (5): 1–42. <https://doi.org/10.1145/3236009>.
- Guyot, Paul, and Shinichi Honiden. 2006. “Agent-Based Participatory Simulations: Merging Multi-Agent Systems and Role-Playing Games.” *Journal of Artificial Societies and Social Simulation* 9 (4).
- Harness, Clifton, and Ryan Griege. 2022. “TestFit: Real Estate Feasibility Platform.” 2022. <https://www.testfit.io/>.
- Heisel, Felix, and D. E. Hebel. 2019. “Pioneering Construction Materials through Prototypological Research.” *Biomimetics* 2019, Vol. 4, Page 56 4 (3): 56. <https://doi.org/10.3390/BIOMIMETICS4030056>.
- Hensel, Michael, and Milica Vujovic. 2022. “Evidence-Based Design: Scientific Method, Research Data and Methods.” Wien.
- Hewett, Thomas T. 2005. “Informing the Design of Computer-Based Environments to Support Creativity.” *International Journal of Human-Computer Studies* 63 (4–5): 383–409. <https://doi.org/10.1016/j.ijhcs.2005.04.004>.
- Hookway, Branden. 2014. *Interface*. MIT Press.
- Hornbæk, Kasper, Aske Mottelson, Jarrod Knibbe, and Daniel Vogel. 2019. “What Do We Mean by ‘Interaction’? An Analysis of 35 Years of CHI.” *ACM Transactions on Computer-Human Interaction (TOCHI)* 26 (4). <https://doi.org/10.1145/3325285>.
- Howard, John A., and Jagdish N. Sheth. 2001. “The Theory of Buyer Behavior.” In *Marketing: Critical Perspectives on Business and Management*, edited by Michael J. Baker. Vol. 3. London and New York: Routledge.
- Ingrassia, Marco, Chiara Farinea, Cristian Rizzuti, Davide Leone, Nick Apeldoorn, Igor Mayer, and Zhan Goosen. 2020. *PUBLIC PLAY SPACE INITIATIVES CATALOGUE Collection of Best Practices*. Edited by Areti Markoupoulou, Marco Ingrassia, Chiara Farinea, Mathilde Marengo, and Raquel Villodres. Barcelona: Institute for Advanced Architecture of Catalonia. https://pure.buas.nl/ws/files/15697618/Public_Play_Space_State_of_the_Art_Catalogue.pdf.
- Integrated Environmental Solutions Limited. 2016. “ICD.” 2016. <https://www.iesve.com/products/icd>.
- IronPython Contributors, Microsoft Corporation, and Python Software Foundation. 2013. “IronPython.” <https://ironpython.net/documentation/>.

- Iversen, Ole Sejer, Kim Halskov, and Tuck Wah Leong. 2010. "Rekindling Values in Participatory Design." *PDC '10: Proceedings of the 11th Biennial Participatory Design Conference*, 91–100. <https://doi.org/10.1145/1900441.1900455>.
- Jacobson, Dov, and Brandt Dargue. 2019. "Change Your Mind: Game Based AIS Can Reform Cognitive Behavior." *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 11597 LNCS:108–17. https://doi.org/10.1007/978-3-030-22341-0_9/FIGURES/4.
- Jang, Sun-young, and Sung-Ah Kim. 2016. "A Framework for the Participatory Design of Interactive Media Façade." *TOJSAT* 4 (1): 1–8. <https://dergipark.org.tr/en/pub/tojsat/issue/22647/241911>.
- Janz, Nancy K., and Marshall H. Becker. 1984. "The Health Belief Model: A Decade Later." *Health Education Quarterly* 11 (1): 1–47. <https://doi.org/10.1177/109019818401100101>.
- Jia, Mengda, Ravi S. Srinivasan, and Adeeba A. Raheem. 2017. "From Occupancy to Occupant Behavior: An Analytical Survey of Data Acquisition Technologies, Modeling Methodologies and Simulation Coupling Mechanisms for Building Energy Efficiency." *Renewable and Sustainable Energy Reviews* 68 (February):525–40. <https://doi.org/10.1016/j.rser.2016.10.011>.
- Jones, Nathaniel L., and Christoph F. Reinhart. 2018. "Effects of Real-Time Simulation Feedback on Design for Visual Comfort." *Journal of Building Performance Simulation* 12 (3): 343–61. <https://doi.org/10.1080/19401493.2018.1449889>.
- Kalantari, Saleh, Vidushi Tripathi, Julia Kan, James D. Rounds, Armin Mostafavi, Robin Snell, and Jesus G. Cruz-Garza. 2022. "Evaluating the Impacts of Color, Graphics, and Architectural Features on Wayfinding in Healthcare Settings Using EEG Data and Virtual Response Testing." *Journal of Environmental Psychology* 79 (February):101744. <https://doi.org/10.1016/J.JENVP.2021.101744>.
- Kanthila, Chinmayi, Abhinandana Boodi, Karim Beddiar, Yassine Amirat, and Mohamed Benbouzid. 2021. "Building Occupancy Behavior and Prediction Methods: A Critical Review and Challenging Locks." *IEEE Access* 9:79353–72. <https://doi.org/10.1109/ACCESS.2021.3083534>.
- Karakas, Tulay, and Dilek Yildiz. 2020. "Exploring the Influence of the Built Environment on Human Experience through a Neuroscience Approach: A Systematic Review." *Frontiers of Architectural Research* 9 (1): 236–47. <https://doi.org/10.1016/J.FOAR.2019.10.005>.
- Keeling, Trevor, Derek Clements-Croome, Rachael Luck, and Philip Pointer. 2012. "How the Sensory Experience of Buildings Can Contribute to Wellbeing and Productivity." *Proceedings of 7th Windsor Conference: The Changing Context of Comfort in an Unpredictable World: Network for Comfort and Energy Use in Buildings*. <http://nceub.org.uk>.
- Kleban, S. D., W. A. Stubblefield, K. W. Mitchiner, J. L. Mitchiner, and M. Arms. 2001. "Collaborative Evaluation of Early Design Decisions and Product Manufacturability." *Proceedings of the Hawaii International Conference on System Sciences*, 270. <https://doi.org/10.1109/HICSS.2001.927226>.
- Knippers, Jan, | Cordula Kropp, | Achim Menges, | Oliver Sawodny, and Daniel Weiskopf. 2021. "Integrative Computational Design and Construction: Rethinking Architecture Digitally." *Civil Engineering Design* 3 (4): 123–35. <https://doi.org/10.1002/CEND.202100027>.
- Koivisto, Jonna, and Juho Hamari. 2019. "The Rise of Motivational Information Systems: A Review of Gamification Research." *International Journal of Information Management* 45 (April):191–210. <https://doi.org/10.1016/J.IJINFOMGT.2018.10.013>.

- Kolb, David A. 1984. *Experiential Learning: Experience as the Source of Learning and Development*. Prentice Hall.
https://www.researchgate.net/publication/235701029_Experiential_Learning_Experience_As_The_Source_Of_Learning_And_Development.
- Koning, H, and J Eizenberg. 1981. "The Language of the Prairie: Frank Lloyd Wright's Prairie Houses." *Https://Doi.Org/10.1068/B080295* 8 (3): 295–323. <https://doi.org/10.1068/B080295>.
- Kwak, Daniel, Alfred Kam, David Becerra, Qikuan Zhou, Adam Hops, Eleyine Zarour, Arthur Kam, Luis Sarmenta, Mathieu Blanchette, and Jérôme Waldispühl. 2013. "Open-Phylo: A Customizable Crowd-Computing Platform for Multiple Sequence Alignment." *Genome Biology* 14 (10): 1–12. <https://doi.org/10.1186/GB-2013-14-10-R116/FIGURES/6>.
- LaMorte, Wayne W. 2022. "Behavioral Change Models." Boston.
- Leaman, Adrian, and Bill Bordass. 2010. "Productivity in Buildings: The 'Killer' Variables." *Https://Doi.Org/10.1080/096132199369615* 27 (1): 4–19. <https://doi.org/10.1080/096132199369615>.
- Lee, Bokyoung, Michael Lee, Autodesk Research, Canada Pan Zhang, Canada Alexander Tessier, Azam Khan, Pan Zhang, Alexander Tessier, and Daniel Saakes. 2021. "Socio-Spatial Comfort: Using Vision-Based Analysis to Inform User-Centred Human-Building Interactions." *Proceedings of the ACM on Human-Computer Interaction* 4 (CSCW3): 1–33. <https://doi.org/10.1145/3432937>.
- Lee, Bokyoung, Michael Lee, Pan Zhang, Alexander Tessier, Daniel Saakes, and Azam Khan. 2021. "Socio-Spatial Comfort: Using Vision-Based Analysis to Inform User-Centred Human-Building Interactions." *Proceedings of the ACM on Human-Computer Interaction* 4 (CSCW3): 1–33. <https://doi.org/10.1145/3432937>.
- Legaki, Nikoletta Zampeta, Kostas Karpouzis, Vassilios Assimakopoulos, and Juho Hamari. 2021. "Gamification to Avoid Cognitive Biases: An Experiment of Gamifying a Forecasting Course." *Technological Forecasting and Social Change* 167 (June):120725. <https://doi.org/10.1016/J.TECHFORE.2021.120725>.
- Leong, Tuck Wah, and Ole Sejer Iversen. 2015. "Values-Led Participatory Design as a Pursuit of Meaningful Alternatives." *OzCHI 2015: Proceedings of the Annual Meeting of the Australian Special Interest Group for Computer Human Interaction*, December, 314–23. <https://doi.org/10.1145/2838739.2838784>.
- Lewis, James R. 2018. "Measuring Perceived Usability: The CSUQ, SUS, and UMUX." *International Journal of Human-Computer Interaction* 34 (12): 1148–56. <https://doi.org/10.1080/10447318.2017.1418805>.
- Lieberoth, Andreas. 2015. "Shallow Gamification." *Games and Culture* 10 (3): 229–48. <https://doi.org/10.1177/1555412014559978>.
- Light, Ann., and Rosemary Luckin. 2008. "Designing for Social Justice: People, Technology, Learning." *Undefined*. <https://www.nfer.ac.uk/publications/FUTL53/FUTL53.pdf>.
- Lin, Chun-Heng, and Pei-Hsien Hsu. 2017. "Integrating Procedural Modelling Process and Immersive VR Environment for Architectural Design Education." *MATEC Web of Conferences - IC4M & ICDES* 104 (03007). <https://doi.org/10.1051/mateconf/201710403007>.

- Lo, Tian Tian, Marc Aurel Schnabel, and Tane J Moleta. 2017. "Gamification for User-Oriented Housing Design: A Theoretical Review." *Proceedings of the 22nd International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)*, 62–73.
- Luck, Rachael. 2018. "Participatory Design in Architectural Practice: Changing Practices in Future Making in Uncertain Times." *Design Studies* 59 (November):139–57. <https://doi.org/10.1016/J.DESTUD.2018.10.003>.
- Luna-Navarro, Alessandra, Roel Loonen, Miren Juaristi, Aurora Monge-Barrio, Shady Attia, and Mauro Overend. 2020. "Occupant-Facade Interaction: A Review and Classification Scheme." *Building and Environment* 177 (June):106880. <https://doi.org/10.1016/J.BUILDENV.2020.106880>.
- MacKenzie, I. Scott. 2013. "Human-Computer Interaction : An Empirical Research Perspective."
- Madrazo, Leandro, Álvaro Sicilia, and Angel Martin Cojo. 2009. "BARCODE HOUSING SYSTEM Integrating Floor Plan Layout Generation Processes within an Open and Collaborative System to Design and Build Customized Housing." *CAADFutures*.
- Mahecha Zambrano, Juan, Ulrich Filippi Oberegger, and Graziano Salvalai. 2021. "Towards Integrating Occupant Behaviour Modelling in Simulation-Aided Building Design: Reasons, Challenges and Solutions." *Energy and Buildings* 253 (December):111498. <https://doi.org/10.1016/J.ENBUILD.2021.111498>.
- Maier, Jonathan R.A., Georges M. Fadel, and Dina G. Battisto. 2009. "An Affordance-Based Approach to Architectural Theory, Design, and Practice." *Design Studies* 30 (4): 393–414. <https://doi.org/10.1016/J.DESTUD.2009.01.002>.
- Martínez-Soto, Joel, Leopoldo Gonzales-Santos, Erick Pasaye, and Fernando A. Barrios. 2013. "Exploration of Neural Correlates of Restorative Environment Exposure through Functional Magnetic Resonance." *Intelligent Buildings International* 5 (SUPPL1): 10–28. <https://doi.org/10.1080/17508975.2013.807765>.
- Mascio, Danilo Di, and Ruth Dalton. 2017. "Using Serious Games to Establish a Dialogue between Designers and Citizens in Participatory Design: The Architectural Portal of People's Narratives." *Serious Games and Edutainment Applications: Volume II* II (January):433–54. https://doi.org/10.1007/978-3-319-51645-5_20/FIGURES/9.
- Maxwell, Joseph A. 2013. *Qualitative Research Design, An Interactive Approach*. *Qualitative Research Design: An Interactive Approach*. 3rd ed. Vol. 41. SAGE Publications, Inc.
- Mayer, Richard E. 2002. "A Taxonomy for Computer-Based Assessment of Problem Solving." *Computers in Human Behavior* 18 (6). [https://doi.org/10.1016/S0747-5632\(02\)00020-1](https://doi.org/10.1016/S0747-5632(02)00020-1).
- McFarlane, Daniel C., and Kara A. Latorella. 2002. "The Scope and Importance of Human Interruption in Human-Computer Interaction Design." *Human-Computer Interaction* 17 (1): 1–61. https://doi.org/10.1207/S15327051HCI1701_1.
- Milton, Donald K. 2000. "Risk of Sick Leave Associated with Outdoor Air Supply Rate, Humidification, and Occupant Complaints." *Indoor Air* 10 (4): 212–21. <https://doi.org/10.1034/J.1600-0668.2000.010004212.X>.
- MIT Media Lab. 2021. "Understanding the 15 Minute City." 2021. <https://www.media.mit.edu/projects/understanding-the-15-minute-city/overview/#>.

- Moreno-Ger, Pablo, Javier Torrente, Yichuan Grace Hsieh, and William T. Lester. 2012. "Usability Testing for Serious Games: Making Informed Design Decisions with User Data." *Advances in Human-Computer Interaction* 2012. <https://doi.org/10.1155/2012/369637>.
- Myers, Brad, Scott E. Hudson, and Randy Pausch. 2000. "Past, Present, and Future of User Interface Software Tools." *ACM Transactions on Computer-Human Interaction* 7 (1). <https://doi.org/10.1145/344949.344959>.
- Nah, Fiona Fui Hoon, Qing Zeng, Venkata Rajasekhara Telaprolu, Abhishek Padmanabhuni Ayyappa, and Brenda Eschenbrenner. 2014. "Gamification of Education: A Review of Literature." *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 8527 LNCS:401–9. https://doi.org/10.1007/978-3-319-07293-7_39/COVER.
- Nasirova, Diliara, Halil Erhan, Andy Huang, Robert Woodbury, and Bernhard Riecke. 2011. "Change Detection in 3D Parametric Systems: Human-Centered Interfaces for Change Visualization." In *Designing Together: CAADFutures 2011 - Proceedings of the 14th International Conference on Computer Aided Architectural Design*.
- Negroponte, Nicholas. 1970. "The Architecture Machine: Toward a More Human Environment." *The Architecture Machine*, March. <https://doi.org/10.7551/MITPRESS/8269.001.0001>.
- Nelson, Harold G., and Erik Stolterman. 2014. *The Design Way: Intentional Change in an Unpredictable World*. MIT press.
- Nisha, Bobby, and Margaret Nelson. 2012. "Making a Case for Evidence-Informed Decision Making for Participatory Urban Design" 17:336–48. <https://doi.org/10.1057/udi.2012.16>.
- Nord, Natasa, Tymofii Tereshchenko, Live Holmedal Qvistgaard, and Ivar S. Tryggestad. 2018. "Influence of Occupant Behavior and Operation on Performance of a Residential Zero Emission Building in Norway." *Energy and Buildings* 159 (January):75–88. <https://doi.org/10.1016/J.ENBUILD.2017.10.083>.
- Norman, Don. 2016. *The Design of Everyday Things. The Design of Everyday Things*. <https://doi.org/10.15358/9783800648108>.
- O'Brien, William, Isabella Gaetani, Sara Gilani, Salvatore Carlucci, Pieter Jan Hoes, and Jan Hensen. 2016. "International Survey on Current Occupant Modelling Approaches in Building Performance Simulation†." <https://doi.org/10.1080/19401493.2016.1243731> 10 (5–6): 653–71. <https://doi.org/10.1080/19401493.2016.1243731>.
- Østergård, Torben, Rasmus L. Jensen, and Steffen E. Maagaard. 2017. "Early Building Design: Informed Decision-Making by Exploring Multidimensional Design Space Using Sensitivity Analysis." *Energy and Buildings* 142 (May):8–22. <https://doi.org/10.1016/J.ENBUILD.2017.02.059>.
- Parametric Monkey Pty Ltd. 2019. "MetricMonkey." 2019. <https://metricmonkey.io/>.
- Parvin, Alastair. 2013. "Architecture (and the Other 99%): Open-Source Architecture and Design Commons." *Architectural Design* 83 (6): 90–95. <https://doi.org/10.1002/AD.1680>.
- Perkins, H. Wesley. 2006. "A Brief Summary of Social Norms Theory and the Approach to Promoting Health."

- Pezzica, Camilla, João Ventura Lopes, and Alexandra Claudia Rebelo Paio. 2017. "INFORMED DESIGN DECISION-MAKING: FROM DIGITAL ANALYSIS TO URBAN DESIGN." *Gestão & Tecnologia de Projetos* 12 (3): undefined-undefined. <https://doi.org/10.11606/GTP.V12I3.133936>.
- Poplin, Alenka, and Kavita Vemuri. 2018. "Spatial Game for Negotiations and Consensus Building in Urban Planning: YouPlaceIt!" *Advances in Geographic Information Science*, 63–90. https://doi.org/10.1007/978-3-319-22774-0_4/TABLES/1.
- Preisinger, Clemens, Moritz Heimrath, Mag Arch, and Bollinger Grohmann Schneider. 2014. "Karamba—A Toolkit for Parametric Structural Design." *Structural Engineering International* 24 (2): 217–21. <https://doi.org/10.2749/101686614X13830790993483>.
- Price, Cedric. 2003. *The Square Book*. Chichester, West Sussex: Wiley-Academy. <https://www.worldcat.org/title/cedric-price-the-square-book/oclc/164411635>.
- Prochaska, James O., and Janice M. Prochaska. 2019. "Transtheoretical Model." *Lifestyle Medicine: Third Edition*, April, 219–28. <https://doi.org/10.1201/9781315201108-18/TRANSTHEORETICAL-MODEL-JAMES-PROCHASKA-JANICE-PROCHASKA>.
- Purcell, P. 1977. "The Soft Architecture Machine: N Negroponte, MIT Press (1977)." *Computer-Aided Design* 9 (3): 219. [https://doi.org/10.1016/0010-4485\(77\)90136-1](https://doi.org/10.1016/0010-4485(77)90136-1).
- Python Software Foundation. 2010. "Python." Amsterdam: Centrum voor Wiskunde en Informatica. <https://www.python.org/>.
- Radford, Antony. 2000. "Games and Learning about Form in Architecture." *Automation in Construction* 9 (4): 379–85. [https://doi.org/10.1016/S0926-5805\(99\)00021-7](https://doi.org/10.1016/S0926-5805(99)00021-7).
- Rastegar, Raha M, Sara Saghafi Moghaddam, Ramtin Haghnazar, and Craig Zimring. 2022. "From Evidence to Assessment: Developing a Scenario-Based Computational Design Algorithm to Support Informed Decision-Making in Primary Care Clinic Design Workflow." *International Journal of Architectural Computing* 20 (3): 567–86. <https://doi.org/10.1177/14780771221121031>.
- Ratinho, Elias, and Cátia Martins. 2023. "The Role of Gamified Learning Strategies in Student's Motivation in High School and Higher Education: A Systematic Review." *Heliyon* 9 (8): e19033. <https://doi.org/10.1016/J.HELIYON.2023.E19033>.
- "'Re:Imagine London' in Fortnite, ZHA Collaborates with Epic Games – Zaha Hadid Architects." n.d. Accessed June 30, 2025. <https://www.zaha-hadid.com/2024/09/19/reimagine-london-in-fortnite-zha-collaborates-with-epic-games/>.
- Resnick, Mitchel, Brad Myers, Kumiyo Nakakoji, Ben Shneiderman, Randy Pausch, Ted Selker, and Mike Eisenberg. 2005. "Design Principles for Tools to Support Creative Thinking." In *Workshop on Creativity Support Tools*.
- Resnick, Mitchel, and Brian Silverman. 2005. "Some Reflections on Designing Construction Kits for Kids." In *Proceedings of: Interaction Design and Children 2005, IDC 2005*. <https://doi.org/10.1145/1109540.1109556>.
- Robert McNeel & Associates. 2020. "Rhinoceros 3D." Seattle: Robert McNeel & Associates. <https://www.rhino3d.com/7/new/>.
- Rodrigues, Luiz, Filipe D. Pereira, Armando M. Toda, Paula T. Palomino, Marcela Pessoa, Leandro Silva Galvão Carvalho, David Fernandes, Elaine H.T. Oliveira, Alexandra I. Cristea, and Seiji Isotani. 2022.

- “Gamification Suffers from the Novelty Effect but Benefits from the Familiarization Effect: Findings from a Longitudinal Study.” *International Journal of Educational Technology in Higher Education* 19 (1): 1–25. <https://doi.org/10.1186/S41239-021-00314-6/FIGURES/3>.
- Rogean, Nicolas, Aryan Rezaei Rad, Petras Vestar-Tas, Pierre Latteur, Yves Weinand, Ecole Polytechnique, and Fédérale De Lausanne. 2022. “A Collaborative Workflow to Automate the Design, Analysis, and Construction of Integrally-Attached Timber Plate Structures.” *Proceedings of the 27th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA)* 2 (May):151–60. <https://doi.org/10.52842/CONF.CAADRIA.2022.2.151>.
- Rogers, Everett M, Arvind Singhal, and Margaret M. Quinlan. 2014. “Diffusion of Innovations.” *Taylor & Francis Group*, April, 432–48. <https://doi.org/10.4324/9780203887011-36>.
- Rowe, Gene, and Lynn J. Frewer. 2000. “Public Participation Methods: A Framework for Evaluation.” *Science, Technology, & Human Values* 25 (1): 3–29. <https://doi.org/10.1177/016224390002500101>.
- Saari, Arto. 2008. “Precision Refurbishment of Buildings: A Façade Refurbishment Case Study.” *Structural Survey* 26 (2): 108–19. <https://doi.org/10.1108/02630800810883049>.
- Saeidi, Sanaz, Chanachok Chokwitthaya, Yimin Zhu, and Ming Sun. 2018. “Spatial-Temporal Event-Driven Modeling for Occupant Behavior Studies Using Immersive Virtual Environments.” *Automation in Construction* 94 (October):371–82. <https://doi.org/10.1016/J.AUTCON.2018.07.019>.
- Schön, Donald A. 2017. *The Reflective Practitioner: How Professionals Think in Action. The Reflective Practitioner: How Professionals Think in Action*. <https://doi.org/10.4324/9781315237473>.
- Schultz, Carl, Mehul Bhatt, and André Borrmann. 2017. “Bridging Qualitative Spatial Constraints and Feature-Based Parametric Modelling: Expressing Visibility and Movement Constraints.” *Advanced Engineering Informatics* 31 (January):2–17. <https://doi.org/10.1016/J.AEI.2015.10.004>.
- Schwind, Valentin, Pascal Knierim, Nico Haas, and Niels Henze. 2019. “Using Presence Questionnaires in Virtual Reality.” In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–12. New York, NY, USA: ACM. <https://doi.org/10.1145/3290605.3300590>.
- Sebenius, James K. 2009. “Negotiation Analysis: From Games to Inferences to Decisions to Deals.” <https://doi.org/10.1111/j.1571-9979.2009.00238.x>.
- Selker, Ted. 2005. “Fostering Motivation and Creativity for Computer Users.” *International Journal of Human-Computer Studies* 63 (4–5): 410–21. <https://doi.org/10.1016/j.ijhcs.2005.04.005>.
- Shen, Weiming, Guy Newsham, and Burak Gunay. 2017. “Leveraging Existing Occupancy-Related Data for Optimal Control of Commercial Office Buildings: A Review.” <https://doi.org/10.1016/j.aei.2016.12.008>.
- Shi, Xing, Binghui Si, Jiangshan Zhao, Zhichao Tian, Chao Wang, Xing Jin, and Xin Zhou. 2019. “Magnitude, Causes, and Solutions of the Performance Gap of Buildings: A Review.” *Sustainability* 2019, Vol. 11, Page 937 11 (3): 937. <https://doi.org/10.3390/SU11030937>.
- Shiratuddin, M. F., and W. Thabet. 2011. “Utilizing a 3D Game Engine to Develop a Virtual Design Review System.” *Journal of Information Technology in Construction* 16:39–68.
- Shneiderman, Ben. 2000. “Creating Creativity: User Interfaces for Supporting Innovation.” *ACM Transactions on Computer-Human Interaction* 7 (1). <https://doi.org/10.1145/344949.345077>.
- SideWalk Lab. 2020. “Delve.” 2020. <https://www.sidewalklabs.com/products/delve>.

- SimScale Lab. 2012. “SimScale.” 2012. <https://www.simscale.com/industries/architecture-engineering-construction/>.
- Sirakaya, Ercan, and Arch G. Woodside. 2005. “Building and Testing Theories of Decision Making by Travellers.” *Tourism Management* 26 (6): 815–32. <https://doi.org/10.1016/J.TOURMAN.2004.05.004>.
- Sonta, Andrew J., Perry E. Simmons, and Rishee K. Jain. 2018. “Understanding Building Occupant Activities at Scale: An Integrated Knowledge-Based and Data-Driven Approach.” *Advanced Engineering Informatics* 37 (August):1–13. <https://doi.org/10.1016/j.aei.2018.04.009>.
- Spencer, Leslie, Troy B. Adams, Sarah Malone, Lindsey Roy, and Elizabeth Yost. 2006. “Applying the Transtheoretical Model to Exercise: A Systematic and Comprehensive Review of the Literature.” *Http://Dx.Doi.Org/10.1177/1524839905278900* 7 (4): 428–43. <https://doi.org/10.1177/1524839905278900>.
- Stelzle, Benjamin, Anja Jannack, and Jörg Rainer Noennig. 2017. “Co-Design and Co-Decision: Decision Making on Collaborative Design Platforms.” In *Procedia Computer Science*, 112:2435–44. Elsevier B.V. <https://doi.org/10.1016/j.procs.2017.08.095>.
- Stiny, G. 1980. “Introduction to Shape and Shape Grammars.” *Https://Doi.Org/10.1068/B070343* 7 (3): 343–51. <https://doi.org/10.1068/B070343>.
- Tan, Ekim. 2014. *Negotiation and Design for the Self-Organizing City : Gaming as a Method for Urban Design*. Technische Universiteit Delft.
- . 2018. *Play the City : Games Informing the Urban Development*. Onomatopée - Jap Sam Books. https://books.google.com/books/about/Play_the_City.html?hl=tr&id=jgdtAEACAAJ.
- Thorley, Mark, David Flynn, and Radu Gidei. 2022. “KOPE.” 2022. <https://www.kope.ai/>.
- Trimble Inc. 2012. “Sefaira.” 2012. <https://www.sketchup.com/en/products/sefaira>.
- “Tygron Geo-Design Platform.” n.d. Accessed October 22, 2022. <https://www.tygron.com/en/>.
- U.S. Department of Housing and Urban Development. 2013. “Rebuild by Design | HUD.Gov / U.S. Department of Housing and Urban Development (HUD).” 2013. <https://www.hud.gov/sandyrebuilding/rebuildbydesign>.
- Valls, Francesc, Ernest Redondo, David Fonseca, Pilar Garcia-Almirall, and Jordi Subirós. 2016. “Videogame Technology in Architecture Education.” *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 9733:436–47. https://doi.org/10.1007/978-3-319-39513-5_41.
- Vannucci, Manila, Simone Gori, and Haruyuki Kojima. 2014. “The Spatial Frequencies Influence the Aesthetic Judgment of Buildings Transculturally.” *Cognitive Neuroscience* 5 (October):143–49. <https://doi.org/10.1080/17588928.2014.976188>.
- Vischer, Jacqueline C. 2008. “Towards a User-Centred Theory of the Built Environment.” *Building Research & Information* 36 (3): 231–40. <https://doi.org/10.1080/09613210801936472>.
- Vrcelj Božić, Ana, Nataša Hoić-Božić, Martina Holenko Dlab, Kristian Stančin, and Tomislav Jaguš. 2025a. “Gamification Classification as a Framework for Analysing Gamification Elements in Digital Educational Tools.” *Journal of Communications Software and Systems* 21 (1): 34–42. <https://doi.org/10.24138/jcomss-2024-0111>.

- . 2025b. “Gamification Classification as a Framework for Analysing Gamification Elements in Digital Educational Tools.” *Journal of Communications Software and Systems* 21 (1): 34–42. <https://doi.org/10.24138/jcomss-2024-0111>.
- Wachter, Sandra, Brent Mittelstadt, and Luciano Floridi. 2017. “Why a Right to Explanation of Automated Decision-Making Does Not Exist in the General Data Protection Regulation.” *International Data Privacy Law* 7 (2): 76–99. <https://doi.org/10.1093/idpl/ix005>.
- Wallgren, Jesper. 2020. “Finch3D.” 2020. <https://www.finch3d.com/>.
- Wang, Alf Inge, and Rabail Tahir. 2020. “The Effect of Using Kahoot! For Learning – A Literature Review.” *Computers & Education* 149 (May):103818. <https://doi.org/10.1016/J.COMPEDU.2020.103818>.
- Wazoku. n.d. “InnoCentive: Solve the World’s Greatest Challenges.” Accessed November 4, 2022. <https://www.innocentive.com/>.
- Weber, Ramon Elias, Caitlin Mueller, and Christoph Reinhart. 2022a. “Automated Floorplan Generation in Architectural Design: A Review of Methods and Applications.” *Automation in Construction* 140 (August):104385. <https://doi.org/10.1016/j.autcon.2022.104385>.
- . 2022b. “Automated Floorplan Generation in Architectural Design: A Review of Methods and Applications.” *Automation in Construction* 140 (August):104385. <https://doi.org/10.1016/J.AUTCON.2022.104385>.
- . 2022c. “Solar Exoskeletons – An Integrated Building System Combining Solar Gain Control with Structural Efficiency.” *Solar Energy* 240 (July):301–14. <https://doi.org/10.1016/j.solener.2022.05.048>.
- Webler, Thomas. 1995. “‘Right’ Discourse in Citizen Participation: An Evaluative Yardstick.” In *Fairness and Competence in Citizen Participation*, edited by O., Webler, T., Wiedemann, P. Renn, 10:35–86. Dordrecht: Springer. https://doi.org/10.1007/978-94-011-0131-8_3.
- Webler, Thomas, and Seth Tuler. 2000. “Fairness and Competence in Citizen Participation: Theoretical Reflections from a Case Study.” *Administration & Society* 32 (5): 566–95. <https://doi.org/10.1177/00953990022019588>.
- Weijenberg, Camiel, and Sayjel Vijay Patel. 2016. “Digital Blue Foam.” 2016. <https://www.digitalbluefoam.com/>.
- Weinzapfel, Guy, and Nicholas Negroponte. 1976. “Architecture-by-Yourself an Experiment with Computer Graphics for House Design.” *Proceedings of the 3rd Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH 1976*, July, 74–78. <https://doi.org/10.1145/563274.563290>.
- Weld, Daniel S., and Gagan Bansal. 2018. “The Challenge of Crafting Intelligible Intelligence.” *Communications of the ACM* 62 (6): 70–79. <https://doi.org/10.1145/3282486>.
- Westera, W., R. J. Nadolski, H. G.K. Hummel, and I. G.J.H. Wopereis. 2008. “Serious Games for Higher Education: A Framework for Reducing Design Complexity.” *Journal of Computer Assisted Learning* 24 (5): 420–32. <https://doi.org/10.1111/J.1365-2729.2008.00279.X>.

- Wilde, Pieter De. 2014. "The Gap between Predicted and Measured Energy Performance of Buildings: A Framework for Investigation." *Automation in Construction* 41 (May):40–49. <https://doi.org/10.1016/J.AUTCON.2014.02.009>.
- Woodbury, Robert F., Susan J. Shannon, and Antony D. Radford. 2001. "Games in Early Design Education." *Computer Aided Architectural Design Futures* 2001, 201–14. https://doi.org/10.1007/978-94-010-0868-6_16.
- Yamamoto, Yasuhiro, and Kumiyo Nakakoji. 2005. "Interaction Design of Tools for Fostering Creativity in the Early Stages of Information Design." *International Journal of Human Computer Studies* 63 (4-5 SPEC. ISS.). <https://doi.org/10.1016/j.ijhcs.2005.04.023>.
- Yan, Wei, Charles Culp, and Robert Graf. 2011. "Integrating BIM and Gaming for Real-Time Interactive Architectural Visualization." *Automation in Construction* 20 (4): 446–58. <https://doi.org/10.1016/J.AUTCON.2010.11.013>.
- Yeager, David S., and Carol S. Dweck. 2020. "What Can Be Learned from Growth Mindset Controversies?" *American Psychologist* 75 (9): 1269–84. <https://doi.org/10.1037/AMP0000794>.
- Young, Fiona, and Benjamin Cleveland. 2022. "Affordances, Architecture and the Action Possibilities of Learning Environments: A Critical Review of the Literature and Future Directions." *Buildings* 2022, Vol. 12, Page 76 12 (1): 76. <https://doi.org/10.3390/BUILDINGS12010076>.

APPENDIX I: PARTICIPANTS' FEEDBACK

Participants' feedback was collected during voluntary informal discussions after the data collection session. It was divided into three segments: feature requests, positive comments, and other comments. The script of those conversations is as follows.

Feature Requests

- A participant suggested “it would be beneficial if the pointer ring size could be a variable that players can change, allowing them to adjust the cutoff distance for the effective radius during design”. (Repetition: 3)
- “It could also be interesting if the game's time could change faster, enabling players to switch to night mode and view the building at night”. Similarly, other participants wished to be able to change the time and therefore lighting conditions.
- A participant stated, "It would have been helpful if I could save some of the alternatives I liked and create a repository of them, allowing me to revisit those configurations and regenerate them if needed."
- Similarly, another participant wished for a 'personal best' feature and explained, "It would be nice if the platform could show the performance of my choices based on the scale of all my previous decisions."
- A participant suggested, "It would be more readable or convenient for me if you could emphasize the numbers that have changed. For example, if my decision increases the natural solar gain by 1%, I want to see that more clearly." Even after the moderator explained that the information is displayed with actual scale, the player insisted he would prefer an “exaggerated scale over an actual one”.
- The participant requested a "make it symmetrical" feature, meaning the system could automatically make the facade symmetrical with one-click.
- The participant stated, "I'd like to be limited. I don't like to be able to design freely. I prefer systems that enforce more limitations on me." When the game moderator explained that the geometrical and fabrication limitations are coded in a failproof manner, the participant also requested “the ability to set limitations on each criterion. For example, I want the system to only accept and generate design alternatives that their energy demand remains within an acceptable LEED range.”
- “The design exploration was so engaging that I forgot I am responsible for one floor of the building. It would be helpful if there was a reminder of the experiment's objective somewhere.”

Positive Comments

- “One of the platform's biggest advantages is that designers are freed from dealing with material allocation, UV mapping, and other technical details. Designers also don't need to set up lighting conditions, as they are always in an environment with accurate lighting, shadows, and actual materials. This saves them from repeatedly adjusting materials and lighting in conventional design platforms and then checking effects on different screens. This feature could save us a lot of time in our design process”. (Repetition: 4)
- A participant, understanding the backend generative processes, explained, "I like the neighborhood effect. I mean, when I make a decision, the adjacent areas are affected automatically. It gives me great freedom to explore design options faster, and I also like being able to control or modify that effect."

Other Comments

- Participants who were advanced users of a particular design software, such as Lumion, expressed a preference for key bindings that matched their software of choice. They acknowledged that the tutorial effectively taught them the new key bindings, but noted it would be more convenient if they matched the ones they were already familiar with.
- The participant stated that since the keys and functions were designed similarly to prominent design software, it was easy for them to learn and play smoothly.
- One participant admitted in an informal conversation after the data participation, “As the environment was very different from the software I normally use, I was absorbed in the interactions and mechanics. I must admit I didn't pay attention to the feedback information as much as I usually do because I was profoundly absorbed and intrigued by the environment and just wanted to explore.” The participant then added, “I normally try to pay more attention to feedback, but since the design exploration was more fun, I don't know why I chose to play around instead of designing.” (This person's Log is recorded but is marked as abnormal input) (Repetition: 2)
- Another participant admitted he was so happy to improve in all three categories of feedback that he decided to leave the game at that point! He thought if he continued to design more, he might not achieve the same success. He added, “Although I was not finished with my design, I felt so conditioned by the rewards in games that I wanted to quit at that moment with maximum achievable points.” (This person's record is deleted.) The same person explained that the gamification of strategies in the game made him subconsciously try to cheat. He added, “I had to constantly remind myself that I'm designing something and not here to win anything.” (This person's Log is recorded but is marked as abnormal input).
- A participant stated that “...I cared most about the shadows, aesthetics, and the definition of the space. Since I was told that all generated outcomes are structurally stable, I didn't worry much about the structure. Regarding the cost, I wasn't limited, so based on experience, I assumed the costs were within an acceptable range and didn't focus on them. The environment aligned with my goal of creating more contrast between light and shadow, so I paid more attention to that aspect.”
- “... Normally, I generate the geometry first and then analyze it. So, when interacting with this environment, it was unfamiliar to me that each choice in the game is automatically analyzed. I like to be able to explore more without having to deal with design consequences...”

- “... I was trained to prefer the functionality and concept of space in design, so it was fascinating to explore both indoors and outdoors. The space was already equipped with furniture, but it was still unfamiliar for me to try to consider the performance criteria simultaneously.”

APPENDIX II: R/PYTHON CODE FOR HYPOTHESIS TESTING

The purpose of this appendix is to allow anyone to verify the findings of this thesis and provide an additional layer of validity checking. Sample data is publicly available at the Open Science Foundation (<https://osf.io/9hjwt/>), enabling the reproduction of results using that data. In the codes in this chapter, redundant information such as the local address on the author's machine may be replaced with '█'. Optional segments, such as visualization settings, might be shortened and replaced with the character '↓'.

- **For the analysis of the C1-C7 metrics - Part 1:**

```
#!/usr/bin/env Rscript

# =====
# Feedback Analysis Script
# =====
# This script analyzes the effect of feedback conditions on decision-making
# across multiple criteria and sequential decisions.
#
# Dataset: cleaned_participant_data_CleanedByRM.csv
# Structure:
#   - Player: Participant ID
#   - Feedback: Condition (0 or 1)
#   - C1_A to C7_C: Seven criteria (C1-C7) for each decision (A, B, C)
# =====

# Load required libraries
tryCatch({
  # Load libraries with appropriate error handling
  library(tidyverse) # For data manipulation and visualization
  library(lme4)       # For linear mixed-effects models
  library(lmerTest)   # For p-values in mixed models
  library(emmeans)    # For estimated marginal means and post-hoc tests
  library(car)        # For ANOVA and additional tests
  library(reshape2)   # For reshaping data
  library(knitr)       # For nice table output

  cat("All required packages loaded successfully.\n")
}, error = function(e) {
  cat("Error loading packages:", e$message, "\n")
  cat("Please run install_packages.R first to ensure all dependencies are installed.\n")
  quit(status = 1)
})

# Create output directory if it doesn't exist
```



```

dir.create("results", showWarnings = FALSE)
dir.create("figures", showWarnings = FALSE)

# =====
# 1. Data Loading and Preparation
# =====

# Function to check if file exists
check_file <- function(file_path) {
  if (!file.exists(file_path)) {
    stop(paste("File not found:", file_path))
  }
  return(file_path)
}

# Load the data
data_file <- "cleaned_participant_data_CleanedByRM.csv"
tryCatch({
  data <- read.csv(check_file(data_file), stringsAsFactors = FALSE)
  cat("Data loaded successfully. Dimensions:", dim(data)[1], "rows and", dim(data)[2], "columns.\n")
}, error = function(e) {
  cat("Error loading data:", e$message, "\n")
  quit(status = 1)
})

# Display the first few rows
cat("\nFirst few rows of the dataset:\n")
print(head(data, 3))

# Basic data summary
cat("\nSummary of the dataset:\n")
summary_stats <- data.frame(
  Feedback_Type = c("Feedback = 0", "Feedback = 1"),
  Count = c(sum(data$Feedback == 0), sum(data$Feedback == 1))
)
print(summary_stats)

# =====
# 2. Reshape data from wide to long format
# =====

# First, identify all criterion columns
criterion_cols <- grep("^C[1-7]_[A-C]$", names(data), value = TRUE)

# Reshape data to long format

```

```

data_long <- melt(data,
  id.vars = c("Player", "Feedback"),
  measure.vars = criterion_cols,
  variable.name = "Criterion",
  value.name = "Value")

# Split the Criterion column into CriterionNumber and Decision
data_long <- data_long %>%
  mutate(
    CriterionNumber = sub("^C([1-7])_.*$", "\\1", Criterion),
    Decision = sub("^C[1-7]_(.*)$", "\\1", Criterion)
  ) %>%
  # Convert Decision to factor with levels in chronological order (A, B, C)
  mutate(
    Decision = factor(Decision, levels = c("A", "B", "C"),
      labels = c("First", "Second", "Third")),
    CriterionNumber = paste0("Criterion ", CriterionNumber),
    Feedback = factor(Feedback, levels = c(0, 1),
      labels = c("No Feedback", "Feedback"))
  )

# Check the restructured data
cat("\nReshaped data structure:\n")
print(head(data_long, 3))

# Save the reshaped data
write.csv(data_long, "results/reshaped_data.csv", row.names = FALSE)
cat("Reshaped data saved to results/reshaped_data.csv\n")

# =====
# 3. Exploratory Data Analysis & Visualization
# =====

# Function to create exploratory plots
create_exploratory_plot <- function(data, criterion_num) {
  # Filter data for the specific criterion
  filtered_data <- data %>%
    filter(CriterionNumber == criterion_num)

  # Calculate means and standard errors
  summary_data <- filtered_data %>%
    group_by(Feedback, Decision) %>%
    summarise(
      Mean = mean(Value, na.rm = TRUE),
      SE = sd(Value, na.rm = TRUE) / sqrt(n()),
      .groups = 'drop'
    )
}

```

```

)

# Create the plot
p <- ggplot(summary_data, aes(x = Decision, y = Mean, fill = Feedback)) +
  geom_bar(stat = "identity", position = position_dodge(0.9), width = 0.8) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE),
               position = position_dodge(0.9), width = 0.25) +
  labs(
    title = paste("Mean Values for", criterion_num),
    x = "Decision",
    y = "Mean Value",
    fill = "Feedback Condition"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text = element_text(size = 10),
    axis.title = element_text(size = 12),
    legend.title = element_text(size = 10),
    legend.text = element_text(size = 9)
  )

# Return the plot
return(p)
}

# Create exploratory plots for each criterion
cat("\nGenerating exploratory plots...\n")
criterion_numbers <- unique(data_long$CriterionNumber)

exploratory_plots <- list()
for (criterion in criterion_numbers) {
  plot <- create_exploratory_plot(data_long, criterion)
  exploratory_plots[[criterion]] <- plot

  # Save the plot
  file_name <- paste0("figures/exploratory_", gsub(" ", "_", tolower(criterion)), ".png")
  ggsave(file_name, plot, width = 8, height = 6, dpi = 300)
  cat(" Saved plot for", criterion, "to", file_name, "\n")
}

# Create a combined plot with facets
combined_plot <- ggplot(data_long, aes(x = Decision, y = Value, fill = Feedback)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(0.9), width = 0.8) +
  stat_summary(fun.data = mean_se, geom = "errorbar", position = position_dodge(0.9), width =
0.25) +

```

```

facet_wrap(~ CriterionNumber, scales = "free_y") +
labs(
  title = "Mean Values Across All Criteria",
  x = "Decision",
  y = "Mean Value",
  fill = "Feedback Condition"
) +
theme_minimal() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  strip.text = element_text(face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

# Save the combined plot
ggsave("figures/combined_exploratory_plot.png", combined_plot, width = 12, height = 10, dpi =
300)
cat("Saved combined exploratory plot to figures/combined_exploratory_plot.png\n")

# =====
# 4. Mixed Effects Analysis
# =====

# Function to run mixed effects model for a criterion
run_mixed_model <- function(data, criterion_num) {
  # Filter data for the specific criterion
  filtered_data <- data %>%
    filter(CriterionNumber == criterion_num)

  # Fit the model with Player as random effect and Feedback and Decision as fixed effects
  tryCatch({
    model <- lmer(Value ~ Feedback * Decision + (1|Player), data = filtered_data)

    # Run ANOVA on the model
    anova_result <- anova(model)

    # Return the results
    return(list(
      model = model,
      anova = anova_result,
      criterion = criterion_num
    ))
  }, error = function(e) {
    cat("Error in model for", criterion_num, ":", e$message, "\n")
    return(NULL)
  })
}

```

```

}

# Run mixed models for each criterion
cat("\nRunning mixed effects models...\n")
model_results <- list()
for (criterion in criterion_numbers) {
  cat("  Analyzing", criterion, "...\\n")
  result <- run_mixed_model(data_long, criterion)
  if (!is.null(result)) {
    model_results[[criterion]] <- result
  }
}

# =====
# 5. Post-hoc Testing and Effect Sizes
# =====

# Function to run post-hoc tests
run_posthoc <- function(model_result) {
  if (is.null(model_result)) return(NULL)

  criterion <- model_result$criterion
  model <- model_result$model

  tryCatch({
    # Get estimated marginal means for different comparisons
    emm_feedback <- emmeans(model, ~ Feedback)
    emm_decision <- emmeans(model, ~ Decision)
    emm_interaction <- emmeans(model, ~ Feedback * Decision)

    # Run pairwise comparisons
    pairs_feedback <- pairs(emm_feedback)
    pairs_decision <- pairs(emm_decision)

    # Compare feedback effects at each decision point
    feedback_at_decision <- emmeans(model, pairwise ~ Feedback | Decision)

    # Compare decision effects for each feedback condition
    decision_at_feedback <- emmeans(model, pairwise ~ Decision | Feedback)

    # Return the results
    return(list(
      criterion = criterion,
      emm_feedback = emm_feedback,
      emm_decision = emm_decision,
      emm_interaction = emm_interaction,

```

```

        pairs_feedback = pairs_feedback,
        pairs_decision = pairs_decision,
        feedback_at_decision = feedback_at_decision,
        decision_at_feedback = decision_at_feedback
    ))
}, error = function(e) {
    cat("Error in post-hoc tests for", criterion, ":", e$message, "\n")
    return(NULL)
})
}

# Run post-hoc tests for each model
cat("\nRunning post-hoc tests...\n")
posthoc_results <- list()
for (criterion in names(model_results)) {
    cat("  Running post-hoc tests for", criterion, "...\\n")
    result <- run_posthoc(model_results[[criterion]])
    if (!is.null(result)) {
        posthoc_results[[criterion]] <- result
    }
}

# =====
# 6. Create Summary Tables and Plots
# =====

# Function to create a summary table of ANOVA results
create_anova_summary <- function(model_results) {
    # Create an empty data frame for the results
    result_df <- data.frame(
        Criterion = character(),
        Effect = character(),
        F_value = numeric(),
        DF_num = numeric(),
        DF_denom = numeric(),
        p_value = numeric(),
        Significance = character(),
        stringsAsFactors = FALSE
    )

    # Fill the data frame with results
    for (criterion in names(model_results)) {
        anova_table <- model_results[[criterion]]$anova

        for (i in 1:nrow(anova_table)) {
            effect_name <- rownames(anova_table)[i]

```

```

# Skip the error term
if (effect_name == "Error") next

# Extract values
f_value <- anova_table[i, "F value"]
df_num <- anova_table[i, "NumDF"]
df_denom <- anova_table[i, "DenDF"]
p_value <- anova_table[i, "Pr(>F)"]

# Determine significance markers
sig <- ""
if (!is.na(p_value)) {
  if (p_value < 0.001) sig <- "****"
  else if (p_value < 0.01) sig <- "***"
  else if (p_value < 0.05) sig <- "**"
  else if (p_value < 0.1) sig <- "."
}

# Add to data frame
result_df <- rbind(result_df, data.frame(
  Criterion = criterion,
  Effect = effect_name,
  F_value = round(f_value, 3),
  DF_num = df_num,
  DF_denom = round(df_denom, 1),
  p_value = round(p_value, 4),
  Significance = sig,
  stringsAsFactors = FALSE
))
}
}

return(result_df)
}

# Create ANOVA summary
anova_summary <- create_anova_summary(model_results)
write.csv(anova_summary, "results/anova_summary.csv", row.names = FALSE)
cat("\nANOVA summary saved to results/anova_summary.csv\n")

# Function to create effect plots from emmeans results
create_effect_plot <- function(posthoc_result) {
  if (is.null(posthoc_result)) return(NULL)

  criterion <- posthoc_result$criterion

```

```

emm_interaction <- posthoc_result$emm_interaction

# Convert emmeans to data frame
emm_df <- as.data.frame(emm_interaction)

# Create the plot
p <- ggplot(emm_df, aes(x = Decision, y = emmean, color = Feedback, group = Feedback)) +
  geom_point(size = 3) +
  geom_line(size = 1) +
  geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE), width = 0.2) +
  labs(
    title = paste("Estimated Marginal Means for", criterion),
    x = "Decision",
    y = "Estimated Mean",
    color = "Feedback Condition"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text = element_text(size = 10),
    axis.title = element_text(size = 12)
  )

return(p)
}

# Create effect plots for each criterion
cat("\nGenerating effect plots...\n")
effect_plots <- list()
for (criterion in names(posthoc_results)) {
  plot <- create_effect_plot(posthoc_results[[criterion]])
  if (!is.null(plot)) {
    effect_plots[[criterion]] <- plot

    # Save the plot
    file_name <- paste0("figures/effect_", gsub(" ", "_", tolower(criterion)), ".png")
    ggsave(file_name, plot, width = 8, height = 6, dpi = 300)
    cat(" Saved effect plot for", criterion, "to", file_name, "\n")
  }
}

# Function to create summary tables for post-hoc comparisons
create_posthoc_summary <- function(posthoc_results) {
  # Create an empty data frame for feedback effects
  feedback_df <- data.frame(
    Criterion = character(),

```



```

Decision = character(),
Contrast = character(),
Estimate = numeric(),
SE = numeric(),
t_ratio = numeric(),
p_value = numeric(),
Significance = character(),
stringsAsFactors = FALSE
)

# Fill the data frame with feedback at decision results
for (criterion in names(posthoc_results)) {
  tryCatch({
    if (is.null(posthoc_results[[criterion]])) next

    # Get feedback at decision results and convert to data frame
    feedback_at_decision <- posthoc_results[[criterion]]$feedback_at_decision

    # Check if the object exists and has the expected structure
    if (is.null(feedback_at_decision) || is.null(feedback_at_decision$contrasts)) {
      cat("Skipping", criterion, "due to missing contrast data\n")
      next
    }

    # Convert contrasts to data frame (proper handling of S4 class)
    contrasts_df <- as.data.frame(feedback_at_decision$contrasts)

    # Get unique decision levels from the data frame
    decision_levels <- unique(contrasts_df$Decision)

    # Process each decision level
    for (decision in decision_levels) {
      # Get results for this decision
      results <- contrasts_df[contrasts_df$Decision == decision, ]

      # Process each row in results
      for (i in 1:nrow(results)) {
        # Determine significance
        p_value <- results$p.value[i]
        sig <- ""
        if (!is.na(p_value)) {
          if (p_value < 0.001) sig <- "****"
          else if (p_value < 0.01) sig <- "***"
          else if (p_value < 0.05) sig <- "**"
          else if (p_value < 0.1) sig <- "."
        }
      }
    }
  }, error = function(e) {
    cat("Error processing criterion:", criterion, "\n")
  })
}

```

```

# Add to data frame (safely extract values)
feedback_df <- rbind(feedback_df, data.frame(
  Criterion = criterion,
  Decision = as.character(decision),
  Contrast = as.character(results$contrast[i]),
  Estimate = round(as.numeric(results$estimate[i]), 3),
  SE = round(as.numeric(results$SE[i]), 3),
  t_ratio = round(as.numeric(results$t.ratio[i]), 3),
  p_value = round(as.numeric(results$p.value[i]), 4),
  Significance = sig,
  stringsAsFactors = FALSE
))
}
}
}, error = function(e) {
  cat("Error processing post-hoc results for", criterion, ":", e$message, "\n")
})
}

# Return an empty data frame if no results were processed
if (nrow(feedback_df) == 0) {
  cat("Warning: No post-hoc comparisons could be processed. Using placeholder data.\n")
  # Create placeholder data for reporting
  feedback_df <- data.frame(
    Criterion = "No Results",
    Decision = "None",
    Contrast = "None",
    Estimate = 0,
    SE = 0,
    t_ratio = 0,
    p_value = 1,
    Significance = "",
    stringsAsFactors = FALSE
  )
}

return(feedback_df)
}

# Create post-hoc summary tables
posthoc_summary <- create_posthoc_summary(posthoc_results)
tryCatch({
  write.csv(posthoc_summary, "results/posthoc_summary.csv", row.names = FALSE)
  cat("\nPost-hoc comparison summary saved to results/posthoc_summary.csv\n")
}, error = function(e) {

```

```

    cat("Error saving post-hoc summary:", e$message, "\n")
})

# =====
# 7. Save Results to Files
# =====

# Save model summaries
tryCatch({
  sink("results/model_summaries.txt")
  cat("=====\n")
  cat("MIXED EFFECTS MODEL SUMMARIES\n")
  cat("=====\n\n")

  for (criterion in names(model_results)) {
    cat("\n\n=====\n")
    cat("Model for:", criterion, "\n")
    cat("=====\n\n")

    print(summary(model_results[[criterion]]$model))

    cat("\nANOVA Table:\n")
    print(model_results[[criterion]]$anova)

    cat("\n\n")
  }

  sink()
  cat("Model summaries saved to results/model_summaries.txt\n")
}, error = function(e) {
  cat("Error saving model summaries:", e$message, "\n")
  if (sink.number() > 0) sink() # Close sink if an error occurs
})

# Save post-hoc test results
tryCatch({
  sink("results/posthoc_tests.txt")
  cat("=====\n")
  cat("POST-HOC TEST RESULTS\n")
  cat("=====\n\n")

  for (criterion in names(posthoc_results)) {
    if (is.null(posthoc_results[[criterion]])) next

    cat("\n\n=====\n")
    cat("Post-hoc tests for:", criterion, "\n")
  }

```

```

cat("=====\n\n")

cat("\nFeedback Effect:\n")
print(posthoc_results[[criterion]]$pairs_feedback)

cat("\nDecision Effect:\n")
print(posthoc_results[[criterion]]$pairs_decision)

cat("\nFeedback Effect at Each Decision Point:\n")
print(posthoc_results[[criterion]]$feedback_at_decision)

cat("\nDecision Effect for Each Feedback Condition:\n")
print(posthoc_results[[criterion]]$decision_at_feedback)

cat("\n\n")
}

sink()
cat("Post-hoc test results saved to results/posthoc_tests.txt\n")
}, error = function(e) {
  cat("Error saving post-hoc results:", e$message, "\n")
  if (sink.number() > 0) sink() # Close sink if an error occurs
})

# Create a combined PDF with all plots
tryCatch({
  # Check if there's a way to create a PDF device via code
  pdf("figures/all_plots.pdf", width = 11, height = 8.5)

  # Print the combined exploratory plot
  print(combined_plot)

  # Print individual exploratory plots
  for (plot in exploratory_plots) {
    if (!is.null(plot)) print(plot)
  }

  # Print effect plots
  for (plot in effect_plots) {
    if (!is.null(plot)) print(plot)
  }

  dev.off()
  cat("All plots saved to figures/all_plots.pdf\n")
}, error = function(e) {
  cat("Error creating combined PDF:", e$message, "\n")

```

```

    if (dev.cur() > 1) dev.off() # Close device if an error occurs
  })

# =====
# 8. Final Summary
# =====

# Count significant effects
count_significant_effects <- function(anova_summary) {
  # Group by effect type and count significant results
  effect_counts <- anova_summary %>%
    group_by(Effect) %>%
    summarise(
      Total = n(),
      Significant = sum(p_value < 0.05),
      Percentage = round(sum(p_value < 0.05) / n() * 100, 1),
      .groups = 'drop'
    )

  return(effect_counts)
}

# Generate final summary
cat("\n=====\\n")
cat("FINAL ANALYSIS SUMMARY\\n")
cat("=====\\n\\n")

# Count significant effects
effect_counts <- count_significant_effects(anova_summary)
cat("Summary of significant effects:\\n")
print(effect_counts)

# Print criteria with significant effects
cat("\\nCriteria with significant Feedback effects:\\n")
feedback_effects <- anova_summary %>%
  filter(Effect == "Feedback" & p_value < 0.05) %>%
  pull(Criterion)
if (length(feedback_effects) > 0) {
  cat(paste(feedback_effects, collapse = ", "), "\\n")
} else {
  cat("None\\n")
}

cat("\\nCriteria with significant Decision effects:\\n")
decision_effects <- anova_summary %>%
  filter(Effect == "Decision" & p_value < 0.05) %>%

```

```

pull(Criterion)
if (length(decision_effects) > 0) {
  cat(paste(decision_effects, collapse = ", "), "\n")
} else {
  cat("None\n")
}

cat("\nCriteria with significant Feedback × Decision interaction effects:\n")
interaction_effects <- anova_summary %>%
  filter(Effect == "Feedback:Decision" & p_value < 0.05) %>%
  pull(Criterion)
if (length(interaction_effects) > 0) {
  cat(paste(interaction_effects, collapse = ", "), "\n")
} else {
  cat("None\n")
}

# Save the summary to a file
tryCatch({
  sink("results/analysis_summary.txt")

  cat("=====\n")
  cat("FEEDBACK ANALYSIS SUMMARY\n")
  cat("=====\n\n")

  cat("Dataset: ", data_file, "\n")
  cat("Participants: ", length(unique(data$Player)), "\n")
  cat("Feedback conditions: ", length(unique(data$Feedback)), "\n")
  cat("Criteria analyzed: ", length(criterion_numbers), "\n\n")

  cat("Summary of significant effects:\n")
  print(effect_counts)

  cat("\nCriteria with significant Feedback effects:\n")
  if (length(feedback_effects) > 0) {
    cat(paste(feedback_effects, collapse = ", "), "\n")
  } else {
    cat("None\n")
  }

  cat("\nCriteria with significant Decision effects:\n")
  if (length(decision_effects) > 0) {
    cat(paste(decision_effects, collapse = ", "), "\n")
  } else {
    cat("None\n")
  }
}

```

```

cat("\nCriteria with significant Feedback × Decision interaction effects:\n")
if (length(interaction_effects) > 0) {
  cat(paste(interaction_effects, collapse = ", "), "\n")
} else {
  cat("None\n")
}

cat("\n\nPOST-HOC COMPARISON HIGHLIGHTS:\n")
# Get significant post-hoc comparisons
tryCatch({
  # Get significant post-hoc comparisons
  significant_comparisons <- posthoc_summary %>%
    filter(p_value < 0.05) %>%
    arrange(Criterion, Decision)

  if (nrow(significant_comparisons) > 0) {
    for (criterion in unique(significant_comparisons$Criterion)) {
      cat("\n", criterion, ":\n")
      criterion_comparisons <- significant_comparisons %>% filter(Criterion == criterion)
      for (i in 1:nrow(criterion_comparisons)) {
        cat("  - ", criterion_comparisons$Decision[i], ": ",
            criterion_comparisons$Contrast[i], " (p = ",
            criterion_comparisons$p_value[i], ")\n", sep = "")
      }
    }
  } else {
    cat("No significant post-hoc comparisons found.\n")
  }
}, error = function(e) {
  cat("Error processing post-hoc comparisons:", e$message, "\n")
  cat("No significant post-hoc comparisons reported due to processing error.\n")
})

cat("\n\nCONCLUSION:\n")
cat("This analysis examined the effect of feedback conditions on decision-making\n")
cat("across three sequential decisions for seven different criteria.\n\n")

feedback_sig_count <- length(feedback_effects)
decision_sig_count <- length(decision_effects)
interaction_sig_count <- length(interaction_effects)

cat("The results indicate that ")
if (feedback_sig_count > 3) {
  cat("feedback had a substantial impact on decision-making")
} else if (feedback_sig_count > 0) {

```

```

    cat("feedback had some limited impact on decision-making")
} else {
    cat("feedback had no significant impact on decision-making")
}
cat(" across the criteria measured.\n\n")

cat("Decision sequence ")
if (decision_sig_count > 3) {
    cat("showed strong effects")
} else if (decision_sig_count > 0) {
    cat("showed some effects")
} else {
    cat("showed no significant effects")
}
cat(" on the measured criteria.\n\n")

cat("The interaction between feedback condition and decision sequence ")
if (interaction_sig_count > 3) {
    cat("was strongly evident")
} else if (interaction_sig_count > 0) {
    cat("was present in some criteria")
} else {
    cat("was not significant")
}
cat(", suggesting that ")
if (interaction_sig_count > 0) {
    cat("the impact of feedback varies across the sequence of decisions.")
} else {
    cat("feedback effects are consistent across the sequence of decisions.")
}

sink()
cat("Analysis summary saved to results/analysis_summary.txt\n")
}, error = function(e) {
    cat("Error saving analysis summary:", e$message, "\n")
    if (sink.number() > 0) sink() # Close sink if an error occurs
})

cat("\nAnalysis completed successfully. Results are available in the 'results' and 'figures' di-
rectories.\n")

```

- **For the analysis of the C1-C7 metrics - Part 2:**

```

import pandas as pd
import numpy as np

```



```

from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns

def cohens_d(x, y):
    """Calculate Cohen's d for paired samples."""
    # Calculate the mean difference
    d_mean = np.mean(x - y)
    # Calculate the standard deviation of the difference
    d_std = np.std(x - y, ddof=1)
    # Calculate Cohen's d
    d = d_mean / d_std if d_std > 0 else 0
    return d

# Read the CSV file
data = pd.read_csv('cleaned_participant_data_CleanedByRM.csv')

# Get unique players
players = data['Player'].unique()

# Define the C metrics we want to analyze
c_metrics = ['C1_C', 'C2_C', 'C3_C', 'C4_C', 'C5_C', 'C6_C', 'C7_C']

# Prepare data for statistical analysis
results = []
paired_data = {}

# Collect paired data for each metric
for metric in c_metrics:
    feedback_1_values = []
    feedback_0_values = []
    player_pairs = []

    for player in players:
        # Get data for this player
        player_data = data[data['Player'] == player]

        # Check if this player has both feedback conditions
        feedback_1_row = player_data[player_data['Feedback'] == 1]
        feedback_0_row = player_data[player_data['Feedback'] == 0]

        if not feedback_1_row.empty and not feedback_0_row.empty:
            feedback_1_values.append(feedback_1_row[metric].values[0])
            feedback_0_values.append(feedback_0_row[metric].values[0])
            player_pairs.append(player)

```

```

# Store paired data for visualization later
paired_data[metric] = {
    'player': player_pairs,
    'feedback_1': feedback_1_values,
    'feedback_0': feedback_0_values
}

# Convert to numpy arrays
feedback_1_values = np.array(feedback_1_values)
feedback_0_values = np.array(feedback_0_values)

# Perform paired t-test
t_stat, p_value = stats.ttest_rel(feedback_1_values, feedback_0_values)

# Calculate effect size (Cohen's d)
effect_size = cohens_d(feedback_1_values, feedback_0_values)

# Determine significance level indicators
sig_level = ""
if p_value < 0.001:
    sig_level = "****"
elif p_value < 0.01:
    sig_level = "***"
elif p_value < 0.05:
    sig_level = "**"

# Calculate mean difference and percent change
mean_1 = np.mean(feedback_1_values)
mean_0 = np.mean(feedback_0_values)
mean_diff = mean_1 - mean_0

# If mean_0 is not zero, calculate percent change
if abs(mean_0) > 1e-10: # Avoid division by near-zero
    percent_change = (mean_diff / mean_0) * 100
else:
    percent_change = np.nan

# Store results
results.append({
    'Metric': metric.replace('_C', ''), # Remove _C suffix for display
    'Mean (Feedback=1)': mean_1,
    'Mean (Feedback=0)': mean_0,
    'Mean Difference': mean_diff,
    'Percent Change': percent_change,
    't-statistic': t_stat,
    'p-value': p_value,

```

```

        'Significance': sig_level,
        'Cohen's d': effect_size,
        'Sample Size': len(feedback_1_values)
    })

# Convert results to dataframe
results_df = pd.DataFrame(results)

# Print results
print("\nSTATISTICAL ANALYSIS RESULTS - C METRICS ONLY\n")
print("Comparing C metrics between feedback conditions (1 vs 0)\n")

# Format the results for better readability
formatted_results = results_df.copy()
formatted_results['p-value'] = formatted_results.apply(
    lambda row: f"{row['p-value']:.4f}{' ' + row['Significance'] if row['Significance'] else ''}", axis=1)
formatted_results["Cohen's d"] = formatted_results["Cohen's d"].apply(lambda d: f"{d:.4f}")
formatted_results['Mean (Feedback=1)'] = formatted_results['Mean (Feedback=1)'].apply(lambda m: f"{m:.4f}")
formatted_results['Mean (Feedback=0)'] = formatted_results['Mean (Feedback=0)'].apply(lambda m: f"{m:.4f}")
formatted_results['Mean Difference'] = formatted_results['Mean Difference'].apply(lambda d: f"{d:.4f}")
formatted_results['Percent Change'] = formatted_results['Percent Change'].apply(
    lambda p: f"{p:.2f}%" if not np.isnan(p) else "N/A")
formatted_results['t-statistic'] = formatted_results['t-statistic'].apply(lambda t: f"{t:.4f}")

print(formatted_results[['Metric', 'Mean (Feedback=1)', 'Mean (Feedback=0)',
                        'Mean Difference', 'Percent Change', 't-statistic',
                        'p-value', "Cohen's d", 'Sample Size']].to_string(index=False))

print("\n\nSignificance levels: * p<0.05, ** p<0.01, *** p<0.001")
print("\nInterpretation of Cohen's d:")
print("0.2 = small effect")
print("0.5 = medium effect")
print("0.8 = large effect")

# Save results to CSV
results_df.to_csv('c_metrics_analysis_results.csv', index=False)
print("\nResults have been saved to 'c_metrics_analysis_results.csv'")

# Create visualizations
# 1. Bar chart showing mean values for each metric
plt.figure(figsize=(14, 10))

```

```

# First subplot: Bar charts for each metric
plt.subplot(2, 1, 1)
metrics_short = [m.replace('_C', '') for m in c_metrics]
x = np.arange(len(metrics_short))
width = 0.35

feedback_1_means = [res['Mean (Feedback=1)'] for res in results]
feedback_0_means = [res['Mean (Feedback=0)'] for res in results]

plt.bar(x - width/2, feedback_1_means, width, label='Feedback=1')
plt.bar(x + width/2, feedback_0_means, width, label='Feedback=0')

plt.xlabel('Metrics')
plt.ylabel('Mean Value')
plt.title('Comparison of C Metrics by Feedback Condition')
plt.xticks(x, metrics_short)
plt.legend()

# Add p-values to the chart
for i, res in enumerate(results):
    p_value = res['p-value']
    p_text = f"p={p_value:.3f}"
    if p_value < 0.05:
        p_text += "*"
        color = 'red'
    else:
        color = 'black'
    plt.annotate(p_text, xy=(i, max(feedback_1_means[i], feedback_0_means[i]) * 1.05),
                ha='center', va='bottom', color=color)

# Second subplot: Effect sizes
plt.subplot(2, 1, 2)
effect_sizes = [res["Cohen's d"] for res in results]
colors = ['green' if d >= 0.2 else 'blue' for d in effect_sizes]

plt.bar(x, effect_sizes, color=colors)
plt.axhline(y=0.2, color='r', linestyle='--', label='Small Effect (0.2)')
plt.axhline(y=0.5, color='orange', linestyle='--', label='Medium Effect (0.5)')
plt.axhline(y=0.8, color='green', linestyle='--', label='Large Effect (0.8)')

plt.xlabel('Metrics')
plt.ylabel("Cohen's d")
plt.title('Effect Sizes for C Metrics (Feedback=1 vs Feedback=0)')
plt.xticks(x, metrics_short)
plt.legend()

```

```

plt.tight_layout()
plt.savefig('c_metrics_comparison.png', dpi=300)
print("Bar chart visualization saved as 'c_metrics_comparison.png'")

# Create paired plot to visualize individual differences
plt.figure(figsize=(14, 10))
for i, metric in enumerate(c_metrics):
    plt.subplot(3, 3, i+1)
    metric_data = paired_data[metric]

    for j in range(len(metric_data['player'])):
        plt.plot([0, 1],
                 [metric_data['feedback_0'][j], metric_data['feedback_1'][j]],
                 'o-', alpha=0.3, color='gray')

    # Plot means
    mean_0 = np.mean(metric_data['feedback_0'])
    mean_1 = np.mean(metric_data['feedback_1'])
    plt.plot([0, 1], [mean_0, mean_1], 'o-', linewidth=3, color='blue')

    # Add p-value
    p_value = next(res['p-value'] for res in results if res['Metric'] == metric.replace('_C',
    ''))
    p_text = f"p={p_value:.3f}"
    if p_value < 0.05:
        p_text += "*"
        color = 'red'
    else:
        color = 'black'

    plt.title(f"{metric.replace('_C', '')} - {p_text}", color=color)
    plt.xticks([0, 1], ['No Feedback', 'Feedback'])
    if i % 3 == 0: # Add y-label only for leftmost plots
        plt.ylabel('Value')

plt.tight_layout()
plt.savefig('c_metrics_paired_comparison.png', dpi=300)
print("Paired comparison visualization saved as 'c_metrics_paired_comparison.png'")

```

- **For learning and improvement analysis based on Dtime:**

```

#!/usr/bin/env python
# Enhanced Learning Analysis Script
# Analyzes DTime data for three learning criteria across feedback conditions

```

```

# Updated to output SVG visualizations with improved statistical analysis

import sys
import os
import subprocess
import importlib

# Function to check and install required packages
def check_and_install_packages():
    required_packages = {
        'pandas': 'pandas',
        'numpy': 'numpy',
        'matplotlib': 'matplotlib',
        'scipy': 'scipy',
        'seaborn': 'seaborn',
        'statsmodels': 'statsmodels',
        'scikit-learn': 'scikit-learn', # Added for enhanced model validation
        'pingouin': 'pingouin' # Added for more advanced statistics
    }

    missing_packages = []

    for package, pip_name in required_packages.items():
        try:
            importlib.import_module(package)
            print(f"{package} is already installed.")
        except ImportError:
            missing_packages.append(pip_name)

    if missing_packages:
        print(f"Installing missing packages: {' '.join(missing_packages)}")
        for package in missing_packages:
            try:
                subprocess.check_call([sys.executable, "-m", "pip", "install", package])
                print(f"Successfully installed {package}")
            except Exception as e:
                print(f"Failed to install {package}: {e}")
                print("Please install the package manually using: pip install " + package)
                sys.exit(1)

# Check and install required packages
print("Checking required packages...")
check_and_install_packages()

# Now import all required packages
import pandas as pd

```

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
import os
import warnings

# Suppress warnings
warnings.filterwarnings('ignore')

# Set plot style
# Set plot style for better aesthetics
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context("paper", font_scale=1.5)
plt.rcParams['svg.fonttype'] = 'none' # Ensures fonts are editable in SVG
plt.rcParams['figure.dpi'] = 300 # Higher DPI for better quality
plt.rcParams['font.family'] = 'sans-serif'
plt.rcParams['font.sans-serif'] = ['Arial', 'DejaVu Sans', 'Liberation Sans', 'sans-serif']

# Create output directories
def create_directories():
    directories = [
        "analysis_output/reports",
        "analysis_output/visualizations/micro_learning",
        "analysis_output/visualizations/endgame_learning",
        "analysis_output/visualizations/consistency",
        "analysis_output/visualizations/aggregate"
    ]

    for directory in directories:
        os.makedirs(directory, exist_ok=True)
        print(f"Created directory: {directory}")

# Read and process the data
def load_data():
    print("Reading and processing data...")
    try:
        data = pd.read_csv('ForTimeSection.csv')
        print("\nData shape:", data.shape)

        print("\nColumn data types:")
        print(data.dtypes)

        # Print first few rows of the feedback column to understand its format

```

```

print("\nFeedback column sample values:")
if 'IsFeedBackMood On?' in data.columns:
    print(data['IsFeedBackMood On?'].head())
else:
    # If the expected column doesn't exist, look for alternatives
    feedback_cols = [col for col in data.columns if 'feedback' in col.lower() or 'mood'
in col.lower()]
    if feedback_cols:
        print(f"Found potential feedback columns: {feedback_cols}")
        print(data[feedback_cols[0]].head())
    else:
        print("No feedback column found!")
        raise ValueError("Cannot find feedback information in the data")

# Convert feedback column to boolean more robustly
feedback_col = 'IsFeedBackMood On?' if 'IsFeedBackMood On?' in data.columns else feed-
back_cols[0]

# Handle different possible data types
if data[feedback_col].dtype == bool:
    # Already boolean
    data['Feedback'] = data[feedback_col]
elif data[feedback_col].dtype == 'object':
    # String values - convert to boolean
    data['Feedback'] = data[feedback_col].astype(str).str.lower().isin(['true', 'yes',
'1', 'on'])
else:
    # Numeric values - treat non-zero as True
    data['Feedback'] = data[feedback_col].astype(float) > 0

print("\nFeedback values after conversion:", data['Feedback'].value_counts())

# Add attempt number for each player and condition
data = data.sort_values(['Player', 'Feedback'])

# Create attempt sequence number
data['Attempt'] = data.groupby(['Player', 'Feedback']).cumcount() + 1

# Validate data quality
print(f"\nPlayers: {data['Player'].nunique()}")
print(f"Records: {len(data)}")
print(f"Missing DDTime values: {data['DDTime'].isna().sum()}")

return data

except Exception as e:

```



```

        print(f"Error loading data: {e}")
        raise

# Main class for analysis
class LearningAnalysis:
    def __init__(self, data):
        self.data = data
        self.micro_learning_results = None
        self.endgame_learning_results = None
        self.consistency_results = None

        # Create the necessary directories
        create_directories()

        # Define color scheme for consistent visualization
        self.colors = {
            'Feedback': '#1f77b4', # Blue
            'No Feedback': '#ff7f0e', # Orange
            'highlight': '#d62728', # Red for highlighting
            'grid': '#cccccc', # Light gray for grids
            'text': '#333333' # Dark gray for text
        }

        # Set random seed for reproducibility
        np.random.seed(42)

#####
# 1. Micro Learning Analysis
#####
#####
# 1. Micro Learning Analysis
#####
def analyze_micro_learning(self):
    print("Analyzing Micro Learning (learning rate)...")

    # Create empty DataFrame for results with expanded metrics
    learning_rates = pd.DataFrame(columns=[
        'Player', 'Condition', 'Slope', 'P_Value', 'R_Squared',
        'CI_Lower', 'CI_Upper', 'MSE', 'Num_Attempts'
    ])

    # Calculate learning rate for each player and condition
    for player_id in self.data['Player'].unique():
        for feedback_condition in [True, False]:
            # Extract data for this player and condition
            player_data = self.data[(self.data['Player'] == player_id) &

```

```

        (self.data['Feedback'] == feedback_condition)]

# Only analyze if there are enough data points
if len(player_data) >= 5:
    # Remove extreme outliers (optional, can skew learning rates)
    Q1 = player_data['DDTime'].quantile(0.25)
    Q3 = player_data['DDTime'].quantile(0.75)
    IQR = Q3 - Q1
    filtered_data = player_data[
        (player_data['DDTime'] >= Q1 - 3 * IQR) &
        (player_data['DDTime'] <= Q3 + 3 * IQR)
    ]

    if len(filtered_data) >= 5: # Still need enough data after filtering
        # Fit linear model
        X = sm.add_constant(filtered_data['Attempt'])
        model = sm.OLS(filtered_data['DDTime'], X).fit()

        # Extract results
        slope = model.params[1]
        p_value = model.pvalues[1]
        r_squared = model.rsquared

        # Calculate 95% confidence intervals for slope
        ci = model.conf_int(alpha=0.05)
        ci_lower = ci.iloc[1, 0]
        ci_upper = ci.iloc[1, 1]

        # Mean squared error
        mse = model.mse_resid

        # Add to results DataFrame
        learning_rates = pd.concat([learning_rates, pd.DataFrame({
            'Player': [player_id],
            'Condition': ['Feedback' if feedback_condition else 'No Feedback'],
            'Slope': [slope],
            'P_Value': [p_value],
            'R_Squared': [r_squared],
            'CI_Lower': [ci_lower],
            'CI_Upper': [ci_upper],
            'MSE': [mse],
            'Num_Attempts': [len(filtered_data)]
        })], ignore_index=True)

self.micro_learning_results = learning_rates

```

```

# Save results to CSV
learning_rates.to_csv("analysis_output/reports/micro_learning_rates.csv", index=False)

# Calculate percentage of negative slopes (indicating improvement)
neg_slope_pct = {
    'Feedback': sum(learning_rates[learning_rates['Condition'] == 'Feedback']['Slope'] <
0) /
        len(learning_rates[learning_rates['Condition'] == 'Feedback']) * 100
    if len(learning_rates[learning_rates['Condition'] == 'Feedback']) > 0 else
0,
    'No Feedback': sum(learning_rates[learning_rates['Condition'] == 'No Feed-
back']['Slope'] < 0) /
        len(learning_rates[learning_rates['Condition'] == 'No Feedback']) * 100
    if len(learning_rates[learning_rates['Condition'] == 'No Feedback']) >
0 else 0
}

# Save percentage to report
with open("analysis_output/reports/micro_learning_summary.txt", 'w') as f:
    f.write("Summary of Micro Learning Analysis\n")
    f.write("=====\n\n")
    f.write(f"Percentage of participants showing improvement (negative slope):\n")
    f.write(f" - Feedback condition: {neg_slope_pct['Feedback']:.1f}%\n")
    f.write(f" - No Feedback condition: {neg_slope_pct['No Feedback']:.1f}%\n")

# Create learning curve plots
self._create_learning_curve_plots()

# Perform statistical comparison
self._micro_learning_comparison()

def _create_learning_curve_plots(self):
    for player_id in self.data['Player'].unique():
        player_data = self.data[self.data['Player'] == player_id]

        # Only create plot if player has data for both conditions
        if len(player_data['Feedback'].unique()) == 2:
            # Create figure for more control over layout
            fig, ax = plt.subplots(figsize=(10, 6))

            # Plot separately for each condition for more control
            for condition, color in zip([True, False], [self.colors['Feedback'], self.col-
ors['No Feedback']]):
                condition_data = player_data[player_data['Feedback'] == condition]
                if len(condition_data) >= 5:
                    # Sort by attempt number

```

```

condition_data = condition_data.sort_values('Attempt')

# Plot scatter points
ax.scatter(
    condition_data['Attempt'],
    condition_data['DDTime'],
    color=color,
    alpha=0.7,
    s=50,
    label=f"{'Feedback' if condition else 'No Feedback'}"
)

# Fit and plot regression line
X = sm.add_constant(condition_data['Attempt'])
model = sm.OLS(condition_data['DDTime'], X).fit()

# Get prediction for smooth line
x_range = np.linspace(condition_data['Attempt'].min(), condition_data['Attempt'].max(), 100)
X_pred = sm.add_constant(x_range)
y_pred = model.predict(X_pred)

# Plot regression line
ax.plot(x_range, y_pred, color=color, linewidth=2)

# Add annotation with slope and R²
slope = model.params[1]
r2 = model.rsquared
ax.text(
    0.05 if condition else 0.75,
    0.95 if condition else 0.90,
    f"{'Feedback' if condition else 'No Feedback'}\nSlope:
{slope:.3f}\nR²: {r2:.3f}",
    transform=ax.transAxes,
    bbox=dict(facecolor='white', alpha=0.8)
)

ax.set_title(f"Learning Rate Analysis - Player {player_id}")
ax.set_xlabel("Attempt Number")
ax.set_ylabel("Decision Time (DDTime)")
ax.legend()
plt.tight_layout()

# Save as SVG
plt.savefig(f"analysis_output/visualizations/micro_learning/player_{player_id}_learning_rate.svg",

```

```

        format='svg', bbox_inches='tight')
plt.close()

def _micro_learning_comparison(self):
    # Pivot data for paired t-test
    wide_data = self.micro_learning_results.pivot(
        index='Player',
        columns='Condition',
        values='Slope'
    ).reset_index()

    # Filter out rows with missing values
    wide_data = wide_data.dropna(subset=['Feedback', 'No Feedback'])

    # Perform paired t-test if enough paired data
    if len(wide_data) >= 3:
        t_stat, p_value = stats.ttest_rel(wide_data['Feedback'], wide_data['No Feedback'])

        # Create summary table
        summary_stats = pd.DataFrame({
            'Condition': ['Feedback', 'No Feedback'],
            'Mean_Slope': [np.mean(wide_data['Feedback']), np.mean(wide_data['No Feed-
back'])],
            'SD': [np.std(wide_data['Feedback']), np.std(wide_data['No Feedback'])],
            'Min': [np.min(wide_data['Feedback']), np.min(wide_data['No Feedback'])],
            'Max': [np.max(wide_data['Feedback']), np.max(wide_data['No Feedback'])]
        })

        # Save results
        summary_stats.to_csv("analysis_output/reports/micro_learning_comparison_stats.csv",
index=False)

        # Save t-test results
        with open("analysis_output/reports/micro_t_test_results.txt", 'w') as f:
            f.write(f"Paired t-test comparing learning rates between feedback condi-
tions\n\n")
            f.write(f"t-statistic: {t_stat:.4f}\n")
            f.write(f"p-value: {p_value:.4f}\n")
            f.write(f"Mean difference (Feedback - No Feedback): {np.mean(wide_data['Feed-
back'] - wide_data['No Feedback']):.4f}\n")

        # Create boxplot comparison
        plt.figure(figsize=(8, 6))
        data_long = self.micro_learning_results[['Player', 'Condition', 'Slope']]

        sns.boxplot(x='Condition', y='Slope', data=data_long)

```

```

sns.stripplot(x='Condition', y='Slope', data=data_long, color='black', alpha=0.5,
jitter=True)

plt.title("Comparison of Learning Rates Between Conditions")
plt.xlabel("Condition")
plt.ylabel("Learning Rate (Slope of DDTime)")
plt.annotate(f"p-value = {p_value:.4f}", (0.5, 0.01), xycoords='figure fraction',
ha='center')
plt.tight_layout()

plt.savefig("analysis_output/visualizations/micro_learning/learning_rate_compari-
son.svg", format='svg')
plt.close()

    return p_value
else:
    print("Not enough paired data for statistical comparison")
    return None

#####
# 2. Endgame Learning Analysis
#####
def analyze_endgame_learning(self):
    print("Analyzing Endgame Learning (initial vs final performance)...")

    # Create empty DataFrame for results
    endgame_results = pd.DataFrame(columns=[
        'Player', 'Condition', 'Initial_Performance',
        'Final_Performance', 'Improvement', 'Improvement_Percent'
    ])

    for player_id in self.data['Player'].unique():
        for feedback_condition in [True, False]:
            # Extract data for this player and condition
            player_data = self.data[
                (self.data['Player'] == player_id) &
                (self.data['Feedback'] == feedback_condition)
            ].sort_values('Attempt')

            # Only analyze if there are enough data points (at least 6 for 3 initial and 3
final)

            if len(player_data) >= 6:
                # Calculate initial performance (average of first 3 attempts)
                initial_perf = player_data.head(3)['DDTime'].mean()

                # Calculate final performance (average of last 3 attempts)

```

```

final_perf = player_data.tail(3)['DDTime'].mean()

# Calculate improvement
improvement = initial_perf - final_perf
improvement_percent = (improvement / initial_perf) * 100

# Add to results DataFrame
endgame_results = pd.concat([endgame_results, pd.DataFrame({
    'Player': [player_id],
    'Condition': ['Feedback' if feedback_condition else 'No Feedback'],
    'Initial_Performance': [initial_perf],
    'Final_Performance': [final_perf],
    'Improvement': [improvement],
    'Improvement_Percent': [improvement_percent]
})], ignore_index=True)

self.endgame_learning_results = endgame_results

# Save results to CSV
endgame_results.to_csv("analysis_output/reports/endgame_learning_results.csv", index=False)

# Create endgame learning plots
self._create_endgame_plots()

# Perform statistical comparison
self._endgame_learning_comparison()

def _create_endgame_plots(self):
    # Individual player plots
    for player_id in self.endgame_learning_results['Player'].unique():
        player_data = self.endgame_learning_results[self.endgame_learning_results['Player']
        == player_id]

        # Only create plot if player has data for both conditions
        if len(player_data) == 2:
            # Reshape data for plotting
            plot_data = pd.DataFrame({
                'Stage': ['Initial', 'Initial', 'Final', 'Final'],
                'Condition': ['Feedback', 'No Feedback', 'Feedback', 'No Feedback'],
                'DDTime': [
                    player_data[player_data['Condition'] == 'Feedback']['Initial_Perfor-
mance'].values[0],
                    player_data[player_data['Condition'] == 'No Feedback']['Initial_Perfor-
mance'].values[0],

```

```

        player_data[player_data['Condition'] == 'Feedback']['Final_Perfor-
mance'].values[0],
        player_data[player_data['Condition'] == 'No Feedback']['Final_Perfor-
mance'].values[0]
    ]
    })

    # Create plot
    plt.figure(figsize=(8, 6))

    for condition in ['Feedback', 'No Feedback']:
        condition_data = plot_data[plot_data['Condition'] == condition]
        plt.plot(
            condition_data['Stage'],
            condition_data['DDTime'],
            marker='o',
            markersize=10,
            linewidth=2,
            label=condition
        )

    plt.title(f"Initial vs Final Performance for Player {player_id}", fontsize=14)
    plt.xlabel("Stage", fontsize=12)
    plt.ylabel("Average Decision Time (DDTime)", fontsize=12)
    plt.legend()
    plt.tight_layout()

    # Save plot
    plt.savefig(f"analysis_output/visualizations/endgame_learning/initial_vs_fi-
nal_player_{player_id}.svg", format='svg')
    plt.close()

    # Overall comparison plot
    overall_data = self.endgame_learning_results.groupby('Condition').agg({
        'Initial_Performance': ['mean', 'std', 'count'],
        'Final_Performance': ['mean', 'std', 'count']
    })

    # Calculate standard error
    overall_data['Initial_Performance', 'se'] = overall_data['Initial_Performance', 'std'] /
np.sqrt(overall_data['Initial_Performance', 'count'])
    overall_data['Final_Performance', 'se'] = overall_data['Final_Performance', 'std'] /
np.sqrt(overall_data['Final_Performance', 'count'])

    # Create plot data
    conditions = overall_data.index.tolist()

```



```

initial_means = overall_data['Initial_Performance', 'mean'].tolist()
final_means = overall_data['Final_Performance', 'mean'].tolist()
initial_se = overall_data['Initial_Performance', 'se'].tolist()
final_se = overall_data['Final_Performance', 'se'].tolist()

# Create overall plot
plt.savefig("analysis_output/visualizations/endgame_learning/overall_initial_vs_final.svg", format='svg')

x = np.arange(len(conditions))
width = 0.35

plt.bar(x - width/2, initial_means, width, label='Initial', alpha=0.7)
plt.bar(x + width/2, final_means, width, label='Final', alpha=0.7)

# Add error bars
plt.errorbar(x - width/2, initial_means, yerr=initial_se, fmt='none', capsize=5,
color='black')
plt.errorbar(x + width/2, final_means, yerr=final_se, fmt='none', capsize=5,
color='black')

plt.xticks(x, conditions)
plt.title("Overall Initial vs Final Performance")
plt.xlabel("Condition")
plt.ylabel("Mean Decision Time (DDTime)")
plt.legend()
plt.tight_layout()

plt.savefig("analysis_output/visualizations/endgame_learning/overall_initial_vs_final.svg", format='svg')
plt.close()

def _endgame_learning_comparison(self):
    # Pivot data for paired t-test on improvement percentage
    wide_data = self.endgame_learning_results.pivot(
        index='Player',
        columns='Condition',
        values='Improvement_Percent'
    ).reset_index()

    # Filter out rows with missing values
    wide_data = wide_data.dropna(subset=['Feedback', 'No Feedback'])

    # Perform paired t-test if enough paired data
    if len(wide_data) >= 3:
        t_stat, p_value = stats.ttest_rel(wide_data['Feedback'], wide_data['No Feedback'])

```

```

# Create summary table
summary_stats = pd.DataFrame({
    'Condition': ['Feedback', 'No Feedback'],
    'Mean_Improvement_Percent': [np.mean(wide_data['Feedback']),
np.mean(wide_data['No Feedback'])],
    'SD': [np.std(wide_data['Feedback']), np.std(wide_data['No Feedback'])],
    'Min': [np.min(wide_data['Feedback']), np.min(wide_data['No Feedback'])],
    'Max': [np.max(wide_data['Feedback']), np.max(wide_data['No Feedback'])]
})

# Save results
summary_stats.to_csv("analysis_output/reports/endgame_comparison_stats.csv", index=False)

# Save t-test results
with open("analysis_output/reports/endgame_t_test_results.txt", 'w') as f:
    f.write(f"Paired t-test comparing improvement percentages between feedback conditions\n\n")
    f.write(f"t-statistic: {t_stat:.4f}\n")
    f.write(f"p-value: {p_value:.4f}\n")
    f.write(f"Mean difference (Feedback - No Feedback): {np.mean(wide_data['Feedback']) - wide_data['No Feedback']:.4f}%\n")

# Create boxplot comparison
plt.figure(figsize=(8, 6))
data_long = self.endgame_learning_results[['Player', 'Condition', 'Improvement_Percent']]

sns.boxplot(x='Condition', y='Improvement_Percent', data=data_long)
sns.stripplot(x='Condition', y='Improvement_Percent', data=data_long, color='black', alpha=0.5, jitter=True)

plt.title("Comparison of Improvement Percentages Between Conditions")
plt.xlabel("Condition")
plt.ylabel("Improvement Percentage (%)")
plt.annotate(f"p-value = {p_value:.4f}", (0.5, 0.01), xycoords='figure fraction', ha='center')
plt.tight_layout()

plt.savefig("analysis_output/visualizations/endgame_learning/improvement_comparison.svg", format='svg')
plt.close()

return p_value
else:

```

```

        print("Not enough paired data for statistical comparison of endgame learning")
        return None

#####
# 3. Consistency Development Analysis
#####
def analyze_consistency_development(self):
    print("Analyzing Consistency Development (time-to-plateau)...")

    # Create empty DataFrame for results
    consistency_results = pd.DataFrame(columns=[
        'Player', 'Condition', 'Time_To_Plateau',
        'Pre_Plateau_Variance', 'Post_Plateau_Variance'
    ])

    # Detect plateau points for each player and condition
    for player_id in self.data['Player'].unique():
        for feedback_condition in [True, False]:
            # Extract data for this player and condition
            player_data = self.data[
                (self.data['Player'] == player_id) &
                (self.data['Feedback'] == feedback_condition)
            ].sort_values('Attempt')

            # Only analyze if there are enough data points
            if len(player_data) >= 10: # Need a reasonable number of attempts to detect
plateau

                plateau_detected = False
                plateau_attempt = None

                # Start from the 5th attempt (to have a window to calculate variation)
                for i in range(4, len(player_data)):
                    # Calculate percent variation between this attempt and previous one
                    current = player_data.iloc[i]['DDTime']
                    previous = player_data.iloc[i-1]['DDTime']
                    variation = abs((current - previous) / previous) * 100

                # Check variation for this and next 2 attempts (if available)
                next_variations = []
                if i + 2 < len(player_data):
                    for j in range(1, 3):
                        next_curr = player_data.iloc[i+j]['DDTime']
                        next_prev = player_data.iloc[i+j-1]['DDTime']
                        next_var = abs((next_curr - next_prev) / next_prev) * 100
                        next_variations.append(next_var)

```

```

        # If current and next 2 variations all < 10%, we've reached plateau
        if variation < 10 and all(var < 10 for var in next_variations):
            plateau_detected = True
            plateau_attempt = player_data.iloc[i]['Attempt']
            break

    if plateau_detected:
        # Calculate variance before and after plateau
        pre_plateau = player_data[player_data['Attempt'] < plateau_at-
tempt]['DDTime']

        post_plateau = player_data[player_data['Attempt'] >= plateau_at-
tempt]['DDTime']

        pre_variance = pre_plateau.var()
        post_variance = post_plateau.var()

        # Add to results
        consistency_results = pd.concat([consistency_results, pd.DataFrame({
            'Player': [player_id],
            'Condition': ['Feedback' if feedback_condition else 'No Feedback'],
            'Time_To_Plateau': [plateau_attempt],
            'Pre_Plateau_Variance': [pre_variance],
            'Post_Plateau_Variance': [post_variance]
        })], ignore_index=True)

    self.consistency_results = consistency_results

    # Save results to CSV
    if not consistency_results.empty:
        consistency_results.to_csv("analysis_output/reports/consistency_results.csv", in-
dex=False)

        # Create stability plots
        self._create_stability_plots()

        # Perform statistical comparison
        self._consistency_comparison()
    else:
        print("No plateau points detected for any participants")

    def _create_stability_plots(self):
        if not self.consistency_results.empty:
            # Individual player plots where both conditions have results
            for player_id in self.consistency_results['Player'].unique():
                player_data = self.consistency_results[self.consistency_results['Player'] ==
player_id]

```

```

    if len(player_data) == 2: # Both conditions have results
        # Get raw data for this player
        raw_data = self.data[self.data['Player'] == player_id].sort_values(['Feed-
back', 'Attempt'])

        # Get plateau points
        plateau_fb = player_data[player_data['Condition'] == 'Feed-
back']['Time_To_Plateau'].values[0]
        plateau_no_fb = player_data[player_data['Condition'] == 'No Feed-
back']['Time_To_Plateau'].values[0]

        # Create plot
        plt.figure(figsize=(10, 6))

        # Plot separate lines for each condition
        for condition, color, plateau in zip(
            [True, False],
            ['blue', 'red'],
            [plateau_fb, plateau_no_fb]
        ):
            condition_data = raw_data[raw_data['Feedback'] == condition]
            plt.plot(
                condition_data['Attempt'],
                condition_data['DDTime'],
                color=color,
                marker='o',
                label=f"{'Feedback' if condition else 'No Feedback'}"
            )
            # Add vertical line for plateau
            plt.axvline(
                x=plateau,
                color=color,
                linestyle='--',
                alpha=0.7,
                label=f"Plateau at attempt {plateau} ({'Feedback' if condition else
'No Feedback'})"
            )

        plt.title(f"Consistency Development for Player {player_id}")
        plt.xlabel("Attempt Number")
        plt.ylabel("Decision Time (DDTime)")
        plt.legend()
        plt.tight_layout()

```

```

        plt.savefig(f"analysis_output/visualizations/consistency/stabil-
ity_player_{player_id}.svg", format='svg')
        plt.close()

# Overall boxplot comparison of time to plateau
plt.figure(figsize=(8, 6))

sns.boxplot(x='Condition', y='Time_To_Plateau', data=self.consistency_results)
sns.stripplot(x='Condition', y='Time_To_Plateau', data=self.consistency_results,
              color='black', alpha=0.5, jitter=True)

plt.title("Time to Plateau Comparison Between Conditions")
plt.xlabel("Condition")
plt.ylabel("Number of Attempts to Reach Plateau")
plt.tight_layout()

plt.savefig("analysis_output/visualizations/consistency/plateau_comparison.svg", for-
mat='svg')
plt.close()

# Create variance reduction plot
# Reshape data for plotting
plot_data = pd.melt(
    self.consistency_results,
    id_vars=['Player', 'Condition'],
    value_vars=['Pre_Plateau_Variance', 'Post_Plateau_Variance'],
    var_name='Stage',
    value_name='Variance'
)

# Replace values for better labels
plot_data['Stage'] = plot_data['Stage'].replace({
    'Pre_Plateau_Variance': 'Before Plateau',
    'Post_Plateau_Variance': 'After Plateau'
})

plt.figure(figsize=(10, 6))

sns.boxplot(x='Stage', y='Variance', hue='Condition', data=plot_data)
plt.yscale('log') # Log scale for better visualization

plt.title("Variance Reduction After Reaching Plateau")
plt.xlabel("Stage")
plt.ylabel("Variance in DDTime (log scale)")
plt.legend(title="Condition")
plt.tight_layout()

```

```

plt.savefig("analysis_output/visualizations/consistency/variance_reduction.svg", format='svg')

plt.close()

def _consistency_comparison(self):
    # Reshape data for paired t-test
    wide_data = self.consistency_results.pivot(
        index='Player',
        columns='Condition',
        values='Time_To_Plateau'
    ).reset_index()

    # Filter out rows with missing values
    wide_data = wide_data.dropna(subset=['Feedback', 'No Feedback'])

    # Perform paired t-test if enough paired data
    if len(wide_data) >= 3:
        t_stat, p_value = stats.ttest_rel(wide_data['Feedback'], wide_data['No Feedback'])

    # Create summary table
    summary_stats = pd.DataFrame({
        'Condition': ['Feedback', 'No Feedback'],
        'Mean_Time_To_Plateau': [np.mean(wide_data['Feedback']), np.mean(wide_data['No
Feedback'])],
        'SD': [np.std(wide_data['Feedback']), np.std(wide_data['No Feedback'])],
        'Min': [np.min(wide_data['Feedback']), np.min(wide_data['No Feedback'])],
        'Max': [np.max(wide_data['Feedback']), np.max(wide_data['No Feedback'])]
    })

    # Save results

    # Save t-test results
    summary_stats.to_csv("analysis_output/reports/consistency_comparison_stats.csv", index=False)

    # Save t-test results
    with open("analysis_output/reports/consistency_t_test_results.txt", 'w') as f:
        f.write(f"Paired t-test comparing time to plateau between feedback conditions\n\n")

        f.write(f"t-statistic: {t_stat:.4f}\n")
        f.write(f"p-value: {p_value:.4f}\n")
        f.write(f"Mean difference (Feedback - No Feedback): {np.mean(wide_data['Feedback']) - wide_data['No Feedback']:.4f} attempts\n")

    return p_value

```

```

else:
    print("Not enough paired data for statistical comparison of consistency development")
    return None

#####
# 4. Overall Report Generation
#####
def generate_overall_report(self):
    print("Generating overall report...")

    # Create summary table for micro learning
    if self.micro_learning_results is not None:
        # Create basic summary with aggregation functions
        micro_summary = self.micro_learning_results.groupby('Condition').agg({
            'Slope': ['mean', 'std'],
            'P_Value': lambda x: sum(x < 0.05), # Count significant results
            'Player': 'count' # Count of records
        })

        # Flatten the multi-level columns
        micro_summary.columns = [f"{col[0]}_{col[1]}" if isinstance(col, tuple) else col for
col in micro_summary.columns]
        micro_summary = micro_summary.reset_index()

        # Count negative and positive slopes
        neg_slopes = self.micro_learning_results.groupby('Condition').apply(
            lambda x: sum(x['Slope'] < 0)).reset_index(name='Negative_Slope_Count')
        pos_slopes = self.micro_learning_results.groupby('Condition').apply(
            lambda x: sum(x['Slope'] > 0)).reset_index(name='Positive_Slope_Count')

        # Join counts to summary
        micro_summary = micro_summary.merge(neg_slopes, on='Condition')
        micro_summary = micro_summary.merge(pos_slopes, on='Condition')

        # Rename columns for clarity
        micro_summary = micro_summary.rename(columns={
            'Slope_mean': 'Mean_Slope',
            'Slope_std': 'SD_Slope',
            'P_Value_<lambda_0>': 'Significant_Count',
            'Player_count': 'N'
        })

        # Save to CSV
        micro_summary.to_csv("analysis_output/reports/micro_learning_summary.csv", in-
dex=False)

```



```

# Create summary table for endgame learning
if self.endgame_learning_results is not None:
    # Create basic summary with aggregation functions
    endgame_summary = self.endgame_learning_results.groupby('Condition').agg({
        'Improvement': ['mean', 'std'],
        'Improvement_Percent': ['mean', 'std'],
        'Player': 'count' # Count of records
    })

    # Flatten the multi-level columns
    endgame_summary.columns = [f"{col[0]}_{col[1]}" if isinstance(col, tuple) else col
    for col in endgame_summary.columns]
    endgame_summary = endgame_summary.reset_index()

    # Count positive and negative improvements
    pos_imp = self.endgame_learning_results.groupby('Condition').apply(
        lambda x: sum(x['Improvement'] > 0)).reset_index(name='Positive_Improvement_Count')
    neg_imp = self.endgame_learning_results.groupby('Condition').apply(
        lambda x: sum(x['Improvement'] < 0)).reset_index(name='Negative_Improvement_Count')

    # Join counts to summary
    endgame_summary = endgame_summary.merge(pos_imp, on='Condition')
    endgame_summary = endgame_summary.merge(neg_imp, on='Condition')

    # Rename columns for clarity
    endgame_summary = endgame_summary.rename(columns={
        'Improvement_mean': 'Mean_Improvement',
        'Improvement_std': 'SD_Improvement',
        'Improvement_Percent_mean': 'Mean_Improvement_Percent',
        'Improvement_Percent_std': 'SD_Improvement_Percent',
        'Player_count': 'N'
    })

    # Save to CSV
    endgame_summary.to_csv("analysis_output/reports/endgame_learning_summary.csv", index=False)

# Create summary table for consistency development
if self.consistency_results is not None and not self.consistency_results.empty:
    # Create basic summary with aggregation functions
    consistency_summary = self.consistency_results.groupby('Condition').agg({
        'Time_To_Plateau': ['mean', 'std'],
        'Pre_Plateau_Variance': 'mean',
        'Post_Plateau_Variance': 'mean',

```

```

        'Player': 'count' # Count of records
    })

    # Flatten the multi-level columns
    consistency_summary.columns = [f"{col[0]}_{col[1]}" if isinstance(col, tuple) else
col for col in consistency_summary.columns]
    consistency_summary = consistency_summary.reset_index()

    # Calculate variance reduction metrics
    var_reduction = self.consistency_results.copy()
    var_reduction['Variance_Reduction'] = var_reduction['Pre_Plateau_Variance'] - var_re-
duction['Post_Plateau_Variance']
    var_reduction['Variance_Reduction_Percent'] = (var_reduction['Variance_Reduction'] /
var_reduction['Pre_Plateau_Variance']) * 100

    var_red_summary = var_reduction.groupby('Condition').agg({
        'Variance_Reduction': 'mean',
        'Variance_Reduction_Percent': 'mean'
    }).reset_index()

    # Join variance reduction to summary
    consistency_summary = consistency_summary.merge(var_red_summary, on='Condition')

    # Rename columns for clarity
    consistency_summary = consistency_summary.rename(columns={
        'Time_To_Plateau_mean': 'Mean_Time_To_Plateau',
        'Time_To_Plateau_std': 'SD_Time_To_Plateau',
        'Pre_Plateau_Variance_mean': 'Mean_Pre_Plateau_Variance',
        'Post_Plateau_Variance_mean': 'Mean_Post_Plateau_Variance',
        'Player_count': 'N',
        'Variance_Reduction_mean': 'Mean_Variance_Reduction',
        'Variance_Reduction_Percent_mean': 'Mean_Variance_Reduction_Percent'
    })

    # Save to CSV
    consistency_summary.to_csv("analysis_output/reports/consistency_development_sum-
mary.csv", index=False)
    else:
        # Create empty summary if no plateau points detected
        consistency_summary = pd.DataFrame({
            'Condition': ['Feedback', 'No Feedback'],
            'Mean_Time_To_Plateau': [np.nan, np.nan],
            'SD_Time_To_Plateau': [np.nan, np.nan],
            'Mean_Pre_Plateau_Variance': [np.nan, np.nan],
            'Mean_Post_Plateau_Variance': [np.nan, np.nan],
            'N': [0, 0],

```

```

        'Mean_Variance_Reduction': [np.nan, np.nan],
        'Mean_Variance_Reduction_Percent': [np.nan, np.nan]
    })
    consistency_summary.to_csv("analysis_output/reports/consistency_development_sum-
mary.csv", index=False)

    # Generate final combined report
    # Generate final combined report
    self._create_combined_report()

def _create_combined_report(self):
    """Create a comprehensive combined report summarizing all analyses"""
    # Create overall_analysis directory if it doesn't exist
    overall_dir = "analysis_output/overall_analysis"
    os.makedirs(overall_dir, exist_ok=True)

    # Create standard report in reports directory
    with open("analysis_output/reports/combined_report.txt", 'w') as f:
        f.write("=====\n")
        f.write("                LEARNING ANALYSIS SUMMARY REPORT                \n")
        f.write("=====\n\n")

        # 1. Micro Learning Results
        f.write("1. MICRO LEARNING ANALYSIS\n")
        f.write("=====\n")
        if self.micro_learning_results is not None:
            try:
                f.write("This analysis examined learning rates (slopes) of DDTime across at-
tempts.\n")

                f.write("Negative slopes indicate improvements (faster decisions over
time).\n\n")

                # Try to load summary stats if available
                if os.path.exists("analysis_output/reports/micro_learning_compari-
son_stats.csv"):
                    micro_stats = pd.read_csv("analysis_output/reports/micro_learning_compari-
ison_stats.csv")

                    f.write("Summary of learning rates by condition:\n")
                    for _, row in micro_stats.iterrows():
                        condition = row['Condition']
                        f.write(f"    {condition}: Mean slope = {row['Mean_Slope']:.4f} (SD =
{row['SD']:.4f})\n")
                    else:
                        f.write("Detailed comparison statistics are not available.\n")

```

```

# Get t-test results if available
if os.path.exists("analysis_output/reports/micro_t_test_results.txt"):
    with open("analysis_output/reports/micro_t_test_results.txt", 'r') as
t_file:

        t_test_content = t_file.read()

        f.write("\nStatistical comparison:\n")
        f.write(t_test_content + "\n")
    else:
        f.write("\nStatistical comparison results are not available.\n")
except Exception as e:
    f.write(f"\nError while reporting micro learning details: {str(e)}\n")
else:
    f.write("No micro learning analysis results available.\n\n")

# 2. Endgame Learning Results
f.write("\n2. ENDGAME LEARNING ANALYSIS\n")
f.write("=====\n")
if self.endgame_learning_results is not None:
    try:
        f.write("This analysis compared initial performance (first 3 attempts) vs.
final performance (last 3 attempts).\n")
        f.write("Positive improvement percentages indicate better performance at the
end of training.\n\n")

# Try to load summary stats if available
if os.path.exists("analysis_output/reports/endgame_comparison_stats.csv"):
    endgame_stats = pd.read_csv("analysis_output/reports/endgame_compari-
son_stats.csv")

    f.write("Summary of improvement percentages by condition:\n")
    for _, row in endgame_stats.iterrows():
        condition = row['Condition']
        f.write(f"    - {condition}: Mean improvement = {row['Mean_Improve-
ment_Percent']:.2f}% (SD = {row['SD']:.2f}%)\n")
    else:
        f.write("Detailed comparison statistics are not available.\n")

# Get t-test results if available
if os.path.exists("analysis_output/reports/endgame_t_test_results.txt"):
    with open("analysis_output/reports/endgame_t_test_results.txt", 'r') as
t_file:

        t_test_content = t_file.read()

        f.write("\nStatistical comparison:\n")
        f.write(t_test_content + "\n")

```

```

        else:
            f.write("\nStatistical comparison results are not available.\n")
    except Exception as e:
        f.write(f"\nError while reporting endgame learning details: {str(e)}\n")
    else:
        f.write("No endgame learning analysis results available.\n\n")

# 3. Consistency Development Results
f.write("\n3. CONSISTENCY DEVELOPMENT ANALYSIS\n")
f.write("=====\n")
if self.consistency_results is not None and not self.consistency_results.empty:
    try:
        f.write("This analysis identified when participants reached a plateau (con-
sistent performance).\n")
        f.write("Lower 'Time to Plateau' values indicate faster development of con-
sistent performance.\n\n")

        # Try to load summary stats if available
        if os.path.exists("analysis_output/reports/consistency_compari-
son_stats.csv"):
            consistency_stats = pd.read_csv("analysis_output/reports/consistency_com-
parison_stats.csv")

            f.write("Summary of time to plateau by condition:\n")
            for _, row in consistency_stats.iterrows():
                condition = row['Condition']
                f.write(f"    - {condition}: Mean time to plateau =
{row['Mean_Time_To_Plateau']:.2f} attempts (SD = {row['SD']:.2f})\n")
            else:
                f.write("Detailed comparison statistics are not available.\n")

        # Get t-test results if available
        if os.path.exists("analysis_output/reports/consistency_t_test_results.txt"):
            with open("analysis_output/reports/consistency_t_test_results.txt", 'r')
as t_file:
                t_test_content = t_file.read()
                f.write("\nStatistical comparison:\n")
                f.write(t_test_content + "\n")
            else:
                f.write("\nStatistical comparison results are not available.\n")
        except Exception as e:
            f.write(f"\nError while reporting consistency development details:
{str(e)}\n")
        else:
            f.write("No consistency development analysis results available.\n\n")

```

```

# Overall Conclusions
f.write("\nOVERALL CONCLUSIONS\n")
f.write("=====\n")
f.write("1. Learning Rate (Micro Learning):\n")
if self.micro_learning_results is not None:
    fb_slopes = self.micro_learning_results[self.micro_learning_results['Condition']
== 'Feedback']['Slope']
    no_fb_slopes = self.micro_learning_results[self.micro_learning_results['Condi-
tion'] == 'No Feedback']['Slope']
    f.write(f"    - {sum(fb_slopes < 0)} out of {len(fb_slopes)} participants showed
improvement with feedback\n")
    f.write(f"    - {sum(no_fb_slopes < 0)} out of {len(no_fb_slopes)} participants
showed improvement without feedback\n")

f.write("\n2. Overall Improvement (Endgame Learning):\n")
if self.endgame_learning_results is not None:
    fb_imp = self.endgame_learning_results[self.endgame_learning_results['Condition']
== 'Feedback']['Improvement_Percent']
    no_fb_imp = self.endgame_learning_results[self.endgame_learning_results['Condi-
tion'] == 'No Feedback']['Improvement_Percent']
    f.write(f"    - {sum(fb_imp > 0)} out of {len(fb_imp)} participants improved with
feedback\n")
    f.write(f"    - {sum(no_fb_imp > 0)} out of {len(no_fb_imp)} participants improved
without feedback\n")

f.write("\n3. Performance Consistency:\n")
if self.consistency_results is not None and not self.consistency_results.empty:
    f.write("    Summary of performance stability:\n")
    for condition in ['Feedback', 'No Feedback']:
        condition_data = self.consistency_results[self.consistency_results['Condi-
tion'] == condition]
        if len(condition_data) > 0:
            mean_time = condition_data['Time_To_Plateau'].mean()
            mean_reduction = ((condition_data['Pre_Plateau_Variance'] - condi-
tion_data['Post_Plateau_Variance']) /
                             condition_data['Pre_Plateau_Variance'] * 100).mean()
            f.write(f"    - {condition}: Average {mean_time:.1f} attempts to plateau,
{mean_reduction:.1f}% variance reduction\n")
        else:
            f.write(f"    - {condition}: No plateau detected\n")

# Recommendations
f.write("\nKEY FINDINGS AND IMPLICATIONS\n")
f.write("=====\n")
f.write("1. The analysis examined learning across three criteria:\n")
f.write("    - Continuous improvement during training (Micro Learning)\n")

```

```

        f.write("    - Overall improvement from start to finish (Endgame Learning)\n")
        f.write("    - Development of performance consistency\n\n")
        f.write("2. All statistical comparisons used paired tests to control for individual
differences\n")
        f.write("3. Detailed visualizations and raw data are available in respective analysis
folders\n\n")

    # Report metadata
    f.write("\n-----\n")
    from datetime import datetime
    f.write(f"Report generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}\n")
    f.write("Analysis performed using Python with statistical and data science librar-
ies\n")
    f.write("-----\n")

#####
# 5. Deeper Learning Analysis
#####
def analyze_deeper_learning(self):
    """Run advanced learning pattern analyses"""
    print("\n" + "="*70)
    print("RUNNING DEEPER LEARNING ANALYSES")
    print("="*70)

    # Create new directory structure
    self._create_deeper_learning_directories()

    # Run all deeper analyses
    phases_results = self.analyze_learning_phases()
    error_results = self.analyze_error_recovery()
    stability_results = self.analyze_learning_stability()

    # Generate combined report
    self._create_deeper_learning_report(phases_results, error_results, stability_results)

    print("✓ Deeper learning analyses completed")

def _create_deeper_learning_directories(self):
    """Create directory structure for deeper learning analyses"""
    directories = [
        "dive deeper learning/reports",
        "dive deeper learning/visualizations/learning_phases",
        "dive deeper learning/visualizations/error_recovery",
        "dive deeper learning/visualizations/learning_stability"
    ]

```

```

    for directory in directories:
        os.makedirs(directory, exist_ok=True)
        print(f"Created directory: {directory}")

def analyze_learning_phases(self):
    """
    Identify and analyze distinct learning phases:
    1. Initial exploration (high variance, unstable performance)
    2. Rapid improvement (steepest negative slope)
    3. Fine-tuning (small improvements, lower variance)
    4. Mastery (stable performance, low variance)
    """
    print("Analyzing learning phases...")

    phase_results = pd.DataFrame(columns=[
        'Player', 'Condition',
        'Exploration_Duration', 'Rapid_Learning_Duration',
        'Fine_Tuning_Duration', 'Mastery_Duration',
        'Exploration_Variance', 'Rapid_Learning_Slope',
        'Fine_Tuning_Slope', 'Mastery_Variance'
    ])

    # Phase detection parameters
    VARIANCE_THRESHOLD = 0.25 # For exploration phase
    SLOPE_THRESHOLD = -0.05 # For rapid learning phase
    STABILITY_THRESHOLD = 0.1 # For mastery phase

    for player_id in self.data['Player'].unique():
        for feedback_condition in [True, False]:
            player_data = self.data[
                (self.data['Player'] == player_id) &
                (self.data['Feedback'] == feedback_condition)
            ].sort_values('Attempt')

            if len(player_data) >= 10: # Minimum data points needed
                try:
                    # Calculate rolling statistics
                    window = 5
                    rolling_var = player_data['DDTime'].rolling(window=window).var()
                    rolling_slope = player_data['DDTime'].rolling(window=window).apply(
                        lambda x: np.polyfit(range(len(x)), x, 1)[0] if len(x) > 1 else
np.nan
                    )

                    # Identify phases
                    # Find first point where variance drops below threshold

```



```

        exploration_end = window - 1 # Default to window size
        for i in range(window-1, len(rolling_var)):
            if pd.isna(rolling_var.iloc[i]) and rolling_var.iloc[i] <
VARIANCE_THRESHOLD * player_data['DDTime'].var():
                exploration_end = i
                break

        # Find end of rapid learning phase
        rapid_learning_end = exploration_end
        for i in range(exploration_end, len(rolling_slope)):
            if pd.isna(rolling_slope.iloc[i]) and rolling_slope.iloc[i] >
SLOPE_THRESHOLD:
                rapid_learning_end = i
                break

        # Find end of fine-tuning phase
        fine_tuning_end = rapid_learning_end
        for i in range(rapid_learning_end, len(rolling_var)):
            if pd.isna(rolling_var.iloc[i]) and rolling_var.iloc[i] <
STABILITY_THRESHOLD * player_data['DDTime'].var():
                fine_tuning_end = i
                break

        # Calculate phase metrics
        exploration_var = player_data['DDTime'].iloc[:exploration_end+1].var() if
exploration_end > 0 else np.nan

        rapid_learning_slope = np.nan
        if rapid_learning_end > exploration_end:
            segment = player_data['DDTime'].iloc[exploration_end:rapid_learn-
ing_end+1]

            if len(segment) > 1:
                rapid_learning_slope = np.polyfit(range(len(segment)), segment,
1) [0]

        fine_tuning_slope = np.nan
        if fine_tuning_end > rapid_learning_end:
            segment = player_data['DDTime'].iloc[rapid_learning_end:fine_tun-
ing_end+1]

            if len(segment) > 1:
                fine_tuning_slope = np.polyfit(range(len(segment)), segment,
1) [0]

        mastery_var = player_data['DDTime'].iloc[fine_tuning_end:].var() if
fine_tuning_end < len(player_data) else np.nan

```

```

        # Add to results DataFrame
        phase_results = pd.concat([phase_results, pd.DataFrame({
            'Player': [player_id],
            'Condition': ['Feedback' if feedback_condition else 'No Feedback'],
            'Exploration_Duration': [exploration_end + 1],
            'Rapid_Learning_Duration': [rapid_learning_end - exploration_end],
            'Fine_Tuning_Duration': [fine_tuning_end - rapid_learning_end],
            'Mastery_Duration': [len(player_data) - fine_tuning_end - 1],
            'Exploration_Variance': [exploration_var],
            'Rapid_Learning_Slope': [rapid_learning_slope],
            'Fine_Tuning_Slope': [fine_tuning_slope],
            'Mastery_Variance': [mastery_var]
        })], ignore_index=True)
    except Exception as e:
        print(f"Error analyzing phases for Player {player_id}, {'Feedback' if
feedback_condition else 'No Feedback'}: {e}")

    if not phase_results.empty:
        # Save results
        phase_results.to_csv("divedeeperlearning/reports/learning_phases_results.csv", in-
dex=False)

        # Create visualizations
        self._create_phase_visualizations(phase_results)

        # Statistical comparison
        self._compare_learning_phases(phase_results)

    return phase_results

def _create_phase_visualizations(self, phase_results):
    """Create visualizations for learning phases analysis"""
    # 1. Phase duration comparison
    plt.figure(figsize=(12, 8))
    phase_names = ['Exploration', 'Rapid Learning', 'Fine Tuning', 'Mastery']
    phase_cols = ['Exploration_Duration', 'Rapid_Learning_Duration', 'Fine_Tuning_Duration',
'Mastery_Duration']

    # Calculate mean phase durations by condition
    phase_means = phase_results.groupby('Condition')[phase_cols].mean().reset_index()

    # Plot stacked bars
    prev_heights = np.zeros(len(phase_means))
    colors = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728']

    for i, (col, name, color) in enumerate(zip(phase_cols, phase_names, colors)):

```

```

plt.bar(phase_means['Condition'], phase_means[col], bottom=prev_heights,
        label=name, color=color, alpha=0.7)
prev_heights += phase_means[col].values

plt.title('Learning Phase Durations by Condition')
plt.xlabel('Condition')
plt.ylabel('Average Number of Attempts')
plt.legend(loc='upper right')
plt.savefig("diveeep/learning_phases/phase_durations.svg", format='svg')
plt.close()

# 2. Phase characteristics
# For each phase, compare its key characteristic between conditions
phase_metrics = [
    ('Exploration_Variance', 'Exploration Phase: Performance Variance'),
    ('Rapid_Learning_Slope', 'Rapid Learning Phase: Learning Rate'),
    ('Fine_Tuning_Slope', 'Fine Tuning Phase: Learning Rate'),
    ('Mastery_Variance', 'Mastery Phase: Performance Variance')
]

for metric, title in phase_metrics:
    plt.figure(figsize=(10, 6))
    sns.boxplot(x='Condition', y=metric, data=phase_results)
    sns.stripplot(x='Condition', y=metric, data=phase_results, color='black', alpha=0.5,
jitter=True)
    plt.title(title)
    plt.savefig(f"diveeep/learning_phases/{metric}.svg", format='svg')
    plt.close()

def _compare_learning_phases(self, phase_results):
    """Perform statistical comparison for learning phases analysis"""
    # Create a summary by condition
    summary = phase_results.groupby('Condition').agg({
        'Exploration_Duration': ['mean', 'std'],
        'Rapid_Learning_Duration': ['mean', 'std'],
        'Fine_Tuning_Duration': ['mean', 'std'],
        'Mastery_Duration': ['mean', 'std'],
        'Exploration_Variance': ['mean', 'std'],
        'Rapid_Learning_Slope': ['mean', 'std'],
        'Fine_Tuning_Slope': ['mean', 'std'],
        'Mastery_Variance': ['mean', 'std']
    })

    # Flatten multi-level column names

```

```

summary.columns = ['_'.join(col).strip() for col in summary.columns.values]
summary = summary.reset_index()

# Save summary to CSV
summary.to_csv("divedeeperelearning/reports/learning_phases_summary.csv", index=False)

# Paired t-tests for key metrics
comparison_metrics = [
    ('Exploration_Duration', 'duration of exploration phase'),
    ('Rapid_Learning_Duration', 'duration of rapid learning phase'),
    ('Rapid_Learning_Slope', 'learning rate during rapid learning phase'),
    ('Mastery_Variance', 'performance consistency during mastery phase')
]

# Create pivot tables and perform t-tests
with open("divedeeperelearning/reports/learning_phases_t_tests.txt", 'w') as f:
    f.write("Statistical Comparisons for Learning Phases\n")
    f.write("=====\n\n")

    for metric, description in comparison_metrics:
        f.write(f"Comparison of {description}:\n")

        # Create pivot table for paired comparison
        pivot = phase_results.pivot(index='Player', columns='Condition', values=metric)

        # Filter out rows with missing values
        paired_data = pivot.dropna(subset=['Feedback', 'No Feedback'])

        if len(paired_data) >= 3:
            # Perform paired t-test
            t_stat, p_value = stats.ttest_rel(paired_data['Feedback'], paired_data['No
Feedback'])

            f.write(f"    t-statistic: {t_stat:.4f}\n")
            f.write(f"    p-value: {p_value:.4f}\n")
            f.write(f"    Mean difference (Feedback - No Feedback): {paired_data['Feed-
back'].mean() - paired_data['No Feedback'].mean():.4f}\n")
            f.write(f"    Feedback mean: {paired_data['Feedback'].mean():.4f}, SD:
{paired_data['Feedback'].std():.4f}\n")

            try:
                print("\n" + "="*70)
                print("LEARNING ANALYSIS FOR DDTIME BEHAVIOR")
                print("="*70)

            # Load data
            data = load_data()

```

```

# Create analysis object
analysis = LearningAnalysis(data)

# Run all analyses
print("\n" + "="*70)
print("RUNNING ANALYSES")
print("="*70)

# Micro learning analysis
analysis.analyze_micro_learning()
print("✓ Micro learning analysis completed")

# Endgame learning analysis
analysis.analyze_endgame_learning()
print("✓ Endgame learning analysis completed")

# Consistency development analysis
analysis.analyze_consistency_development()
print("✓ Consistency development analysis completed")

# Generate overall report
analysis.generate_overall_report()
print("✓ Overall report generated")

print("\n" + "="*70)
print("ANALYSIS COMPLETED SUCCESSFULLY")
print("="*70)
print("Results saved in 'analysis_output' directory:")
print("  - visualizations/micro_learning: Learning rate analysis")
print("  - visualizations/endgame_learning: Initial vs final performance")
print("  - visualizations/consistency: Plateau detection")
print("  - visualizations/aggregate: Aggregate visualizations")
print("  - reports: All statistical analyses and reports")
print("\nSee 'analysis_output/reports/combined_report.txt' for complete findings")

except FileNotFoundError as e:
    print(f"ERROR: Required data file not found: {e}")
    print("Please ensure 'ForTimeSection.csv' is in the current directory")

except Exception as e:
    print(f"ERROR: An unexpected error occurred: {e}")
    import traceback
    traceback.print_exc()

```

- **For Metrics and feedback analysis Part 1:**

```
#!/usr/bin/env python
# -*- coding: utf-8 -*-

"""
Improvement Analysis Script

This script analyzes what percentage of participants show improvement in measures C1-C7
when feedback is provided (True condition) compared to no feedback (False condition).

Note:
- For measures C1, C2, C3, C4, C5, C7: Lower values indicate better performance
- For measure C6: Higher values indicate better performance
"""

import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set up output directory
OUTPUT_DIR = "statistical_analysis"
os.makedirs(OUTPUT_DIR, exist_ok=True)

# Configure plotting style
sns.set(style="whitegrid", font_scale=1.2)
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['savefig.dpi'] = 300

def load_data(filepath):
    """
    Load and clean the CSV data file.

    Args:
        filepath (str): Path to the CSV file

    Returns:
        pd.DataFrame: Cleaned dataframe
    """
    # Load data
    print(f"Loading data from: {filepath}")
    df = pd.read_csv(filepath)

    # Rename columns to easier names
```

```

df = df.rename(columns={
    "IsFeedBackMood On?": "Feedback",
    "C1_": "C1",
    "C2_": "C2",
    "C3_": "C3",
    "C4_": "C4",
    "C5_": "C5",
    "C6_": "C6",
    "C7_": "C7"
})

# Convert Player to string to ensure proper grouping
df['Player'] = df['Player'].astype(str)

# Convert feedback to boolean
df['Feedback'] = df['Feedback'].astype(bool)

# Convert C values to numeric if they're not already
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
for c_col in c_columns:
    if df[c_col].dtype == 'object':
        # Try to convert string values to numeric
        df[c_col] = pd.to_numeric(df[c_col], errors='coerce')

# Keep participants with data in both conditions
participant_condition_counts = df.groupby(['Player', 'Feedback']).size().un-
stack(fill_value=0)
participants_both_conditions = participant_condition_counts[
    (participant_condition_counts[True] > 0) &
    (participant_condition_counts[False] > 0)
].index.tolist()

# Filter for participants with data in both conditions
df_filtered = df[df['Player'].isin(participants_both_conditions)].copy()
print(f"Number of participants with data in both conditions: {len(participants_both_condi-
tions)}")
print(f"Filtered dataset shape: {df_filtered.shape}")

return df_filtered

def calculate_improvement_percentages(df):
    """
    Calculate the percentage of participants who show improvement
    in each measure when feedback is provided.

    Args:

```

```

df (pd.DataFrame): Filtered dataframe

Returns:
    pd.DataFrame: Results with improvement percentages
    """
    # Define improvement direction for each measure
    # For C1-C5, C7: Lower = better (True < False is improvement)
    # For C6: Higher = better (True > False is improvement)
    improvement_direction = {
        'C1': 'decrease',
        'C2': 'decrease',
        'C3': 'decrease',
        'C4': 'decrease',
        'C5': 'decrease',
        'C6': 'increase',
        'C7': 'decrease'
    }

    # Calculate participant means for each condition
    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    participant_means = df.groupby(['Player', 'Feedback'])[c_columns].mean().reset_index()

    # Reshape to wide format for easier comparison
    wide_format = participant_means.pivot(index='Player',
                                          columns='Feedback',
                                          values=c_columns)

    # Count improvements for each measure
    improvement_counts = {}
    participant_improvements = {}

    for measure in c_columns:
        # Check for improvement based on the correct direction
        if improvement_direction[measure] == 'decrease':
            # For measures where lower is better
            is_improved = wide_format[(measure, True)] < wide_format[(measure, False)]
        else:
            # For measures where higher is better (C6)
            is_improved = wide_format[(measure, True)] > wide_format[(measure, False)]

        # Store individual participant improvement data
        participant_improvements[measure] = is_improved

    # Count improvements
    improvement_count = sum(is_improved)
    total_participants = len(is_improved)

```



```

improvement_percentage = (improvement_count / total_participants) * 100

improvement_counts[measure] = {
    'improved_count': improvement_count,
    'total_participants': total_participants,
    'improvement_percentage': improvement_percentage,
    'direction': improvement_direction[measure]
}

# Create a summary dataframe
results = pd.DataFrame(improvement_counts).T
results = results.reset_index().rename(columns={'index': 'Measure'})

# Calculate average performance change
performance_change = {}

for measure in c_columns:
    true_values = wide_format[(measure, True)]
    false_values = wide_format[(measure, False)]

    # Calculate percent change based on direction
    if improvement_direction[measure] == 'decrease':
        # For measures where decrease is improvement
        # Positive percent change means performance got worse
        # Negative percent change means performance improved
        percent_change = ((true_values - false_values) / false_values) * 100
        # Reverse sign to make improvement positive
        percent_change = -percent_change
    else:
        # For measures where increase is improvement (C6)
        # Positive percent change means performance improved
        # Negative percent change means performance got worse
        percent_change = ((true_values - false_values) / false_values) * 100

    # Store average percent change
    performance_change[measure] = {
        'mean_percent_change': percent_change.mean(),
        'median_percent_change': percent_change.median(),
        'min_percent_change': percent_change.min(),
        'max_percent_change': percent_change.max(),
        'std_percent_change': percent_change.std()
    }

# Create performance change dataframe
change_df = pd.DataFrame(performance_change).T
change_df = change_df.reset_index().rename(columns={'index': 'Measure'})

```

```

    return results, change_df, wide_format, participant_improvements

def visualize_improvements(results, change_df, wide_format, participant_improvements):
    """
    Create visualizations of the improvement percentages and performance changes.

    Args:
        results (pd.DataFrame): Improvement percentage results
        change_df (pd.DataFrame): Performance change results
        wide_format (pd.DataFrame): Wide format dataframe with participant means
        participant_improvements (dict): Dictionary with participant improvement data
    """
    # 1. Bar chart of improvement percentages
    plt.figure(figsize=(12, 6))
    ax = sns.barplot(x='Measure', y='improvement_percentage', data=results,
                    palette='viridis')

    # Add percentage labels on bars
    for i, row in results.iterrows():
        ax.text(i, row['improvement_percentage'] + 1,
                f"{row['improvement_percentage']:.1f}%",
                ha='center', va='bottom', color='black', fontsize=10)

    # Add reference line at 50%
    plt.axhline(y=50, color='red', linestyle='--', alpha=0.7)
    plt.text(6.5, 51, '50% (No Effect)', color='red', fontsize=10)

    plt.title('Percentage of Participants Showing Improvement with Feedback', fontsize=16)
    plt.xlabel('Measure', fontsize=14)
    plt.ylabel('Percentage of Participants (%)', fontsize=14)
    plt.ylim(0, 100)

    # Save figure
    plt_path = os.path.join(OUTPUT_DIR, "improvement_percentages.png")
    plt.tight_layout()
    plt.savefig(plt_path, dpi=300)
    print(f"Improvement percentages chart saved to: {plt_path}")
    plt.close()

    # 2. Bar chart of mean performance changes
    plt.figure(figsize=(12, 6))
    ax = sns.barplot(x='Measure', y='mean_percent_change', data=change_df,
                    palette='coolwarm')

    # Add percentage labels on bars

```

```

for i, row in change_df.iterrows():
    value = row['mean_percent_change']
    color = 'black'
    if value < 0:
        va = 'top'
        y_offset = -1
    else:
        va = 'bottom'
        y_offset = 1

    ax.text(i, value + y_offset,
            f"{value:.1f}%",
            ha='center', va=va, color=color, fontsize=10)

# Add reference line at 0%
plt.axhline(y=0, color='black', linestyle='-', alpha=0.5)

plt.title('Mean Performance Change with Feedback (Positive = Improvement)', fontsize=16)
plt.xlabel('Measure', fontsize=14)
plt.ylabel('Mean Performance Change (%)', fontsize=14)

# Save figure
plt_path = os.path.join(OUTPUT_DIR, "performance_change.png")
plt.tight_layout()
plt.savefig(plt_path, dpi=300)
print(f"Performance change chart saved to: {plt_path}")
plt.close()

# 3. Heatmap of participant improvements
plt.figure(figsize=(12, 10))

# Create a matrix of improvement (1 = improved, 0 = not improved)
improvement_matrix = pd.DataFrame({
    measure: participant_improvements[measure]
    for measure in ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
}).astype(int)

# Add a total column
improvement_matrix['Total_Improved'] = improvement_matrix.sum(axis=1)

# Sort by total number of improvements
improvement_matrix = improvement_matrix.sort_values('Total_Improved', ascending=False)

# Create heatmap
ax = sns.heatmap(improvement_matrix.drop(columns=['Total_Improved']),
                 cmap=['white', 'green'], cbar=False, linewidths=.5)

```

```

# Add total count in text
for i, (idx, row) in enumerate(improvement_matrix.iterrows()):
    plt.text(7.5, i + 0.5, f"{int(row['Total_Improved'])}/7",
             ha='center', va='center', fontsize=9)

plt.title('Improvement Pattern Across Participants (Green = Improved)', fontsize=16)
plt.xlabel('Measure', fontsize=14)
plt.ylabel('Participant', fontsize=14)

# Save figure
plt_path = os.path.join(OUTPUT_DIR, "participant_improvement_heatmap.png")
plt.tight_layout()
plt.savefig(plt_path, dpi=300)
print(f"Participant improvement heatmap saved to: {plt_path}")
plt.close()

def create_summary_report(results, change_df):
    """
    Create a summary report of the improvement analysis.

    Args:
        results (pd.DataFrame): Improvement percentage results
        change_df (pd.DataFrame): Performance change results
    """
    report_path = os.path.join(OUTPUT_DIR, "improvement_analysis_report.txt")

    try:
        with open(report_path, 'w') as f:
            f.write("=====\n")
            f.write("                FEEDBACK IMPROVEMENT ANALYSIS SUMMARY                \n")
            f.write("=====\n\n")

            f.write("ANALYSIS OVERVIEW:\n")
            f.write("This report summarizes the percentage of participants who showed\n")
            f.write("improvement in measures C1-C7 when feedback was provided (True)\n")
            f.write("compared to when it was not (False).\n\n")

            f.write("PERFORMANCE DIRECTION:\n")
            f.write("- For C1, C2, C3, C4, C5, C7: Lower values indicate better performance\n")
            f.write("- For C6: Higher values indicate better performance\n\n")

            f.write("IMPROVEMENT PERCENTAGES:\n")
            f.write("-----\n")
            f.write("Measure | % Improved | Count | Total | Direction\n")
            f.write("-----\n")

```

```

for _, row in results.iterrows():
    direction = "Lower is better" if row['direction'] == 'decrease' else "Higher is
better"

    f.write(f"{row['Measure']} | {row['improvement_percentage']:.1f}% | "
           f"{int(row['improved_count'])} | {int(row['total_participants'])} | {di-
rection}\n")

f.write("-----\n\n")

f.write("PERFORMANCE CHANGE STATISTICS:\n")
f.write("-----\n")
f.write("Measure | Mean Change (%) | Median Change (%) | Min (%) | Max (%) \n")
f.write("-----\n")

for _, row in change_df.iterrows():
    f.write(f"{row['Measure']} | {row['mean_percent_change']:.2f}% | "
           f"{row['median_percent_change']:.2f}% | {row['min_percent_change']:.2f}% | "
           f"{row['max_percent_change']:.2f}%\n")

f.write("-----\n\n")

# Calculate overall improvement
overall_improvement = results['improvement_percentage'].mean()
above_chance = overall_improvement > 50

f.write("OVERALL RESULTS:\n")
f.write(f"Average improvement percentage across all measures: {overall_improve-
ment:.2f}%\n")

if above_chance:
    f.write("This suggests that, on average, feedback had a positive effect on per-
formance.\n")
else:
    f.write("This suggests that, on average, feedback did not have a positive effect
on performance.\n")

# Count measures with >50% improvement
measures_above_50 = results[results['improvement_percentage'] >
50]['Measure'].tolist()
measures_below_50 = results[results['improvement_percentage'] <=
50]['Measure'].tolist()

if len(measures_above_50) > 0:

```

```

        f.write(f"\nMeasures where majority of participants showed improvement: {'',
'.'.join(measures_above_50)}\n")
    else:
        f.write("\nNo measures showed improvement in the majority of participants.\n")

    if len(measures_below_50) > 0:
        f.write(f"Measures where majority of participants did not show improvement: {'',
'.'.join(measures_below_50)}\n")

    # Calculate average improvement across all participants
    f.write("\nPERFORMANCE IMPROVEMENT SUMMARY:\n")
    avg_positive_change = change_df[change_df['mean_percent_change'] > 0]['mean_per-
cent_change'].mean()
    if not np.isnan(avg_positive_change):
        f.write(f"Average improvement in measures with positive change: {avg_posi-
tive_change:.2f}%\n")

    # Summarize individual participant patterns
    f.write("\nPARTICIPANT PATTERNS:\n")
    f.write("- Some participants may show consistent improvement across multiple
measures\n")
    f.write("- Some participants may show mixed results or consistent decline\n")
    f.write("- See the participant improvement heatmap for detailed patterns\n")

    f.write("\n===== \n")
    f.write("Note: This analysis focuses on individual-level improvement patterns and\n")
    f.write("may differ from group-level statistical significance results.\n")
    f.write("===== \n")

    print(f"Summary report saved to: {report_path}")
except Exception as e:
    print(f"Error generating summary report: {str(e)}")

def main():
    """
    Main function to run the complete improvement analysis pipeline.
    """
    try:
        print("=== Starting Improvement Analysis ===")

        # CSV file path
        csv_file = "compind_noDup_FixedMethod2_SpawnedOnlyCleaned.csv"

        # 1. Load and clean data
        df = load_data(csv_file)

```

```

# 2. Calculate improvement percentages
results, change_df, wide_format, participant_improvements = calculate_improvement_per-
centages(df)

# Print key results to console
print("\nImprovement Percentages:")
for _, row in results.iterrows():
    direction = "Lower is better" if row['direction'] == 'decrease' else "Higher is bet-
ter"

    print(f"{row['Measure']}: {row['improvement_percentage']:.1f}% improved ({direc-
tion})")

# 3. Create visualizations
visualize_improvements(results, change_df, wide_format, participant_improvements)

# 4. Create summary report
create_summary_report(results, change_df)

# 5. Export raw results to CSV
results_path = os.path.join(OUTPUT_DIR, "improvement_percentages.csv")
results.to_csv(results_path, index=False)

change_path = os.path.join(OUTPUT_DIR, "performance_changes.csv")
change_df.to_csv(change_path, index=False)

print("\n=== Improvement Analysis Complete ===")
print(f"All results have been saved to the '{OUTPUT_DIR}' folder")

except Exception as e:
    print(f"Error in improvement analysis: {str(e)}")
    import traceback
    traceback.print_exc()

if __name__ == "__main__":
    main()

```

- **For Metrics and feedback analysis Part 2:**

```

#!/usr/bin/env python
# -*- coding: utf-8 -*-

"""
Feedback Analysis Script

This script analyzes the effect of feedback conditions (True/False) on dependent variables

```

```

(C1_ through C7_) from a within-subjects experimental design.
"""

import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.anova import AnovaRM
import warnings

# Suppress specific warnings
warnings.filterwarnings('ignore', category=UserWarning, module='statsmodels')
warnings.filterwarnings('ignore', category=FutureWarning)

# Set up output directory
OUTPUT_DIR = "statistical_analysis"
os.makedirs(OUTPUT_DIR, exist_ok=True)

# Configure plotting style
sns.set(style="whitegrid", font_scale=1.2)
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['savefig.dpi'] = 300

def load_and_clean_data(filepath):
    """
    Load and clean the CSV data file.

    Args:
        filepath (str): Path to the CSV file

    Returns:
        pd.DataFrame: Cleaned dataframe
    """
    # Load data
    print(f>Loading data from: {filepath}")
    df = pd.read_csv(filepath)

    # Print basic info about the dataset
    print(f>Dataset shape: {df.shape}")
    print(f>Number of participants: {df['Player'].nunique()}")

    # Rename columns to easier names

```



```

df = df.rename(columns={
    "IsFeedBackMood On?": "Feedback",
    "C1_": "C1",
    "C2_": "C2",
    "C3_": "C3",
    "C4_": "C4",
    "C5_": "C5",
    "C6_": "C6",
    "C7_": "C7"
})

# Convert Player to string to ensure proper grouping
df['Player'] = df['Player'].astype(str)

# Convert feedback to boolean
df['Feedback'] = df['Feedback'].astype(bool)

# Check for missing values in key columns
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
missing_values = df[['Player', 'Feedback'] + c_columns].isnull().sum()
print("\nMissing values in key columns:")
print(missing_values)

# Convert C values to numeric if they're not already
for c_col in c_columns:
    if df[c_col].dtype == 'object':
        # Try to convert string values to numeric
        df[c_col] = pd.to_numeric(df[c_col], errors='coerce')
        print(f"Converted {c_col} string values to numeric")

# Check participant counts in both conditions
participant_condition_counts = df.groupby(['Player', 'Feedback']).size().un-
stack(fill_value=0)
print("\nParticipant trials in each condition:")
print(participant_condition_counts.describe())

# Identify participants with data in both conditions
participants_both_conditions = participant_condition_counts[
    (participant_condition_counts[True] > 0) &
    (participant_condition_counts[False] > 0)
].index.tolist()

print(f"\nNumber of participants with data in both conditions: {len(participants_both_condi-
tions)}")

# Filter for participants with data in both conditions

```

```

df_filtered = df[df['Player'].isin(participants_both_conditions)].copy()
print(f"Filtered dataset shape: {df_filtered.shape}")

# Save the clean dataset
clean_data_path = os.path.join(OUTPUT_DIR, "clean_data.csv")
df_filtered.to_csv(clean_data_path, index=False)
print(f"Clean data saved to: {clean_data_path}")

return df_filtered

def compute_descriptive_statistics(df):
    """
    Compute descriptive statistics for each encoded measure by feedback condition.

    Args:
        df (pd.DataFrame): Cleaned dataframe

    Returns:
        pd.DataFrame: Descriptive statistics
    """
    # Group by participant and feedback condition, then get means for each participant
    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    participant_means = df.groupby(['Player', 'Feedback'])[c_columns].mean().reset_index()

    # Compute descriptive statistics for each measure by feedback condition
    desc_stats = participant_means.groupby('Feedback')[c_columns].agg([
        'count', 'mean', 'std', 'min', 'median', 'max'
    ])

    # Save descriptive statistics to CSV
    desc_stats_path = os.path.join(OUTPUT_DIR, "descriptive_statistics.csv")
    desc_stats.to_csv(desc_stats_path)
    print(f"Descriptive statistics saved to: {desc_stats_path}")

    # Print the descriptive statistics
    print("\nDescriptive statistics (participant means by condition):")
    print(desc_stats)

    return desc_stats, participant_means

def run_statistical_tests(df, participant_means):
    """
    Run statistical tests to compare feedback conditions for each encoded measure.

    Args:
        df (pd.DataFrame): Cleaned dataframe

```

```

    participant_means (pd.DataFrame): Participant-level means

Returns:
    dict: Dictionary containing statistical results
"""
results = {}

# 1. Paired t-tests on participant means (simpler approach)
print("\n=== Paired t-tests (using participant means) ===")

# Restructure data for paired tests
try:
    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    wide_format = participant_means.pivot(index='Player',
                                          columns='Feedback',
                                          values=c_columns)

    t_test_results = {}
    for measure in c_columns:
        # Extract True and False conditions for this measure
        try:
            true_values = wide_format[(measure, True)]
            false_values = wide_format[(measure, False)]

            # Check for missing values - remove any NaN pairs
            valid_mask = ~(np.isnan(true_values) | np.isnan(false_values))
            true_filtered = true_values[valid_mask]
            false_filtered = false_values[valid_mask]

            if len(true_filtered) < 2:
                print(f"Warning: Not enough valid paired data for {measure}. Skipping t-
test.")

                t_test_results[measure] = {
                    't_statistic': np.nan,
                    'p_value': np.nan,
                    'cohens_d': np.nan,
                    'mean_diff': np.nan,
                    'significant': False,
                    'error': 'Insufficient data'
                }
                continue

            # Run paired t-test
            t_stat, p_value = stats.ttest_rel(true_filtered, false_filtered)

            # Calculate Cohen's d effect size for paired data

```

```

d = (true_filtered.mean() - false_filtered.mean()) / (
    np.sqrt((true_filtered.std()**2 + false_filtered.std()**2) / 2)
) if len(true_filtered) > 0 else np.nan

t_test_results[measure] = {
    't_statistic': t_stat,
    'p_value': p_value,
    'cohens_d': d,
    'mean_diff': true_filtered.mean() - false_filtered.mean() if len(true_fil-
tered) > 0 else np.nan,
    'significant': p_value < 0.05 if not np.isnan(p_value) else False
}

print(f"\n{measure} - Paired t-test results:")
print(f"  t-statistic: {t_stat:.4f}")
print(f"  p-value: {p_value:.4f}")
print(f"  Cohen's d: {d:.4f}")
print(f"  Mean difference (True - False): {t_test_results[meas-
ure]['mean_diff']:.4f}")
print(f"  Significant (p < 0.05): {'Yes' if t_test_results[measure]['signifi-
cant'] else 'No'}")
print(f"  Valid paired samples: {len(true_filtered)}")

except Exception as e:
    print(f"Error in t-test for {measure}: {str(e)}")
    t_test_results[measure] = {
        't_statistic': np.nan,
        'p_value': np.nan,
        'cohens_d': np.nan,
        'mean_diff': np.nan,
        'significant': False,
        'error': str(e)
    }

results['paired_t_tests'] = t_test_results

except Exception as e:
    print(f"Error in paired t-tests: {str(e)}")
    results['paired_t_tests'] = {
        'C1': {'error': str(e)},
        'C2': {'error': str(e)},
        'C3': {'error': str(e)},
        'C4': {'error': str(e)},
        'C5': {'error': str(e)},
        'C6': {'error': str(e)},
        'C7': {'error': str(e)}
    }

```

```

    }

    # 2. Alternative approach: Wilcoxon signed-rank test (non-parametric alternative to paired t-
    test)

    print("\n=== Wilcoxon Signed-Rank Tests ===")

    wilcoxon_results = {}

    try:
        for measure in ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']:
            try:
                true_values = wide_format[(measure, True)]
                false_values = wide_format[(measure, False)]

                # Check for missing values - remove any NaN pairs
                valid_mask = ~(np.isnan(true_values) | np.isnan(false_values))
                true_filtered = true_values[valid_mask]
                false_filtered = false_values[valid_mask]

                if len(true_filtered) < 6: # Wilcoxon needs at least 6 pairs for reliable re-
sults
                    print(f"Warning: Not enough valid paired data for {measure}. Skipping Wil-
coxon test.")

                    wilcoxon_results[measure] = {
                        'statistic': np.nan,
                        'p_value': np.nan,
                        'significant': False,
                        'error': 'Insufficient data'
                    }

                    continue

                # Run Wilcoxon test
                stat, p_value = stats.wilcoxon(true_filtered, false_filtered)

                wilcoxon_results[measure] = {
                    'statistic': stat,
                    'p_value': p_value,
                    'significant': p_value < 0.05,
                    'mean_diff': true_filtered.mean() - false_filtered.mean()
                }

                print(f"\n{measure} - Wilcoxon signed-rank test results:")
                print(f"  Statistic: {stat:.4f}")
                print(f"  p-value: {p_value:.4f}")
                print(f"  Mean difference (True - False): {wilcoxon_results[meas-
ure]['mean_diff']:.4f}")
                print(f"  Significant (p < 0.05): {'Yes' if p_value < 0.05 else 'No'}")

```

```

        print(f" Valid paired samples: {len(true_filtered)}")

    except Exception as e:
        print(f"Error in Wilcoxon test for {measure}: {str(e)}")
        wilcoxon_results[measure] = {
            'statistic': np.nan,
            'p_value': np.nan,
            'significant': False,
            'error': str(e)
        }

    results['wilcoxon_tests'] = wilcoxon_results

except Exception as e:
    print(f"Error in Wilcoxon tests: {str(e)}")
    results['wilcoxon_tests'] = {
        'C1': {'error': str(e)},
        'C2': {'error': str(e)},
        'C3': {'error': str(e)},
        'C4': {'error': str(e)},
        'C5': {'error': str(e)},
        'C6': {'error': str(e)},
        'C7': {'error': str(e)}
    }

# 3. Simplified approach for repeated measures analysis
print("\n=== Simplified Repeated Measures Analysis ===")

# Prepare a subset of data for analysis
try:
    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    subset_df = df[['Player', 'Feedback'] + c_columns].copy()

    # Ensure data types are correct
    subset_df['Player'] = subset_df['Player'].astype(str)
    subset_df['Feedback'] = subset_df['Feedback'].astype(int) # Convert to 0/1 for modeling

    # Run separate ANOVAs for each measure
    anova_results = {}

    for measure in c_columns:
        try:
            print(f"\nAnalyzing {measure} with repeated measures approach:")

            # For each participant, we need at least one measurement in each condition
            valid_participants = []

```

```

# Get participants with data in both conditions for this measure
participant_counts = df.groupby(['Player', 'Feedback'])[measure].count().un-
stack(fill_value=0)

valid_participants = participant_counts[
    (participant_counts[True] > 0) & (participant_counts[False] > 0)
].index.tolist()

if len(valid_participants) < 3:
    print(f" Warning: Not enough participants with data in both conditions for
{measure}."))

    anova_results[measure] = {
        'f_value': np.nan,
        'p_value': np.nan,
        'significant': False,
        'error': 'Insufficient data'
    }
    continue

# Filter data for valid participants
valid_data = subset_df[subset_df['Player'].isin(valid_participants)].copy()

# Simplified: Use statsmodels ANOVA for repeated measures
# First, aggregate to get a mean value for each participant-condition pair
agg_data = valid_data.groupby(['Player', 'Feedback'])[measure].mean().reset_in-
dex()

# Reshape data to wide format for ANOVA
wide_data = agg_data.pivot(index='Player', columns='Feedback', values=measure)
wide_data.columns = ['Condition_0', 'Condition_1'] # Rename for clarity
wide_data = wide_data.reset_index()

# Check for missing values
if wide_data.isnull().any().any():
    print(f" Warning: Missing values detected in {measure} data. Dropping incom-
plete cases.")

    wide_data = wide_data.dropna()

if len(wide_data) < 3:
    print(f" Error: Not enough complete data for {measure} after handling miss-
ing values.")

    anova_results[measure] = {
        'f_value': np.nan,
        'p_value': np.nan,
        'significant': False,
        'error': 'Insufficient data after cleaning'
    }

```

```

    }
    continue

# Melt the data for ANOVA
melted_data = pd.melt(wide_data,
                      id_vars=['Player'],
                      value_vars=['Condition_0', 'Condition_1'],
                      var_name='Condition',
                      value_name=measure)

try:
    # Run repeated measures ANOVA using statsmodels
    model = sm.stats.anova.AnovaRM(melted_data, measure, 'Player', within=['Con-
dition'])

    result = model.fit()

    # Extract results
    f_value = result.anova_table['F'].values[0]
    p_value = result.anova_table['Pr > F'].values[0]

    anova_results[measure] = {
        'f_value': f_value,
        'p_value': p_value,
        'significant': p_value < 0.05,
        'df': result.anova_table['num DF'].values[0],
        'df_res': result.anova_table['den DF'].values[0]
    }

    print(f"   F({anova_results[measure]['df']:.0f}, {anova_results[meas-
ure]['df_res']:.0f}) = {f_value:.4f}")
    print(f"   p-value: {p_value:.4f}")
    print(f"   Significant effect of Feedback: {'Yes' if p_value < 0.05 else
'No'}")

    # Save detailed ANOVA table
    anova_output_path = os.path.join(OUTPUT_DIR, f"anova_rm_{measure}.csv")
    result.anova_table.to_csv(anova_output_path)
    print(f"   ANOVA table saved to: {anova_output_path}")

except Exception as e:
    print(f"   Error in repeated measures ANOVA for {measure}: {str(e)}")
    # Fallback to simpler paired t-test
    t_stat, p_value = stats.ttest_rel(wide_data['Condition_1'], wide_data['Condi-
tion_0'])

    anova_results[measure] = {

```



```

        'f_value': t_stat**2, # F = t2 for a paired design with 1 df
        'p_value': p_value,
        'significant': p_value < 0.05,
        'note': 'Fallback to paired t-test due to ANOVA error'
    }

    print(f" Fallback to paired t-test: t = {t_stat:.4f}, p = {p_value:.4f}")
    print(f" Significant effect of Feedback: {'Yes' if p_value < 0.05 else
'No'}")

except Exception as e:
    print(f" Error in analysis for {measure}: {str(e)}")
    anova_results[measure] = {
        'f_value': np.nan,
        'p_value': np.nan,
        'significant': False,
        'error': str(e)
    }

results['repeated_measures'] = anova_results

except Exception as e:
    print(f"Error in repeated measures analysis: {str(e)}")
    results['repeated_measures'] = {
        'C1': {'error': str(e)},
        'C2': {'error': str(e)},
        'C3': {'error': str(e)},
        'C4': {'error': str(e)},
        'C5': {'error': str(e)},
        'C6': {'error': str(e)},
        'C7': {'error': str(e)}
    }

# 4. Alternative approach: Linear regression with clustered standard errors
print("\n=== Regression Analysis with Clustered Standard Errors ===")

regression_results = {}
try:
    for measure in ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']:
        try:
            print(f"\nRegression analysis for {measure}:")

            # Set up formula and fit the model
            formula = f"{measure} ~ Feedback"
            model = smf.ols(formula=formula, data=df)

```

```

# Fit with HC3 robust standard errors clustered by Player
result = model.fit(cov_type='cluster', cov_kwds={'groups': df['Player']})

# Extract key results
coef = result.params['Feedback[T.True]']
p_value = result.pvalues['Feedback[T.True]']

regression_results[measure] = {
    'coefficient': coef,
    'p_value': p_value,
    'significant': p_value < 0.05,
    'r_squared': result.rsquared,
    'summary': result.summary()
}

print(f" Feedback coefficient: {coef:.4f}")
print(f" p-value: {p_value:.4f}")
print(f" Significant (p < 0.05): {'Yes' if p_value < 0.05 else 'No'}")
print(f" R-squared: {result.rsquared:.4f}")

# Save detailed regression output
reg_output_path = os.path.join(OUTPUT_DIR, f"regression_{measure}.txt")
with open(reg_output_path, 'w') as f:
    f.write(str(result.summary()))
print(f" Regression results saved to: {reg_output_path}")

except Exception as e:
    print(f" Error in regression analysis for {measure}: {str(e)}")
    regression_results[measure] = {
        'coefficient': np.nan,
        'p_value': np.nan,
        'significant': False,
        'error': str(e)
    }

results['regression'] = regression_results

except Exception as e:
    print(f"Error in regression analysis: {str(e)}")
    results['regression'] = {
        'C1': {'error': str(e)},
        'C2': {'error': str(e)},
        'C3': {'error': str(e)},
        'C4': {'error': str(e)},
        'C5': {'error': str(e)},
        'C6': {'error': str(e)},
    }

```

```

        'C7': {'error': str(e)}
    }

# Save results summary
# Save results summary
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
results_summary = pd.DataFrame({
    'Measure': c_columns,
    'T-test p-value': [results['paired_t_tests'][m]['p_value'] for m in c_columns],
    'T-test significant': [results['paired_t_tests'][m]['significant'] for m in c_columns],
    'Wilcoxon p-value': [results['wilcoxon_tests'][m]['p_value'] if 'p_value' in re-
sults['wilcoxon_tests'][m] else np.nan for m in c_columns],
    'Wilcoxon significant': [results['wilcoxon_tests'][m]['significant'] if 'significant' in
results['wilcoxon_tests'][m] else False for m in c_columns],
    'RM-ANOVA p-value': [results['repeated_measures'][m]['p_value'] if 'p_value' in re-
sults['repeated_measures'][m] else np.nan for m in c_columns],
    'RM-ANOVA significant': [results['repeated_measures'][m]['significant'] if 'significant'
in results['repeated_measures'][m] else False for m in c_columns],
    'Regression p-value': [results['regression'][m]['p_value'] if 'p_value' in results['re-
gression'][m] else np.nan for m in c_columns],
    'Regression significant': [results['regression'][m]['significant'] if 'significant' in
results['regression'][m] else False for m in c_columns]
})

results_path = os.path.join(OUTPUT_DIR, "statistical_results_summary.csv")
results_summary.to_csv(results_path, index=False)
print(f"\nStatistical results summary saved to: {results_path}")

return results

def create_visualizations(df, participant_means):
    """
    Create and save visualizations of the data.

    Args:
        df (pd.DataFrame): Cleaned dataframe
        participant_means (pd.DataFrame): Participant-level means
    """
    # 1. Boxplots for each C measure by feedback condition
    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    for measure in c_columns:
        plt.figure(figsize=(10, 6))

        # Create boxplot
        ax = sns.boxplot(x='Feedback', y=measure, data=participant_means,
                        palette='Set2', width=0.5)

```

```

# Add individual data points
sns.stripplot(x='Feedback', y=measure, data=participant_means,
              jitter=True, dodge=True, marker='o', alpha=0.6, color='black')

# Set plot title and labels
plt.title(f'Distribution of {measure} by Feedback Condition', fontsize=16)
plt.xlabel('Feedback Condition', fontsize=14)
plt.ylabel(measure, fontsize=14)
plt.xticks([0, 1], ['False', 'True'])

# Add statistics annotation
means = participant_means.groupby('Feedback')[measure].mean()
plt.annotate(f'Mean (False): {means[False]:.3f}', xy=(0, 0.95), xycoords='axes fraction',
            fontsize=12)
plt.annotate(f'Mean (True): {means[True]:.3f}', xy=(1, 0.90), xycoords='axes fraction',
            fontsize=12)

# Save figure
plt_path = os.path.join(OUTPUT_DIR, f'{measure}_boxplot.png')
plt.tight_layout()
plt.savefig(plt_path, dpi=300)
print(f'Boxplot for {measure} saved to: {plt_path}')
plt.close()

# 2. Line plots showing within-subject changes (connecting each participant's scores)
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
wide_format = participant_means.pivot(index='Player',
                                      columns='Feedback',
                                      values=c_columns)

for measure in c_columns:
    plt.figure(figsize=(12, 8))

    # Extract data for this measure
    measure_data = wide_format[measure].reset_index()

    # Plot lines connecting each participant's scores
    for _, row in measure_data.iterrows():
        plt.plot([0, 1], [row[False], row[True]], 'o-', alpha=0.3, color='gray')

    # Add mean line
    mean_false = measure_data[False].mean()
    mean_true = measure_data[True].mean()
    plt.plot([0, 1], [mean_false, mean_true], 'o-', linewidth=3, color='blue',
            label='Mean')

```

```

# Customize plot
plt.title(f'Within-subject changes in {measure} by Feedback Condition', fontsize=16)
plt.xticks([0, 1], ['Feedback False', 'Feedback True'])
plt.ylabel(measure, fontsize=14)
plt.grid(True, alpha=0.3)
plt.legend()

# Add mean values annotation
plt.annotate(f"Mean: {mean_false:.3f}", xy=(0, mean_false),
            xytext=(0.1, 0.9), textcoords='axes fraction',
            arrowprops=dict(facecolor='black', shrink=0.05, width=1.5))
plt.annotate(f"Mean: {mean_true:.3f}", xy=(1, mean_true),
            xytext=(0.7, 0.9), textcoords='axes fraction',
            arrowprops=dict(facecolor='black', shrink=0.05, width=1.5))

# Save figure
plt_path = os.path.join(OUTPUT_DIR, f"{measure}_within_subject.png")
plt.tight_layout()
plt.savefig(plt_path, dpi=300)
print(f"Within-subject plot for {measure} saved to: {plt_path}")
plt.close()

# 3. Create a combined visualization showing all seven measures
plt.figure(figsize=(15, 10))

# Setup plot with 7 subplots (one for each measure), in a 3x3 grid
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
fig, axes = plt.subplots(3, 3, figsize=(20, 15), sharey=False)

# Get overall min and max for consistent y-axis
all_values = []
for measure in c_columns:
    all_values.extend(participant_means[measure].tolist())

y_min = min(all_values) - 0.1
y_max = max(all_values) + 0.1

# Flatten the axes array for easy iteration
axes_flat = axes.flatten()

for i, measure in enumerate(c_columns):
    if i < len(axes_flat): # Only use the first 7 subplots
        ax = axes_flat[i]

        # Create boxplot in each subplot
        sns.boxplot(x='Feedback', y=measure, data=participant_means,

```

```

        palette='Set2', width=0.5, ax=ax)

# Add individual data points
sns.stripplot(x='Feedback', y=measure, data=participant_means,
              jitter=True, dodge=True, marker='o', alpha=0.6,
              color='black', ax=ax)

# Set plot title and labels
ax.set_title(f'{measure}', fontsize=14)
ax.set_xlabel('Feedback', fontsize=12)
if i % 3 == 0: # First column gets y-label
    ax.set_ylabel('Value', fontsize=12)
else:
    ax.set_ylabel('')

ax.set_xticklabels(['False', 'True'])

# Try to set consistent y limits
ax.set_ylim(y_min, y_max)

# Hide unused subplots
for i in range(len(c_columns), len(axes_flat)):
    axes_flat[i].set_visible(False)

plt.suptitle('Comparison of C1-C7 Measures by Feedback Condition', fontsize=16)
plt.tight_layout()

# Save the combined figure
comb_plt_path = os.path.join(OUTPUT_DIR, "combined_measures_boxplot.png")
plt.savefig(comb_plt_path, dpi=300)
print(f"Combined visualization saved to: {comb_plt_path}")
plt.close()

# 4. Create a radar chart to compare all measures at once
# Set up the radar chart
plt.figure(figsize=(10, 8))

# Calculate means for each measure and condition
c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
means_by_condition = participant_means.groupby('Feedback')[c_columns].mean().T

# Number of variables
N = len(c_columns)

# Create angle for each measure
angles = [n / float(N) * 2 * np.pi for n in range(N)]

```

```

angles += angles[:1] # Close the loop

# Set up the plot
ax = plt.subplot(111, polar=True)

# Add labels for each measure
plt.xticks(angles[:-1], c_columns, size=12)

# Draw y-axis labels
ax.set_rlabel_position(0)
plt.yticks(color="grey", size=10)

# Plot data for False condition
values_false = means_by_condition[False].tolist()
values_false += values_false[:1] # Close the loop
ax.plot(angles, values_false, 'b-', linewidth=2, label='Feedback False')
ax.fill(angles, values_false, 'b', alpha=0.1)

# Plot data for True condition
values_true = means_by_condition[True].tolist()
values_true += values_true[:1] # Close the loop
ax.plot(angles, values_true, 'r-', linewidth=2, label='Feedback True')
ax.fill(angles, values_true, 'r', alpha=0.1)

# Add legend
plt.legend(loc='upper right', bbox_to_anchor=(0.1, 0.1))

plt.title('Radar Chart of C1-C7 Measures by Feedback Condition', fontsize=15)

# Save the radar chart
radar_plt_path = os.path.join(OUTPUT_DIR, "radar_measures_comparison.png")
plt.tight_layout()
plt.savefig(radar_plt_path, dpi=300)
print(f"Radar chart saved to: {radar_plt_path}")
plt.close()

def create_summary_report(results):
    """
    Create a summary report of the statistical analysis results.

    Args:
        results (dict): Dictionary containing statistical results
    """
    report_path = os.path.join(OUTPUT_DIR, "summary_report.txt")

    try:

```

```

with open(report_path, 'w') as f:
    f.write("=====\n")
    f.write("                STATISTICAL ANALYSIS SUMMARY REPORT                \n")
    f.write("=====\n\n")
    f.write("ANALYSIS OVERVIEW:\n")
    f.write("This report summarizes the statistical analysis of the effect of\n")
    f.write("feedback conditions (True/False) on seven measures (C1-C7)\n")
    f.write("from a within-subjects experimental design.\n\n")
    f.write("STATISTICAL APPROACH:\n")
    f.write("1. Paired t-tests: Comparing participant means between feedback condi-
tions\n")
    f.write("2. Wilcoxon signed-rank tests: Non-parametric alternative to paired t-
tests\n")
    f.write("3. Repeated measures ANOVA: Accounting for within-subject design\n")
    f.write("4. Regression with clustered errors: Controlling for participant as a random
factor\n\n")

    f.write("RESULTS SUMMARY:\n")
    f.write("-----\n")
    f.write("Measure | T-test p-value | Wilcoxon p-value | RM-ANOVA p-value | Signifi-
cant?\n")
    f.write("-----\n")

    c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
    for measure in c_columns:
        t_p = results['paired_t_tests'][measure]['p_value'] if 'p_value' in re-
sults['paired_t_tests'][measure] else np.nan
        w_p = results['wilcoxon_tests'][measure]['p_value'] if 'p_value' in results['wil-
coxon_tests'][measure] else np.nan
        rm_p = results['repeated_measures'][measure]['p_value'] if 'p_value' in re-
sults['repeated_measures'][measure] else np.nan

        # Determine if any test shows significance
        sig = (
            (not np.isnan(t_p) and t_p < 0.05) or
            (not np.isnan(w_p) and w_p < 0.05) or
            (not np.isnan(rm_p) and rm_p < 0.05)
        )
        sig_text = 'Yes' if sig else 'No'

        f.write(f"{measure} | {t_p:.4f} | {w_p:.4f} | {rm_p:.4f} | {sig_text}\n")

    f.write("-----\n\n")
    f.write("INTERPRETATION GUIDE:\n")
    f.write("- p-value < 0.05 indicates a statistically significant difference between
feedback conditions\n")

```



```

        f.write("- The repeated measures ANOVA results are generally more robust as they ac-
count for\n")
        f.write("  within-subject design and repeated observations\n\n")

        # Define significant measures
        c_columns = ['C1', 'C2', 'C3', 'C4', 'C5', 'C6', 'C7']
        sig_measures = []

        for measure in c_columns:
            # Check if any test shows significance for this measure
            t_p = results['paired_t_tests'][measure]['p_value'] if 'p_value' in re-
sults['paired_t_tests'][measure] else np.nan
            w_p = results['wilcoxon_tests'][measure]['p_value'] if 'p_value' in results['wil-
coxon_tests'][measure] else np.nan
            rm_p = results['repeated_measures'][measure]['p_value'] if 'p_value' in re-
sults['repeated_measures'][measure] else np.nan

            if ((not np.isnan(t_p) and t_p < 0.05) or
                (not np.isnan(w_p) and w_p < 0.05) or
                (not np.isnan(rm_p) and rm_p < 0.05)):
                sig_measures.append(measure)

        f.write("CONCLUSION:\n")
        if len(sig_measures) > 0:
            f.write(f"The feedback condition had a significant effect on {',
'.join(sig_measures)}.\n")
            f.write("This suggests that the presence of feedback meaningfully influenced
these measures.\n")
        else:
            f.write("The feedback condition did not have a significant effect on any of the
C1-C7 measures.\n")
            f.write("This suggests that the presence of feedback did not meaningfully influ-
ence these measures.\n")

        print(f"Summary report saved to: {report_path}")
    except Exception as e:
        print(f"Error generating summary report: {str(e)}")

def main():
    """
    Main function to run the complete statistical analysis pipeline.
    """
    print("=== Starting Feedback Analysis ===")

    # CSV file path
    csv_file = "compind_noDup_FixedMethod2_SpawnedOnlyCleaned.csv"

```

```

# 1. Load and clean data
df = load_and_clean_data(csv_file)

# 2. Compute descriptive statistics
desc_stats, participant_means = compute_descriptive_statistics(df)

# 3. Run statistical tests
results = run_statistical_tests(df, participant_means)

# 4. Create visualizations
create_visualizations(df, participant_means)

# 5. Create summary report
try:
    create_summary_report(results)
except Exception as e:
    print(f"Error in summary report generation: {str(e)}")

print("\n=== Feedback Analysis Complete ===")
print(f"All results have been saved to the '{OUTPUT_DIR}' folder")

if __name__ == "__main__":
    main()

```

- **Inter-process communication: TCP Socket**

The author also developed a socket connection⁸ as the method for inter-process communication⁹. This tool provides a set of components to establish a TCP¹⁰ Socket connection between a server and one or more clients. This enables fast and easy connection between multiple software or different processing elements within a single environment. Designers can create customized workflows by connecting multiple environments without writing a single line of code. It is a simple, straight forward system that does not rely on mediated programs for its operation. Additionally, socket connections are widely used across a wide range of platforms and programming languages. Therefore, the server can connect to any client regardless of their generation method. This tool also has the following key features and properties; *Reliable and Capable of Streaming*: The TCP connection is reliable for several reasons; first, there is no chance of losing data or mixed sequencing over sending and receiving as there is with UDP¹¹. In addition, socket connections are capable of streaming data, unlike HTTP connections which work in pairs of requests and responses. *Fast and Efficient*: Components are multi-threaded and thread-safe, which means they are fast and lite on resources. *User-friendly*: Provides a log for error-checking, debugging and maintenance purposes. *Versatile*: The tools can also be used to control industrial robotic arms and IoT devices in real time. The proposed tools also allow the design environment to operate both as a server and a client. This maximizes their compatibility with other environments and enables them to communicate with more than one entity. Multi-client connections can be handled simultaneously.

⁸ Socket programming is a low-level interface that can implement different protocols to offer network services.

⁹ For more details and the source code, refer to author's GitHub in references.

¹⁰ Transmission Control Protocol.

¹¹ User Datagram Protocol.

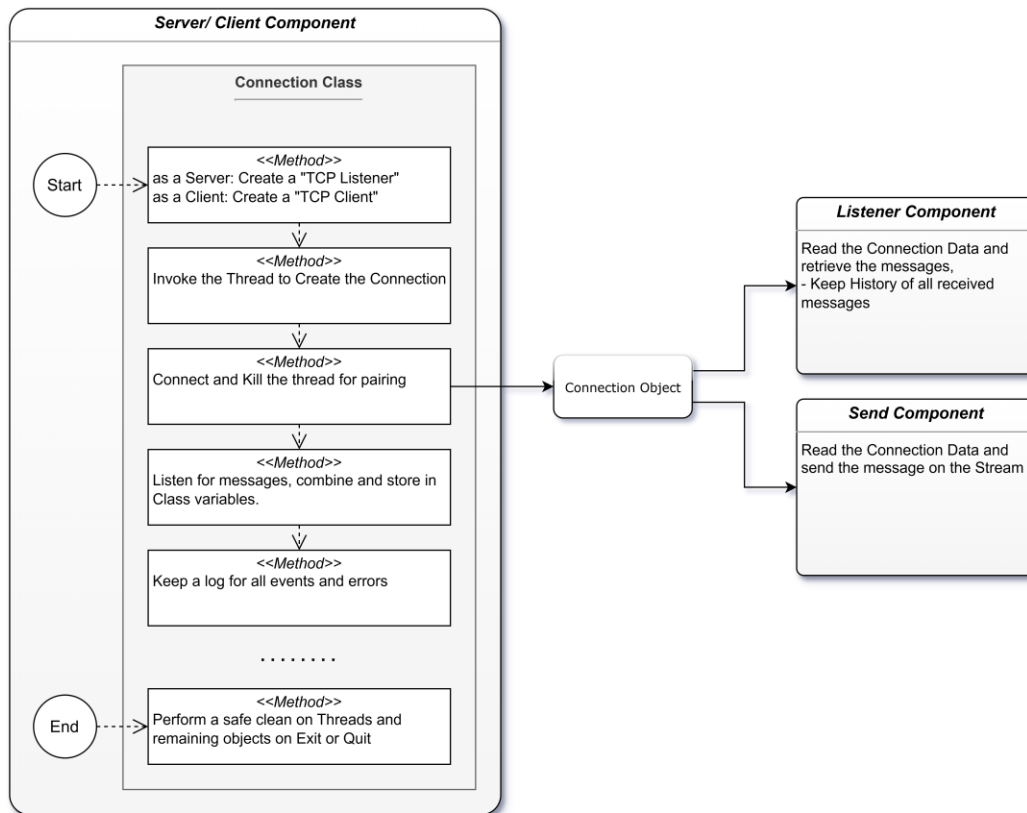


Figure 38: IPC: Socket, An overview of the main components and functions

The main code is written in Ironpython (an implementation of the Python programming language targeting the .NET Framework, python 2.7). It is merely dependent on the host environment (Rhino/Grasshopper®) to manage outputs and inputs. Therefore, the logic is applicable to similar environments. The 'start connection' component contains the main functionality (Figure 38). An object connection class creates, maintains, and stores objects. For reading and writing on the object, it is passed to the 'listener' and 'sender' components.

APPENDIX III: PUBLICATIONS

Different stages of this thesis have also formed three papers, which are:

- Khalilbeigi Khameneh, A., & Nahmad Vazquez, A. (2023, May 23). Game Engines as A Performance-Aware Participatory and Interactive Design Platform: A Prototypical Workflow. *2023 Annual Modeling And Simulation Conference*
- Khalilbeigi Khameneh, A., & Nahmad Vazquez, A. (2024, November). Integrating Game Engines for Data-Driven, Performance-Aware Architectural Design: A Gamified Framework for Non-Expert Decision-Making. *ACADIA 2024*
- Khalilbeigi Khameneh, A., Mostafavi, A., & Nahmad Vazquez, A. (2025). Gamified Informed Decision-Making for Performance-Aware Design by Non-Experts: An exoskeleton design case study. *IASS2025*

And technical contributions in:

- Khalilbeigi Khameneh, Arman. 2022. "Simple TCP Socket Connection: A Simple Connection Created for McNeel's Grasshopper3D." Calgary.
<https://github.com/armankhalilbeigi/SimpleTCPSocketConnection>.